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TITLE 5—ADMINISTRATIVE PERSONNEL

Chapter I—Civil Service Commission

PART 6—EXCEPTIONS FROM THE COMPETITIVE SERVICE

SUBVERSIVE ACTIVITIES CONTROL BOARD

Under authority of § 6.1 (a) of Executive Order 9830, the Commission has approved the exception from the competitive service of certain positions in the Subversive Activities Control Board. Effective upon publication in the FEDERAL REGISTER, a new § 6.153 is added as follows:

§ 6.153 *Subversive Activities Control Board.* (a) One confidential secretary—administrative assistant to the Chairman.

(b) Five assistants to Board Members.

(c) Four confidential secretaries to Board Members.

(R. S. 1753, sec. 2, 22 Stat. 403; 5 U. S. C. 631, 633. E. O. 9830, Feb. 24, 1947, 12 F. R. 1259; 3 CFR, 1947 Supp. E. O. 9973, June 28, 1948, 13 F. R. 3600; 3 CFR, 1948 Supp.)

UNITED STATES CIVIL SERVICE COMMISSION.

[SEAL] HARRY B. MITCHELL, *Chairman.*

[F. R. Doc. 50-10030; Filed, Nov. 9, 1950; 8:48 a. m.]

TITLE 7—AGRICULTURE

Chapter IX—Production and Marketing Administration (Marketing Agreements and Orders), Department of Agriculture

PART 933—ORANGES, GRAPEFRUIT AND TANGERINES GROWN IN FLORIDA

APPROVAL OF BUDGET OF EXPENSES AND FIXING OF RATE OF ASSESSMENT FOR 1950–1951 FISCAL PERIOD

On October 17, 1950, notice of proposed rule making was published in the FEDERAL REGISTER (15 F. R. 6934) regarding the budget of expenses and the fixing of the rate of assessment for the 1950–1951 fiscal period under Marketing Agreement No. 84, as amended, and Order No. 33, as amended (7 CFR Part 933), regulating the handling of oranges, grapefruit, and tangerines grown in the State of Florida. This regulatory program is effective pursuant to the Agricultural Marketing Agreement Act of 1937, as amended. After consideration of all relevant matters presented, including the proposals set forth in the aforesaid notice, which were submitted by the Growers Administrative Committee (established pursuant to the amended marketing agreement and order), it is hereby found and determined that:

§ 933.204 *Budget of expenses and rate of assessment for the 1950–1951 fiscal period.* (a) The expenses necessary to be incurred by the Growers Administrative Committee established pursuant to the provisions of the aforesaid amended marketing agreement and order, for the maintenance and functioning, during the fiscal period beginning August 1, 1950, and ending July 31, 1951, both dates inclusive, of the Growers Administrative Committee and the Shippers Advisory Committee, established under the aforesaid amended marketing agreement and order, will amount to $128,000 and the rate of assessment to be paid by each handler shall be four mills ($0.004) per standard packed box of fruit shipped by such handler during the said fiscal period; and such rate of assessment is hereby approved as each handler's pro rata share of the aforesaid expenses.

(b) As used in this section, the terms "standard packed box," "handler," "shipped," and "fruit" shall have the same meaning as is given to each such term in said amended marketing agreement and order.

(Sec. 5, 49 Stat. 753, as amended; 7 U. S. C. and Sup., 608c)

Done at Washington, D. C., this 7th day of November 1950, to be effective 30 days after the date of publication hereof in the FEDERAL REGISTER.

[SEAL] CHARLES F. BRANNAN, *Secretary of Agriculture.*

[F. R. Doc. 50-10059; Filed, Nov. 9, 1950; 8:49 a. m.]

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DETERMINATION RELATIVE TO EXPENSES AND FIXING OF RATE OF ASSESSMENT FOR 1950–1951 FISCAL PERIOD

On October 17, 1950, notice of proposed rule making was published in the FEDERAL REGISTER (15 F. R. 6934) regarding the expenses and rate of assessment for the 1950–1951 fiscal period under Marketing Agreement No. 96, as amended, and Order No. 55, as amended (7 CFR Part 955) regulating the handling of grapefruit grown in the State of Arizona; in Imperial County, California; and in that part of Riverside County, California, situated south and east of the San Gorgonio Pass. This regulatory program is effective pursuant to the Agricultural Marketing Agreement Act of 1937, as amended. After consideration of all relevant matters presented, including the proposals set forth in the aforesaid notice which were submitted by the Administrative Committee (established pursuant to the amended marketing agreement and order), it is hereby found and determined that:

§ 955.204 *Expenses and rate of assessment for the 1950–1951 fiscal period.* (a) The expenses necessary to be incurred by the Administrative Committee, established pursuant to the provisions of the aforesaid amended marketing agreement and order, for its maintenance and functioning during the fiscal period beginning August 1, 1950, will amount to $18,750; and the rate of assessment to be paid by each handler who first ships grapefruit shall be one and one-quarter cents ($0.0125) per standard box of fruit shipped by such handler as the first handler thereof during the said fiscal period. Such rate of assessment is hereby fixed as each such handler's pro rata share of the aforesaid expenses.

(b) As used in this section, "handler," "ship," "fruit," "fiscal period," and "standard box" shall each have the same meaning as is given to each such term in said amended marketing agreement and order.

(Sec. 5, 49 Stat. 753, as amended; 7 U. S. C. and Sup., 608c)

Done at Washington, D. C., this 7th day of November 1950, to be effective 30 days after the date of publication hereof in the FEDERAL REGISTER.

[SEAL] CHARLES F. BRANNAN, *Secretary of Agriculture.*

[F. R. Doc. 50-10058; Filed, Nov. 9, 1950; 8:49 a. m.]

PART 987—IRISH POTATOES IN MAINE

APPROVAL OF BUDGET OF EXPENSES AND FIXING RATE OF ASSESSMENT

Notice of proposed rule making regarding rules and regulations relative to a proposed budget and rate of assessment, to be made effective under Marketing Agreement No. 108 and Order No. 87 (7 CFR 987.1 et seq.), regulating the handling of Irish potatoes grown in the State of Maine was published in the FEDERAL REGISTER (15 F. R. 6672). This regulatory program is effective under the Agricultural Marketing Agreement Act of 1937, as amended (48 Stat. 31, as amended; 7 U. S. C. 601 et seq.; 61 Stat. 202, 707; 62 Stat. 1247; 63 Stat. 1051). After consideration of all relevant matters presented, including the rules and regulations set forth in the aforesaid notice, which rules and regulations were adopted and submitted for approval by the State of Maine Potato Committee (established pursuant to said agreement and order), the following rules and regulations are hereby approved.

§ 987.203 *Budget of expenses and rate of assessment.* (a) The expenses necessary to be incurred by the State of Maine Potato Committee, established pursuant to Marketing Agreement No. 108 and Order No. 87, to enable such committee to perform its functions pursuant to the provisions of the aforesaid marketing agreement and order and regulations duly issued thereunder, during the fiscal period ending June 30, 1951, will amount to $40,000.00.

(b) The rate of assessment, to be paid by each handler who first ships potatoes during such fiscal year, shall be seventy cents per railroad car; sixty cents per truckload of a net weight of 25,000 pounds or more; forty-five cents per truckload of at least 10,000 pounds but less than 25,000 pounds net weight; fifteen cents per truckload of less than 10,000 pounds net weight; and $0.00125 per hundredweight for other units, whether by floor lot, bin, or warehouse.

(c) Terms used in this section shall have the same meaning as when used in Marketing Agreement No. 108 and Order No. 87.

(Sec. 5, 49 Stat. 753, as amended; 7 U. S. C. and Sup., 608c)

Done at Washington, D. C., this 7th day of November 1950, to become effective 30 days after publication hereof in the FEDERAL REGISTER.

[SEAL] CHARLES F. BRANNAN, *Secretary of Agriculture.*

[F. R. Doc. 50-10096; Filed, Nov. 9, 1950; 8:53 a. m.]

TITLE 14—CIVIL AVIATION

Chapter I—Civil Aeronautics Board

Subchapter A—Civil Air Regulations

[Supp. 7, Amdt. 54]

PART 60—AIR TRAFFIC RULES

DANGER AREA ALTERATIONS

The danger area alterations appearing hereinafter have been coordinated with the civil operators involved, the Army, the Navy, and the Air Force, through the Air Coordinating Committee, Airspace Subcommittee, and are adopted when indicated in order to promote safety of the flying public. Compliance with the notice, procedures, and effective date provisions of section 4 of the Administrative Procedure Act would be impracticable and contrary to the public interest, and therefore is not required. Title 14, § 60.13–1 is amended as follows:

1. The Guadalupe Mountains, New Mexico, area, published on June 23, 1949, in 14 F. R. 3393, and amended on March 17, 1950, in 15 F. R. 1510, is revised to read:

Name and location (chart)	Description by geographical coordinates	Designated altitudes	Time of designation	Using agency
GUADALUPE MOUNTAINS (Roswell chart).	Beginning at lat. 32°18'00" N, long. 105°03'00" W; due S to lat. 32°09'45" N; due W to long. 105°23'00" W; NE to lat. 32°18'00" N, long. 105°20'00" W; due E to lat. 32°18'00" N, long. 105° 08'00" W, point of beginning.	Surface to 50,000 feet.	Daylight hours, 7 days a week.	8th Air Force, Biggs AFB, El Paso, Tex.

2. The Matagorda Island, Texas, area, published on April 21, 1949, in 14 F. R. 1913, is revised to read:

Name and location (chart)	Description by geographical coordinates	Designated altitudes	Time of designation	Using agency
MATAGORDA ISLAND (San Antonio Chart).	Area I: Beginning at lat. 28°51'30" N, long. 96°26'30" W; SE to a point 3 nautical miles from the shoreline at lat. 28°49'10" N, long. 96°24'30" W; southwesterly paralleling the shoreline of Matagorda Peninsula at a distance of 3 nautical miles to lat. 28°20'00" N, long. 96°14'40" W; NW to lat. 28°27'30" N, long. 96°16'30" W; NE along the shoreline of the Matagorda Peninsula to lat. 28°51'30" N, long. 96°26'30" W, point of beginning; excluding that portion which lies within a 3-mile radius of Matagorda Club Airport at lat. 28°27'15" N, long. 96°18'00" W.	Surface to 50,000 feet.	Daylight hours, 7 days a week.	8th Air Force, Carswell AFB, Fort Worth, Tex.
	Area II: Beginning at lat. 28°20'00" N, long. 96°28'30" W; SE to lat. 28°18'45" N, long. 96°27'30" W; SSE to lat. 28°16'15" N, long. 96°26'50" W; SW to a point 3 nautical miles from the shoreline at lat. 28°15'40" N, long. 96°27'15" W; southwesterly paralleling the shoreline of Matagorda Island at a distance of 3 nautical miles to lat. 28°12'30" N, long. 96°33'20" W; due W to long. 96°34'20" W; NW to lat. 28°15'30" N, long. 96°37'00" W; NE to lat. 28°20'00" N. long. 96°28'30" W, point of beginning.	do	do	Do.

3. The Water Island (St. Thomas), Virgin Islands, area, published on April 8, 1950, in 15 F. R. 2024, is deleted.

4. The West End (St. Thomas), Virgin Islands, area, published on April 8, 1950, in 15 F. R. 2024, is deleted.

(Sec. 205, 52 Stat. 984, as amended; 49 U. S. C. 425. Interpret or apply sec. 601, 52 Stat. 1007, as amended; 49 U. S. C. and Sup., 551)

This amendment shall become effective on November 10, 1950.

[SEAL] LEONARD W. JURDEN,
Acting Administrator of Civil Aeronautics.

[F. R. Doc. 50–10051; Filed, Nov. 9, 1950; 8:48 a. m.]

[Supp. 7, Amdt. 55]

PART 60—AIR TRAFFIC RULES

DANGER AREA ALTERATIONS

The danger area alterations appearing hereinafter have been coordinated with the civil operators involved, the Army, the Navy, and the Air Force, through the Air Coordinating Committee, Airspace Subcommittee, and are adopted when indicated in order to promote safety of the flying public. Compliance with the notice, procedures, and effective date provisions of section 4 of the Administrative Procedure Act would be impracticable and contrary to the public interest, and therefore is not required. Title 14, § 60.13–1 is amended as follows:

1. The Fort Ord, California, area, published on April 21, 1949, in 14 F. R. 1913, is amended by changing the "Designated Altitudes" column to read: "Surface to 3,000 feet."

2. A Fort Sill, Oklahoma, temporary area is added to read:

Name and location (chart)	Description by geographical coordinates	Designated altitudes	Time of designation	Using agency
FORT SILL (Oklahoma City Chart).	N boundary: lat. 34°47'00" N; S boundary: lat. 34°38'00" N; E boundary: long. 98°21'00" W; W boundary: long. 98°26'00" W.	Surface to unlimited.	Continuous, Nov. 13, 1950, through Nov. 18, 1950.	Artillery School, Fort Sill, Okla.

(Sec. 205, 52 Stat. 984, as amended; 49 U. S. C. 425. Interprets or applies sec. 601, 52 Stat. 1007, as amended; 49 U. S. C. and Sup., 551)

This amendment shall become effective on November 13, 1950.

[SEAL] LEONARD W. JURDEN,
Acting Administrator of Civil Aeronautics.

[F. R. Doc. 50–10052; Filed, Nov. 9, 1950; 8:48 a. m.]

[Supp. 3, Amdt. 4]

PART 60—AIR TRAFFIC RULES

STANDARD INSTRUMENT APPROACH PROCEDURES

Under section 205 (a) of the Civil Aeronautics Act of 1938, as amended, the Administrator of Civil Aeronautics is authorized to make and amend such rules, regulations, and procedure as are necessary to carry out the provisions of, and to perform and exercise his powers and duties under, the act. Under section 601 of the Civil Aeronautics Act of 1938, as amended, the Civil Aeronautics Board is empowered to delegate to the Administrator of Civil Aeronautics the authority to prescribe rules, regulations, and standards which promote safety of flight in air commerce. Under §§ 42.55 (b), 42.56, and 60.46 of the Civil Air Regulations, the Civil Aeronautics Board has authorized the Administrator of Civil Aeronautics to prescribe standard instrument approach procedures, including ceiling and visibility minimums.

Acting pursuant to the foregoing statutes and regulations, standard instrument approach procedures were prescribed. Those procedures are hereby amended. This amendment is made effective without delay, in order to promote safety of the flying public. Compliance with the notice, procedures, and effective date provisions of section 4 of the Administrative Procedure Act would be impracticable and contrary to the public interest, and therefore is not required.

1. The low frequency range procedures prescribed in § 60.46–4 are amended to read:

Low Frequency Range Procedures

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
ABERDEEN, S. DAK. Aberdeen Arpt. 329 kc; ABR SBRAZ-DTV	N—2,800′ (Jamestown Range) SE—3,000′ (Watertown Range) S—2,500′ (Huron Range) NW—3,200′ (S crs Bismarck)	None	S	10 mi.—2,400′ E side S crs 15 mi.—2,400′ E side S crs 20 mi.—2,400′ E side S crs 25 mi.—2,400′ E side S crs	1,900	341	2.6	1,300	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,800′ on N crs within 25 mi. *Runway No. 35.
ABILENE, TEX. Abilene Mun. Arpt. No. 2. 379 kc; ABI SBRAZ-DTV	N—Min. en route altitude N—2,500′ (Phantom Hill FM) (final) E—3,500′ (Fort Worth Range) S—3,800′ (E crs San Angelo) W—4,000′ (Big Spring Range)	None	N	10 mi.—3,000′ W side N crs 15 mi.—3,000′ W side N crs 20 mi.—3,000′ W side N crs 25 mi.—3,000′ W side N crs	2,500	180	2.3	1,757	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 1.5 2.0 1.0	Climb to 3,800′ on S crs within 10 mi.*, or alternate procedure (when directed by ATC), turn left, climb to 8,000′ on E crs within 25 mi. *Caution: 2,754′ MSL radio tower, 11 mi. S of airport.
Abilene Mun. Arpt. No. 1	(MAKE LET-DOWN TO ABILENE MUN. ARPT. NO. 2. FLY VFR FROM ABILENE MUN. ARPT. NO. 2 TO ABILENE MUN. ARPT. NO. 1)													
ACOMITA, N. MEX. CAA Int. Field 371 kc; ACO BMLZ-DTV	N—Min. en route altitude E—10,000′ (Albuquerque Range) S—Min. en route altitude W—11,000′ (Zuni Range)	None	E	10 mi.—9,000′ S side E crs 15 mi.—9,000′ S side E crs 20 mi.—9,000′ S side E crs 25 mi.—9,000′ S side E crs	7,600	254	1.9	6,575	R A T	1,000 1,500 1,000	2.0 3.0 2.0	1,000 1,500 1,000	3.0 4.0 3.0	Make left turn climbing to 11,000′ on S crs within 25 mi.
ADVANCE, MO.	(NO ASSOCIATED AIRPORT)													
AKRON, COLO. CAA Int. Field 288 kc; AKO SBMRAZ-DTV	N—6,000′ (E crs Cheyenne) E—5,600′ (Hayes Center Range) S—6,000′ (E crs Thurman VAR) W—6,600′ (Denver Range)	None	N	10 mi.—5,700′ W side N crs 15 mi.—5,700′ W side N crs 20 mi.—5,700′ W side N crs 25 mi.—5,700′ W side N crs	5,200	154	3.6	4,585	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 6,000′ on S crs within 25 mi. Note: Heavy soft use with caution.
AKRON, OHIO Akron Arpt. 362 kc; AKR SBMRLZ-DTV	NE—2,500′ (E crs Cleveland) NE—2,400′ (Akron LFM) SE—2,500′ (S crs Youngstown) SW—2,400′ (E crs Columbus) NW—2,500′ (W crs Cleveland)	None	SW	10 mi.—2,300′ S side SW crs 15 mi.—2,400′ S side SW crs 20 mi.—2,400′ S side SW crs 25 mi.—2,400′ S side SW crs	1,860	45	.4	1,051	R (R) A T	700 600 800 400	1.5 1.0 2.0 1.0	700 600 800 400	1.5 1.5 2.0 1.0	Climb to 2,500′ on NE crs.
ALBANY, GA. Albany Arpt. 254 kc; ABY SBMRAZ-DTV	N—1,600′ (SW crs Macon) E—1,600′ (Alma Range) S—1,400′ (Tallahassee Range) W—1,600′ (NE crs Dothan)	None	E	10 mi.—1,400′ N side E crs 15 mi.—1,400′ N side E crs 20 mi.—1,400′ N side E crs 25 mi.—1,400′ N side E crs	1,200	270	4.2	196	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,600′ on W crs.
Turner AFB	N—1,600′ (SW crs Macon) E—1,600′ (Alma Range) S—1,400′ (Tallahassee Range) W—1,600′ (NE crs Dothan)	To 1,400′ on S crs.	S	10 mi.—1,400′ E side S crs 15 mi.—1,400′ E side S crs 20 mi.—1,400′ E side S crs 25 mi.—1,400′ E side S crs	1,000	9	3.7	215	R A T	500 800 300	2.0 2.0 2.0	800 800 300	2.0 2.0 2.0	Climb to 1,600′ on N crs.
ALBANY, N. Y. Albany Arpt. 263 kc; ALB SBRAZ-DTV (Procedure No. 1)	N—4,500′ (Burlington Range) N—3,000′ (Glens Falls Rbn) N—1,600′ (Saratoga Springs FM) N—1,100′ (Round Lake FM) (final) E—5,500′ (N crs Westfield) E—3,000′ (Grafton FM) S—3,000′ (Poughkeepsie Range) S—2,200′ (Coxsackie FM) S—1,700′ (Delmar FM) W—2,700′ (Utica Range) W—1,800′ (Schenectady FM)	N crs to 1,800′ within 15 mi	N	10 mi—1,600′ W side N crs 15 mi—1,800′ W side N crs 20 mi—1,800′ W side N crs 25 mi—1,800′ W side N crs	1,100	195	2.8	282	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,000′ on S crs, turn right, continue climb and proceed out W crs climbing to 2,700′, or alternate procedure (when directed by ATC), climb to 5,000′ on S crs. *Runway No. 19.
(Procedure No. 2)	N—4,500′ (Burlington Range) N—2,200′ (Saratoga Springs FM) E—5,500′ (N crs Westfield) E—3,000′ (Grafton FM) S—3,000′ (Poughkeepsie Range) S—1,700′ (Coxsackie FM) (final) W—2,700′ (Utica Range) W—1,800′ (Schenectady FM)	N crs to 1,800′ within 15 mi.	S	10 mi—2,200′ E side S crs 15 mi—2,200 E side S crs 20 mi—2,200′ E side S crs 25 mi—2,200′ E side S crs	1,700 (over Delmar FM)	15 (from Delmar FM)	5.4	282	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 3,000′ on N crs within 25 mi., or alternate procedure (when directed by ATC), climb to 2,700′ on W crs. *Runway No. 1 only.
Schenectady Co Arpt.	(MAKE LET-DOWN TO ALBANY ARPT. FLY VFR FROM ALBANY ARPT. TO SCHENECTADY CO. ARPT.)													
Troy Arpt.	(MAKE LET-DOWN TO ALBANY ARPT. FLY VFR FROM ALBANY ARPT. TO TROY ARPT.)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
ALBUQUERQUE, N. MEX. Kirtland AFB 230 kc; ABQ SBRAZ-DTV	N—Min. en route altitude E—11,000' (S crs Otto) S—10,000' (Hot Springs Range) S—6,300' (Peralta FM) (final) W—10,000' (Acomita Range)	None	S	10 mi.—8,000' W side S crs 15 mi.—8,000' W side S crs 20 mi.—8,000' W side S crs 25 mi.—8,000' W side S crs	6,300	354	3.1	5,330	R A T	800 800 500	2.0 2.0 2.0	800 800 500	2.0 2.0 2.0	Turn left, climb to 10,000' on W crs.
ALEXANDRIA, LA. Alexandria Arpt. 261 kc; AEX SBMRAZ-DTV	NE—Min. en route altitude SE—1,500' (Baton Rouge Range) SW—1,500' (N crs Lake Charles) NW—1,500' (S crs Shreveport)	None	SE	10 mi.—1,500' E side SE crs 15 mi.—1,500' E side SE crs 20 mi.—1,500' E side SE crs 25 mi.—1,500' E side SE crs	900	310	5.7	88	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,300' on NW crs within 25 mi.
ALEXANDRIA, MINN. Alexandria Arpt. 289 kc; AXN BMLZ-DTV	N—Min. en route altitude SE—2,600' (Minneapolis Range) S—Min. en route altitude NW—2,800' (Fargo Range)	None	N	10 mi.—2,600' W side N crs 15 mi.—2,600' W side N crs 20 mi.—2,600' W side N crs 25 mi.—2,600' W side N crs	2,200	172	1.9	1,423	R A T	600 800 300	1.5 2.0 1.0	700 800 300	2.0 2.0 1.0	Climb to 2,600' on S crs. 1,590' MSL stack, 1,585' MSL tank—1.2 mi. NNE of arpt. 1,573' MSL tower—1.5 mi. NW of arpt.
ALICE, TEX. Alice Arpt. 224 kc; ALI SBRAZ-DTV	N—1,700' (S crs San Antonio) E—1,500' (Corpus Christi Range) S—1,400' (Brownsville Range) W—1,800' (Laredo Range)	None	W	10 mi.—1,400' S side W crs 15 mi.—1,400' S side W crs 20 mi.—1,400' S side W crs 25 mi.—1,400' S side W crs	1,000	72	3.6	177	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on E crs.
ALLENTOWN, PA. Allentown-Bethlehem-Easton Arpt. 320 kc; ABL SBMRAZ-DTV	NE—2,700' (NW crs La Guardia) NE—1,500' (W crs Newark) (final). E—2,500' (SW crs Newark) SW—2,500' (E crs Harrisburg) W—4,000' (S crs Williamsport) W—3,000' (Slatington FM)	None	NE	10 mi.—2,600' E side NE crs 15 mi.—2,600' E side NE crs 20 mi.—2,700' E side NE crs 25 mi.—2,700' E side NE crs	1,500	192	2.7	392	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	600 600 800 300	2.0 2.0 2.0 1.0	Turn right and climb to 2,500' on SW crs. 1,600' terrain on N side NE crs.
ALMA, GA. CAA Int. Field 242 kc; AMG SBRAZ-DTV	NE—Min. en route altitude SE—1,500' (Jacksonville Range) SW—Min. en route altitude NW—1,500' (Macon Range)	None	NW	10 mi.—1,300' W side NW crs 15 mi.—1,300' W side NW crs 20 mi.—1,300' W side NW crs 25 mi.—1,300' W side NW crs	800	144	2.2	200	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 1,500' on right side SE crs.
ALTOONA, PA. Blair Co. Arpt. 338 kc; AOO SBMRAZ-DTV	N—4,500' (S crs Philipsburg) E—4,000' (S crs Philipsburg) S—4,700' (SE crs Pittsburgh) W—4,500' (E crs Pittsburgh)	None	N	10 mi.—3,600' W side N crs 15 mi.—3,600' W side N crs 20 mi.—3,600' W side N crs 25 mi.—3,600' W side N crs	3,100	203	1.3	1,495	R A T	1,000 1,500 1,000	2.0 3.0 1.0	1,000 1,500 1,000	3.0 3.0 1.0	Climb to 4,700' on S crs.
AMARILLO, TEX. Amarillo Air Term. 251 kc; AMA SBRAZ-DTV (Procedure No. 1)	N—Min. en route altitude E—4,700' NE crs Clarendon) S—5,000' (Lubbock Range) W—5,200' (Tucumcari Range) W—4,500' (Soney FM) (final)	None	W	10 mi.—5,000' S side W crs 15 mi.—5,000' S side W crs 20 mi.—5,000' S side W crs 25 mi.—5,000' S side W crs	4,500	84	2.0	3,604	R (R) A T	500 500 800 300	1.5 *1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 4,700' on E crs, or alternate procedure (when directed by ATC), climb to 5,000' on S crs.
(Procedure No. 2)	N—Min. en route altitude E—4,700' (NE crs Clarendon) E—4,200' (NW crs Clarendon) (final) S—5,000' (Lubbock Range) W—5,200' (Tucumcari Range)	None	E	10 mi.—4,700' S side E crs 15 mi.—4,700' S side E crs 20 mi.—4,700' S side E crs 25 mi.—4,700' S side E crs	4,200 (over Amarillo LFM)	257 (from Amarillo LFM)	1.9	3,604	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 5,200' on W crs, or alternate procedure (when directed by ATC), climb to 5,000' on N crs within 25 mi.
ANDREWS (Camp Springs), MD. Andrews AFB 379 kc; ADW SBMRAZ-DV	NE—1,500' (S crs Baltimore) S—1,500' (SE crs Quantico) SW—1,500' (SW crs Washington) N—1,500' (NE crs Washington)	S crs to 1,500'	S	10 mi.—1,500' E side S crs 15 mi.—1,500' E side S crs 20 mi.—1,500' E side S crs 25 mi.—1,500' E side S crs	1,200	6	7.5	279	R A T	600 1,000 300	2.0 3.0 1.0	600 1,000 300	3.0 3.0 2.0	Climb to 1,500' on NE crs. CAUTION: Acft to remain E of N and S courses of Andrews rng during let-down.
ARCATA, CALIF. Humboldt Co. Arpt. 209 kc; NZM SMRLZ-D	NE—On top—not over 10,000' SE—On top—not over 10,000' SW—On top—not over 10,000' NW—On top—not over 10,000'	On NW crs to 1,500' within 10 mi.	NW	10 mi.—1,500' S side NW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	820	4	2.7	217	R A T	500 1,000 500	2.0 2.0 1.0	500 1,000 500	2.0 2.0 1.0	If not contact over rng, make left turn and shuttle on NW crs to on top. NOTE: Not for general use; for emergency use only or by prior arrangement with station.
ARCOLA, VA.	(NO ASSOCIATED AIRPORT)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
ATLANTA, GA. Atlanta Arpt. 266 kc; ATL SBRAZ-DTXV	NE—2,800' (Spartanburg Range) NE—2,200' (Stone Mountain FM) SE—2,000' (Macon Range) SE—1,600' (Jonesboro FM) (final) SW—2,500' (Marwell Range) NW—4,000' (Chattanooga Range) NW—2,500' (Smyrna FM)	None	SE	10 mi.—2,000' E side SE crs 15 mi.—2,000' E side SE crs 20 mi.—2,000' E side SE crs 25 mi.—2,000' E side SE crs	1,600	316	1.8	1,024	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn left, climb to 2,500' and proceed to Campbellton range via E crs Campbellton, or alternate procedure (when directed by ATC), climb to 3,500' on *NW crs Atlanta, then proceed to Campbellton range. *Runway No. 33.
Atlanta NAS 323 kc; NCQ MRLZ	N—Min. en route altitude E—3,000' (NE crs Atlanta) S—3,000' (NE crs Atlanta) W—3,000' (NW crs Atlanta)	None	W	10 mi.—2,500' S side W crs 15 mi.—3,000' S side W crs 20 mi.—3,000' S side W crs 25 mi.—3,000' S side W crs	1,700	86	3.2	1,002	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	700 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 3,000' on E crs within 25 mi.
ATLANTIC CITY, N. J. Atlantic City NAS 206 kc; NBB SMRLZ	N—1,500' (NE crs Millville) E—1,500' (SE crs Lakehurst) S—1,500' (SE crs Millville) W—1,500' (SE crs Millville)	None	S	10 mi.—1,500' E side S crs 15 mi.—1,500' E side S crs 20 mi.—1,500' E side S crs 25 mi.—1,500' E side S crs	800	354	3.8	70	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on N crs, make procedure turn right (E) within 8 mi. *Runway No. 35.
AUGUSTA, GA. Daniel Field 385 kc; AGS SBRAZ-DTV	NE—1,600' (W crs Columbia) E—1,500' (Charleston Range) SW—Min. en route altitude W—2,600' (NE crs Atlanta)	None	SW	10 mi.—1,600' S side SW crs 15 mi.—1,600' S side SW crs 20 mi.—1,600' S side SW crs 25 mi.—1,600' S side SW crs	1,100	46	2.1	427	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,600' on NE crs.
AUGUSTA, MAINE Augusta State Arpt. 221 kc; AUG BMRLZ-DTV	NE—1,600' (SW crs Bangor) E—Min. en route altitude SW—1,800' (Portland Range) W—2,000' (SW crs Bangor)	None	SW	10 mi.—1,800' S side SW crs 15 mi.—1,800' S side SW crs 20 mi.—1,800' S side SW crs 25 mi.—1,800' S side SW crs	1,300	81	2.0	357	R* A* T*	600 800 400	1.0 2.0 1.0	600 800 300	1.5 2.0 1.0	Climb to 1,600' on NE crs. *Minimums apply to acft. with stall speeds of 75 mph or less. NOTE: Night operations not authorized on NE/SW runway.
AUSTIN, TEX. Mueller Arpt. 281 kc; AUS SBMRLZ-DTV	NE—2,000' (SE crs Waco) SE—1,900' (W crs Richmond) SW—2,500' (N crs San Antonio) NW—2,300' (NE crs K lly) 25 NW—1,740' (Lake Travis FM) (final)	None	NW	10 mi.—2,200' W side NW crs 15 mi.—2,300' W side NW crs 20 mi.—2,300' W side NW crs 25 mi.—2,300' W side NW crs	1,740	117	1.8	631	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,900' on SE crs.
Bergstrom AFB	NE—2,000' (SE crs Waco) SE—1,900' (W crs Richmond) SW—2,500' (N crs San Antonio) NW—2,300' (NE crs Kelly) NW—1,740' (Lake Travis FM) (final)	None	NW	10 mi.—2,200' W side NW crs 15 mi.—2,300' W side NW crs 20 mi.—2,300' W side NW crs 25 mi.—2,300' W side NW crs	1,740	147	5.6	515	R A T	700 1,000 300	2.0 3.0 2.0	700 1,000 300	3.0 3.0 3.0	Turn to heading of 90° and proceed out SE crs climbing to 1,900'.
Baile Field	(MAKE LET-DOWN TO MUELLER AIRPORT. FLY VFR FROM MUELLER AIRPORT TO HAILE FIELD)							725	R S A T	1,000 NA 1,000 1,000	1.0 3.0 1.0	1,000 NA 1,000 1,000	2.0 3.0 2.0	
BAKER, OREG. Baker Arpt. 221 kc; BKE SBRAZ-DTV	NE—Min. en route altitude SE—9,000' (Boise Range) SW—Min. en route altitude NW—10,000' (Pendleton Range)	On SE crs to 8,000' within 20 mi.	NW	10 mi.—7000' E side NW crs 15 mi.—9,000' E side NW crs 20 mi.—NA 25 mi.—NA	5,060	111	1.8	3,368	R A T	1,700 1,700 1,000	1.0 2.0 1.0	1,700 1,700 1,000	2.0 2.0 2.0	If not contact over range, immediately make 180° left turn and climb to 9,000' on NW crs within 10 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
BAKERSFIELD, CALIF. Bakersfield-Kern Co. Arpt. No. 1 229 kc; BFL SBMRAZ-DTV	NE—Min. enroute altitude SE—10,000' (Newhall Range) SE—6,000' (Lebec FM) SW—Min. enroute altitude NW—3,000' (Fresno Range) NW—1,600' (Famoss FM) (final)	None	NW	10 mi.—2,000' W side NW crs 15 mi.—2,000' W side NW crs 20 mi.—2,000' W side NW crs 25 mi.—2,000' W side NW crs	1,600	150	1.5	515	R A T	700 800 300	2.0 2.0 1.0	700 800 300	2.0 2.0 1.0	Turn right (W) and climb to 3,000' on NW crs, or alternate procedure (when directed by ATC), climb to 3,000' on SW crs within 15 mi. NOTE: Take-offs or landings not authorized on Runway 7/25.
Shafter-Kern Co. Arpt.	(MAKE LET-DOWN TO BAKERSFIELD-KERN CO. AIRPORT NO. 1. FLY VFR FROM BAKERSFIELD-KERN CO. AIRPORT NO. 1 TO SHAFTER-KERN CO. ARPT.)							425	R S A T	1,000 NA 1,000 1,000	1.0 3.0 1.0	1,000 NA 1,000 1,000	2.0 3.0 2.0	
BALTIMORE, MD. Baltimore Arpt. 257 kc; BAL SBMRLZ-DTV	N—1,600' (W crs Philadelphia) E—Min. en route altitude S—1,500' (SE crs Washington) S-800' (Severna Park FM) (final) W—2,000' (NE crs Arcola) W—1,500' (NE crs Washington)	None	S	10 mi.—1,500' W side S crs 15 mi.—1,500' W side S crs 20 mi.—1,500' W side S crs 25 mi.—1,500' W side S crs	800	15	2.4	14	R (R) S A T	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	600 600 600 800 400	1.5 1.5 1.5 2.0 1.5	Climb to 1,600' on N crs.
BANGOR, MAINE Dow AFB 229 kc; BGR SBRAZ-DTV	NE—2,500' (S crs Houlton) SE—Min. en route altitude SW—2,300' (NE crs Augusta) NW—1,700' (S crs Millinocket) NW—1,200' (East Corinth FM) (final)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—2,000' W side NW crs 20 mi.—2,000' W side NW crs 25 mi.—2,000' W side NW crs	1,200	180	2.2	192	R (R) S A T	600 600 600 1,000 300	1.5 1.0 1.0 3.0 1.0	700 600 600 1,000 300	1.5 1.5 1.5 3.0 1.0	Climb to 2,500' on SE crs.
BARKSDALE (Shreveport), LA. Barksdale AFB 379 kc; BAD SBMRAZ	NE—1,500' (E crs Shreveport) SE—Min. en route altitude SW—1,500' (S crs Shreveport) NW—1,500' (E crs Shreveport)	None	SE	6 mi.—1,500' E side SE crs (Elmgrove FM) 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,000	323	4.4	167	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 2.0	500 500 500 1,000 300	1.5 1.5 1.5 3.0 3.0	Turn immediately to 115° heading and climb to 1,500' on NE crs within 25 mi.
BATON ROUGE, LA. Harding Field 347 kc; BTR BMRLZ-DTV	NE—Min. en route altitude SE—1,500' (W crs New Orleans) SW—1,500' (Lafayette Rbn) NW—1,500' (Alexandria Range)	None	NW	10 mi.—1,100' W side NW crs 15 mi.—1,200' W side NW crs 20 mi.—1,200' W side NW crs 25 mi.—1,200' W side NW crs	700	126	3.4	70	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500' on SE crs within 25 mi.
BATTLE CREEK, MICH. Kellogg Field 272 kc; BTL SBMLZ-DTV	N—2,200' (SE crs Grand Rapids) E—2,300' (SE crs Lansing) S—2,000' (E crs South Bend) W—2,200' (N crs Goshen)	None	S	10 mi.—2,000' E side S crs 15 mi.—2,000' E side S crs 20 mi.—2,000' E side S crs 25 mi.—2,000' E side S crs	1,600	3	3.2	936	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,200' on N crs. *Runway No. 36.
BATTLE MOUNTAIN, NEV.	(NO INSTRUMENT LET-DOWNS AUTHORIZED)													
BEAUMONT, TEX. Jefferson Co. Arpt. 368 kc; BUJ SBMRLZ-DTV	N—Mn. enroute altitude E—1,300' (Lake Charles Range) S—Min. enroute altitude W—1,600' (Houston Range)	None	N	10 mi.—1,400' W side N crs 15 mi.—1,400' W side N crs 20 mi.—1,400' W side N crs 25 mi.—1,400' W side N crs	900	160	3.6	15	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500' on S crs.
Beaumont Arpt.	(MAKE LET-DOWN TO JEFFERSON CO. AIRPORT. FLY VFR FROM JEFFERSON CO. AIRPORT TO BEAUMONT AIRPORT)							31	R S A T	1,000 NA 1,000 1,000	1.0 3.0 1.0	1,000 NA 1,000 1,000	2.0 3.0 2.0	
BELLINGHAM, WASH. Bellingham Arpt. 391 kc; BLI SBRAZ-DTV	NE—10,000' (Princeton Range) S—4,000' (Everett Range) SW—3,500' (E crs Patricia Bay) NW—1,500' (Vancouver Range)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—1,500' W side NW crs 25 mi.—1,500' W side NW crs	1,000	115	2.9	158	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	700 700 800 300	2.0 1.0 2.0 1.0	Turn right and climb to1,500' on NW crs.
BIG SPRING, TEX. Big Spring Arpt. 326 kc; BGS SBRAZ-DTV	E—4,000' (Abilene Range) SE—4,000' (SW crs San Angelo) W—4,000' (NW crs Midland) W—3,200' (Stanton FM) (final) NW—4,500' (S crs Lubbock)	None	W	10 mi.—4,000' S side W crs 15 mi.—4,000' S side W crs 20 mi.—4,000' S side W crs 25 mi.—4,000' S side W crs	3,200	70	3.9	2,595	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 4,000' on E crs within 25 mi., or alternate procedure (when directed by ATC), turn right, climb to 4,000' on SE crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
BILLINGS, MONT. Billings Arpt. 400 kc; BIL SBRAZ-DTV	NE—5,000' (Miles City Range) NE—4,500' (Nibbe FM) (final) SE—8,000' (Sheridan Range) W—9,000' (Livingston Range) W—5,500' (Park City FM) N—8,000' (E crs Lewistown) N—6,000' (Lavina FM)	None	NE	10 mi.—5,000' N side NE crs 15 mi.—5,000' N side NE crs 20 mi.—5,000' N side NE crs 25 mi.—5,000' N side NE crs	4,500	256	1.3	3612	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 5,500' on W crs within 15 mi.
BIRMINGHAM, ALA. Birmingham Arpt. 224 kc; BHM SBRAZ-DTV	N—2,500' (SW crs Chattanooga) N—1,800' (Bradford FM) (final) E—4,000' (Campbellton Range) E—2,600' (Eden FM) S—2,700' (W crs Maxwell) SW—2,500' (Meridian Range)	None	N	10 mi.—2,500' W side N crs 15 mi.—2,500' W side N crs 20 mi.—2,500' W side N crs 25 mi.—2,500' W side N crs	1,800	178	3.0	643	R (R) A T	900 800 1000 300	1.5 1.0 2.0 1.0	900 800 1000 300	1.5 1.0 2.0 1.0	Climb to 2,700' on S crs, or alternate procedure (when directed by ATC), turn left and climb to 2,600' on E crs within 25 mi.
BISMARCK, N. DAK. Bismarck Arpt. 260 kc; BIS SBRAZ-DTV	N—3,400' (SE crs Minot) E—3,400' (Jamestown Range) S—Min. enroute altitude W—3,800' (Dickinson Range) W—3,300' (Judson FM)	None	E	10 mi.—3,400' S side E crs 15 mi.—3,400' S side E crs 20 mi.—3,400' S side E crs 25 mi.—3,400' S side E crs	2,500	266	2.1	1653	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 3,300' on W crs to Judson FM. CAUTION: 2,413' msl radio tower located 4 mi N and 8 mi E of Bismarck range.
BLACKSTONE, VA. Blackstone Arpt. 326 kc; BKT BMLZ-DTV	NE—1,800' (SW crs Richmond) SE—2,000' (SW crs Richmond) SW—2,300' (S crs Lynchburg) NW—3,000' (NE crs Lynchburg)	None	NW	10 mi.—1,800' W side NW crs 15 mi.—1,800' W side NW crs 20 mi.—1,800' W side NW crs 25 mi.—1,800' W side NW crs	1,300	133	5.5	440	R A T	500 800 500	2.0 2.0 1.0	500 800 500	3.0 2.0 2.0	Climb to 2,000' on SE crs.
BLYTHE, CALIF. Blythe Arpt. 251 kc; BLH SBMRAZ-DTV	N—6,000' (Needles Range) E—6,000' (N crs Gila Bend) S—5,000' (Yuma Range) W—8,000' (N crs Thermal)	None	S	10 mi.—2,000' W side S crs 15 mi.—3,000' W side S crs 20 mi.—4,000' W side S crs 25 mi.—4,000' W side S crs	1,200	268	4.1	397	R A T	800 800 800	2.0 2.0 2.0	800 800 800	2.0 2.0 2.0	If not contact over range, turn left and climb to 5,000' on S crs.
BLYTHEVILLE, ARK.	(NO PROCEDURE APPROVED)													
BOISE, IDAHO Boise Air Terminal 350 kc; BOI SBRAZ-DTV	NE—Min. enroute altitude SE—9,000' (W crs Burley) SE—7,700' (Mountain Home FM) SW—Min. enroute altitude NW—9,000' (Baker Range) NW—5,500' (Payette FM) NW—3,800' (Eagle FM) (final)	None	NW	10 mi.—4,000' W side NW crs 15 mi.—5,000' W side NW crs 20 mi.—5,000' W side NW crs 25 mi.—5,000' W side NW crs	3,800	116	2.3	2,858	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	700 600 800 300	1.5 1.0 2.0 1.0	Turn right and climb to 4,000' on NW crs within 10 mi. High terrain 8 mi. N of NW crs and 10 mi. S of range on SE crs.
BOSTON, MASS. Logan Arpt. 382 kc; BOS SBRAZ-DTV (Procedure No. 1)	N—1,500' (SE crs Concord) N—800' (Peabody FM) (final) E—1,500' (NE crs Squantum) SW—1,700' (N crs Providence) W—1,700' (N crs Providence)	None	N	10 mi.—1,300' W side N crs 15 mi.—1,500' W side N crs 20 mi.—1,500' W side N crs 25 mi.—1,500' W side N crs	800	238	1.4	19	R (R) A T	600 *500 800 300	1.5 1.0 2.0 1.0	600 *500 800 300	1.5 1.0 2.0 1.0	Make climbing left turn to 1,500' on E crs, or alternate procedure (when directed by ATC), climb to 1,700' on SW crs. *600-1 required when circling W of airport.
(Procedure No. 2)	N—1,500' (SE crs Concord) N—1,300' (Peabody FM) E—1,500' (NE crs Squantum) SW—1,700' (N crs Providence) SW—1,200' (Dedham FM) (final) W—1,700' (N crs Providence)	None	SW	10 mi.—1,700' S side SW crs 15 mi.—1,700' S side SW crs 20 mi.—1,700' S side SW crs 25 mi.—1,700' S side SW crs	1,200 (over South Boston LFM)	58 (from South Boston LFM)	1.9	19	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	600 600 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on N crs, or alternate procedure (when directed by ATC), make climbing right turn to 1,500' on E crs.
BOWLING GREEN, KY. Bowling Green-Warren Co. Arpt. 368 kc; BWG SBMRAZ-DTV	NE—2,200' (S crs Louisville) SE—2,200' (N crs Smithville) SW—2,000' (NW crs Nashville) NW—2,000' (NW crs Nashville)	None	SE	10 mi.—1,900' N side SE crs 15 mi.—2,200' N side SE crs 20 mi.—2,200' N side SE crs 25 mi.—2,200' N side SE crs	1,400	296	3.8	540	R S A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,000' on NW crs. CAUTION: 860' msl tower, 1.5 mi NNW of arpt. NOTE: Minimums apply to all aircraft.
BOZEMAN, MONT. Gallatin Fld. 329 kc; BZN SBMRAZ-DTV	N—9,000' (SE crs Helena) E—11,100' (NW crs Livingston) S—Min. en route altitude S—10,000' (W crs Livingston) W—9,000' (Whitehall Range)	On N crs to 9,000'	S	10 mi.—8,500' W side S crs 15 mi.—9,500' W side S crs 20 mi.—NA 25 mi.—NA	6,000	219	.78	4,461	R A T	1,500 1,500 1,500	2.0 2.0 2.0	1,500 1,500 1,500	3.0 2.0 2.0	Turn left, climbing to 9,000' on W crs within 20 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)		Ceiling and visibility minimums: Day Ceiling (ft.)	Day Visibility (mi.)	Night Ceiling (ft.)	Night Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
BRIDGEPORT, CONN. Bridgeport Arpt. 221 kc; BDE MRLWZ	NE—2,000' (SE crs Hartford) SE—1,200' (NE crs Mitchel) SW—1,500' (E crs La Guardia) NW—2,000' (NE crs Newark)	None	NE	10 mi.—1,500' N side NE crs 15 mi.—1,500' N side NE crs 20 mi.—2,000' N side NE crs 25 mi.—2,000' N side NE crs	1,000	231	3.7	9	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,500' on SW crs.
BROWNSVILLE, TEX. International Arpt. 388 kc; BRO SBRAZ-DTV	N—1,600' (Alice Range) N—800' (Los Fresnos FM) (final) E—Min. en route altitude S—Min. en route altitude W—1,500' (SE Lersaredo)	None	N	10 mi.—1,200' W side N crs 15 mi.—1,200' W side N crs 20 mi.—1,200' W side N crs 25 mi.—1,200' W side N crs	800	157	2.3	22	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,200' on S crs within 25 mi. CAUTION: 274' MSL tower, 3 mi. WSW of rng.
Mauldin Arpt.	MAKE LET-DOWN TO INTERNATIONAL AIRPORT. FLY VFR FROM INTERNATIONAL AIRPORT TO MAULDIN AIRPORT)							22	R S A T	1,000 NA 1,000 1,000	1.0 3.0 1.0	1,000 NA 1,000 1,000	2.0 3.0 2.0	
BRYAN, TEX. Bryan AFB 212 kc; BYT BMRLZ-DTV	NE—Min. en route altitude SE—1,600' (Houston Range) SW—1,500' (SE crs Austin) NW—2,000' (Waco Range)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—1,500' W side NW crs 25 mi.—1,500' W side NW crs	800	126	4.2	265	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		Climb to 1,600' on SE crs.
BUFFALO, N. Y. Buffalo Arpt. 260 kc; BUF SBRAZ-DTXV (Procedure No. 1)	NE—2,000' (W crs Rochester) NE—1,300' (Wolcottsville FM) (final) E—2,300' (S crs Rochester) E—1,700' (E Pembroke FM) SW—2,000' (E crs Clear Creek) W—2,100' (NE crs Clear Creek)	None	NE	10 mi.—1,700' N side NE crs 15 mi.—1,700' N side NE crs 20 mi.—2,000' N side NE crs 25 mi.—2,500' N side NE crs	1,300	214	.9	711	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,000' on SW crs.
(Procedure No. 2)	NE—2,000' (W crs Rochester) NE—1,800' (Wolcottsville FM) E—2,300' (S crs Rochester) E—1,700' (E Pembroke FM) SW—2,000' (E crs Clear Creek) SW—1,500' (Angola FM) (final) W—2,100' (NE crs Clear Creek)	None	SW	10 mi.—2,000' S side SW crs 15 mi.—2,000' S side SW crs 20 mi.—2,000' S side SW crs 25 mi.—2,000' S side SW crs	1,500 (over Cheektowaga LFM)	50 (from Cheektowaga LFM)	2.7	711	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000' on NE crs.
BURBANK, CALIF. Lockheed Air Term. 248kc; BUR SBML-DTV	NE—Min. en route altitude* SE—5,000'* (NE crs Long Beach) SW—5,000'* (W crs Los Angeles) NW—5,000'* (SW crs Newhall) (Simi Int.) NW—4,750'* (Simi Int.) (final) NW—1,500'* (Chatsworth Rbn) (final) (Burbank approaches normally start from primary holding fix at Simi Int.)	None Sliding scale for tops at Simi Int. Top / Ceil 5,000' / 600' D 700' N 5,500' / 900' 6,000' / 1,300' 6,500' / 1,700' 7,000' / 2,100' 7,500' / 2,500'	NW	(W of Chatsworth Rbn) 10 mi.—5,000'* S side NW crs 15 mi.—5,000'* S side NW crs 20 mi.—5,000'* S side NW crs 25 mi.—5,000'* S side NW crs	1,500	93	3.0	763	R S A T	#900 #600 #500 300	1.5 1.0 2.0 1.0	#900 #700 #500 300	2.0 1.5 2.0 1.0	If visual contact not established within 1.5 mi. after passing rng, climb to 5,000' on SE crs. *All procedures, including procedure turn, must be on top and not below 5,000'. After procedure turn, or after leaving Simi Int., cross Chatsworth Rbn inbound at 4,750'. #Sliding scale based on tops reported at Simi Int. NOTE: Straight-in approach to runway No. 7.
San Fernando Valley Airport	(MAKE LET-DOWN TO LOCKHEED AIR TERMINAL. FLY VFR FROM LOCKHEED AIR TERMINAL TO SAN FERNANDO VALLEY AIRPORT)							800	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	Minimums are reported at Lockheed Air Term. If lower minimums are encountered en route from Lockheed Air Term. to San Fernando Valley Arpt., acft are to return to Lockheed Air Term.
BURLEY, IDAHO Burley Arpt. 272 kc; BYI SBRAZ-DTPV	NE—Min. en route altitude NE—7,000' (W crs Pocatello) E—11,000' (Malad City Range) SW—12,000' (NW crs Lucin) W—7,000' (SE crs Boise)	None	NE	10 mi.—6,000' W side NE crs 15 mi.—6,000' W side NE crs 20 mi.—6,000' W side NE crs 25 mi.—6,000' W side NE crs	5,150	198	2.6	4,150	R A T	600 800 500	1.0 2.0 1.0	600 800 500	2.0 2.0 2.0	Make right turn, climb to 7,000' on W crs within 25 mi. NOTE: Arpt restricted to acft with stall speeds of 75 mph or less.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
BURLINGTON, IOWA Burlington Arpt. 326 kc; BRL SBRAZ-DTV	E—2,000' (S crs Moline) S—Min. en route altitude SW—2,000' (Kirksville Range) N—1,900' (W crs Moline)	None	S	10 mi.—1,800' E side S crs 15 mi.—1,800' E side S crs 20 mi.—2,000' E side S crs 25 mi.—2,000' E side S crs	1,300	357	1.9	608	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,900' on N crs. *Runway No. 36.
BURLINGTON, VT. Burlington Arpt. 323 kc; BTV SBRAZ-DTV	N—3,500' (E crs Montreal) SE—5,000' (SW crs Montpelier) SE—2,600' (Huntington FM) S—4,500' (Albany Range) S—2,000' (Vergennes FM) NW—1,500' (Montreal Range) NW—1,000' (Grand Isle FM) (final)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—2,500' W side NW crs 25 mi.—2,500' W side NW crs	1,000	151	2.7	334	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.0 2.0 1.0	Turn right and climb to 4,500' on S crs, or alternate procedure (when directed by ATC), turn right, climb to 1,500' on NW crs. *Runway No. 15.
BUTTE, MONT., Butte Arpt. 233 kc; BTM BMRLZ-DTV (Procedure No. 1)	N—9,500' (E crs Drummond) SE—10,000' (Whitehall Range) S—11,500' (Dillon Range) NW—12,700' (SW crs Drummond)	None	N	10 mi.—9,000' E side N crs* 15 mi.—9,000' E side N crs* 20 mi.—NA 25 mi.—NA	9,000	99	16.1	5,554	R A T	2,500 2,500 1,500	1.5 1.5 1.0	2,500 2,500 1,500	2.0 2.0 2.0	If not contact over range, climb to 10,000' on SE crs within 15 mi. *Procedure turns to W not authorized due to high terrain.
(Procedure No. 2) (Using Whitehall Range) 284 kc; HIA SBRAZ—DTV	N—10,500' (Helena Range) E—9,000' (Bozeman Range) S—10,500' (Dillon Range) NW—10,000' (Butte Range)	None	NW crs Whitehall	None	9,500 (over Homestake FM)	275 (from Homestake FM)	4.0	5,554	R A T	2,500 2,500 1,500	1.5 1.5 1.0	2,500 2,500 1,500	2.0 2.0 2.0	Proceed to Butte range, climbing to 10,000' before reaching there.
CARLSBAD, N. MEX. Carlsbad Arpt. 311 kc; CNM BMRLZ-DTV	NE—5,000' (W crs Hobbs) SE—5,000' (W crs Wink) SW—10,800' (E crs Salt Flat) NW—5,500' (SW crs Roswell)	SE crs to 5,000' within 25 mi.	SE	10 mi.—5,000' E side SE crs 15 mi.—5,000' E side SE crs 20 mi.—5,000' E side SE crs 25 mi.—5,000' E side SE crs	4,200	322	3.5	3276	R (R) S A T	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	600 500 500 800 300	2.5 2.0 2.0 2.0 1.0	Climb to 5,000' on NE crs within 25 mi.
CASPER, WYO. Natrona Co. Arpt. 269 kc; CPR SBRAZ-DTV	N—7,500' (SE crs Sheridan) E—Min. en route altitude E—6,000' (Parkerton FM) (final) S—11,000' (E crs Sinclair) W—Min. en route altitude	None	E	10 mi.—6,500' N side E crs 15 mi.—7,000' N side E crs 20 mi.—7,000' N side E crs 25 mi.—7,000' N side E crs	6,000	256	7.7	5347	R A T	500 800 300	2.0 2.0 1.0	500 800 300	2.0 2.0 1.0	Climb to 7,500' on N crs within 25 mi.
CHANUTE (Rantoul), ILL. Chanute AFB 365 kc; RAN *SBMRAZ	NE—1,900' (NW crs La Fayette) E—2,000' (SW crs La Fayette) SW—2,100' (NW crs Effingham) W—2,000' (NE crs Springfield)	None	SW	10 mi.—2,100' W side SW crs 15 mi.—2,100' W side SW crs 20 mi.—2,100' W side SW crs 25 mi.—2,100' W side SW crs	1,600	43	3.4	737	R (R) S A T	500 500 500 1000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	1.5 1.5 1.5 3.0 1.0	Climb to 1,900' on NE crs.
CHANUTE, KANS. Chanute Arpt. 284 kc; CNU SBRAZ-DTV	N—2,300' (SW crs Kansas City) E—2,200' (N crs Joplin) S—2,200' (NE crs Tulsa) W—2,600' (NE crs Wichita)	None	E	10 mi.—2,100' N side E crs 15 mi.—2,100' N side E crs 20 mi.—2,100' N side E crs 25 mi.—2,100' N side E crs	1,600	304	2.5	980	R A T	*500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,600' on W crs. *All minimums approved for acft with stall speed of 75 mph or less only.
CHARLESTON, S. C. Charleston Arpt. 329 kc; CHS SBRAZ-DTV	N—1,200' (Florence Range) E—Min. en route altitude SW—1,300' (Savannah Range) W—1,500' (Augusta Range) W—600' (Summerville FM) (final)	None	W	10 mi.—1,200' S side W crs 15 mi.—1,200' S side W crs 20 mi.—1,500' S side W crs 25 mi.—1,500' S side W crs	600	110	2.0	45	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on E crs, or alternate procedure, (when directed by ATC), turn right and climb to 1,300' on SW crs.
CHARLESTON, W. VA. Kanawha Co. Arpt. 400 kc; CHW SBMRAZ-DTV	N—Min. en route altitude E—6,000' (N crs Roanoke) S—5,000' (SE crs Huntington) W—2,500' (NE crs Huntington)	On W crs descending to 2,500' within 25 mi.	W	10 mi.—2,500' N side W crs 15 mi.—2,500' N side W crs 20 mi.—2,500' N side W crs 25 mi.—2,500' N side W crs	1,980	116	10.7	985	R A T	1,000 1,500 300	2.0 2.0 1.0	1,500 1,500 300	2.0 2.0 1.0	If not contact over rng, turn left and climb to 2,500' on W crs. Night operations NA on NW/SE rwy. Maintain 1,600' on night approach until lined up with rwy. Climb straight ahead to 1,600' before turning on all take-offs or unaccomplished landings.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
CHARLOTTE, N. C. Douglas Field 212 kc; CLT SBMLZ-DTV	N—2,500' (SW crs Greensboro) E—Min. en route altitude S—Min. en route altitude S—1,620' (Fort Mill FM) (final)* W—2,800' (NE crs Spartanburg)	None	S	10 mi.—2,100' W side S crs 15 mi.—2,100' W side S crs 20 mi.—2,100' W side S crs 25 mi.—2,100' W side S crs	*1,620	359	1.7	748	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,500' on N crs, or alternate procedure (when directed by ATC), turn right and climb to 2,200' on E crs within 25 mi. *1,116' MSL tower, 4 mi. SE of rng and 2.3 mi. E of S crs.
Cannon Arpt.	(MAKE LET-DOWN TO DOUGLAS FIELD. FLY VFR FROM DOUGLAS FIELD TO CANNON AIRPORT)							738	R S A T	1,000 NA 1,000 1,000	1.0 2.0 1.0	1,000 NA 1,000 1,000	2.0 3.0 2.0	
CHATTANOOGA, TENN. Lovell Field 341 kc; CHA SBRAZ-DTV	NE—3,000' (W crs Knoxville) NE—1,800' (Daisy FM) (final) SE—4,000' (Atlanta Range) SW—4,000' (N crs Birmingham) NW—4,000' (Nashville Range)	None	NE	10 mi.—2,500' E side NE crs 15 mi.—2,500' E side NE crs 20 mi.—2,500' E side NE crs 25 mi.—2,500' E side NE crs	1,800	203	3.0	678	R (R) A T	700 500 800 300	2.0 2.0 2.0 1.0	700 700 800 300	2.0 2.0 2.0 1.0	Climb to 4,000' on SE crs, or alternate procedure (when directed by ATC), climb to 4,000' on SW crs of Chattanooga within 25 mi.
CHERRY POINT, N. C. Cherry Point MCAS 220 kc; NKT SBMRLZ	NE—Min. en route altitude SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude NW—1,500' (New Bern FM)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—1,500' W side NW crs 25 mi.—1,500' W side NW crs	1,000	141	4.0	29	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500' on SE crs.
CHEYENNE, WYO. Cheyenne Arpt. 326 kc; CYS SBRAZ-DTV (Procedure No. 1)	N—7,500' (NE crs Laramie) E—7,300' (SW crs Scottsbluff) S—7,500' (Denver Range) W—10,500' (NE crs Laramie) W—8,000' (Silver Crown FM)	On E crs to 8,000' within 25 mi.	W	10 mi.—8,000'* N side W crs** 15 mi.—NA 20 mi.—NA 25 mi.—NA	6,800	67	.9	6,156	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 7,500' on S. crs., or alternate procedure (when directed by ATC), climb to 7,500' on N crs. *Procedure turn on S side W crs NA due to high terrain. **Procedure turn must be completed within 10 mi. account high terrain to W.
(Procedure No. 2)	N—7,500' (NE crs Laramie) E—7,300' (SW crs Scottsbluff) E—7,000' (Hillsdale FM) (final) S—7,500' (Denver Range) W—10,500' (NE crs Laramie) W—8,000' (Silver Crown FM)	On E crs to 7,000' within 25 mi.	E	10 mi.—7,000' N side E crs 15 mi.—7,000' N side E crs 20 mi.—7,000' N side E crs 25 mi.—7,000' N side E crs	7,000 (over LOM)*	256 (from LOM)	6.1	6,156	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 7,500' on S crs, or alternate procedure (when directed by ATC), climb to 7,500' on N crs. *LOM must be received on ADF. #Runway No. 26.
CHICAGO, ILL. Chicago-Midway Arpt. 350 kc; CHI SBRAZ-DTXV (Procedure No. 1)	NE—2,300' (N crs So. Bend) SE—2,000' (W crs Goshen) W—2,300' (SE crs Rockford) NW—2,500' (E crs Rockford) NW—1,500' (Wilson Int.) (final)	None	NW	10 mi.—2,000' W side NW crs 15 mi.—2,500' W side NW crs 20 mi.—2,500' W side NW crs 25 mi.—2,500' W side NW crs	1,500	152	2.2	618	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.0 2.0 1.0	Climb to 2,500' on N & S crs of Harvey, proceeding S to Momence Int; or alternate procedure (when directed by ATC), climb to 2,500' on SE crs of Chicago, proceeding to Thornton Int. or: Make left turn, climbing to 2,500' on NE crs of Chicago, proceeding to Lake Shore Int. *Rwys 13 R & L and 18 R & L.
(Procedure No. 2)	NE—2,300' (N crs So. Bend) SE—2,000' (W crs Goshen) SE—1,500' (E crs Harvey) (final) W—2,300' (SE crs Rockford) NW—2,500' (E crs Rockford)	None	SE	10 mi.—2,000' E side SE crs 15 mi.—2,000' E side SE crs 20 mi.—2,000' E side SE crs 25 mi.—2,000' E side SE crs	1,500 (Over Ashburn LFM)	332 (From Ashburn LFM)	2.5	618	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Make right turn and climb to 2,500' on NE crs of Chicago or alternate procedure (when directed by ATC), climb to 2,500' on W crs of Chicago, proceeding to Downers Grove Int. *Rwys 36 R & L and 31 L.
CHINCOTEAGUE, VA. Chincoteague NAF 227 kc; NKZ SMRLZ	N—1,500' (SE crs Washington) E—1,500' (NE crs Norfolk) S—1,500' (NE crs Norfolk) W—1,500' (SE crs Patuxent Riv.)	None	W	10 mi.—1,500' S side W crs 15 mi.—1,500' S side W crs 20 mi.—1,500' S side W crs 25 mi.—1,500' S side W crs	800	98	4.5	38	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on E crs. *Runway No. 10.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
CINCINNATI, OHIO Lunken Arpt. 335 kc; CIN SBRAZ-DTV	NE—2200′ (E crs Patterson) NE—1,600′ (Loveland FM) (final) SE—2,300′ (S crs Columbus) SE—2,100′ (Mt. Orab FM) SW—2,400′ (Louisville Range) NW—2,300′ (E crs Indianapolis)	None	NE	10 mi.—2,100′ N side NE crs 15 mi.—2,600′ N side NE crs 20 mi.—2,600′ N side NE crs 25 mi.—2,600′ N side NE crs	1,600	227	5.2	488	R (R) S* A T	900 800 800 1,100 400	1.5 1.0 1.0 2.0 1.0	900 800 800 1,100 600	1.5 1.5 1.5 2.0 1.0	Climb to 2300′ on SE crs. *Runway No. 24. NOTE: Official weather available only between 0800 and 1630 est.
CLARENDON, TEX. Clarendon Arpt. 223 kc; CDT MRLWZ	NE—4,200′ (E crs Amarillo) SE—3,900′ (Wichita Falls Range) SW—Min. en route altitude NW—4,700′ (E crs Amarillo)	None	NE	10 mi.—4,200′ N side NE crs 15 mi.—4,200′ N side NE crs 20 mi.—4,200′ N side NE crs 25 mi.—4,200′ N side NE crs	3,660	199	2.0	2,885	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 4,600′ on SW crs with in 25 mi.
CLEVELAND, OHIO Cleveland Arpt. 344 kc; CLE SBRAZ-DTXV	E—2,600′ (NE crs Akron) SE—2,600′ (Akron Range) W—1,900′ (NW crs Akron) W—1,400′ (Elyria FM) (final) NW—1,900′ (Windsor Range) NW—1,900′ (Avon Lake FM)	None	W	10 mi.—1,900′ S side W crs 15 mi.—1,900′ S side W crs 20 mi.—1,900′ S side W crs 25 mi.—1,900′ S side W crs	1,400	100	.7	789	R (R) A T	500 800 800 300	1.5 1.0 2.0 1.0	500 800 800 300	1.5 1.0 2.0 1.0	Climb to 2,600′ on E crs. NOTE: New E W runway authorized for daylight operations only.
COCHISE, ARIZ. CAA Int. Field 272 kc; CIE MRAWZ	E—12,000′ (Rodeo Range) E—10,000′ (Hilltop FM) SE—10,000′ (SW crs Rodeo) W—10,000′ (SE crs Tucson) N—13,000′ (NE crs Tucson)	To 7,000′ on SE crs within 20 mi.	SE	10 mi.—6,000′ E side SE crs 15 mi.—7,000′ E side SE crs 20 mi.—7,000′ E side SE crs 25 mi.—9,000′ E side SE crs	5,000	253	2.2	4,317	R A T	700 1,000 (BCOB) 500	2.0 2.0 2.0	700 1,000 (BCOB) 500	2.0 2.0 2.0	If not contact over range, turn right and shuttle to 9,000′ on SE crs within 25 mi. 7,500 terrain on S side of W crs within 10 mi.
COEUR D'ALENE, IDAHO. Coeur D'Alene Air Terminal 347 kc; COE BMRLZ-DTV	N—Min. en route altitude E—9,000′ (Mullan Pass Range) S—Min. en route altitude W—7,000′ (Spokane Range)	None	S	10 mi.—6,000′ E side S crs 15 mi.—6,000′ E side S crs 20 mi.—6,000′ E side S crs 25 mi.—6,000′ E side S crs	4,500	321	5.2	2,310	R A T	1,500 1,500 500	2.0 3.0 1.0	1,500 1,500 500	3.0 3.0 1.0	Climb to 6,000′ on N crs within 25 mi. Procedure turn not authorized below 6,000′.
Weeks Field	(MAKE LET-DOWN TO COEUR D'ALENE AIR TERMINAL. FLY VFR FROM COEUR D'ALENE AIR TERMINAL TO WEEKS FIELD)													
COLORADO SPRINGS, COLO. Peterson Field 224 kc; COS MRLZ-D	N—8,900′ (Denver Range) N—7,500′ (Black Forest FM) (final) E—Min. en route altitude S—*8,000′ (Pueblo Range) W—Min. en route altitude	None	N	10 mi.—8,700′ E side N crs 15 mi.—8,700′ E side N crs 20 mi.—8,700′ E side N crs 25 mi.—8,700′ E side N crs	8,300	166	3.9	6,172	R (R) A T	800 800 800 300	1.5 1.0 2.0 1.0	800 800 800 300	2.0 2.0 2.0 2.0	Climb to 8,000′ on S crs within 25 mi. *Climb to be made to 8,700′ before proceeding out N crs to accomplish lowering down thru procedure. 600′ minimum ceiling if final approach made from Black Forest FM.
COLUMBIA, MO. Columbia Arpt. 379 kc; CBI SBRAZ-DTV	N—2,200′ (Kirksville Range) E—2,000′ (St. Louis Range) S—2,000′ (NW crs Vichy) W—2,200′ (NE crs Kansas City)	None	W	10 mi.—1,800′ S side W crs 15 mi.—1,800′ S side W crs 20 mi.—1,800′ S side W crs 25 mi.—1,800′ S side W crs	1,300	89	3.9	778	R A T	500 800 300	1.0 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,000′ on E crs. Procedure authorized only for acft having stall speeds of 75 mph or less.
COLUMBIA, S. C. Columbia Arpt. 227 kc; CAE SBRAZ-DTV	E—1,500′ (SW crs Florence) SE—1,500′ (NW crs Charleston) W—1,600′ (NE crs Augusta) NW—Min. en route altitude	None	E	10 mi.—1,500′ N side E crs 15 mi.—1,500′ N side E crs 20 mi.—1,500′ N side E crs 25 mi.—1,500′ N side E crs	1,200	268	7.3	244	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,600′ on W crs.
Congaree Air Base	E—1,500′ (SW crs Florence) SE—1,500′ (NW crs Charleston) W—1,600′ (NE crs Augusta) NW—Min. en route altitude	None	W	10 mi.—1,500′ S side W crs 15 mi.—1,500′ S side W crs 20 mi.—1,500′ S side W crs 25 mi.—1,500′ S side W crs	1,200	99	10.7	237	R A T	700 1,000 500	2.0 3.0 2.0	700 1,000 500	3.0 3.0 3.0	Climb to 1,500′ on E crs.
Owens Field	E—1,500′ (SW crs Florence) SE—1,500′ (NW crs Charleston) W—1,600′ (NE crs Augusta) NW—Min. en route altitude	None	SE	10 mi.—1,500′ E side SE crs 15 mi.—1,500 'E side SE crs 20 mi.—1,500′ E side SE crs 25 mi.—1,500′ E side SE crs	1,000	330	1.9	192	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 1,500′ on NW crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
COLUMBUS, N. MEX. CAA Int. Field 284 kc; CUS SBMRAZ-DTV	N—Min. en route altitude E—8,500′ (El Paso Range) S—Min. en route altitude W—9,000′ (Rodeo Range)	None	E	10 mi.—6,500′ N side E crs 15 mi.—6,500′ N side E crs 20 mi.—6,500′ N side E crs 25 mi.—6,500′ N side E crs	5,500	202	1.9	4,024	R A T	1,000 1,500 1,000	2.0 3.0 2.0	1,000 1,500 1,000	3.0 4.0 3.0	Climb to 9,000′ on W crs within 25 mi.
COLUMBUS, OHIO Port Columbus Arpt. 391 kc; CMH SBRAZ-DTV (Procedure No. 1)	NE—2,600′ (W crs Cleveland) NE—2,500′ (Mansfield Rbn) NE—2,300′ (Mt. Liberty FM) E—2,600′ (SE crs Cleveland) E—2,300′ (Newark FM) S—2,300′ (NW crs Huntington) W—2,300′ (N crs Patterson) W—1,820′ (Hilliard FM) (final)	None	W	10 mi.—2,300′ S side W crs 15 mi.—2,300′ S side W crs 20 mi.—2,300′ S side W crs 25 mi.—2,300′ S side W crs	1,820	79	1.9	817	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,300′ on E crs within 25 mi.
(Procedure No. 2)	NE—2,600′ (W crs Cleveland) NE—2,500′ (Mansfield Rbn) NE—2,300′ (Mt. Liberty FM) E—2,600′ (SE crs Cleveland) E—1,800′ (Newark FM) (final) S—2,300′ (NW crs Huntington) W—2,300′ (N crs Patterson)	None	E	10 mi.—2,000′ N side E crs 15 mi.—2,200′ N side E crs 20 mi.—2,300′ N side E crs 25 mi.—2,300′ N side E crs	1,800 (over Summit Sta. LFM)	264 (from Summit Sta. LFM)	3.0	817	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,300′ on W crs.
Lockbourne AFB	NE—2,500′ (W crs Cleveland) E—2,500′ (SE crs Cleveland) S—2,500′ (NW crs Huntington) W—2,300′ (N crs Patterson)	None	NE	10 mi.—2,500′ N side NE crs 15 mi.—2,500′ N side NE crs 20 mi.—2,500′ N side NE crs 25 mi.—2,500′ N side NE crs	2,500	182	11.7	774	R A T	800 1,000 300	2.0 3.0 2.0	800 1,000 300	3.0 3.0 3.0	Climb to 2,500′ on S crs.
CONCORD, N. H. Concord Arpt. 261 kc; CON BMRLZ-DTV	NE—2,500′ (Portland Range) SE—2,000′ (N crs Boston) SW—5,200′ (N crs Westfield) NW—5,000′ (SW crs Montpelier)	None	SE	10 mi.—2,000′ W side SE crs 15 mi.—2,000′ W side SE crs 20 mi.—2,000′ W side SE crs 25 mi.—2,000′ W side SE crs	1,400	323	1.9	345	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.0 2.0 1.0	Climb to 5,000′ on NW crs.
CORPUS CHRISTI, TEX. Cliff Maus Field 382 kc; CRP SBMRLZ-DTV	NE—1,400′ (N crs Corpus Christi NAS) SE—Not authorized SW—1,400′ (S crs Alice) NW—1,700′ (S crs San Antonio) NW—840′ (Odem FM) (final)	None	NW	10 mi.—1,300′ W side NW crs 15 mi.—1,300′ W side NW crs 20 mi.—1,300′ W side NW crs 25 mi.—1,300′ W side NW crs	840	135	2.9	42	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Make right turn and climb to 1,400′ on SW crs within 25 mi. *Runway No. 12.
Corpus Christi NAS 248 kc; NGP SMRLZ	N—2,500′ (NE crs Corpus Christi) E—Min. en route altitude S—Min. en route altitude W—2,500′ (S crs Alice)	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	800	350	3.3	19	R (R) S A T	500 500 500 NA 300	1.5 1.0 1.0 1.0	500 500 500 NA 300	1.5 1.5 1.5 1.0	Turn right and climb to 1,800′ on E crs.
Corpus Christi NAS Seaplane Base	N—2,500′ (NE crs Corpus Christi) E—Min. en route altitude S—Min. en route altitude W—2,500′ (S crs Alice)	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	800	350	5.0	S. L.	R (R) S A T	500 500 500 NA 300	1.5 1.0 1.0 1.0	500 500 500 NA 300	1.5 1.5 1.5 1.0	Turn right and climb to 1,500′ on E crs.
Waldron NAAS	N—2,500′ (NE crs Corpus Christi) E—Min. en route altitude S—Min. en route altitude W—2,500′ (S crs Alice)	None	E	10 mi.—1,500′ S side E crs 15 mi.—1,500′ S side E crs 20 mi.—1,500′ S side E crs 25 mi.—1,500′ S side E crs	800	250	1.0	25	R S A T	(*) (*) (*) (*)		(*) (*) (*) (*)		Turn left and climb to 1,500′ on S crs. *This airfield is inactive; procedure is for daylgt emergency use only.
CRAIG (SELMA), ALA. Craig AFB 272 kc; SEM BMRLZ	NE—2,000′ (S crs Birmingham) SE—2,000′ (SW crs Maxwell) SW—2,000′ (N crs Pensacola) NW—2,000′ (SW crs Birmingham) NW—1,200′ (W crs Maxwell)	SW crs	SE	10 mi.—1,200′ E side SE crs 15 mi.—1,700′ E side SE crs 20 mi.—1,700′ E side SE crs 25 mi.—1,700′ E side SE crs	1,000	320	2.5	166	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	1.5 1.5 1.5 3.0 1.0	Climb to 2,000′ on NW crs within 25 mi.
CRESTVIEW, FLA. Crestview Arpt. 350 kc; CEW SBRAZ-DTV	N—Min. en route altitude E—1,400′ (SE crs Dothan) S—1,400′ (Eglin Range) W—1,200′ (N crs Pensacola)	None	E	10 mi—1,200′ N side E crs 15 mi—1,200′ N side E crs 20 mi—1,200′ N side E crs 25 mi—1,200′ N side E crs	800	263	2.6	274	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,200′ on W crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)		Ceiling and visibility minimums: Day Ceiling (ft.)	Day Visibility (mi.)	Night Ceiling (ft.)	Night Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
CROSS CITY, FLA. Cross City Arpt. 269 kc; CTY SBMRAZ-DTV	NE—1,200′ (W crs Jacksonville) SE—1,200′ (N crs Tampa) SW—Min. en route altitude NW—1,200′ (E crs Tallahassee)	None	SE	10 mi.—1,100′ E side SE crs 15 mi.—1,100′ E side SE crs 20 mi.—1,100′ E side SE crs 25 mi.—1,100′ E side SE crs	600	315	2.7	42	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,200′ on NW crs. *Runway No. 31.
CUT BANK, MONT. Cut Bank Arpt. 263 kc; CTB SBMRAZ-DTV	E—Min. en route altitude SE—6,500′ (Great Falls Range) W—12,000′ (E crs Kalispell) N—6,500′ (Lethbridge Range)	None	SE	10 mi.—5,500′ E side SE crs 15 mi.—5,500′ E side SE crs 20 mi.—5,500′ E side SE crs 25 mi.—5,500′ E side SE crs	4,800	327	3.1	3,900	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	2.0 2.0 2.0 1.0	Climb to 6,500′ on N crs within 17 mi.
DAGGETT, CALIF. CAA Int. Field 365 kc; DAG SBRAZ-DTV	N—Min. en route altitude E—9,000′ (Needles Range) SE—Min. en route altitude W—6,000′ (Palmdale Range)	None	W	10 mi.—6,000′ N side W crs 15 mi.—6,000′ N side W crs 20 mi.—6,000′ N side W crs 25 mi.—6,000′ N side W crs	3,900	75	.0	1,927	R A T	2,000 2,000 2,000	4.0 4.0 4.0	2,000 2,000 2,000	4.0 4.0 4.0	If not contact over range, climb to 9,000′ on E crs.
DALLAS, TEXAS Love Field 341 kc; DAL SBMRAZ-DTV	N—2,000′ (N crs Ft. Worth) N—1,700′ (E crs Ft. Worth) E—2,000′ (NW crs Tyler) S—1,900′ (Waco Range) S—1,380′ (Duncanville FM) (final) S—1,200′ (Dallas FM) (final) W—2,100′ (S crs Ft. Worth)	None	S	10 mi.—1,900′ E side S crs 15 mi.—1,900′ E side S crs 20 mi.—1,900′ E side S crs 25 mi.—1,900′ E side S crs	**1,380	356	2.2	483	R (R) #S A T	500 500 *500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 2,000′ on N crs within 25 mi., or alternate procedure (when directed by ATC), turn right, climb to 2,000′ on track of 45°, intersect and proceed E on E crs Ft. Worth. *When Dallas FM not received, regular minima required. **If Dallas FM received, altitude over range on final approach is 1,200′. #Runway No. 36.
Dallas NAS (Hensley)	N—2,000′ (N crs Ft. Worth) N—1,700′ (E crs Ft. Worth) E—2,000′ (NW crs Tyler) S—1,900′ (Waco Range) S—1,380′ (Duncanville FM) (final) S—1,200′ (Dallas FM) (final) W—2,100′ (S crs Ft. Worth)	None	S	10 mi.—1,900′ E side S crs 15 mi.—1,900′ E side S crs 20 mi.—1,900′ E side S crs 25 mi.—1,900′ E side S crs	#1,380	*356	*2.2	495	R A T	600 800 300	2.0 2.0 1.0	600 800 300	3.0 2.0 2.0	If not contact over Love Fld, climb to 2,000′ on N crs. If unable to maintain visual reference en route to Dallas NAS, or if landing not accomplished, climb to 2,000′ on heading of 211° and advise Dallas AC. *To Love Fld; fly by visual reference from Love Fld to Dallas NAS—211°, 8.2 mi. #If Dallas FM rec'd, alt. over rng is 1,200′.
DAVENPORT, IOWA Davenport Arpt.	(SEE MOLINE, ILL.)													
DAYTON, OHIO Dayton Arpt.* 243 kc; DAY SBMLZ-DTV	N—2,100′ (W crs Columbus) E—2,100′ (N crs Patterson) S—2,300′ (NW crs Cincinnati) W—2,300′ (E crs Indianapolis) W—1,700′ (Verona FM) (final)	None	W	10 mi.—2,200′ S side W crs 15 mi.—2,200′ S side W crs 20 mi.—2,200′ S side W crs 25 mi.—2,200′ S side W crs	1,700	90	3.2	1,007	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,100′ on N crs. *Runway No. 9.
DAYTONA BEACH, FLA. Daytona Beach Arpt. 236 kc; DAB SBMLZ-DTV	N—Min. en route altitude E—1,200′ (NE crs Orlando) S—1,500′ (Orlando Range) W—Min. en route altitude	None	W	10 mi.—1,100′ S side W crs 15 mi.—1,200′ S side W crs 20 mi.—1,200′ S side W crs 25 mi.—1,200′ S side W crs	600	101	2.1	34	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,200′ on E crs. *Runway No. 10.
DELTA, UTAH Delta Arpt. 212 kc; DTA SBRAZ-DTV	NE—11,000′ (Fairfield Range) SE—Min. en route altitude SW—9,000′ (Milford Range) NW—Min. en route altitude	None	SW	10 mi.—7,000′ W side SW crs 15 mi.—7,000′ W side SW crs 20 mi.—8,000′ W side SW crs 25 mi.—8,000′ W side SW crs	5,800	25	3.5	4,755	R A T	800 800 500	2.0 2.0 1.0	800 800 500	2.0 2.0 2.0	Climb to 11,000′ on NE crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
DENVER, COLO. Stapleton Field 379 kc; DEN SBMRAZ-DTXV (Procedure No. 1)	N—7,500′ (Cheyenne Range) N—6,300′ (Dacono FM) N—5,900′ (Henderson FM) (final) E—6,600′ (Akron Range) E—6,300′ (Watkins FM) S—8,900′ (Colorado Springs Range). S—7,000′ (Franktown FM) (N bound only)* W—Min. enroute altitude W—10,000′ (Superior FM) (E bound only).	On N crs within 25 mi.†	N	10 mi.—6,300′ E side N crs** 15 mi.—6,300′ E side N crs** 20 mi.—6,300′ E side N crs** 25 mi.—6,300′ E side N crs**	5,900	162	2.4	5,325	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 8,900′ on S crs, or alternate procedure (when directed by ATC), climb to 8,500′ intercepting an ADF track of 120° mag. outbound from Denver LF rng within 25 mi. *6,715′ MSL, hill, 6 mi. NW of Franktown. **All procedure turns on N crs of Denver will be made to E on account of high terrain to the W. CAUTION: 5,579′ and 5,813′ MSL twrs, 6 and 8 mi. W of N crs. †When directed by ATC, shuttle approved on S crs between rng and Aurora FM, or on E crs within 25 mi. #Runway No. 17.
(Procedure No. 2)	N—7,500′ (Cheyenne Range) N—6,300′ (Daconno or Henderson FM)* E—6,600′ (Akron Range)* E—6,300′ (Watkins FM)* S—8,900′ (Colorado Springs Range) S—7,000′ (Franktown FM) (N bound only)** W—Min. en route altitude W—10,000′ (Superior FM) (E bound only)	On N crs within 25 mi.†	S	10 mi.—7,000′ E side S crs # 15 mi.—NA 20 mi.—NA 25 mi.—NA	7,000 (Over Aurora FM)	335 (From Aurora FM)	6.5	5,325	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 8,600′ on E crs, or alternate procedure (when directed by ATC), climb to 8,500′ intercepting an ADF track of 120° mag. outbound from Denver LF range within 25 mi. *If contact not estab. at Denver range, climb to 8,900′ on S crs. **CAUTION: 6,715′ MSL hill, 6 mi. NW of Franktown. #All procedure turns must be accomplished within 10 mi. on account of high terrain to the S. †When directed by ATC, shuttle approved on S crs between range and Aurora FM, or on E crs within 25 mi.
Lowry AFB	N—7,500′ (Cheyenne Range) N—6,300′ (Daconno FM) N—5,900′ (Henderson FM) (final) E—6,600′ (Akron Range) E—6,300′ (Watkins FM) S—8,900′ (Colorado Springs Range) S—7,000′ (Franktown (FM) (N bound only)** W—Min. en route altitude W—10,000′ (Superior FM) (E bound only)	N crs within 25 mi.†	N	10 mi.—6,300′ E side N crs* 15 mi.—6,300′ E side N crs* 20 mi.—6,300′ E side N crs* 25 mi.—6,300′ E side N crs*	5,900	166	5.0	5,420	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 2.0	500 500 500 800 300	1.5 1.0 1.5 2.0 3.0	Climb to 8,900′ on S crs, or alternate procedure (when directed by ATC), climb to 8,500′, intersecting an ADF track of 120° mag. outbound from Denver LF range within 25 mi. *All procedure turns on N crs Denver will be made to E on account of high terrain to W. **6,715′ MSL hill, 6 mi. NW of Franktown. †When directed by ATC, shuttle approved on S crs between range and Aurora FM, or on E crs within 25 mi. #Runway No. 16.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
DENVER, COLO. Denver NAS (Buckley)	(MAKE LET-DOWN TO STAPLETON FIELD. FLY VFR FROM STAPLETON FIELD TO DENVER NAS)													
DES MOINES, IOWA Des Moines Arpt. 212 kc; DSM SBRAZ-DTV	N—2,500′ (SW crs LaCrosse) E—2,200′ (N crs Burlington) S—2,500′ (NW crs Kirksville) S—1,600′ (Indianola FM) (final) W—2,500′ (Omaha Range)	None	S	10 mi.—2,100′ E side S crs* 15 mi.—2,100′ E side S crs* 20 mi.—2,200′ E side S crs* 25 mi.—2,200′ E side S crs*	*1,600	344	2.2	957	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,800′ on N crs, or alternate procedure (when directed by ATC), climb to 2,500′ on W crs. *If procedure turn is accomplished beyond 10 mi., descent to final approach alt NA until after passing Indianola FM. #Runway No. 35.
DETROIT, MICH. Detroit-Wayne Major Arpt. 288 kc; RML SBRAZ-DTV	E—2,300′ (SW crs Windsor) SE—2,300′ (SW crs Windsor) W—2,000′ (SE crs Lansing) W—1,500′ (Willow Run LOM) (final) N—2,200′ (E crs Lansing)	None	W	10 mi.—1,900′ S side W crs 15 mi.—2,000′ S side W crs 20 mi.—2,000′ S side W crs 25 mi.—2,000′ S side W crs	1,500	99	2.5	634	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,300′ on E crs.
Willow Run Arpt	E—2,300′ (SW crs Windsor) SE—2,300′ (SW crs Windsor) W—2,000′ (SE crs Lansing) N—2,200′ (E crs Lansing)	None	E	10 mi.—2,300′ N side E crs 15 mi.—2,300′ N side E crs 20 mi.—2,300′ N side E crs 25 mi.—2,300′ N side E crs	1,800	274	6.2	716	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000′ on W crs. #Runways No. 27L & 27R.
Detroit City Arpt.	(SEE WINDSOR, ONT.)													
DICKINSON, N. DAK. Dickinson Arpt. 308 kc; DIK SBMRAZ-DTV	N—Min. en route altitude E—3,800′ (Bismarck Range) S—Min. en route altitude W—4,200′ (Mile City Range)	None	N	10 mi.—3,800′ W side N crs 15 mi.—4,000′ W side N crs 20 mi.—4,000′ W side N crs 25 mi.—4,000′ W side N crs	3,260	167	2.1	2,589	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.5 2.0 1.0	Climb to 4,000′ on S crs within 20 mi.
DILLON, MONT. Dillon Arpt. 379 kc; DLN BMRLZ-DTV	NE—10,500′ (Whitehall Range) E—Min. en route altitude S—11,500′ (Dubois Range) W—min. en route altitude	None	NE	10 mi.—8,000′ W side NE crs 15 mi.—9,000′ W side NE crs 20 mi.—9,000′ W side NE crs 25 mi.—9,000′ W side NE crs	7,240	169	2.1	5,238	R A T	2,000 2,000 1,000	3.0 3.0 1.0	2,000 2,000 1,000	3.0 3.0 2.0	Turn right and climb to 11,500′ on NE crs within 25 mi.
DOTHAN, ALA. Dothan Arpt. 233 kc; DHN SBRAZ-DTV	NE—Min. en route altitude SE—1,400′ (E crs Crestview) SW—1,500′ (Crestview Range) NW—1,600′ (E crs Maxwell)	None	SW	10 mi.—1,400′ S side SW crs 15 mi.—1,400′ S side SW crs 20 mi.—1,400′ S side SW crs 25 mi.—1,400′ S side SW crs	900	47	3.7	330	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500′ on NE crs within 25 mi. *Runway No. 5.
Napier Field	(MAKE LET-DOWN TO DOTHAN ARPT. FLY VFR FROM DOTHAN ARPT TO NAPIER FIELD)													
DOUGLAS, ARIZ. Bisbee-Douglas Intn'l Arpt. 379 kc; DUG BMRLZ-DTV	NE—11,000′ (SW crs Rodeo) SE—Min. en route altitude SW—Min. en route altitude NW—10,000′ (W crs Cochise)	SE crs to 9,000′	NW	10 mi.—7,000′ N side NW crs 15 mi.—7,000′ N side NW crs 20 mi.—9,000′ N side NW crs 25 mi.—9,000′ N side NW crs	6,250	120	3.5	4,158	R A T	2,100 2,100 500	2.5 3.0 1.0	2,100 2,100 500	3.5 3.5 2.0	If not contact over range, climb to 10,000′ on NW crs. No standby equipment.
DRUMMOND, MONT. CAA Int. Field 296 kc; DRU SBMRAZ-DTV	NE—Min. en route altitude E—9,000′ (N crs Butte) SW—10,500′ (NW crs Butte) W—9,000′ (Missoula Range)	None	NE	10 mi.—9,000′ W side NE crs 15 mi.—9,000′ W side NE crs 20 mi.—9,000′ W side NE crs 25 mi.—9,000′ W side NE crs	7,700	185	1.3	4,300	R A T	3,000 3,000 1,500	2.0 2.0 2.0	3,000 3,000 1,500	3.0 3.0 2.0	Climb to 9,000′ on SW crs within 15 mi.
DUBOIS, IDAHO Dubois Arpt. 388 kc; DBS SBRAZ-DTV	N—11,500′ (Dillon Range) E—Min. en route altitude S—7,500′ (Idaho Falls Range) W—Min. enroute altitude	To 7,000′ on S. crs	S	10 mi.—7,000′ E side S crs 15 mi.—7,000′ E side S crs 20 mi.—7,000′ E side S crs 25 mi.—7,000′ E side S crs	6,000	332	1.7	5,123	R A T	800 800 500	1.0 2.0 1.0	800 800 500	2.0 2.0 2.0	Make immediate left turn and climb to 7,500′ on S crs within 25 mi.
DULUTH, MINN. Duluth Arpt. 221 kc; DLH SBRAZ-DTV	N—Min. en route altitude E—Min. en route altitude S—2,600′ (Minneapolis Range) W—3,100′ (Fargo Range)	None	S	10 mi.—2,600′ E side S crs 15 mi.—2,600′ E side S crs 20 mi.—2,600′ E side S crs 25 mi.—2,600′ E side S crs	2,100	13	3.9	1,430	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,800′ on N crs. *Runway No. 3.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
EFFINGHAM, ILL. CAA Int. Field 275 kc; EFF BMRLZ-DTV	N—Min. en route altitude E—2,000' (Terre Haute Range) S—2,000' (W crs Evansville) W—2,000' (NW crs Scott with E—crs St. Louis)	None	E	10 mi.—1,600' N side E crs 15 mi.—2,000' N side E crs 20 mi.—2,000' N side E crs 25 mi.—2,000' N side E crs	1,400	251	0.0	608	R A T	800 1,000 300	2.0 3.0 2.0	800 1,000 300	2.0 3.0 2.0	Climb to 2,000' on W crs.
EGLIN (VALPARISO), FLA. Eglin AFB 209 kc; VPS BMRLPZ	N—1,500' (SW crs Maxwell) E—1,200' (NW crs Tyndall) S—1,200' (SW crs Tyndall) W—1,300' (NE crs Pensacola)	None	N	10 mi.—1,400' W side N crs 15 mi.—1,400' W side N crs 20 mi.—1,400' W side N crs 25 mi.—1,500' W side N crs	800	178	2.9	71	R A T	500 1,500 300	2.0 3.0 2.0	500 1,500 300	3.0 3.0 3.0	Climb to 1,200' on S crs.
EL CENTRO, CALIF. El Centro NAS 212 kc; ELC SBMRLZ—DTV	E—3,000' (Yuma Range) SE—Min. en route altitude W—9,000' (San Diego Range) NW—9,000' (Thermal Range)	On E crs to 1,500' within 25. mi	E	10 mi.—1,000' N side E crs 15 mi.—1,500' N side E crs 20 mi.—1,500' N side E crs 25 mi.—1,500' N side E crs	560	320	2.8	−43	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.0 1.0	If not contact over rng, make 180° right (N.) turn and climb to 1,500' on E crs within 15 mi.
ELIZABETH CITY, N. C.	(SEE WEEKSVILLE, N. C.)													
ELKINS, W. VA. Elkins Arpt 281 kc; EKN SBMRAZ-DTV	N—5,300' (SW crs Morgantown) E—6,800' (SE crs Morgantown) S—6,000' (N crs Lynchburg) W—5,700' (SW crs Morgantown)	None	N	10 mi.—4,500' W side N crs 15 mi.—4,500 W side N crs 20 mi.—4,500' W side N crs 25 mi.—4,500' W side N crs	4,500	174	2.9	1,988	R A T	2,000 2,000 1,000	2.0 3.0 2.0	NA NA NA		Climb to 6,000' on S crs within 25 mi.
ELKO, NEV. Elko Arpt. 291 kc; EKO SBRAPZ-DTV	NE—12,000' (W crs Lucin) NE—9,000' (Deeth FM) S—Min. en route altitude W—11,000' (Battle Mountain Range) W—7,600' (Carlin FM) (final)	None	W	10 mi.—8,500' N side W crs 15 mi.—8,500' N side W crs 20 mi.—10,000' N side W crs 25 mi.—10,000' N side W crs	7,600	176	.9	5,136	R A T	2,500 2,500 1,000	3.0 3.0 2.0	2,500 2,500 1,000	3.0 3.0 3.0	If not contact over rng, climb to 9,000' on NE crs bet range and Deeth FM making procedure turn on N side of NE crs.
ELLENSBURG, WASH. Bowers Field 385 kc; ELN SBMRAZ-DTV	N.—Min. en route altitude E—7,000' (Ephrata Range) S—5,500' (NW crs Yakima) W—8,000' (Seattle Range)	None	E	10 mi.—5,500' *S side E crs 15 mi.—6,000' *S side E crs 20 mi.—6,000' *S side E crs 25 mi.—6,000' *S side E crs	4,000	279	2.0	1,766	R A T	1,500 1,500 1,000	2.0 3.0 1.0	1,500 1,500 1,000	2.0 3.0 1.0	Make left turn and climb to 7,000' on E crs. Climb to 5,500' within 10 mi. *Procedure turn not authorized on N side of E crs due to high terrain.
ELMIRA, N. Y. Chemung Co. Arpt. 385 kc; ELM SBRAZ-DTV	NE—3,500' (S crs Syracuse) SE—3,200' (N crs Wilkes-Barre) SW—4,500' (W crs Williamsport) NW—3,500' (S crs Rochester)	None	SW	10 mi.—3,400' S side SW crs 15 mi.—3,400' S side SW crs 20 mi.—3,400' S side SW crs 25 mi.—3,400' S side SW crs	2,900	75	1.7	951	R A T	1,300 2,000 1,000	2.0 2.0 2.0	NA NA NA		Climb to 3500' on NE crs.
EL PASO, TEX. El Paso International Arpt. 242 kc; ELP SBRAZ-DTV	N—Min. en route altitude N—5,000' (Newman Rbn) E—8,000' (Salt Flat Range) E—5,000' (Hueco Mountain FM) (final)* S—Min. en route altitude S—5,000' (Clint Rbn) W—8,500' (Columbus Range) W—6,500' (SW crs ILS)	None	E	10 mi.—6,500' S side E crs 15 mi.—6,500' S side E crs 20 mi.—8,000' S side E crs 25 mi.—8,000' S side E crs	5,000	257	4.5	3,936	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Make left turn to 125° mag to intersect S crs; climb to 6,000' on S crs within 25 mi. Maintain 8,000' until 5 mi W of Hueco Mtn. FM. CAUTION: High terrain on N side of E course.
Biggs AFB	(MAKE LET-DOWN TO EL PASO INTERNATIONAL ARPT. FLY VFR FROM EL PASO INTERNATIONAL ARP TO BIGGS AFB.)							3,937	R A T	500 1,500 300	2.0 3.0 2.0	500 1,500 300	2.0 3.0 3.0	
EL TORO, CALIF.	(NO PROCEDURE APPROVED)													
ENTERPRISE, UTAH CAA Int. Field 341 kc; ENP SBMRAZ-DTV	NE—9,000' (Milford Range) SE—Min. en route altitude. SW—10,000' (Las Vegas Range) NW—Min. en route altitude.	None	NE	10 mi.—8,500 W side NE crs 15 mi.—8,500 W side NE crs 20 mi.—8,500' W side NE crs 25 mi.—8,500 'W side NE crs	7,200	25	2.3	5,204	R A T	2,000 2,000 1,000	4.0 4.0 2.0	2,000 2,000 1,000	4.0 4.0 2.0	If not contact over rng, turn left and climb to 9,000' on NE crs. High terrain 3 mi S and W, also 6 mi. E of rng.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
EPHRATA, WASH. Ephrata Arpt. 209 kc; EPH BMRLZ-DTV	N—Min. en route altitude E—5,000′ (Spokane Range) E—3,500′ (Harrington FM) S—Min. en route altitude W—7,000′ (Ellensburg Range)	None	S	10 mi.—2,500′ E side S crs 15 mi.—2,500′ E side S crs 20 mi.—2,500′ E side S crs 25 mi.—2,500′ E side S crs	2,060	336	3.9	1,258	R A T	800 800 300	2.0 2.0 1.0	NA NA NA		Turn right and climb to 3,500′ on E crs within 25 mi.
Larson AFB	(MAKE LET-DOWN TO EPHRATA ARPT. FLY VFR FROM EPHRATA ARPT. TO LARSON AFB.)													
ERIE, PA. Port Erie Arpt. 371 kc; ERI SBMRAZ-DTV	NE—2,500′ (SW crs Buffalo) S—2,500′ (E crs Youngstown) SW—2,200′ (N crs Youngstown) SW—1,500′ (No. Springfield FM) (final). N—2,000′ (E crs Clear Creek)	None	SW	10 mi.—2,000′ S side SW crs 15 mi.—2,000′ S side SW crs 20 mi.—2,000′ S side SW crs 25 mi.—2,000′ S side SW crs	1,500	64	2.0	732	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,500′ on NE crs.
EUGENE, OREG. Mahlon Sweet Arpt. 251 kc; EUG SBRAZ-DTV	N—3,000′ (Portland Range) E—Min. en route altitude S—6,500′ (Medford Range) S—4,000′ (Cottage Grove FM) W—Min. en route altitude	None	N	10 mi.—3,000′ W side N crs 15 mi.—3,000′ W side N crs 20 mi.—3,000′ W side N crs 25 mi.—3,000′ W side N crs	1,500	150	4.5	365	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn right and climb to 3,000′ on N crs within 25 mi. 1,000′ terrain 6 mi. W of N crs, 6 mi. N of range. *Rwy. No. 16.
EVANSVILLE, IND. Evansville Arpt. 233 kc; EVV SBRAZ-DTV	N—1,900′ (Terre Haute Range) N—1,300′ (Princeton FM) (final) E—2,000′ (Louisville Range) S—2,000′ (NW crs Nashville) W—2,000′ (S crs Effingham)	None	N	10 mi.—1,800′ W side N crs 15 mi.—1,900′ W side N crs 20 mi.—1,900′ W side N crs 25 mi.—1,900′ W side N crs	1,300	177	5.7	389	R (R) S* A T	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	600 600 600 800 300	2.0 2.0 1.5 2.0 1.0	Climb to 2,000′ on S crs. *Runway No. 18.
EVERETT, WASH. Paine Field 371 kc; EVE BMRLZ-DTV	N—4,000′ (Bellingham Range) N—3,000′ (SE crs Patricia Bay) E—Min. en route altitude E—2,000′ (NE crs Seattle) S—3,600′ (Seattle Range) W—Min. en route altitude W—2,000′ (NW crs Seattle)	None	S	10 mi.—1,600′ W side S crs 15 mi.—1,600′ W side S crs 20 mi.—2,000′ W side S crs 25 mi.—3,000′ W side S crs	1,400	325	3.1	602	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	700 500 500 800 300	2.0 1.0 1.0 2.0 1.0	Climb to 3,000′ on N crs within 25 mi. Procedure turns on E side of S crs not authorized due to high terrain. *Rwy. No. 34.
FAIRFIELD, UTAH	(NO PROCEDURE AUTHORIZED)													
FAIRFIELD-SUISUN (FAIRFIELD), CALIF. Fairfield-Suisun AFB 248 kc; SUU SBRAZ-D	NE—2,500′ (NW crs Sacramento) S—5,000′ (W crs Stockton) SW—4,000′ (NW crs Oakland) N—4,000′ (Williams Range)	None	NE	10 mi.—2,500′ N side NE crs 15 mi.—2,500′ N side NE crs 20 mi.—2,500′ N side NE crs 25 mi.—2,500′ N side NE crs	1,000	210	4.5	58	R A T	500 1,000 300	1.5 3.0 2.0	500 1,000 300	2.0 3.0 3.0	Climb to 4,000′ on SW crs.
FALLON, NEV. Fallon Arpt. 230 kc; FFN SBRAZ-DTV	NE—Min. en route altitude SE—Min. en route altitude SW—Min. en route altitude NW—9,000′ (NE crs Reno)	NW crs to 6,500′ within 25 mi.	NE	10 mi.—6,500′ N side NE crs 15 mi.—7,000′ N side NE crs 20 mi.—11,000′ N side NE crs 25 mi.—11,000′ N side NE crs	5,500	225	2.4	3,930	R A T	1,000 1,000 500	2.0 2.0 1.0	NA NA NA		Climb to 8,500′ on SW crs within 25 mi.
FARGO, N. DAK. Fargo Arpt. 365 kc; FAR SBRAZ-DTV (Procedure No. 1)	E—3,100′ (Duluth Range) E—1,600′ (Glyndon FM) (final) SE—2,800′ (Alexandria Range) SE—2,300′ (Barnesville FM) W—2,600′ (Jamestown Range) W—2,200′ (Wheatland FM) N—2,300′ (Grand Forks Range)	None	E	10 mi.—2,100′ N side E crs 15 mi.—2,200′ N side E crs 20 mi.—2,300′ N side E crs 25 mi.—2,400′ N side E crs	1,600	278	.9	900	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,600′ on W crs.
(Procedure No. 2)	E—3,100′ (Duluth Range) E—2,[illegible]′ (Glyndon FM) SE—2,800′ (Alexandria Range) SE—2,300′ (Barnesville FM) W—2,600′ (Jamestown Range) W—1,700′ (Wheatland FM) (final) N—2,300′ (Grand Forks Range)	None	W	10 mi.—2,200′ S side W crs 15 mi.—2,200′ S side W crs 20 mi.—2,200′ S side W crs 25 mi.—2,300′ S side W crs	1,700 (over W. Fargo LFM)	83 (from W. Fargo LFM)	3.8	900	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,600′ on E crs.
FAYETTEVILLE, N. C. Grannis Arpt.	(SEE POPE (FT. BRAGG), N. C.)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
FLORENCE, S. C. Florence Arpt. 220 kc; FLO SBMRLZ-DTV	NE—1,400' (Lumberton Int.) SE—Min. en route altitude SE—1,200' (NW crs Myrtle Beach VAR) SW—1,500' (E crs Columbia) NW.—Min. en route altitude	None	SE	10 mi.—1,200' N side SE crs 15 mi.—1,200' N side SE crs 20 mi.—1,200' N side SE crs 25 mi.—1,200' N side SE crs	800	292	1.7	146	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on NW crs within 25 mi.
FLOYD BENNETT, N. Y.	(SEE NEW YORK, N. Y.)													
FORT BRIDGER, WYO. CAA Int. Field 382 kc; FBR SBRAZ-DTV	N—10,000' (E crs Malad City) E—10,000' (Rock Springs Range) S—Min. en route altitude W—12,000' (Ogden Range)	None	E	10 mi.—8,500' N side E crs* 15 mi.—9,000' N side E crs* 20 mi.—9,000' N side E crs* 25 mi.—9,000' N side E crs*	*7,500	242	1.8	7,016	R A T	500 800 300	1.0 2.0 1.0	500 800 300	2.0 2.0 1.0	Turn N and climb to 10,000' on E crs within 25 mi. *If procedure turn is accomplished beyond 10 mi., alt. on final approach is 8,000'. Minimums are for acft having stall speeds of 75 mph or less, NA for acft with higher stall speeds.
FORT JONES, CALIF.	(NO ASSOCIATED AIRPORT)													
FORT MYERS, FLA. Page Field 341 kc; FMY SBRAZ-DTV	NE—Min. en route altitude SE—1,200' (W crs Miami) SW—Min. en route altitude NW—1,500' (Tampa Range)	None	SW	10 mi.—1,200' S side SW crs 15 mi.—1,200' S side SW crs 20 mi.—1,200' S side SW crs 25 mi.—1,200' S side SW crs	700	38	3.7	17	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 1,200' on NE crs.
FORT WAYNE, IND. Baer Field 251 kc; FWA SBMRLZ-DTV	NE—2,300' (E crs Goshen) SE—2,200' (Dayton Range) SW—2,000' (S crs Goshen) SW—1,500' (Markle FM) (final) NW—2,300' (E crs Goshen)	None	SW	10 mi.—2,000' S side SW crs 15 mi.—2,000' S side SW crs 20 mi.—2,000' S side SW crs 25 mi.—2,000' S side SW crs	1,500	51	3.5	801	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,300' on NE crs. *Runway No. 4.
FORT WORTH, TEX. Meacham Field 365 kc; FTW SBRAZ-DTXV (Procedure No. 1)	N—2,000' (S crs Okla. City) N—1,400' (Haslet FM) (final) E—2,200' (N crs Dallas) S—2,500' (NW crs Waco) W—2,500' (Mineral Wells Rbn)	None	N	10 mi.—2,000' W side N crs 15 mi.—2,000' W side N crs 20 mi.—2,000' W side N crs 25 mi.—2,000' W side N crs	1,400	177	2.4	692	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,000' on S crs prior to crossing Joshua FM. CAUTION: 950' MSL building located on final approach path, 0.6 mi. S of range. *Rwy No. 17.
(Procedure No. 2)	N—2,000' (S crs Okla. City) E—2,200' (N crs Dallas) S—2,000' (NW crs Waco) S—1,500' (Joshua FM) (final) W—2,500' (Mineral Wells Rbn)	None	S	10 mi.—2,000' E side S crs 15 mi.—2,000' E side S crs 20 mi.—2,000' E side S crs 25 mi.—2,000' E side S crs	1,500 (over Ft. Worth LFM)	357 (from Ft. Worth LFM)	2.6	692	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,000' on N crs before crossing Haslet FM. *Rwy No. 35.
Carswell AFB	N—2,000' (S crs Okla. City) N—1,400' (Haslet FM) E—2,200' (N crs Dallas) S—2,000' (NW crs Waco) W—2,500' (Mineral Wells Rbn)	None	N	10 mi.—2,000' W side N crs 15 mi.—2,000' W side N crs 20 mi.—2,000' W side N crs 25 mi.—2,000' W side N crs	1,400	211	6.8	636	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 2.0	Turn to a heading of 214°, climb to 2,000' and contact Ft. Worth Approach Control for further instructions.
FRESNO, CALIF. Chandler Arpt. 344 kc; FNO SBRAZ-DTV (Procedure No. 1)	NE—Min. en route altitude SE—3,000' (Bakersfield Range) SE—800' (Bowles FM) (final) SW—Min. en route altitude W—6,000' (NE crs Salinas) W—3,000 (Los Banos FM) (inbound only)	None	SE	10 mi.—1,500' W side SE crs 15 mi.—1,500' W side SE crs 20 mi.—1,500' W side SE crs 25 mi.—1,500' W side SE crs	800	23	.9	280	R A T	500 800 300	2.0 2.0 1.0	500 800 300	2.0 2.0 1.0	If not contact over range, climb to 2,000' on W crs within 20 mi. NOTE: Minima apply to acft with stall speeds of 75 mph or less.
(Procedure No. 2)	NE—Min. en route altitude SE—3,000' (Bakersfield Range) SE—1,300' (Bowles FM) SW—Min. en route altitude W—6,000' (N E crs Salinas) W—3,000' (Los Banos FM)	None	W	10 mi.—1,500' S side W crs 15 mi.—1,500' S side W crs 20 mi.—1,500' S side W crs 25 mi.—1,500' S side W crs	800	23	.9	280	R A T	500 800 300	2.0 2.0 1.0	500 800 300	2.0 2.0 1.0	If not contact over range, climb to 1,500' on SE crs within 25 mi. NOTE: Minima apply to acft with stall speeds of 75 mph or less.
Furlong Field	(MAKE LET-DOWN TO CHANDLER ARPT. FLY VFR FROM CHANDLER ARPT TO FURLONG FIELD.)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
FRONT ROYAL, VA.	(NO ASSOCIATED AIRPORT)													
GAGE, OKLA. Gage Arpt. 266 kc; GAG SBRAZ-DTV	NE—3,800' (Wichita Range) SE—3,500' (W crs Okla. City) SW—4,500' (E crs Amarillo) NW—4,000' (S crs Garden City)	None	NE	10 mi.—3,600' N side NE crs 15 mi.—3,600' N side NE crs 20 mi.—3,600' N side NE crs 25 mi.—3,600' N side NE crs	3,100	226	2.0	2,223	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 4,500' on SW crs.
GALVESTON, TEX. Galveston Arpt. 263 kc; GLS SBMRLZ-DTV	NE—Min. en route altitude SE—Min. en route altitude SW—Min. en route altitude NW—1,300' (Houston Range) NW—800' (Webster FM) (Final)	None	NW	10 mi.—1,300' W side NW crs 15 mi.—1,300' W side NW crs 20 mi.—1,300' W side NW crs 25 mi.—1,300' W side NW crs	800	125	4.2	7	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,300' on SE crs within 25 mi.
GARDEN CITY, KANS. Garden City Arpt. 257 kc; GCK SBRAZ-DTV	N—Min. en route altitude E—4,000' (Hutchinson Range) S—4,000' (NW crs Gage) W—5,500' (La Junta Range)	None	N	10 mi.—4,000' W side N crs 15 mi.—4,000' W side N crs 20 mi.—4,000' W side N crs 25 mi.—4,000' W side N crs	3,500	168	3.7	2,842	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 4,000' on S crs within 25 mi.
Garden City Mun Arpt.	N—Min. en route altitude E—4,000' (Hutchinson Range) S—4,000' (NW crs Gage) W—5,500' (La Junta Range)	None	N	10 mi.—4,000' W side N crs 15 mi.—4,000' W side N crs 20 mi.—4,000' W side N crs 25 mi.—4,000' W side N crs	3,500	132	7.5	2,895	R A (T)	500 800 300	1.5 2.0 1.0	500 800 300	2.0 2.0 1.0	Climb to 4,000' on S crs.
GILA BEND, ARIZ. Gila Bend AF Aux. 257 kc; GBN SBRAZ-DTV	N—5,000' (W crs Phoenix) E—5,000' (S crs Phoenix) S—Min. en route altitude W—4,000' (Yuma Range)	None	W	10 mi.—3,000' S side W crs 15 mi.—3,000' S side W crs 20 mi.—4,000' S side W crs 25 mi.—4,000' S side W crs	1,600	166	4.0	858	R (R) S* A T	700 700 700 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		If not contact over range, climb to 5,000' on E crs. *Runway No. 17.
GLENVIEW, ILL. Glenview NAS 269 kc; NBU SMLZ	NE—Min. en route altitude SE—2,300' (NE crs Chicago) SW—2,500' (SE crs Rockford) NW—2,000' (E crs Rockford)	None	NW	10 mi.—1,900' N side NW crs 15 mi.—2,000' N side NW crs 20 mi.—2,000' N side NW crs 25 mi.—2,000' N side NW crs	1,400	140	3.0	682	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Turn left and climb to 2,000' on NW crs within 25 mi.
GLYNCO, GA. Glynco NAS	(NO PROCEDURE APPROVED)													
GODMAN (FT. KNOX), KY. Godman AFB 323 kc; FTK BMRLZ	N—2,500' (W crs Louisville) E—2,500' (S crs Louisville) S—2,500' (NE crs Bowling Green) W—Min. en route altitude	None	N	10 mi.—2,500' W side N crs 15 mi.—2,500' W side N crs 20 mi.—2,500' W side N crs 25 mi.—2,500' W side N crs	1,500	172	3.1	753	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,500' on S crs. *Runway No. 17.
GORDONSVILLE, VA. CAA Int. Field 388 kc; GVE SBMRAZ-DTV	NE—3,000' (SW crs Washington) SE—3,000' (Richmond Range) SW—6,000' (N crs Lynchburg) SW—2,500' (W crs Richmond) NW—5,000'—Min. en route altitude	NE crs to 3,000'	SW	10 mi.—2,500' S side SW crs 15 mi.—2,500' S side SW crs 20 mi.—2,500' S side SW crs 25 mi.—NA	2,000	135	2.5	438	R A T	1,000 1,500 800	2.0 3.0 1.0	NA NA NA		Climb to 3,000' on SE crs. 1,820' terrain 5 mi. NW of rng.
GOSHEN, IND. Goshen Arpt. 320 kc; GSH SBRAZ-DTV	N—2,100' (E crs South Bend) E—2,100' (NW crs Ft. Wayne) S—2,100' (NE crs La Fayette) W—2,100' (S crs South Bend)	None	W	10 mi.—2,000' S side W crs 15 mi.—2,100' S side W crs 20 mi.—2,100' S side W crs 25 mi.—2,100' S side W crs	1,500	90	2.7	823	R S* A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2300' on E crs. NOTE: Minima are for acft with stall speeds of 75 mph or less. *Runway No. 9.
GRAND FORKS, N. DAK. Grand Forks Arpt. 388 kc; GFK BMLZ-DTV	N—2,100' (Pembina Range) E—Min. en route altitude S—Min. en route altitude W—Min. en route altitude	None	S	10 mi.—2,000' E side S crs 15 mi.—2,000' E side S crs 20 mi.—2,000' E side S crs 25 mi.—2,000' E side S crs	1,500	350	3.4	836	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,100' on N crs.
GRAND ISLAND, NEBR. Grand Island Arpt. 263 kc; GRI SBMRLZ-DTV	N—Min. en route altitude E—2,900' (N crs Lincoln) S—Min. en route altitude W—4,500' (S crs No. Platte)	None	N	10 mi.—3,000' W side N crs 15 mi.—3,000' W side N crs 20 mi.—3,000' W side N crs 25 mi.—3,000' W side N crs	2,400	165	1.3	1,846	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 3,200' on S crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
GRAND MARAIS, MICH. Grand Marais Arpt. 341 kc; GMI BRL	N—Min. en route altitude E—Min. en route altitude S—2,500' (NW crs Traverse City) W—2,500' (Houghton Range)	None	W	10 mi.—2,400' S side W crs 15 mi.—2,400' S side W crs 20 mi.—2,400' S side W crs 25 mi.—2,400' S side W crs	1,700	111	3.3	838	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500' on E crs. *Runway No. 13.
GRAND RAPIDS, MICH. Kent Co. Arpt. 329 kc; GRR SBMRLZ-DTV	NE—Min. en route altitude SE—2,000' (N crs Battle Creek) SE—1,500' (Alaska FM) (final) SW—2,200' (NE crs Chimgo) NW—1,900' (Muskegon Range)	None	SE	10 mi.—2,000' E side SE crs 15 mi.—2,000' E side SE crs 20 mi.—2,000' E side SE crs 25 mi.—2,000' E side SE crs	1,500	303	2.0	692	R A T	500 800 300	1.5 2.0 1.0	500 800 400	2.0 2.0 1.0	Climb to 1,900' on NW crs.
GREAT FALLS, MONT. Great Falls Arpt. 317 kc; GTF SBRAZ-DTV	NE—Min. en route altitude E—9,000' (Lewistown Range) E—5,500' (Belt FM) SW—8,500' (N crs Helena) SW—5,500' (Cascade FM) SW—4,200' (Cascade FM) (final) NW—6,500' (Cut Bank Range)	None	SW	10 mi.—5,500' S side SW crs 15 mi.—5,500' S side SW crs 20 mi.—6,500' S side SW crs 25 mi.—7,400' S side SW crs	4,200	17	1.5	3,669	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 5,500' on NE crs. within 25 mi. *Runway No. 2.
Great Falls AFB	NE—Min. en route altitude E—9,000' (Lewistown Range) E—5,500' (Belt FM) SW—8,500' (N crs Helena) SW—5,500' (Cascade FM) NW—6,500' (Cut Bank Range)	None	SW	10 mi.—5,500' S side SW crs 15 mi.—5,500' S side SW crs 20 mi.—6,500' S side SW crs 25 mi.—7,400' S side SW crs	4,200	53	9.4	3,485	R A T	700 1,500 300	2.0 3.0 2.0	700 1,500 300	3.0 3.0 3.0	Climb to 5,500' on NE crs within 25 mi.
GREENSBORO, N. C. Greensboro-High Point Arpt. 265 kc; GSO SBRAZ-DTV	NE—2,300' (Blackstone Range) NE—1,800' (Summit Hill FM) (final) SE—Min. en route altitude SW—2,200' (SE crs Winston-Salem) NW—6,000' (Pulaski Range)	None	NE	10 mi.—2,300' N side NE crs 15 mi.—2,300' N side NE crs 20 mi.—2,300' N side NE crs 25 mi.—2,300' N side NE crs	1,800	241	1.4	914	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Make left turn and climb to 2,300' on SE crs within 25 mi; or alternate procedure (when directed by ATC), turn right and climb to 4,000' on NW crs within 25 mi.
GREENVILLE, S. C. Greenville Arpt. 281 kc; GRL SBMLZ-DTV	N—5,000' (NW crs Spartanburg) E—8,000' (SW crs Spartanburg) S—Min. en route altitude S—1,500' (SW crs Spartanburg) (final) W—Min. en route altitude	None	S	10 mi.—2,500' W side S crs 15 mi.—2,500' W side S crs 20 mi.—2,500' W side S crs 25 mi.—2,500' W side S crs	1,800	02	1.9	1,049	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 3,000' on 55° crs within 15 mi. Turn left, intercept N crs of Greenville range, then proceed S.
Greenville AFB	N—5,000' (NW crs Spartanburg) E—8,000' (SE crs Spartanburg) S—4,000' (W crs Columbia) W—Min. en route altitude	To 4,000' between range and 15 mi out on N crs	N	10 mi.—4,000' E side N crs 15 mi.—4,000' E side N crs 20 mi.—4,000' E side N crs 25 mi.—8,000' E side N crs	2,500	199	3.8	932	R A T	500 800 300	2.0 2.0 2.0	500 800 300	3.0 3.0 3.0	Climb to 4,000' on S crs.
GREENWOOD, MISS. Greenwood Mun. Arpt. 281 kc; GRW BMLZ-DTV	N—1,300' (Memphis Range) E—Min. en route altitude S—1,700' (Jackson Range) W—Min. en route altitude	None	E	10 mi.—1,600' N side E crs 15 mi.—1,600' N side E crs 20 mi.—1,600' N side E crs 25 mi.—1,600' N side E crs	1,100	265	1.9	129	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on W crs within 25 mi.
Greenwood Arpt.	N—1,300' (Memphis Range) E—Min. en route altitude S—1,700' (Jackson Range) W—Min. en route altitude	None	W	10 mi.—1,500' S side W crs 15 mi.—1,500' S side W crs 20 mi.—1,500' S side W crs 25 mi.—1,500' S side W crs	900	85	3.9	162	R (R) S A T	500 500 800 800 300	1.5 1.0 1.0 2.0 1.0	500 500 800 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500' on E crs within 25 mi.
HARRISBURG, PA. Harrisburg State Arpt. 242 kc; HAR SBRAZ-DTV	N—3,500' (E crs Philipsburg) E—2,000' (SW crs Allentown) S—2,500' (NE crs Washington) W—4,000' (S crs Philipsburg) W—1,800' (New Kingston FM) (final)	None	W	10 mi.—2,400' S side W crs 15 mi.—3,000' S side W crs 20 mi.—3,800' S side W crs 25 mi.—3,500' S side W crs	1,500	118	1.1	344	R A T	1,000 1,000 500	2.0 2.0 1.0	1,000 1,000 500	2.0 2.0 2.0	Climb to 2,000' on E crs. Landing to N authorized daylight only with ceiling 1,500' or better.
Olmsted AFB (Middletown)	N—3,500' (E crs Philipsburg) E—2,000' (SW crs Allentown) S—2,500' (NE crs Washington) W—4,000' (S crs Philipsburg) W—1,800' (New Kingston FM) (final)	None	W	10 mi.—2,400' S side W crs 15 mi.—3,000' S side W crs 20 mi.—3,500' S side W crs 25 mi.—3,500' S side W crs	1,800	116	6.4	300	R A T	1,300 1,500 500	1.0 2.0 1.0	1,300 1,500 500	2.0 3.0 2.0	Climb to 2000' on heading of 113° to avoid high terrain S of E crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
HARTFORD, CONN. Brainard Arpt. 329 kc; HFD SBMRAZ-DTV	NE—2,000' (E crs Westfield) SE—2,000' (SW crs Providence) SW—2,000' (S crs Westfield) NW—2,500' (S crs Westfield)	None	SE	10 mi.—2,000' E side SE crs 15 mi.—2,000' E side SE crs 20 mi.—2,000' E side SE crs 25 mi.—2,000' E side SE crs	1,620	323	2.3	18	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.0 1.0	Turn right and climb to 2,000' on SE crs.
Rentschler Field	NE—2,000' (E crs Westfield) SE—2,000' (SW crs Providence) SW—2,000' (S crs Westfield) NW—2,500' (S crs Westfield)	None	SE	10 mi.—2,000' E side SE crs 15 mi.—2,000' E side SE crs 20 mi.—2,000' E side SE crs 25 mi.—2,000' E side SE crs	1,420	61	2.4	45	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.0 1.0	Turn right and climb to 2,000, on SE crs.
HARVEY, ILL. Rubinkam Arpt.	(NO PROCEDURE ESTABLISHED BECAUSE OF HEAVY TRAFFIC IN CHICAGO AREA).													
HAYES CENTER, NEBR.	(NO PROCEDURE APPROVED)													
HELENA, MONT., Helena Arpt. 371 kc; HLN SBMRAZ-DTV	N—9,500' (SW crs Great Falls) SE—10,000' (NW crs Livingston) SE—9,000' (NW crs Bozeman) SE—*8,500' (Winston FM) S—10,500' (Whitehall Range) W—9,000' (N crs Butte) W—8,500' (McDonald Pass FM)	To 8,500' on SE crs within 25 mi.	SE	10 mi.—7,500' N side SE crs 15 mi.—8,500' N side SE crs 20 mi.—8,500' N side SE crs 25 mi.—8,500 N side SE crs	5,800	252	1.9	3,882	R A T	1,500 1,500 800	2.0 3.0 1.0	1,500 1,500 1,000	3.0 3.0 1.5	Make right turn climbing to 8,500' on SE crs within 10 mi. *Maintain at least 8,500' until 6 mi past Winston FM.
HOBBS, N. MEX. Lea Co. Arpt. 293 kc; HOB BMRLZ-DTV	N—5,500' (W crs Lubbock) E—5,000' (S crs Lubbock) S—5,000' (Wink Range) W—5,500' (SE crs Roswell)	None	N	10 mi.—5,000' W side N crs 15 mi.—5,000' W side N crs 20 mi.—5,000' W side N crs 25 mi.—5,000' W side N crs	4,500	168	8.4	3,639	R A T	600 800 300	1.5 2.0 1.0	600 800 300	2.0 2.0 1.0	Climb to 5,000' on S crs within 25 mi.
Airbase City Public	N—5,500' (W crs Lubbock) E—5,000' (S crs Lubbock) S—5,000' (Wink Range) W—5,500' (SE crs Roswell)	None	N	10 mi.—5,000' W side N crs 15 mi.—5,000' W side N crs 20 mi.—5,000' W side N crs 25 mi.—5,000' W side N crs	4,500	168	2.8	3,767	R (R) S A T	600 600 600 1,000 (BCOB) 300	1.5 1.0 1.0 2.0 1.0	600 600 600 1,000 (BCOB) 300	2.0 2.0 2.0 2.0 1.0	Climb to 5,000' on S crs.
HOMESTEAD, FLA. Homestead Arpt. 266 kc; HST SMRLZ-DTV	NE—1,400' (Miami Range) E—1,400' (SE crs Miami) SW—1,400' (E crs Key West) W—Min. en route altitude	None	W	10 mi.—1,200' S side W crs 15 mi.—1,200' S side W crs 20 mi.—1,200' S side W crs 25 mi.—1,200' S side W crs	800	87	5.1	7	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,400' on E crs.
HOT SPRINGS, N. MEX. Hot Springs Arpt. 344 kc; HSP SBRAZ-DTV	N—10,000' (Albuquerque Range) E—Min. en route altitude S—10,000' (W crs El Paso) W—Min. en route altitude	None	N	10 mi.—8,000' E side N crs 15 mi.—8,000' E side N crs 20 mi.—8,000' E side N crs 25 mi.—8,000' E side N crs	6,340	238	16.6	4,850	R (R) A T	1,500 1,500 2,000 500	1.5 1.0 3.0 1.0	NA NA NA NA		If not contact over range, climb to 10,000' on S crs.
HOUGHTON, MICH. Houghton Co. Arpt. 227 kc; CMX BRL-DT	N—Min. en route altitude E—2,500' (Grand Marais Range) S—Min. en route altitude W—Min. en route altitude	None	N	10 mi.—2,500' W side N crs 15 mi.—2,500' W side N crs 20 mi.—2,500' W side N crs 25 mi.—2,500' W side N crs	2,000	197	3.8	1,092	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,900' on S crs.
HOULTON, MAINE Houlton Arpt. 272 kc; HUL BMRLZ-DTV	N—3,000' (E crs Sprague ville) E—2,000' (NW crs Blissville) S—2,200' (E crs Millinocket) W—2,500' (S crs Presque Isle)	None	N	10 mi.—2,000' E side N crs 15 mi.—2,000' E side N crs 20 mi.—2,700' E side N crs 25 mi.—2,700' E side N crs	1,500	195	3.1	493	R A T	800 1,000 300	1.5 2.0 1.0	800 1,000 300	2.0 2.0 1.0	Climb to 2,200' on S crs.
HOUSTON, TEX. Houston, Arpt. 332 kc; HOU SBRAZ-DTV (Procedure No. 1)	E—1,600' (Beaumont Range) SE—1,300' (Galveston Range) SE—700' (Webster Fm) (final) SW—1,500' (Palacios Range) SW—1,200' (Arcola FM) NW—1,600' (Bryan Range) NW—1,300' (Houston FM)	None	SE	10 mi.—1,100' E. side SE crs 15 mi.—1,100' E side SE crs 20 mi.—1,100' E side SE crs 25 mi.—1,100' E side SE crs	700	309	2.2	50	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,600' on NW crs. NOTE: Take-offs on Rnwy 26 and landings on Rnwys 8 and 26 not authorized.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	*	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night			
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)		
HOUSTON, TEX.—Con. (Procedure No. 2)	E—1,600' (Beaumont Range) SE—1,300' (Galveston Range) SE—1,100' (Webster FM) SW—1,500' (Palacios Range) SW—1,200' (Arcola FM) NW—1,600' (Bryan Range) NW—1,200' (NE crs Richmond) (final)	None	NW	10 mi.—1,600' W side NW crs 15 mi.—1,600' W side NW crs 20 mi.—1,600' W side NW crs 25 mi.—1,600' W side NW crs	1,200 (over Houston FM)	129 (from Houston FM)	8.8	80	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,300' on SE crs. NOTE: Take-offs on Rnwy 26 and landings on Rnwys 8 and 26 not authorized.
Ellington AFB	E—1,600' (Beaumont Range) SE—1,300' (Galveston Range) SW—1,500' (Palacios Range) SW—1,200' (Arcola FM) NW—1,600' (Bryan Range) NW—700' (Houston FM) (final)	NW crs	NW	10 mi.—1,600' W side NW crs 15 mi.—1,600' W side NW crs 20 mi.—1,600' W side NW crs 25 mi.—1,600' W side NW crs	700	91	4.9	39	R A T	500 800 300	2.0 2.0 2.0	500 800 300	3.0 2.0 3.0	Climb to 1,600' on a heading of 100° mag. NOTE: Maintain 1,600' inbound on final until crossing Houston FM.
HUNTINGTON, W. VA. Huntington Arpt. Chesapeake, Ohio 236 kc; HTW BMRLZ-DTV	E—2,500' (W crs Charleston) SE—4,000' (S crs Charleston) W—Min. en route altitude NW—2,300' (S crs Columbus)	None	W	10 mi.—2,100' S side W crs 15 mi.—2,100' S side W crs 20 mi.—2,100' S side W crs 25 mi.—2,100' S side W crs	1,600	75	2.3	560	R A T	1,000 1,500 1,000	2.0 3.0 1.0	NA NA NA		Climb to 2,500' on E crs.
HURON, S. DAK. Huron Arpt. 391 kc; HON BMRLZ-DTV	NE—2,900' (Watertown Range) SE—2,800' (Sioux Falls Range) SW—2,500' (E crs Pierre) SW—2,000' (E crs Pierre) (Final) NW—Min. en route altitude	None	SW	10 mi.—2,500' S side SW crs* 15 mi.—2,600' S side SW crs* 20 mi.—3,000' S side SW crs* 25 mi.—3,000' S side SW crs*	*2,000	48	2.8	1,287	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 1.5 2.0 1.0	Climb to 2,900' on NE crs within 25 mi. *If procedure turn is accomplished beyond 20 mi., final approach alt. inbound to Int. SW Huron and E Pierre is 2,500'.
HUTCHINSON, KANS. Hutchinson Arpt. 227 kc; HUT SBMRLZ-DTV	N—Min. en route altitude E—2,700' (N crs Wichita) S—2,800' (SW crs Wichita) W—4,000' (Garden City Range)	None	S	10 mi.—2,800' E side S crs 15 mi.—2,800' E side S crs 20 mi.—2,800' E side S crs 25 mi.—2,800' E side S crs	2,300	350	3.1	1,542	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 3,100' on N ck within 25 mi. *Rwy No. 35.
IDAHO FALLS, IDAHO Idaho Falls, Arpt. 359 kc; IDA SBMRAZ-DTV	NE—8,000' (E crs Dubois) SE—Min. en route altitude SW—7,500' (Pocatello Range) NW—Min. en route altitude	None	NE	10 mi.—6,600' W side NE crs 15 mi.—6,600' W side NE crs 20 mi.—7,500' W side NE crs 25 mi.—7,500' W side NE crs	5,320	198	2.5	4,721	R A T	700 800 300	2.0 2.0 1.0	700 800 300	2.5 2.0 1.0	Climb to 7,500' on SW crs.
INDIANAPOLIS, IND. Weir Cook Arpt. 266 kc; IND SBRAZ-DTV	E—2,300' (NW crs Cincinnati) S—2,200' (W crs Louisville) W—2,000' (Terre Haute Range) W—1,500' (Clayton FM) (Final) NW—2,100' (NE crs La Fayette)	None	W	10 mi.—2,000' S side W crs 15 mi.—2,000' S side W crs 20 mi.—2,000' S side W crs 25 mi.—2,000' S side W crs	1,500	75	3.2	796	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,300' on E crs. *Rwy No. 9.
Stout Field	E—2,300' (NW crs Cincinnati) S—2,200' (W crs Louisville) W—2,000' (Terre Haute Range) W—1,500' (Clayton FM) (final) NW—2,100' (NE crs La Fayette)	None	W	10 mi.—2,000' S side W crs 15 mi.—2,000' S side W crs 20 mi.—2,000' S side W crs 25 mi.—2,000' S side W crs	1,500	75	6.0	714	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	NA NA NA NA		Climb to 2,300' on E crs.
INYOKERN, CALIF. Inyokern NAF 326 kc; NID MRLWZ	N—Min. en route altitude E—8,000' (NW crs Silver Lake) S—7,000' (NE crs Palmdale) W—Min. en route altitude	To 7,000' on S crs within 15 mi.	N	6 mi.—8,500' W side N crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	4,000	169	4.1	2,218	R A T	1,000 NA 1,000	1.5 1.0	1,000 NA 1,000	2.0 1.0	Climb to 7,000' on S crs within 15 mi.
JACKSON, MISS. Hawkins Field 260 kc; JAN SBRAZ-DTV	N—1,700' (Greenwood Range) E—1,800' (Meridian Range) E—1,700' (Pelahatchee FM) S—1,600' (New Orleans Range) W—1,500' (Monroe Range)	None	N	10 mi.—1,700' W side N crs 15 mi.—1,700' W side N crs 20 mi.—1,700' W side N crs 25 mi.—1,700' W side N crs	1,200	193	2.1	343	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 1,600' on S crs, or alternate procedure (when directed by ATC), climb to 1,800' on E crs.
JACKSON, TENN. Jacks Creek CAA Int. Field 332 kc; MKL BMRLZ-DTV	N—Min. en route altitude E—2,500' (Nashville Range) S—2,000' (W crs Muscle Shoals) W—1,800' (Memphis Range)	None	N	10 mi.—1,800' W side N crs 15 mi.—1,800' W side N crs 20 mi.—1,800' W side N crs 25 mi.—1,800' W side N crs	1,500	186	2.6	550	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	1.5 1.5 1.5 3.0 1.0	Climb to 2,000' on S crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
										Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
JACKSONVILLE, FLA. Imeson Arpt. 344 kc; JAX SBRAZ-DTV	N—1,200 (Savannah Range) E—Min. en route altitude E—600′ (Ft. George Is. FM) (final) S—1,200′ (N crs Daytona Beach) W—1,200′ (NE crs Cross City)	None	E	10 mi.—1,100′ N side E crs 15 mi.—1,100′ N side E crs 20 mi.—1,100′ N side E crs 25 mi.—1,100′ N side E crs	600	269	1.7	32	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,200′ on W crs, or alternate procedure (when directed by ATC), turn right and climb to 1,200′ on N crs. *Runway No. 27.
Cecil NAAS	(MAKE LET-DOWN TO IMESON ARPT. FLY VFR FROM IMESON ARPT. TO CECIL NAAS)													
Lee OLF	(MAKE LET-DOWN TO IMESON ARPT. FLY VFR FROM IMESON ARPT. TO LEE OLF)													
Mayport NAAS	(MAKE LET-DOWN TO IMESON ARPT. FLY VFR FROM IMESON ARPT. TO MAYPORT NAAS)													
Jacksonville NAS 213 kc NIP SMRLZ	N—1,600′ (W crs Jacksonville) E—1,200′ (S crs Jacksonville) S—Min. en route altitude W—Min. en route altitude	None	W	10 mi.—1,300′ S side W crs 15 mi.—1,300′ S side W crs 20 mi.—1,300′ S side W crs 25 mi.—1,300′ S side W crs	600	89	3.1	20	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn right and climb to 1,500′ on S crs Jacksonville (NIP) range. *Rwy No. 9.
JAMESTOWN, N. DAK. Jamestown Arpt. 344 kc; JMS SBMRAZ-DTV	N—Min. en route altitude E—2,600′ (Fargo Range) S—2,800′ (Aberdeen Range) W—3,400′ (Bismarck Range)	None	E	10 mi.—2,600′ N side E crs 15 mi.—2,600′ N side E crs 20 mi.—2,600′ N side E crs 25 mi.—2,600′ N side E crs	2,100	253	1.5	1,498	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 3,400′ on W crs.
JOLIET, ILL. Joliet Arpt. 257 kc; JOT SBRAZ-DTV	NE—2300′ (W crs Chicago) E—2,300′ (S crs Chicago) SW—2,000′ (E crs Peoria) W—2,000′ (N crs Peoria)	None	W	10 mi.—1,900′ S side W crs 15 mi.—1,900′ S side W crs 20 mi.—1,900′ S side W crs 25 mi.—1,900′ S side W crs	1,400	126	3.7	580	R A T	500 800 300	1.5 2.0 1.0	500 800 300	2.0 2.0 1.0	Turn right and climb to 2,000′ on SW crs within 25 mi. NOTE: Minimums are for acft having stall speeds of 75 mph or less.
JOPLIN, MO. Joplin Arpt. 400 kc; JLN BMRLZ-DTV	N—2,500′ (SE crs Kansas City) N—1,600′ (E crs Chanute) (final) E—2,500′ (SW crs Springfield) S—2,500′ (NE crs Tulsa) W—2,500′ (S crs Chanute)	None	N	10 mi.—2,000′ W side N crs 15 mi.—2,000′ W side N crs 20 mi.—2,000′ W side N crs 25 mi.—2,000′ W side N crs	1,600	172	6.2	980	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,500′ on S crs within 25 mi. *Rwy No. 17.
KANSAS CITY, MO. Kansas City Arpt. 359 kc; MKC SBRAZ-DTXV	NE—2,300′ (Kirksville Range) NE—2,300′ (W crs Columbia) (Excelsior Springs Int). SE—2,600′ (E crs Olathe) (Belton Int). SW—2,000′ (S crs St. Joseph) (De Soto Int). NW—2,400′ (St. Joseph Range) NW—1,600′ (Linkville FM or Int E crs Topeka VAR) (final).	None	NW	10 mi.—2,200′ E side NW crs 15 mi.—2,200′ E side NW crs 20 mi.—2,200′ E side NW crs 25 mi.—2,200′ E side NW crs	1,600	142	2.9	744	R (R) S# A T	700 500 500 800 *300	1.5 1.0 1.0 2.0 1.0	700 500 500 800 *300	1.5 1.0 1.5 2.0 1.0	Immediately make right turn and climb to 2,200′ on SW crs. *Take-offs to S and SW when weather is below 1,000-3 will assume a heading of at least 210° mag as soon as practicable after take-off and maintain heading until reaching 2,500′ prior to making left turn. CAUTION: 1. 1,430′ MSL obstructions located 2.5 mi. SE of arpt. 2. 902′ MSL cracking plant and 809′ MSL stack, 0.6 mi. NW of approach end of Rwy No. 17. 3. 1,670′ MSL twr, 2.9 mi. S of arpt. #Rwy No. 17.
Fairfax Field (Kansas City, Kans.)	NE—2,300′ (Kirksville Range) NE—2,200′ (W crs Columbia) (Excelsior Springs Int.) SE—2,600′ (E crs Olathe) (Belton Int.) SW—2,000′ (S crs St. Joseph) (DeSoto Int.) NW—2,400′ (St. Joseph Range) NW—1,600′ (Linkville FM or Int. E crs Topeka VAR) (final)	None	NW	10 mi.—2,200′ E side NW crs 15 mi.—2,200′ E side NW crs 20 mi.—2,200′ E side NW crs 25 mi.—2,200′ E side NW crs	1,600	148	.8	746	R (R) A T	700 500 800 300	1.5 1.0 2.0 1.0	700 500 800 300	1.5 1.0 2.0 1.0	Immediately make right climbing turn and climb to 2,200′ on SW crs. CAUTION: SE crs of rng extends over city, 1,430′ MSL obstructions location 2.5 mi SE of arpt. Adhere strictly to pull-up procedure. CAUTION: 902′ MSL cracking plant and 809′ MSL stack, 0.5 mi ESE of approach end of Rnwy 35. 1,670′ MSL tower, 4.7 mi SSE.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
KEESLER (Biloxi), MISS., Keesler Field 391 kc; BIX BMRLZ	NE—1,500' (NW crs Mobile) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	None	NE	10 mi.—1,500' N side NE crs 15 mi.—1,500' N side NE crs 20 mi.—1,500' N side NE crs 25 mi.—1,500' N side NE crs	1,000	210	2.9	26	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on SW crs.
Gulfport Arpt.	(MAKE LET-DOWN TO KEESLER FLD. FLY VFR FROM KEESLER FLD TO GULFPORT AIRPORT)					267	7.6	28	R S A T	1,000 NA 1,000 (BCOB) 300	1.0 3.0 1.0	NA NA NA NA		
KELLY (San Antonio), TEX. Kelly AFB 254 kc; SKF SBRAZ-DTV	NE—2,500' (SW crs Austin) NE—2,200' (W crs San Antonio) SE—2,000' (S crs San Antonio) SW—2,000' (Laredo Range) NW—3,500' (S crs San Angelo) NW—2,500' (Medina FM)	None	NW	10 mi.—2,500' W side NW crs 15 mi.—2,600' W side NW crs 20 mi.—3,000' W side NW crs 25 mi.—3,000' W side NW crs	1,640	107	2.0	675	R A T	700 800 300	2.0 2.0 1.0	700 800 300	2.0 2.0 2.0	Climb to 2,000' on SW crs within 25 mi.
Brooks AFB	(As above except where noted)					97	11.7	594	R A T	700 800 300	2.0 2.0 1.0	700 800 300	2.0 2.0 2.0	Proceed out on a heading of 180° to intersect SE crs Kelly, climbing to 2,000'.
KEY WEST, FLA. Meacham Field 332 kc; EYW SBRAZ-DTV	N—Min. en route altitude E—1,400' (SW crs Homestead) E—1,300' (Stock Is. FM) S—Min. en route altitude W—Min. en route altitude.	None	W	10 mi.—1,300' S side W crs 15 mi.—1,300' S side W crs 20 mi.—1,300' S side W crs 25 mi.—1,300' S side W crs	800	70	1.5	4	R (R) A T	500 500 800 300	2.0 1.5 2.0 1.0	NA NA NA NA		Climb to 1,400' on E crs.
Key West NAS	N—Min. en route altitude E—1,400' (SW crs Homestead) S—Min. en route altitude W—Min. en route altitude	None	W	10 mi.—1,300' S side W crs 15 mi.—1,300' S side W crs 20 mi.—1,300' S side W crs 25 mi.—1,300' S side W crs	800	70	5.8	4	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,400' on E crs.
KIRKSVILLE, MO. Kirksville Arpt. 391 kc; IRK BMRLZ-DTV	NE—2,000' (Burlington Range) SE—2,100' (W crs St. Louis) SW—2,300' (Excelsior Springs Int.) NW—2,500' (S crs Des Moines)	None	SE	10 mi.—2,100' E side SE crs 15 mi.—2,100' E side SE crs 20 mi.—2,100' E side SE crs 25 mi.—2,100' E side SE crs	1,600	317	.9	970	R* A* T*	500 800 300	1.0 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,500' on NW crs within 25 mi. *Restricted to DC-3 and smaller acft except in emergency.
KLAMATH FALLS, OREGON Klamath Falls Arpt. 242 kc; LMT SBRAZ-DTV	N—10,000' (S crs Redmond) N—9,000' (E crs Medford) E—Min. en route altitude S—10,000' (N crs Whitmore S—9,000' (E crs Ft. Jones) W—9,500' (S crs Medford)	None	S	12 mi.—8,000' E side S crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	6,500	338	2.5	4,085	R A T	1,800 1,800 1,500	1.5 2.0 1.0	1,800 1,800 1,500	2.0 2.0 2.0	Turn left and climb to 9,000' on S crs within 15 mi. NOTE: Minimums shall apply to all type acft.
KNOXVILLE, TENN. McGhee-Tyson Arpt. 257 kc; TYS SBRAZ-DTV	NE—5,000' (Tri-City Range) NE—4,000' (Piedmont FM) S—Min. en route altitude S—3,500' (Tallassee FM) W—4,500' (Smithville Range) W—3,000' (Kingston Int.) N—Min. en route altitude N—2,000' (Inskip FM) (final)	None	N	10 mi.—3,000' E side N crs 15 mi.—3,000' E side N crs 20 mi.—3,000' E side N crs 25 mi.—3,000' E side N crs	2,000	192	2.8	989	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Turn right and climb to 4,000 on W crs within 25 mi.
LA CROSSE, WIS. La Crosse Arpt. 271 kc; LSE SBRAZ-DTV	NE—Min. en route altitude SE—2,500' (Lone Rock Range) SW—2,500' (S crs Rochester) NW—2,600' (N crs Rochester)	None	NW	10 mi.—2,500' W side NW crs 15 mi.—2,600' W side NW crs 20 mi.—2,600' W side NW crs 25 mi.—2,600' W side NW crs	1,800	141	5.0	653	R (R) S* A T	600 600 500 1,000 400	1.5 1.0 1.5 3.0 1.0	600 600 500 1,000 400	2.0 2.0 2.0 2.0 1.5	Climb to 2,500' on SE crs. *Runway No. 12.
LA FAYETTE, IND. Purdue Univ. Arpt. 227 kc; LAF BMRLZ-DTV	NE—2,000' (S crs Goshen) SE—2,100' (E crs Indianapolis) SW—1,800' (N crs Terre Haute) NW—1,900' (NE crs Chanute)	None	SW	10 mi.—1,800' S side SW crs 15 mi.—1,800' S side SW crs 20 mi.—2,000' S side SW crs 25 mi.—2,000' S side SW crs	1,400	46	2.4	805	R A T	700 800 300	1.5 2.0 1.0	700 800 300	1.5 2.0 1.0	Climb to 2,000' on NE crs. 900' msl unlighted stacks, 1.5 mi. NE of arpt. NOTE: Minimums apply to DC-3 or smaller acft only.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
LA JUNTA, COLO. La Junta Arpt. 281 kc; LHX SBMRLZ-DTV	NE—Min. en route altitude SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	None	NE	10 mi.—5,400' W side NE crs 15 mi.—5,800' W side NE crs 20 mi.—5,800' W side NE crs 25 mi.—5,800' W side NE crs	4,900	194	2.6	4,238	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 6,000' on NE crs within 25 mi. *Runway No. 19.
LAKE CHARLES, LA. Lake Charles Arpt. 242 kc; LCH SBRAZ-DTV	N—1,500' (Shreveport Range) N—1,300' (SW crs Alexandria) E—1,400' (New Orleans Range) E—1,200' (SW crs Baton Rouge) S—Min. en route altitude W—1,200' (Beaumont Range)	None	S	10 mi.—1,200' E side S crs 15 mi.—1,200' E side S crs 20 mi.—1,200' E side S crs 25 mi.—1,200' E side S crs	800	352	2.4	17	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,300' on N crs within 25 mi.
LAKEHURST, N. J. Lakehurst NAS 520 kc; NEL SBMRLZ	NE—1,500' (SE crs Newark) SE—1,500' (NE crs Millville) SW—1,500' (SE crs McGuire) NW—1,700' (NE crs Philadelphia) NW—700' (Cassville FM) (final)*	None	NW	10 mi.—1,300' W side NW crs 15 mi.—1,300' W side NW crs 20 mi.—1,300' W side NW crs 25 mi.—1,500' W side NW crs	800	157	.7	94	R (R) S A T	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on SE crs. *Descent to field may be started after passing Cassville FM, if FM not received final approach altitude over range is 800'.
LANGLEY, VA. Langley AFB 221 kc; LFI SBMRAPZ-DT	N—1,500' (SW crs Patuxent River) E—1,500' (NE crs Norfolk) S—1,500' (SW crs Norfolk) W—1,500' (SE crs Richmond)	On N crs	W	10 mi.—1,500' N side W crs 15 mi.—1,500' N side W crs 20 mi.—NA 25 mi.—NA	1,000	109	3.5	10	R A T	800 800 300	1.5 2.0 2.0	800 800 300	2.0 2.0 3.0	Turn left and climb to 1,500' and return to range.
LANSING, MICH. Capital City Arpt. 206 kc; LAN SBMRAZ-DTV	E—2,200' (N crs Detroit) SE—2,200' (W crs Detroit) W—2,200' (N crs Battle Creek) NW—Min. en route altitude	None	E	10 mi.—2,200' N side E crs 15 mi.—2,200' N side E crs 20 mi.—2,200' N side E crs 25 mi.—2,200' N side E crs	1,760	280	2.6	858	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 2.0 2.0 1.0	Climb to 2,200' on W crs. N/S rwy not authorized for night landings or take-offs, no range lights.
LARAMIE, WYO. Brees Field 245 kc; LAR SBMRAZ-DTV	NE—10,500' (N crs Cheyenne) SE—11,500' (N crs Denver) SW—Min. en route altitude NW—11,000' (E crs Sinclair) NW—10,500' (Two Rivers Int.)	None	NW	10 mi.—8,500' E side NW crs 15 mi.—8,500' E side NW crs 20 mi.—8,500' E side NW crs 25 mi.—8,500' E side NW crs	8,000	147	2.1	7,273	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 2.0 2.0 1.0	Climb to 11,500' on SE crs.
LAREDO, TEX. Laredo Arpt. 206 kc; LED SBRAZ-DTV	NE—2,000' (Kelly Range) NE—1,800' (SW crs Randolph) SE—1,800' (W crs Brownsville) SW—Min. en route altitude NW—Min. en route altitude	None	NW	10 mi.—1,700' W side NW crs 15 mi.—1,700' W side W crs 20 mi.—1,700' W side NW crs 25 mi.—1,700' W side NW crs	1,200	137	3.1	512	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1800' on SE crs within 25 mi.
LAS VEGAS, NEV. McCarran Field 206 kc; LAS SBMRAZ-DTV	NE—10,000' (Enterprise Range) NE—7,000' (Crystal FM) SE—8,000' (N crs Needle) SW—9,500' (Silver Lake Range) NW—Min. en route altitude	To 7,000' on NE & SW crs within 20 mi. (turns to S)	SW	10 mi.—5,000' S side SW crs 15 mi.—5,000' S side SW crs 20 mi.—7,000' S side SW crs 25 mi.—10,000' S side SW crs	3,670	195	10.2	2,172	R A T	1,500 1,500 800	2.0 2.0 2.0	1,500 1,500 800	2.0 2.0 2.0	If not contact over range, climb to 7,000' on NE crs within 20 mi.
Nellis AFB	NE—10,000' (Enterprise Range) NE—7,000' (Crystal FM) SE—8,000' (N crs Needles) SW—9,500' (Silver Lake Range) NW—Min. en route altitude	To 7,000' on NE & SW crs within 20 mi. (turns to S)	SW	10 mi.—5,000' S side SW crs 15 mi.—5,000' S side SW crs 20 mi.—7,000' S side SW crs 25 mi.—10,000' S side SW crs	3,400	34	1.9	1,890	R A T	1,500 1,500 800	2.0 2.0 2.0	1,500 1,500 800	2.0 2.0 2.0	Climb to 7,000' on NE crs within 20 mi.
Sky Haven Arpt.	(MAKE LET-DOWN TO NELLIS AFB. FLY VFR FROM NELLIS AFB TO SKY HAVEN ARPT.)													
LAS VEGAS, N. M. Las Vegas Arpt. 339 kc; LVS SBMRAZ-DTV	NE—11,000' (S crs Trinidad) SE—9,500' (W crs Tucumcari) SW—10,000' (E crs Otto) NW—Min. en route altitude	None	SW	10 mi.—9,000' E side SW crs 15 mi.—9,000' E side SW crs 20 mi.—9,000' E side SW crs 25 mi.—9,000' E side SW crs	7,800	08	3.2	6,866	R S A T	800 800 300	1.5 2.0 1.0	800 800 300	2.0 2.0 2.0	Climb to 11,000' on right side NE crs within 25 mi.
LAWSON (Columbus), GA., Lawson AFB 335 kc; LSF BMRLZ	NE—Min. en route altitude SE—No approach—Danger area SW—2,000' (NW crs Dothan) NW—2,000' (E crs Maxwell)	None	SW	10 mi.—1,700' E side SW crs 15 mi.—1,700' E side SW crs 20 mi.—1,700' E side SW crs 25 mi.—1,700' E side SW crs	1,000	21	2.6	232	R (R) S A T	700 700 700 800 300	1.5 1.0 1.0 2.0 1.0	700 700 700 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,000' on NE crs. Make all turns on NE crs on W side of crs, danger area in E quadrant.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
LEBO, KANSAS: CAA Int Field 311 kc; LBO BMLZ-DTV	NE—2,300' (S crs Topeka VAR) SE—2,300' (N crs Chanute) SW—2,600' (W crs Chanute) NW—Min. en route altitude	None	SE	10 mi.—2,300' E side SE crs 15 mi.—2,300' E side SE crs 20 mi.—2,300' E side SE crs 25 mi.—2,300' E side SE crs	1,700	315 (Adj to Arpt)	6.0	1,165	R A T	600 800 300	1.0 2.0 1.0	600 800 300	1.5 2.0 1.0	If not contact over range, climb to 2,600' on SW crs. Procedure authorized only for acft having stall speed of 75 mph or less.
LEWISTOWN, MONT. Lewistown Arpt. 353 kc; LWT SBRAZ-DTV	NE—Min. en route altitude E—8,000' (N crs Billings) SW—Min. en route altitude W—9,000' (Great Falls Range) W—8,000' (Stanford FM) (final)	None	W	10 mi.—6,000' S side W crs 15 mi.—6,000' S side W crs 20 mi.—6,000' S side W crs 25 mi.—6,000' S side W crs	5,000	99	2.9	4,177	R (R) A T	800 700 800 300	2.0 2.0 2.0 1.0	800 700 800 300	2.0 2.0 2.0 1.0	Turn right and climb to 6,000' on W crs within 25 mi.
LINCOLN, NEBR. Lincoln Arpt. 385 kc; LNK SBMRLZ-DTV	N—2,500' (W crs Omaha) E—2,500' (SE crs Omaha) S—Min. en route altitude W—3,200' (S crs Grand Island)	None	N	10 mi.—2,400' W side N crs 15 mi.—2,800' W side N crs 20 mi.—2,500' W side N crs 25 mi.—2,500' W side N crs	1,900	170	2.4	1,191	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,600' on S crs within 25 mi., or alternate procedure (when directed by ATC), climb to 3,200' on W crs. *Runway No. 17.
LITTLE ROCK, ARK. Adams Field 353 kc; LIT SBRAZ-DTV	NE—1,600' (N crs Stuttgart) SE—1,500' (SW crs Memphis) SE—800' (Keo FM) (final) SW—1,700' (NE crs Texarkana) NW—3,800' (Ft. Smith Rbn)	None	SE	10 mi.—1,300' E side SE crs 15 mi.—1,300' E side SE crs 20 mi.—1,300' E side SE crs 25 mi.—1,300' E side SE crs	800	315	3.8	257	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 3,000' on NW crs within 25 mi. and return to station, or alternate procedure (when directed by ATC), turn right and climb to 1,600' on NE crs within 25 mi.
LIVINGSTON, MONT.	(NO PROCEDURE APPROVED)													
LONE ROCK, WIS. CAA Int. Field 278 kc; LNR BMLZ-DTV	N—Min. en route altitude. E—2,500' (Madison Range) S—Min. en route altitude. W—Min. en route altitude.	None	W	10 mi.—2,300' S side W crs 15 mi.—2,500' S side W crs 20 mi.—2,500' S side W crs 25 mi.—2,500' S side W crs	2,020	98	.0	717	R A T	1,300 1,300 500	2.0 2.0 1.0	1,300 1,300 800	2.0 2.0 3.0	Climb to 2,500' on E crs. NOTE: Approved only for acft with stall speeds of 75 mph or less.
LONG BEACH, CALIF. Long Beach Arpt. 293 kc; LGB SBMRLZ-DTV	NE—3,000' (E crs Los Angeles) NE—1,500' (La Habra FM) SE—4,000' (N crs San Diego) SE—1,000' (Huntington Bch. FM) (final) SW—Min. en route altitude NW—1,500' (Los Angeles Range)	None	SE	10 mi.—1,500' W side SE crs 15 mi.—1,500' W side SE crs 20 mi.—NA 25 mi.—NA	1,000	298	3.9	56	R (R) S* A T	600 500 500 800 300	2.0 1.0 1.0 2.0 1.0	600 500 500 800 300	2.0 1.0 1.0 2.0 1.0	Make climbing left turn to heading of 250° and intersect S crs of Los Angeles (LF); proceed to San Pedro Int. at minimum altitude of 2,000'. *Runway No. 30.
LONG BEACH, CALIF. Los Alamitos NAS	(PROCEDURE BEING REVISED)													
LOS ANGELES, CALIF. Los Angeles International Arpt. 333 kc; LAX SBRAZ-DTXV	N—7,000' (Newhall Range) N—4,000' (Burbank Range) N—3,000' (Hollywood Hills FM) E—5,000' (Riverside Range) E—3,000' (La Habra FM) E—1,500' (Downey FM) E—1,000' (Los Angeles LOM) (final)* S—3,000' (SW crs Long Beach)** W—Min. en route altitude	None	E	10 mi.—2,000' S side E crs *** 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,000	244	3.9	109	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000' on W crs. *1,500' must be maintained until past Los Angeles LOM. **Destrous all approaches from Long Beach be made via Downey FM. ***Procedure turn within 10 mi, or no farther E than Downey FMH.
Central Arpt.	(MAKE LET-DOWN TO LOS ANGELES INTERNATIONAL AIRPORT. FLY VFR FROM LOS ANGELES INTERNATIONAL AIRPORT TO CENTRAL ARPT.)							86	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	Minimums are reported at Los Angeles Intn'l Arpt. If lower minimums are encountered en route from Los Angeles Intn'l Arpt to the satellite field, the acft is to return to Los Angeles Intn'l.
Culver City Arpt.	(MAKE LET-DOWN TO LOS ANGELES INTERNATIONAL ARPT. FLY VFR FROM LOS ANGELES INTERNATIONAL ARPT. TO CULVER CITY ARPT.)							25	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	(See remarks under Central Arpt.)

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
LOS ANGELES, CALIF.—Continued Gardena Valley Arpt	(MAKE LET-DOWN TO LOS ANGELES INTERNATIONAL ARPT. FLY VFR FROM LOS ANGELES INTERNATIONAL ARPT. TO GARDENA VALLEY ARPT.)							50	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	(See remarks under Central Arpt.)
Hawthorne Arpt.	(MAKE LET-DOWN TO LOS ANGLES INTERNATIONAL ARPT. FLY VFR FROM LOS ANGELES INTERNATIONAL ARPT. TO HAWTHORNE ARPT.)							64	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	(See remarks under Central Arpt.)
Santa Monica Arpt.	(MAKE LET-DOWN TO LOS ANGELES INTERNATIONAL ARPT. FLY VFR FROM LOS ANGELES INTERNATIONAL ARPT TO SANTA MONICA ARPT.)							175	R A T	1,000 1,000 500	1.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	(See remarks under Central Arpt.)
LOUISVILLE, KY. Standiford Field 359 kc; LOU SBMRLZ-DTXV	N—Min. en route altitude E—2,400' (N crs Lexington VAR) E—1,500' (Eastwood FM) (final) S—2,200' (NE crs Bowling Green) S—2,000' (Shepherdsville FM) W—2,000' (Evansville Range)	None	E	10 mi.—2,000' S side E crs 15 mi.—2,400' S side E crs 20 mi.—2,400' S side E crs 25 mi.—2,400' S side E crs	1,500	262	6.1	497	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.5 2.0 1.0	600 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 2,000' on W crs. CAUTION: 1,440' rdo twr on N side E crs, 12 mi. from rng. 1,032' rdo twr on S side E crs, 6.5 mi. from rng. *Runway No. 24.
Bowman Field	N—Min. en route altitude E—2,400' (N crs Lexington VAR) E—1,500' (Eastwood FM) (final) S—2,200' (NE crs Bowling Green) S—2,000' (Shepherdsville FM) W—2,000' (Evansville Range)	None	E	10 mi.—2,000' S side E crs 15 mi.—2,400' S side E crs 20 mi.—2,400' S side E crs 25 mi.—2,400' S side E crs	1,500	266	1.4	549	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,000' on W crs. CAUTION: 1,440' rdo twr on N side E crs, 12 mi. from rng. 1,032' rdo twr on S side E crs. 6.5 mi. from rng. *Runway No. 24.
LOVELOCK, NEV. CAA Int. Field	(NO PROCEDURE APPROVED)													
LUBBOCK, TEX. Lubbock Arpt. 391 kc; LBB BMRLZ-DTV	N—5,000' (Amarillo Range) E—4,500' (Guthrie VAR) S—4,500' (NW crs Big Spring) W—Min. en route altitude	None	W	10 mi.—4,400' S side W crs 15 mi.—4,400' S side W crs 20 mi.—4,700' S side W crs 25 mi.—4,700' S side W crs	4,400	78	11.5	3,256	R A T	700 800 300	2.0 2.0 1.0	700 800 300	2.0 2.0 1.0	If not contact within 4 mi after passing rng, climb to 4,500' on E crs within 25 mi.
LUCIN, UTAH	(NO PROCEDURE APPROVED)													
LYNCHBURG, VA. Preston Glenn Arpt. 200 kc; LYH SBRAZ-DTV (Procedure No. 1)	NE—5,000' (NW crs Richmond) S—2,500' (NE crs Greensboro) W—5,000' (Roanoke Range) N—5,000' (NE crs Roanoke)	None	S	10 mi.—2,500' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	1,940	157	1.9	942	R A T	1,000 1,500 500	2.0 2.0 1.0	1,000 1,500 500	2.0 2.0 1.0	If not contact over range, turn right and climb to 5,000' on NE crs.
(Procedure No. 2)	NE—5,000' (NW crs Richmond) S—2,500' (NE crs Greensboro) W—5,000' (Roanoke Range) N—5,000' (NE crs Roanoke)	None	S	10 mi.—2,500' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	1,740 (Over Oak Grove FM)	345 (From Oak Grove FM)	4.0	942	R (R) S* A T	800 600 600 900 500	1.5 1.0 1.0 2.0 1.0	800 600 600 900 500	1.5 1.0 1.0 2.0 1.0	Turn right and climb to 5,000' on NE crs. *Runway No. 35.
McCHORD (Tacoma), WASH., McChord AFB 272 kc; TCM BMRLZ (Procedure No. 1)	NE—2,500' (SE crs Seattle) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	NW crs within 20 mi.	NW	10 mi.—3,500' S side NW crs 15 mi.—3,500' S side NW crs 20 mi.—3,500' S side NW crs 25 mi.—NA	1,500	26	1.7	286	R A T	1,200 1,200 500	1.0 3.0 1.0	1,200 1,200 600	2.0 3.0 2.0	Proceed out NW crs, climbing to 2,500' within 25 mi. and contact McChord twr for instructions. GCA should be used whenever practicable.
(Procedure No. 2)	NE—2,500' (SE crs Seattle) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	NW crs within 20 mi.	SW	10 mi.—2,000' S side SW crs 15 mi.—2,000' S side SW crs 20 mi.—3,500' S side SW crs 25 mi.—3,500' S side SW crs	1,000	26	1.7	286	R A T	700 1,200 500	1.5 3.0 1.0	700 1,200 600	2.0 3.0 2.0	Proceed out NW crs, climbing to 2,500' within 25 mi. and contact McChord tower for further instructions. NOTE: This procedure authorized only when Ft. Lewis firing rng is inactive. Procedure No. 1 will be used at all other times.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (m.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
McGUIRE (Wrightstown), N. J McGuire AFB 382 kc; WRI MRAWZ	NE—1,700' (Freehold Int.) SE—1,500' (NE crs Millville) SW—2,000' (S crs Philadelphia) NW—2,000' (NE crs N. Phila.)	To 1,500' on SW crs within 20 mi.	SW	10 mi.—1,500' S side SW crs 15 mi.—1,500' S side SW crs 20 mi.—1,500' S side SW crs 25 mi.—NA	1,000	56	3.3	120	R A T	500 800 300	1.5 2.0 2.0	500 800 300	1.5 2.0 3.0	Turn right and climb to 1,500' on SW crs. CAUTION: Range freq is only 3 kcs higher than Floyd Bennett range freq. Range is operating at reduced power of 25 watts.
MACON, GA., Macon Arpt. 209 kc; MCN SBMRAZ-DTV.	NE—Min. en route altitude SE—1,500' (Alma range) SW—1,600' (N crs Albany) NW—2,000' (Atlanta range)	None	NW	10 mi.—1,600' W side NW crs 15 mi.—1,600' W side NW crs 20 mi.—1,600' W side NW crs 25 mi.—1,800' W side NW crs	1,200	143	3.5	351	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on SE crs.
Robins AFB	NE—Min en route altitude SE—1,500' (Alma range) SW—1,600' (N crs Albany) NW—2,000' (Atlanta range)	None	NW	10 mi.—1,600' W side NW crs 15 mi.—1,600' W side NW crs 20 mi.—1,600' W side NW crs 25 mi.—1,800' W side NW crs	1,200	143	8.6	295	R A T	500 800 300	2.0 2.0 1.0	500 800 300	3.0 2.0 2.0	Climb to 1,500' on SE crs.
Smart Arpt.	(MAKE LET-DOWN TO MACON ARPT. FLY VFR FROM MACON ARPT. TO SMART ARPT.)													
MADISON, WIS. Madison Arpt. 600 kc; MSN BMRLZ-DTV.	NE—Min. en route altitude SE—Min. en route altitude. SW—2,400' (NW crs Rockford) NW—3,300' (N crs Lone Rock)	None	SE	10 mi.—2,100' E side SE crs 15 mi.—2,100' E side SE crs 20 mi.—2,100' E side SE crs 25 mi.—2,100' E side SE crs	1,600	314	2.6	859	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 3,300' on NW crs. *Runway No. 31
MALAD CITY, IDAHO, CAA Int. Field 314 kc; MLD SBRAZ-DTV	N—11,000' (Pocatello range) E—12,000' (N crs Ft. Bridger) S—11,000' (Ogden range) W—11,000' (Burley range)	None	W	10 mi.—9,000' S side W crs 12 mi.—9,000' S side W crs 20 mi.—NA 25 mi.—NA	8,303	121	3.5	4,503	R A T	4,000 4,000 4,000	3.0 3.0 3.0	4,000 4,000 4,000	3.0 3.0 3.0	If not contact over range, turn right and climb to 11,000' on W crs within 18 mi.
MARTINSBURG, W. VA. Martinsburg Arpt. 209 kc; MRB SBMRLZ-DTV	NE—3,300' (S crs Philipsburg) E—3,600' (Baltimore range) SW—3,000' (Front Royal range) W—4,000' (S crs Altoona)	None	SW	10 mi.—2,500' E side SW crs 15 mi.—2,500' E side SW crs 20 mi.—2,500' E side SW crs 25 mi.—2,500' E side SW crs	1,500	40	2.6	536	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 3,500' on NE crs.
MAXWELL (Montgomery), ALA. Maxwell Field 362 kc; MXF SBRAZ	N—3,000' (E crs Birmingham) E—1,600' (Lawson Range) SW—2,000' (Mobile Range) W—2,000' (Mulberry Int.)	None	W	10 mi.—1,500' N side W crs 15 mi.—1,500' N side W crs 20 mi.—NA 25 mi.—NA	1,000	85	2.1	167	R A T	500 800 300	2.0 2.0 2.0	500 800 300	3.0 2.0 3.0	Climb to 1,600' on E crs.
Dannelly Field	N—3,000' (E crs Birmingham) E—1,500' (Lawson Range) SW—2,000' (Mobile Range) W—2,000' (Mulberry Int.)	None	N	10 mi.—1,700' W side N crs 15 mi.—1,700' W side N crs 20 mi.—1,700' W side N crs 25 mi.—1,700' W side N crs	1,200	170	5.6	219	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	2.0 2.0 2.0 2.0 1.0	Climb to 2,000' on SW crs. *Runway No. 13
MEDFORD, OREG. Medford Arpt. 263 kc; MFR SBRAZ-DTV	N—6,500' (Eugene Range) N—3,000' (Trail FM) (final) E—Min. en route altitude S—10,000' (Ft. Jones Range) S—8,000' (W crs Klamath Falls) W—Min. en route altitude	None	N	10 mi.—6,000' E side N crs* 14 mi.—6,000' E side N crs* 20 mi.—6,500' E side N crs* 25 mi.—6,500' E side N crs*	3,600	158	2.4	1,329	R A T	1,000 1,000 300	2.0 2.0 1.0	1,000 1,000 300	2.0 2.0 1.0	Make 180° right turn to N crs climbing to 6,500' within 14 mi. *Procedure turns on W side of N crs not authorized due to high terrain.
MELBOURNE, FLA. Melbourne-Eau Gallie 257 kc; MLB Arpt. SBRAZ-DTV	N—1,200' (NE crs Orlando) E—Min. en route altitude S—1,200' (N crs W. Palm Beach) W—Min. en route altitude	None	N	10 mi.—1,200' W side N crs 15 mi.—1,200' W side N crs 20 mi.—1,200' W side N crs 25 mi.—1,200' W side N crs	800	158	2.3	26	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		Climb to 1,200' on S crs. *Rwy No. 16.
Long Range Proving Ground AFB (Cocoa)	N—1,200' (NE crs Orlando) E—Min. en route altitude S—1,200' (N crs W. Palm Beach) W—Min. en route altitude	S crs between rng and point 2 min. out	S	10 mi.—1,200' E side S crs 15 mi.—1,200' E side S crs 20 mi.—1,200' E side S crs 25 mi.—1,200' E side S crs	800	17	6.7	9	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,200' on N crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
MEMPHIS, TENN. Memphis Arpt. 371 kc; MEM SBRAZ-DTV	NE—2,000′ (Jackson Range) S—1,300′ (Greenwood Range) S—800′ (Nesbitt FM) (final) SW—1,600′ (SE crs Little Rock) N—2,000′ (Advance Range)	None	S	10 mi.—1,300′ E side S crs 15 mi.—1,300′ E side S crs 20 mi.—1,300′ E side S crs 25 mi.—1,300′ E side S crs	800	356	2.1	269	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000′ on N crs, or alternate procedure (when directed by ATC), turn right, climb to 2,000′ on NE course. *Runway No. 2.
MERIDIAN, MISS. Key Field 239 kc; MEI SBMRAZ-DTV	NE—2,000′ (Birmingham Range) S—Min. en route altitude W—1,800′ (Jackson Range) N—Min. en route altitude	None	N	10 mi.—1,500′ W side N crs 15 mi.—1,500′ W side N crs 20 mi.—1,500′ W side N crs 25 mi.—1,500′ W side N crs	1,000	154	2.8	297	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,000′ on S crs.
MIAMI, FLA. Miami International Airport 265 kc; MIA SBRAZ-DTV	E—Min. en route altitude SE—1,400′ (E crs Key West) W—1,100′ (SE crs Ft. Myers) W—600′ (Krome FM) (final) N—1,300′ (W crs W. Palm Beach)	None	W	10 mi.—1,100′ S side W crs 15 mi.—1,100′ S side W crs 20 mi.—1,100′ S side W crs 25 mi.—1,100′ S side W crs	600	85	5.8	9	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,400′ on E crs, or alternate procedure (when directed by ATC), turn right, climb to 1,400′ on SE course. *Runway No. 9R.
MIDLAND, TEX. Midland Air Terminal 209 kc; MAF BMRLZ-DTV	NE—4,300′ (Big Spring Range) SE—Min. en route altitude SW—4,500′ (SE crs Wink) NW—4,500′ (W crs Big Spring)	None	SW	10 mi.—4,200′ S side SW crs 15 mi.—4,500′ S side SW crs 20 mi.—4,500′ S side SW crs 25 mi.—4,500′ S side SW crs	5,740	48	3.8	2,867	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 4,000′ on SE crs within 25 mi.
Midland Airpark	(MAKE LET-DOWN TO MIDLAND AIR TERMINAL. FLY VFR FROM MIDLAND AIR TERMINAL TO MIDLAND AIRPARK.)							2,805	R A T	1,000 1,000 300	1.0 3.0 1.0	1,000 1,000 300	2.0 3.0 1.0	
MILES CITY, MONT. Miles City Arpt. 320 kc; MLS SBMRAZ-DTV	N—Min. en route altitude E—4,200′ (Dickinson Range) S—Min. en route altitude W—5,000′ (Billings Range)	None	W	10 mi.—4,500′ S side W crs 15 mi.—4,500′ S side W crs 20 mi.—4,500′ S side W crs 25 mi.—4,500′ S side W crs	3,500	58	2.1	2,631	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 4,200′ on E crs within 20 mi.
MILFORD, UTAH Milford Arpt. 320 kc; MLF BMRLZ-DTV	NE—9,000′ (Delta Range) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	SW crs to 8,500′ within 25 mi.	SW	10 mi.—8,000′ W side SW crs 15 mi.—8,000′ W side SW crs 20 mi.—8,500′ W side SW crs 25 mi.—8,500′ W side SW crs	6,100	33	1.2	5,061	R A T	1,000 1,000 300	4.0 4.0 1.0	1,000 1,000 300	4.0 4.0 1.0	If not contact over range, climb to 9,000′ on NE crs within 25 mi. High terrain E and W of airport.
MILLINOCKET, ME. Millinocket Arpt. 344 kc; MLT SBRAZ-DTV	N—3,500′ (W crs Houlton) E—2,500′ (S crs Houlton) S—2,300′ (NW crs Bangor) W—5,000′ (NW crs Bangor)	None	E	10 mi.—1,800′ N side E crs 15 mi.—2,000′ N side E crs 20 mi.—2,000′ N side E crs 25 mi.—2,000′ N side E crs	1,300	292	1.6	408	R A T	600 800 400	1.5 2.0 1.0	600 800 400	1.5 2.0 1.0	Climb to 2,300′ on S crs.
MILLVILLE, N. J. Millville Arpt. 365 kc; MIV SBRAZ-DTV	NE—1,500′ (SE crs N. Phila.) SE—1,500′ (SW crs Atlantic City VAR) SW—1,500′ (E crs Baltimore) NW—1,800′ (W crs Philadelphia)	None	NW	10 mi.—1,600′ W side NW crs 15 mi.—1,600′ W side NW crs 20 mi.—1,600′ W side NW crs 25 mi.—1,600′ W side NW crs	1,100	144	4.2	68	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 2.0 2.0 1.0	Climb to 1,500′ on SE crs within 25 mi. NOTE: Night operations authorized on Runwy 12/30 only.
MILWAUKEE, WIS. Gen. Mitchell Arpt. 242 kc; MKE SBRAZ-DTV	N—Min. en route altitude E—2,000′ (Muskegon Range) S—2,100′ (E crs Rockford) S—2,000′ (Kenosha FM) S—1,400′ (Franksville FM) (final) W—2,500′ (Madison Range) W—2,300′ (Genesee FM)	None	S	10 mi.—1,900′ E side S crs 15 mi.—1,900′ E side S crs 20 mi.—1,900′ E side S crs 25 mi.—2,000′ E side S crs	1,400	352	2.2	698	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.0 2.0 1.0	Climb to 2,300′ on N crs within 25 mi. *Runways No. 1 and 36.
MINNEAPOLIS, MINN. Minneapolis-St. Paul International Arpt. 266 kc; MSP SBRAZ-DTV	N—2,600′ (Duluth Range) SE—2,400′ (N crs Rochester) SE—1,500′ (Hastings FM) (final) SW—3,000′ (Sioux Falls Range) SW—2,300′ (Jordan FM) NW—2,600′ (Alexandria Range) NW—2,300′ (Hamel FM)	None	SE	10 mi.—2,300′ N side SE crs 15 mi.—2,400′ N side SE crs 20 mi.—2,400′ N side SE crs 25 mi.—2,400′ N side SE crs	1,500	298	3.6	838	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500′ on NW crs within 25 mi. *Runway No. 29.
MINOT, N. D. Port O'Minot Arpt. 209 kc; MOT BMRLZ-DTV	NE—2,800′ (S crs Rivers) SE—2,900′ (N crs Bismarck) SW—Min. en route altitude NW—Min. en route altitude	None	SE	10 mi.—2,900′ E side SE crs 15 mi.—2,900′ E side SE crs 20 mi.—3,400′ E side SE crs 25 mi.—3,400′ E side SE crs	2,400	301	3.1	1,723	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 3,500′ on NW crs. *Runway No. 30.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
MIRAMAR, CALIF. Miramar NAAS 310 kc; NMI MRAWZ	*(To be made on San Diego Range) N—4,000' (SE crs Long Beach) E—9,000' (El Centro Range) E—2,600' (La Mesa FM) SE—Min. en route altitude SE—2,600' (Coronado FM) W—Min. en route altitude	None	S (Miramar Range)	10 mi.—2,500' W side S crs 15 mi.—3,000' W side S crs 20 mi.—3,000' W side S crs 25 mi.—NA (From Miramar Range)	2,000 (over Miramar Range)	325 (from Miramar Range)	3.2	475	R A T	800 1,000 300	1.0 2.0 1.0	800 1,000 300	1.0 2.0 1.0	Climb to 1,500' turning left to 270° mag. Maintain 1,500' until 5 mi. W of N crs San Diego. Contact San Diego approach control for further clearance. *All initial approaches will be made on San Diego range. If cleared by San Diego approach control, these initial approaches are applicable to the Miramar range.
MISSOULA, MONT. Missoula Co. Arpt. 308 kc; MSO SBMRAZ-DTV	N—Min. en route altitude SE—9,000' (Drummond Range) S—Min. en route altitude NW—9,000' (Superior Range) NW—6,200' (Alberton FM) (final)	None	NW	10 mi.—8,000' N side NW crs 15 mi.—8,300' N side NW crs 20 mi.—9,000' N side NW crs 25 mi.—9,000' N side NW crs	6,200	120	1.5	3,203	R A T	3,000 3,000 2,500	2.0 3.0 2.0	3,000 3,000 2,500	2.0 3.0 2.0	Turn right and climb to 9,000' on NW crs shuttling within 10 mi. of range.
MITCHEL (Hempstead) N. Y. Mitchel AFB 227 kc; HEM SBMRAZ-DT	NE—1,500' (E crs LaGuardia) E—1,500' (SW crs Islip VAR) E—800' (Babylon FM) (final) SW—1,500' (E crs Idlewild) W—1,500' (NE crs Idlewild)	None	E	10 mi.—1,500' S side E crs 15 mi.—1,500' S side E crs 20 mi.—NA 25 mi.—NA	800	301	3.2	85	R A T	500 800 300	2.0 2.0 1.0	500 800 300	3.0 2.0 2.0	Climb straight ahead to 800, make immediate right turn to 76° climbing to 1,500', intersect E crs LaGuardia, then proceed to and hold at 1,500' on E crs LaGuardia E of St. James Int. All altitudes must be strictly adhered to.
Roosevelt Fld. (Mineola)	NE—1,500' (E crs LaGuardia) E—1,500' (SW crs Islip VAR) E—800' (Babylon FM) (final) SW—1,500' (E crs Idlewild) W—1,500' (NE crs Idlewild)	None	E	10 mi.—1,500' S side E crs 15 mi.—1,500' S side E crs 20 mi.—NA 25 mi.—NA 25 mi.—1,500' N side NE crs	800	310	4.5	90	R A T T	600 800 300 300	1.5 2.0 1.0 1.0	600 800 300 300	1.5 2.0 1.0 1.0	If not contact over Mitchel AFB, make right climbing turn to 1,500', intercept E crs LaGuardia, then proceed to and hold at 1,500' on E crs of LaGuardia E of St. James Int.
MOBILE, ALA. Bates Fld 248 kc; MOB SBRAZ-DTV	NE—1,500' (N crs Pensacola) (Goodway Int.) SE—1,300' (SW crs Pensacola) W—1,500' (Keesler Range) NW—1,500' (NE crs Keesler)	None	NE	10 mi.—1,500' N side NE crs 15 mi.—1,500' N side NE crs 20 mi.—1,500' N side NE crs	1,200	271	7.5	217	R A	500 800	2.0 2.0	500 800	2.0 2.0	Climb to 1,500 on W crs.
Brookley AFB	NE—1,500' (N crs Pensacola) (Goodway Int.) SE—1,300' (SW crs Pensacola) W—1,500' (Keesler Range) NW—1,500' (NE crs Keesler)	None	NW	10 mi.—1,400' W side NW crs 15 mi.—1,400' W side NW crs 20 mi.—1,500' W side NW crs 25 mi.—1,500' W side NW crs	1,000	145	3.2	26	R A T	500 800 300	2.0 2.0 1.0	500 800 300	3.0 2.0 1.0	Climb to 1,300' on SE crs.
MOFFETT (Mountain View), CALIF. Moffett NAS 323 kc; NUQ SMRLZ	NE—5,000' (SE crs Oakland) SE—5,000' (Fresno Range) SE—4,000' (Morgan Hill FM) SE—2,500' (Evergreen FMH) SW—6,000' (NW crs Salinas) NW—6,000' (SW crs San Francisco)	NW crs to 4,000' within 10 mi. (Procedure turn to N)	SE	10 mi.—1,500' S side SE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	600	323	1.6	39	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.4 1.0	Climb to 1,500' on NW crs, make procedure turn to N climbing to cross range at 2000'. CAUTION: (1) 640' msl radio tower approx 284°, 4 mi. from fld. (2) High tension lines approx 0.5 mi N of runway. NOTE: This procedure not authorized for civil acft.
MOLINE, ILL. Quad-City Arpt. 224 kc; MLI SBRAZ-DTV	N—Min. en route altitude E—2,000' (N crs Peoria) S—2,000' (N crs Burlington) W—2,000' (N crs Burlington)	None	W	10 mi.—2,000' S side W crs 15 mi.—2,000' S side W crs 20 mi.—2,000' S side W crs 25 mi.—2,000' S side W crs	*1,500	140	10.6	590	R A T	1,000 1,000 300	2.0 3.0 1.0	1,000 1,000 300	2.0 3.0 1.0	If not contact over range, climb to 2,500' on E crs within 25 mi. *Proceed by VFR from range to arpt. CAUTION: Rdo twrs 1,067' & 1045' msl, 2.5 & 6 mi. respectively, NE & NNE of arpt.

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
MOLINE, ILL.—Con. Davenport Arpt. (Davenport, Iowa)	N—Min. en route altitude E—2,000′ (N crs Peoria) S—2,000′ (E crs Burlington) W—2,000′ (N crs Burlington)	None	W	10 mi.—2,000′ S side W crs 15 mi.—2,000′ S side W crs 20 mi.—2,000′ S side W crs 25 mi.—2,000′ S side W crs	1,300	35	2.5	753	R A T	500 800 300	1.0 2.0 1.0	NA NA NA		Climb to 2,000′ on E crs. NOTE: Minima approved for acft with stall speeds of 75 mph or less only. Weather observations 0730-1630 daily.
MONROE, LA. Selman Field 359 kc; MLU SBMRLZ-DTV	NE—Min. en route altitude E—1,500′ (Jackson Range) SW—Min. en route altitude W—1,500′ (Shreveport Range)	None	SW	10 mi.—1,200′ E side SW crs 15 mi.—1,200′ E side SW crs 20 mi.—1,200′ E side SW crs 25 mi.—1,200′ E side SW crs	700	25	2.8	79	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,400′ on NE crs within 25 mi.
MONTPELIER, VT. Barre-Montpelier Arpt: 257 kc; MPV BMRLZ-DTV	NE—Min. en route altitude SE—Min. en route alternate SW—4,500′ (SE crs Burlington) NW—6,000′ (N crs Burlington)	None	NE	10 mi.—4,000′ N side NE crs 15 mi.—4,000′ N side NE crs 20 mi.—4,000′ N side NE crs 25 mi.—4,000′ N side NE crs	3,000	230	2.7	1,149	R A T	1,400 2,000 800	2.0 3.0 2.0	1,900 2,000 1,400	3.0 3.0 2.0	Climb to 4,500′ on SW crs within 10 mi.
MORGANTOWN, W. VA. Morgantown Arpt. 269 kc; MGW SBMRLZ-DTV	NE—4,000′ (SE crs Pittsburgh) SE—6,000′ (W crs Front Royal) SW—4,000′ (W crs Elkins) SW—2,600′ (N crs Elkins) NW—3,000′ (W crs Pittsburgh)	None	NW	10 mi.—2,600′ W side NW crs 15 mi.—2,600′ W side NW crs 20 mi.—2,600′ W side NW crs 25 mi.—2,600′ W side NW crs	*2,100	173	5.6	1,256	R A T	700 800 300	1.5 2.0 1.0	700 800 300	2.0 2.0 1.0	If not contact over range, climb to 4,000′ on SW crs within 25 mi. *2,295′ terrain 4 mi. SE of arpt.
MUSCLE SHOALS, ALA. Muscle Shoals Arpt. 251 kc; MSL SBRAZ-DTV	N—2,600′ (SW crs Nashville) SE—2,500′ (N crs Birmingham) S—Min. en route altitude W—2,000′ (S crs Jackson)	None	SE	10 mi.—1,800′ N side SE crs 15 mi.—1,800′ N side SE crs. 20 mi.—1,800′ N side SE crs 25 mi.—1,800′ N side SE crs	1,300	282	4.6	546	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 2.0 2.0 1.0	Climb to 2,000′ on W crs.
MUSKEGON, MICH. Muskegon Co. Arpt. 263 kc; MKG SBMLRZ-DTV	NE—Min. en route altitude SE—1,900′ (S W crs Grand Rapids) SW—Min. en route altitude NW—Min. en route altitude	None	SE	10 mi.—1,800′ E side SE crs 15 mi.—1,800′ E side SE crs 20 mi.—1,800′ E side SE crs 25 mi.—1,800′ E side SE crs	1,300	317	2.7	627	R S* A T	500 500 800 300	1.0 1.0 2.0 1.0	500 500 800 300	2.0 1.5 2.0 1.0	Climb to 2,000′ on NW crs. NOTE: Minima apply only to acft with stall speeds of 75 mph or less. *Runway 32.
NASHVILLE, TENN. Berry Field 304 kc; BNA SBRAZ-DTV	NE—2,500′ (NW crs Smithville) NE—1,200′ (Mt. Juliet FM) (final) SE—3,500′ (S crs Smithville) SE—2,500′ (Walterhill FM) SW—2,500′ (Jackson Range) NW—2,000′ (SW crs Bowling Green)	None	NE	10 mi.—2,000′ N side NE crs 15 mi.—2,500′ N side NE crs 20 mi.—3,000′ N side NE crs 25 mi.—3,000′ N side NE crs	1,200	249	2.6	606	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,500′ on SW crs, or alternate procedure (when directed by ATC) turn right, climb to 2,000′ on NW crs.
NEEDLES, CALIF. Needles Arpt. 375 kc; EED BMRLZ-DTV	N—8,000′ (SE crs Las Vegas) E—10,000′ (Prescott Range) S—6,000′ (Blythe Range) W—9,000′ (Daggett Range)	To 5,000′ on S crs within 25 mi.	N	10 mi.—4,000′ W side N crs 15 mi.—5,000′ W side N crs 20 mi.—5,000′ W side N crs 25 mi.—5,000′ W side N crs	2,000	196	1.9	990	R A T	1,000 1,000 1,000	2.0 3.0 2.0	1,000 1,000 1,000	3.0 3.0 3.0	Turn left and climb to 7,000′ on N crs within 25 mi.
NEWARK, N. J. Newark Arpt. 341 kc; EWR SBRAZ-DTV	NE—2,000′ (NW crs LaGuardia) SE—2,000′ (SW crs LaGuardia) SW—1,500′ (E crs Allentown) SW—900′ (Metuchen FM) (final) SW—*800′ (Linden Rbn) (final) W—2,700′ (NE crs Allentown)	None	SW	10 mi.—1,500′ S side SW crs 15 mi.—1,500′ S side SW crs 20 mi.—1,500′ S side SW crs 25 mi.—1,500′ S side SW crs	*800	58	1.1	18	R (R) S A T	500 500 #500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 #500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,000′ on NE crs, make left turn climbing to 2,700′ on heading of 290° before proceeding out W crs; or alternate procedure (when directed by ATC), climb to 1,000′ on NE crs, turn left and proceed to Paterson Rbn, climbing to 2,000′. *Descent to arpt may be started after passing Elizabeth Rbn, if Rbn not rec'd, final approach alt. over rng is 800′. #These minima apply only when Elizabeth Rbn is used. NOTE: Straight-in approach to Rwy 6.
NEWHALL, CALIF.	(NO PROCEDURE APPROVED)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)		Ceiling and visibility minimums: Day Ceiling (ft.)	Day Visibility (mi.)	Night Ceiling (ft.)	Night Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
NEW ORLEANS, LA. Moisant International Arpt. 388 kc; MSY SBRAZ-DTV (Procedure No. 1)	E—1,500' (Keesler Range) E—800' (Lake Shore FM) (final) S—1,300' (Grand Isle Rbn) W—1,400' (SW crs Baton Rouge) N—1,600' (Jackson Range) N-1,400' (Madisonville FM)	None	E	10 mi.—1,400' N side E crs 15 mi.—1,400' N side E crs 20 mi.—1,400' N side E crs 25 mi.—1,400' N side E crs	800	267	3.1	3	R (R) S A T	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,400' on W crs within 25 mi., or alternate procedure (when directed by ATC), climb to 1,300' on S crs within 25 mi.
(Procedure No. 2)	E—1,500' (Keesler Range) E—1,300' (Lake Shore FM) S—1,300' (Grand Isle Rbn) W—1,400' (SW crs Baton Rouge) W—940' (La Place FM) (final) N—1,600' (Jackson Range) N—1,400' (Madisonville FM)	None	W	10 mi.—1,400' S side W crs 15 mi.—1,400' S side W crs 20 mi.—1,400' S side W crs 25 mi.—1,400' S side W crs	940 (Over ILS outer mrkr)	91 (From ILS outer marker)	3.9	3	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on E crs within 25 mi., or alternate procedure (when directed by ATC), climb to 1,300' on S crs.
New Orleans Arpt.	E—1,500' (Keesler Range) E—1,300' (Lake Shore FM) S—1,300' (Grand Isle Rbn) W—1,400' (SW crs Baton Rouge) N—1,600' (Jackson Range) N—1,400' (Madisonville FM)	None	W	10 mi.—1,400' S side W crs 15 mi.—1,400' S side W crs 20 mi.—1,400' S side W crs 25 mi.—1,400' S side W crs	*940 (Over Lake Shore FM)	63 (From Lake Shore FM)	3.2	8	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	NA NA NA NA	----------	Climb to 1,500' on E crs within 25 mi. *If Lake Shore FM not received, descent below 940' not authorized.
New Orleans NAS	E—1,500' (Keesler Range) E—1,300' (Lake Shore FM) S—1,300' (Grand Isle Rbn) W—1,400' (SW crs Baton Rouge) N—1,600' (Jackson Range) N—1,400' (Madisonville FM)	None	W	10 mi.—1,400' S side W crs 15 mi.—1,400' S side W crs 20 mi.—1,400' S side W crs 25 mi.—1,400' S side W crs	900	63	8.1	7	R (R) S A T	500 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	NA NA NA NA NA	----------	If not contact over Lake Shore FM, climb to 1,500' on E course within 25 mi.
NEW YORK, N. Y. LaGuardia Field 209 kc; LGA SBRAZ-DTXV (Procedure No. 1)	NE—1,500' (NE crs Newark) NE—1,000' (Port Chester FM) (final). E—1,500' (NE crs Idlewild) SW—2,300' (SE crs Newark) NW—1,700' (S crs Poughkeepsie)	None	NE	10 mi.—1,500' N side NE crs 15 mi.—1,500' N side NE crs 20 mi.—1,500' N side NE crs 25 mi.—1,900' N side NE crs	*1,000	223	3.2	20	R (R) S# A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on SW crs. *Descent to 800' authorized after passing New Rochelle Rbn; descent to landing minimums authorized only after passing LaGuardia Range. #Runway No. 22.
(Procedure No. 2)	NE—1,500' (NE crs Newark) E—1,500' (NE crs Idlewild) SW—2,300' (SE crs Newark) SW—1,200' (Flatbush FM) (final) NW—1,700' (S crs Poughkeepsie)	None	SW	10 mi.—1,500' S side SW crs 15 mi.—1,500' S side SW crs 20 mi.—1,500' S side SW crs 25 mi.—1,500' S side SW crs	1,200 (Over Maspeth LOM)	43 (From Maspeth LOM)	4.5	20	R (R) S# A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on NE crs. *Descent to landing minimums authorized after passing Maspeth LOM. #Runway No. 4.
Flushing Arpt.	(MAKE LET-DOWN TO LA GUARDIA FLD. FLY VFR FROM LAGUARDIA FLD. TO FLUSHING ARPT.)													
New York International Arpt. 268 kc; IDL SMRA	NE—1,500' (W crs Mitchel) E—1,200' SW crs Mitchel) SW—1,500' (SE crs Lakehurst) SW—700' (SE crs Newark) (final) W—1,500' (SW crs LaGuardia)	None	SW	10 mi.—1,200' E side SW crs 15 mi.—1,200' E side SW crs 20 mi.—1,200' E side SW crs 25 mi.—1,200' E side SW crs	700	38	2.9	12	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn right to a heading of 130° intersecting the SW crs of Mitchel, climb to 1,500' on SW crs Mitchel, or alternate procedure (when directed by ATC), climb to 1,200' on SW crs Idlewild within 25 mi. *Rwy No. 4R.
Floyd Bennett NAS 379 kc; FBT SMLZ-DT	NE—2,000' (W crs Mitchel) E—1,500' (SW crs Mitchel) SW—1,500' (NE crs Philadelphia) W—2,000' (Newark Range) W—*800' (Newark Range) (final)	None	W	10 mi.—1,500' S side W crs 15 mi.—2,000' S side W crs 20 mi.—2,000' S side W crs 25 mi.—2,000' S side W crs	*800	122	1.9	16	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	1.5 3.0 1.0	Climb to 1,500' on E crs. *Descent below 1,200' not authorized until after passing SW crs LaGuardia.
NORFOLK, VA. Norfolk Arpt. 239 kc; ORF SBMRLZ-DTV	NE—1,300' (E crs Langley) SE—Min. en route altitude SW—Min. en route altitude SW—900' (S crs Langley) (final) NW—1,300' (W crs Langley)	None	SW	10 mi.—1,400' S side SW crs 15 mi.—1,400' S side SW crs 20 mi.—1,400' S side SW crs 25 mi.—1,400' S side SW crs	900	44	3.3	25	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,300' on NE crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
NORFOLK, VA.—Con. Norfolk NAS (East) 356 kc; NGU SBMRLZ	N—1,400' (E crs Langley) E—Min. en route altitude S—Min. en route altitude W—1,400' (SE crs Richmond)	None	E	10 mi.—1,400' S side E crs 15 mi.—1,400' S side E crs 20 mi.—1,400' S side E crs 25 mi.—1,400' S side E crs	700	276	2.9	15	R (R) S A T	500 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	Make climbing left turn to 1,400' on E crs.
Norfolk NAS–Chambers Fld.	(MAKE LET-DOWN TO NORFOLK NAS (EAST). FLY VFR FROM NORFOLK NAS (EAST) TO CHAMBERS FLD.)													
NORTH PLATTE, NEBR. Lee Bird Field 226 kc; LBF SBRAZ-DTV	N—Min. en route altitude E—4,200' (Grand Island Range) S—4,200' (E crs Hayes Center) W—5,000' (SE crs Scottsbluff) W—4,300' (N crs Hayes Center)	None	S	10 mi.—4,200' W side S crs 15 mi.—4,200' W side S crs 20 mi.—4,200' W side S crs 25 mi.—4,200' W side S crs	3,700	352	2.2	2,779	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 4,500' on N crs within 25 mi.
OAKLAND, CALIF. Oakland Arpt. 362 kc; OAK SBRAZ-DTXV	NE—5,000' (Sacramento Range) NE—3,000' (Bay Point FM) SE—6,000' (NE crs Salinas) SE—3,500' (Evergreen FM) SE—3,500' (Newark FM) (final) SW—3,000' (San Francisco Range NW—3,000' (SW crs Fairfield-Suisun)	None	SE	10 mi.—1,500' W side SE crs 15 mi.—1,500' W side SE crs 20 mi.—3,500' W side SE crs 25 mi.—3,500' W side SE crs	500	199	.2	5	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	If not contact over rng. climb to 3,500' on NW crs within 25 mi. CAUTION: 3,849' MSL terrain, 5 mi. SE of NE crs, 17 mi. out, and 2,187' MSL, 7 mi. NE of SE crs, 18 mi. out. 162' MSL tank, 0.5 mi. NE of SE crs, 3 mi. out.
Alameda NAS	(MAKE LET-DOWN TO OAKLAND RANGE. FLY VFR FROM OAKLAND RANGE TO ALAMEDA NAS)													
Hayward Arpt.	(MAKE LET-DOWN TO OAKLAND RANGE. FLY VFR FROM OAKLAND RANGE TO HAYWARD ARPT.)													
OGDEN, UTAH Ogden Arpt. 263 kc; OGD SBRAZ-DTV (Procedure No. 1)	N—11,000' (Malad City Range) N—10,000' (Corrine FM) E—12,000' (Ft. Bridger Range) S—12,000' (Fairfield Range) S—11,000' (Riverton FM) S—6,500' (Salt Lake City Range) S—5,250' (Layton FM) (final) W—11,000' (Lucin Range)	N crs 10,000' minimum (Procedure turn W)	S	10 mi.—7,500' W side S crs 15 mi.—7,500' W side S crs 20 mi.—7,500' W side S crs 25 mi.—10,000' W side S crs	5,250	107	3.4	4,455	R A T	*800 *800 400	2.0 2.0 1.0	*800 *1,000 400	3.0 3.0 1.0	If not contact over rang. climb to 9,000' on W crs. *Sliding scale not applicable except that landings are approved when the visibility is 1 mile or more provided that (1) ceiling is unlimited, (2) the restriction to visibility is due to haze or smoke and, (3) top of smoke level is below 7,500' S. L.
(Procedure No. 2)	N—11,000' (Malad City Range) N—5,450' (Corrine FM) (final) E—12,000' (Ft. Bridger Range) S—12,000' (Fairfield Range) S—11,000' (Riverton FM) W—11,000' (Lucin Range)	None	N	10 mi.—11,000' *W side N crs 15 mi.—11,000' *W side N crs 20 mi.—11,000' *W side N crs 25 mi.—11,000' *W side N crs	5,450	107	3.4	4,455	R A T	1,000 1,000 400	2.0 3.0 1.0	1,000 1,000 400	3.0 3.0 1.0	If not contact over range, climb to 9,000' on W crs. *All approaches must start from Corrine FM at 11,000'.
Hill AFB	N—11,000' (Malad City Range) N—5,780' (Corrine FM) (final) E—12,000' (Ft. Bridger Range) S—12,000' (N crs Fairfield) S—6,500' (Salt Lake City Range) W—11,000' (Lucin Range)	None	N	10 mi.—10,500' W side N crs 15 mi.—10,500' W side N crs 20 mi.—10,500' W side N crs 25 mi.—10,500' W side N crs	5,780	124	8.0	4,788	R A T	1,000 1,000 400	2.0 3.0 1.5	1,000 1,000 400	3.0 3.0 2.0	If not contact over range, make immediate right turn climbing to 11,000' on W crs.
OKLAHOMA CITY, OKLA. Will Rogers Fld. 350 kc; OKC SBRAZ-DTV (Procedure No. 1)	N—3,000' (NW crs Tulsa) N—2,500' (W crs Tinker) E—2,700' (SW crs Tulsa) S—2,500' (N crs Ft. Worth) W—2,500' (SE crs Gage) W—2,000' (Mustang FM) (final)	None	W	10 mi.—2,500' S side W crs 15 mi.—2,500' S side W crs 20 mi.—2,500' S side W crs 25 mi.—2,500' S side W crs	2,000	78	1.6	1,283	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 2,700' on E crs within 25 miles, or alternate procedure (when directed by ATC), climb to 3,000' on N crs.
(Procedure No. 2)	N—3,600' (NW crs Tulsa) N—2,500' (W crs Tinker) E—2,700' (SW crs Tulsa) E—2,250' (Newalla FM) (final) S—2,500' (N crs Ft. Worth) W—2,500' (SE crs Gage)	None	E	10 mi.—2,700' N side E crs 15 mi.—2,700' N side E crs 20 mi.—2,700' N side E crs 25 mi.—2,700' N side E crs	2,200 (Over Okla. City LFM)	256 (From Oklahoma City (LFM)	3.2	1,283	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 2,500' on W crs, or alternate procedure (when directed by ATC), climb to 3,000' on N crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
OKLAHOMA CITY—Continued Country Club Arpt.	(MAKE LET-DOWN TO WILL ROGERS FLD. FLY VFR FROM WILL ROGERS FLD. TO COUNTRY CLUB ARPT.)													
Tinker AFB	(MAKE LET-DOWN TO WILL ROGERS FLD. FLY VFR FROM WILL ROGERS FLD. TO TINKER AFB.)													
Tulakes Arpt.	(MAKE LET-DOWN TO WILL ROGERS FLD. FLY VFR FROM WILL ROGERS FLD. TO TULAKES ARPT.)													
OLATHE, KANS. Olathe NAS 371 kc; NUU SMRLZ	N—2,300' (SW crs Kansas City) E—2,300' (SE crs Kansas City) S—Min. en route altitude W—2,300' (SW crs Kansas City)	None	S	10 mi.—2,100' E side S crs 15 mi.—2,100' E side S crs 20 mi.—2,100' E side S crs 25 mi.—2,100' E side S crs	1,600	352	2.9	1,085	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Turn right (East) and climb to 2,100' on S crs within 25 miles. *Runway No. 35.
Gardner Arpt.	(MAKE LET-DOWN TO OLATHE NAS. FLY VFR FROM OLATHE NAS TO GARDNER ARPT.)													
OMAHA, NEBR. Omaha Arpt. 320 kc; OMA SBRAZ-DTV	N—2,300' (Sioux City Range) N—1,800' (California FM) (final) E—2,500' (Des Moines Range) SE—2,500' (St. Joseph Range) W—2,700' (N crs Lincoln) (Weston Int).	None	N	10 mi.—2,300' W side N crs 15 mi.—2,300' W side N crs 20 mi.—2,300' W side N crs 25 mi.—2,300' W side N crs	1,800	139	5.1	982	R (R) S* A T	500 500 800 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 1.5 1.5 2.0 1.0	Climb to 2,500' on SE crs. Upon reaching 2,500', proceed to E crs Omaha on heading of 45°, or alternate procedure (when directed by ATC) climb to 2,500' on SE crs. CAUTION: *1,192' MSL stack located 4 mi. NW of arpt, 1 mi. W of SE crs. 1,260' MSL bluff located 1.5 mi. NE of arpt. *Runway No. 14L.
ORLANDO, FLA. Orlando Mun. Arpt. No. 1 371 kc; ORL SBMRLZ-DTV	NE—1,500' (N crs Melbourne) NE—700' (Oviedo FM) (final) SE—1,500' (Melbourne Range) SW—Min. en route altitude NW—Min. en route altitude	None	NE	10 mi.—1,200' N side NE crs 15 mi.—1,200' N side NE crs 20 mi.—1,200' N side NE crs 25 mi.—1,200' N side NE crs	700	220	3.4	115	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on SW crs, or alternate procedure (when directed by ATC), turn right and climb to 1,500' on NW course, 236' MSL tank, 0.8 mi. W of arpt; 250' MSL stack, 3 mi. SW of arpt on SW crs of range.
Cannon Mills Arpt.	(MAKE LET-DOWN TO ORLANDO MUN. ARPT. NO. 1. FLY VFR FROM ORLANDO MUN. ARPT. NO. 1 TO CANNON MILLS ARPT.)													
Hoequist Arpt.	(MAKE LET-DOWN TO ORLANDO MUN. ARPT. NO. 1. FLY VFR FROM ORLANDO MUN. ARPT. NO. 1 TO HOEQUIST ARPT.)													
Orlando Mun. Arpt. No. 2	(MAKE LET-DOWN TO ORLANDO MUN. ARPT. NO. 1. FLY VFR FROM ORLANDO MUN. ARPT. NO. 1 TO ORLANDO MUN. ARPT. NO. 2)													
OTTO, N. MEX. CAA Int. Fld. 290 kc; OTO BMRLZ-DTV	N—10,000' (Santa Fe Rbn) E—9,000' (SE crs Las Vegas) E—8,500' (SW crs Las Vegas) S—Min. en route altitude W—11,000' (Albuquerque Range)	None	S	10 mi.—8,000' E side S crs 15 mi.—8,000' E side S crs 20 mi.—8,000' E side S crs 25 mi.—8,000' E side S crs	7,220	74	.2	6,226	R A T	1,000 1,000 500	1.0 2.0 1.0	1,000 1,000 500	2.0 2.0 2.0	If not contact over range, climb to 9,500' on E crs. NOTE: Minima apply to aeft with stall speeds of 75 mph or less.
PALACIOS, TEX. Palacios Arpt. 350 kc; PSX BMLZ-DTV	NE—1,500' (Houston Range) SE—Min. en route altitude SW—1,400' (Corpus Christi Range) NW—Min. en route altitude	None	NW	10 mi.—1,200' W side NW crs 15 mi.—1,200' W side NW crs 20 mi.—1,200' W side NW crs 25 mi.—1,200' W side NW crs	700	118	1.6	13	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	NA NA NA NA		Climb to 1,200' on SE crs within 25 mi.
PALMDALE, CALIF. Los Angeles Co. Arpt. 317 kc; PMD SBMRAZ-DTV	NE—8,000' (N crs Daggett) SE—12,000' (N crs Riverside) SW—9,000' (Newhall Range) NW—10,000' (S crs Bakersfield)	To 6,000' on NE crs within 16 mi.	NE	10 mi.—6,000' N side NE crs 15 mi.—5,000' N side NE crs 20 mi.—5,000'* N side NE crs 25 mi.—5,000'* N side NE crs	3,560	214	2.5	2,549	R A T	1,000 1,000 300	2.0 2.0 1.0	1,000 1,000 300	2.0 2.0 1.0	If not contact over range, make 180° right turn (N) and climb to 6,000' on NE crs within 16 mi. High terrain 5 mi. SW of arpt. *Danger area 16 mi. NE. of range-make procedure turn within 15 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
PATUXENT RIVER, MD. Patuxent River NAS 215 kc; NHK SMRLZ	NE—1,500' (SE crs Washington) SE—1,500' (W crs Chincoteague) SW—1,500' (S crs Andrews) SW—1,000' (S crs Andrews) (final) NW—1,500' (SE crs Washington)	None	SW	10 mi.—1,500' S side SW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,000	57	4.4	38	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 400	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on NE crs.
NAS Seaplane Base	NE—1,500' (SE crs Washington) SE—1,500' (W crs Chincoteague) SW—1,500' (S crs Andrews) NW—1,500' (SE crs Washington)	None	NE	10 mi.—1,500' E side NE crs* 15 mi.—1,500' E side NE crs* 20 mi.—1,500' E side NE crs* 25 mi.—1,500' E side NE crs*	1,000 (over Meekins Neck FM)	237 (from Meekins Neck FM)	6.1	SL	R A T	500 800 500	2.0 2.0 2.0	500 800 500	2.0 2.0 2.0	Climb to 1,500' on SW crs. *After passing range, proceed out NE crs for 2 min past Meekins Neck FM before making procedure turn.
PEMBINA, N. DAK. CAA Int. Field 260 kc; PMB SBMRAZ-DTV	N—2,000' (Winnipeg Range) E—Min. en route alternate S—2,100' (Grand Forks Range) W—Min. en route alternate	None	S	10 mi.—1,900' E side S crs 15 mi.—1,900' E side S crs 20 mi.—1,900' E side S crs 25 mi.—1,900' E side S crs	1,400	356	1.6	792	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.0 2.0	Climb to 2,000' on N crs. NOTE: Minima apply to acft with stall speeds of 75 mph or less.
PENDLETON, OREG. Pendleton Arpt. 341 kc; PDT SBRAZ-DTV	E—Min. en route alternate E—2,400' (SW crs Walla Walla) (final) SE—10,000' (Baker Range) SE—6,000' (La Grande FM) SE—4,000' (Cabbage Hill FM) W—4,000' (The Dalles Range) NW—5,000' (Yakima Range)	None	E	10 mi.—3,000' N side E crs 15 mi.—5,000' N side E crs 20 mi.—NA 25 mi.—NA	2,400	258	1.9	1,493	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.0 2.0 1.0	Climb to 4,000' on W crs within 25 mi.
PENSACOLA, FLA. Pensacola Arpt. 226 kc; PNS SBRAZ-DTV	NE—1,300' (Crestview Range) S—Min. en route altitude. SW—Min. en route altitude N—1,300' (W crs Crestview)	None	S	10 mi.—1,100' E side S crs 15 mi.—1,100' E side S crs 20 mi.—1,100' E side S crs 25 mi.—1,100' E side S crs	700	340	2.1	121	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,300' on N crs. *Runway No. 35.
Corry NAAS	NE—1,300' (Crestview Range) S—Min. en route altitude SW—Min. en route altitude N—1,300' (W crs Crestview)	None	NE	10 mi.—1,100' N side NE crs 15 mi.—1,200' N side NE crs 20 mi.—1,200' N side NE crs 25 mi.—1,300' N side NE crs	800	244	6.4	22	R (R) S A T	800 800 800 800 300	1.5 1.0 1.0 2.0 1.0	800 800 800 800 300	1.5 1.0 1.0 2.0 1.0	If not contact within 4.5 mi after passing range, climb to 1,500' on SW crs.
Pensacola NAS Seaplane Base	NE—1,300' (Crestview Range) S—Min. en route altitude SW—Min. en route altitude N—1,300' (W crs Crestview)	None	N	10 mi.—1,100' W side N crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	800 (from point) After contact is estab., fly heading of 221° to SPB.	160 contact	1.7	SL	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 500	1.5 1.0 1.0 2.0 1.0	If visual contact not estab. within 1.7 naut. mi. after passing range, climb to 1,300 on SE crs. CAUTION: Do not turn to heading of 221° to SPB until 1.7 naut. mi. from low cone and visual contact estab.
Saufley NAAS	(MAKE LET-DOWN TO PENSACOLA ARPT. FLY VFR FROM PENSACOLA ARPT TO SAUFLEY NAAS.)													
PEORIA, ILL. Peoria Arpt. 356 kc; PIA SBMRAZ-DTV	N—2,000' (E crs Burlington) E—2,000' (SW crs Joliet) S—2,000' (Springfield Range) W—2,000' (S crs Moline)	None	N	10 mi.—2,000' W side N crs 15 mi.—2,000' W side N crs 20 mi.—2,000' W side N crs 25 mi.—2,000' W side N crs	1,500	178	2.2	659	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 1.5 2.0 1.0	Climb to 2,000' on S crs.
PERRIN (Sherman), TEX. Perrin AFB 333 kc; PNX MRLW	NE—3,000' (NW crs Texarkana) SE—3,000' (E crs Ft. Worth) SW—3,000' (N crs Dallas) NW—3,000' (N crs Ft. Worth)	NE	NE	10 mi.—2,100' N side NE crs 15 mi.—2,100' N side NE crs 20 mi.—2,100' N side NE crs 25 mi.—2,100' N side NE crs	1,500	215 (On field)	.9	753	R A T	800 1,000 300	1.5 2.0 2.0	800 1,000 300	2.0 2.0 3.0	If not contact over range, climb to 3,000' on SE crs. NOTE: Range available 0700-1800 Mon. thru Fri. and on 30 min. call thru Carswell Arwys or Perrin Twr.
PHILADELPHIA, PA. International Arpt. 266 kc; PHL SBMRAZ-DTV	NE—1,800' (SE crs North Philadelphia) S—1,500' (NE crs Millville) W—1,800' (NE crs Washington) W—800' (Boothwyn FM) (final) N—2,000' (SW crs Willow Grove)	None	W	10 mi.—1,500' S side W crs 15 mi.—1,700' S side W crs 20 mi.—1,800' S side W crs 25 mi.—1,800' S side W crs	900	97	3.0	10	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,800' on NE crs. *Rwy No. 9.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
PHILADELPHIA, PA.—Continued North Philadelphia Arpt. 371 kc; PHF MLWZ-DTV (Procedure No. 1)	NE—1,500′ (E crs Allentown) SE—2,000′ (NE crs Philadelphia) SW—1,800′ (N crs Philadelphia) SW—1,600′ (Holmesburg FM) NW—2,000′ (N crs Philadelphia)	None	NE	10 mi.—1,500′ N side NE crs 15 mi.—1,500′ N side NE crs 20 mi.—1,500′ N side NE crs 25 mi.—1,500′ N side NE crs	800	238	2.8	129	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Turn right within 10 mi., climb to 1,500′ on NE crs, or alternate procedure (when directed by ATC), climb to 2,000′ on NW crs. *Rwy 24.
(Procedure No. 2)	NE—1,500′ (E crs Allentown) SE—2,000′ (NE crs Philadelphia) SW—1,800′ (N crs Philadelphia) NW—2,000′ (N crs Philadelphia)	None	SW	10 mi.—1,800′ S side SW crs 15 mi.—1,800′ S side SW crs 20 mi.—1,800′ S side SW crs 25 mi.—1,800′ S side SW crs	1,300 (over Holmesburg FM)	58 (from Holmesburg FM)	2.8	129	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 1,300′ on NE crs, or alternate procedure (when directed by ATC), climb to 2,000′ on NW crs.
PHILIPSBURG, PA. Philipsburg Arpt. 224 kc; PSB SBRAZ-DTV	E—4,000′ (S crs Williamsport) S—4,500′ (E crs Altoona) W—4,000′ (Youngstown Range) NW—Min. en route altitude	None	NW	10 mi.—3,500′ W side NW crs 15 mi.—3,500′ W side NW crs 20 mi.—3,500′ W side NW crs 25 mi.—3,500′ W side NW crs	3,000	169	5.0	1,903	R A T	600 1,000 500	2.0 3.0 1.0	1,000 1,000 500	2.0 3.0 2.0	Climb to 4,500′ on S crs within 25 mi. 2,440′ MSL tower, 3.5 mi. ENE of arpt.
Albert Arpt.	E—4,000′ (S crs Williamsport) S—4,500′ (E crs Altoona) W—4,000′ (Youngstown Range) NW—Min. en route altitude	None	E	10 mi.—3,500′ N side E crs 15 mi.—3,500′ N side E crs 20 mi.—3,500′ N side E crs 25 mi.—3,500′ N side E crs	3,000	283	6.3	1,750	R S A T	600 1,000 500	2.0 2.0 1.0	800 1,000 500	2.0 2.0 1.0	Climb to 4,000′ on W crs.
PHOENIX, ARIZ. Sky Harbor Arpt. 326 kc; PHX SBMRAZ-DTV	N—10,000′ (S crs Prescott) N—5,000′ (NE crs Phoenix VAR) E—Min. en route altitude S—6,000′ (E crs Gila Bend) W—6,000′ (N crs Gila Bend) W—2,600′ (Perryville FM)	None	E	10 mi.—2,600′ S side E crs 15 mi.—2,600′ S side E crs 20 mi.—4,100′ S side E crs 25 mi.—4,100′ S side E crs	2,100	261	1.9	1,120	R A T	500 800 300	1.0 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 6,000′ on W crs. 2,000′ hills 5 mi. S of arpt.
Litchfield Park NAF	(MAKE LET-DOWN TO SKY HARBOR ARPT. FLY VFR FROM SKY HARBOR ARPT. TO LITCHFIELD PARK NAF.)													
PIERRE, S. DAK. Pierre Arpt. 347 kc; PIR SBRAZ-DTV	N—3,200′ (Bismarck Range) E—3,300′ (SW crs Huron) S—Min. en route altitude W—4,400′ (Rapid City Range) W—3,500′ (Philip Rbn)	None	E	10 mi.—2,900′ N side E crs 15 mi.—3,100′ N side E crs 20 mi.—3,100′ N side E crs 25 mi.—3,100′ N side E crs	*2,400	249	3.5	1,742	R A T	500 800 300	1.5 2.0 1.0	500 800 300	2.0 2.0 1.0	Climb to 3,500′ on W crs within 25 mi. *If procedure turn made beyond 10 mi., the final approach alt. is 2,600′.
PITTSBURGH, PA. Allegheny Co. Arpt. 254 kc; PIT SBRAZ-DTV	E—4,500′ (N crs Altoona) E—2,600′ (New Alexandria FM) SE—4,500′ (S crs Altoona) SE—2,500′ (Mt. Pleasant FM) W—2,500′ (SE crs Cleveland) W—2,000′ (Cecil FM) (final) NW—2,500′ (S crs Youngstown)	None	W	10 mi.—2,500′ S side W crs 15 mi.—2,500′ S side W crs 20 mi.—2,500′ S side W crs 25 mi.—2,500′ S side W crs	2,000	85	2.3	1,252	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,800′ on E crs within 25 mi., or alternate procedure (when directed by ATC), climb to 4,000′ on SE crs, and hold NW of Mt. Pleasant FM on SW side of SE crs. *Runway No. 9
POCATELLO, IDAHO Phillips Field 257 kc; PIH BMRLZ-DTV	N—Min. en route altitude E—Min. en route altitude S—11,000′ (Malad City Range) W—7,000′ (N crs Burley)	None	W	10 mi.—6,500′ N side W crs 15 mi.—6,500′ N side W crs 20 mi.—6,500′ N side W crs 25 mi.—6,500′ N side W crs	5,240	326	.5	4,446	R (R) A T	800 500 800 300	1.5 1.0 2.0 1.0	800 500 800 300	2.0 2.0 2.0 1.0	If not contact over range, turn left and climb to 7,000′ on W crs within 25 mi. NOTE. Procedure turn NA on S side of W crs due to high terrain.
POPE (Ft. Bragg), N. C. Pope Field 238 kc; FTB SBMRAZ	NE—1,900′ (S crs Raleigh) S—1,900′ (S crs Raleigh) SW—Min. en route altitude N—Min. en route altitude	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	1,900	355	2.3	220	R A T	600 1,000 500	2.0 3.0 1.0	600 1,000 500	3.0 3.0 2.0	Climb to 1,500′ on N crs 802′ MSL twr, 0.5 SW of range.
Grannis Arpt. (Fayetteville)	NE—1,900′ (S crs Raleigh) S—1,900′ (S crs Raleigh) SW—Min. en route altitude N—Min. en route altitude	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	*980	147	11.2	188	R A T	800 1,000 300	2.0 2.0 1.0	800 1,000 300	2.0 2.0 1.0	If not contact over range, climb to 1,500′ on N crs. *If unable to maintain contact at 980′ MSL, approach shall be discontinued and climb made to 1,500′ on N crs. NOTE. 992′ MSL twr, 0.5 mi. SW of range.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
PORTLAND, MAINE Portland Arpt. 215 kc; PWM BMRLZ-DTV	NE—1,800′ (Augusta Range) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	None	NW	10 mi.—1,500′ W side NW crs 15 mi.—2,000′ W side NW crs 20 mi.—2,000′ W side NW crs 25 mi.—2,500′ W side NW crs	1,000	149	2.2	60	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500′ on SE crs. *Runway No. 14.
PORTLAND, OREG. Portland Arpt. 232 kc; PDX SBRAZ-DTV	N—5,000′ (Toledo Range) N—2,500′ (Woodland FM) N—1,500′ (La Center FM) (final) E—7,000′ (The Dalles Range) E—6,000′ (Stevenson FM) E—4,000′ (Washougal FM) S—8,000′ (Eugene Range) W—Min. en route altitude	None	N	10 mi.—2,000′ W side N crs 15 mi.—2,000′ W side N crs 20 mi.—3,000′ W side N crs 25 mi.—4,000′ W side N crs	1,500	169	3.2	23	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	700 600 800 300	1.5 1.0 2.0 1.0	Climb to 3,000′ on S crs, climbing to 2,000′ within 10 mi. of range.
POUGHKEEPSIE, N.Y. Dutchess Co. Arpt. 236 kc; POU SBRAZ-DTV	N—5,000′ (Albany Range) E—3,000′ (NE crs LaGuardia) S—3,000′ (NW crs LaGuardia) W—3,500′ (NW crs LaGuardia)	S crs to 3,000′ within 10 mi.	S	10 mi.—2,600′ E side S crs 15 mi.—2,600′ E side S crs 20 mi.—2,600′ E side S crs 25 mi.—2,600′ E side S crs	2,100	5	5.2	166	R A T	600 800 300	1.5 2.0 1.0	NA NA NA		Climb to 2,500′ on N crs within 25 mi.
PRESCOTT, ARIZ. Prescott Arpt. 347 kc; PRC SBRAZ-DTV	E—10,000′ (Winslow Range) SE—10,000′ (Phoenix Range) W—10,000′ (Needles Range) NW—Min. en route altitude	None	NW	10 mi.—7,000′ E side NW crs 15 mi.—8,000′ E side NW crs 20 mi.—8,000′ E side NW crs 25 mi.—8,000′ E side NW crs	6,500	142	5.5	5,042	R A T	1,800 1,500 1,000	2.0 3.0 2.0	1,500 1,500 1,000	2.0 3.0 2.0	If not contact within 2 mi. after passing range, turn left and climb to 8,000′ on NW crs within 25 mi.
PRESQUE ISLE, MAINE Presque Isle Arpt. 388 kc; PQI SBRAZ-DT	NE—2,700′ (SE crs Mont Jeli) NE—2,000′ (Limestone FM) E—3,000′ (NW crs Blissville) S—2,500′ (W crs Houlton) W—Min. en route altitude	None	NE	10 mi.—2,000 N side NE crs 15 mi.—2,000 N side NE crs 20 mi.—2,000 N side NE crs 25 mi.—2,000 N side NE crs	1,500	189	3.2	534	R (R) S A T	500 500 500 800 500	1.5 1.0 1.0 2.0 1.0	600 500 500 800 500	2.0 1.5 1.5 2.0 1.0	Climb to 2,700′ on S crs.
SPB Range	(SEE SPRAGUEVILLE, MAINE)													
PROVIDENCE, R. I. Green Arpt. 347 kc; PVD SBMRLZ-DTV	N—1,700′ (SW crs Boston) E—1,500′ (E crs Quonset Point) SW—1,600′ (SE crs Moosup VAR) SW—1,000′ (Wyoming FM) (final) W—1,600′ (SW crs Boston)	None	SW	10 mi.—1,600 W side SW crs 15 mi.—1,700 W side SW crs 20 mi.—1,700 W side SW crs 25 mi.—1,700 W side SW crs	1,000	47	3.3	56	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,700′ on N crs, or alternate procedure (when directed by ATC), climb to 2,700′ on SW crs of Squantum. *Rwy 5.
PUEBLO, COLO. Pueblo Arpt. No. 1 302 kc; PUB SBRAZ-DTV	N—6,000′ (Colorado Springs Range) N—7,000′ (Fountain FM) (S bound only) E—*6,000′ (La Junta Range) S—7,500′ (Trinidad Range) W—Min. en route altitude	None	S	10 mi.—6,600 E side S crs 15 mi.—7,500 E side S crs 20 mi.—7,500 E side S crs 25 mi.—7,500 E side S crs	**5,800	328	1.8	4,853	R A T	500 800 400	1.5 2.0 1.0	500 800 400	2.0 2.0 1.0	Turn right and climb to 6,000′ on E crs within 25 mi., or alternate procedure (when directed by ATC), climb to 8,000′ on N crs. *If contact is not estab. at 6,000′ prior to reaching range, climb to 6,600′ on S crs within 10 mi. for procedure turn. **If procedure turn is accomplished beyond 10 mi. the altitude on final approach is 6,100′.
PULASKI, VA. Loving Field 272 kc; PSK SBMRAZ-DTV	E—6,000′ (Roanoke Range) SE—Min. en route altitude SW—7,000′ (Tri-City Range) NW—Min. en route altitude	None	E	10 mi.—5,000 N side E crs 15 mi.—5,000 N side E crs 20 mi.—5,000 N side E crs 25 mi.—5,500 N side E crs	3,700	247	2.1	2,200	R A T	1,500 2,000 1,000	2.0 2.0 2.0	1,500 2,000 1,000	2.0 2.0 2.0	Climb to 7,000′ on SW crs.
QUANTICO, VA. Quantico MCAS 208 kc; NYG SMRLZ	N—1,800′ (SE crs Arcola) E—1,500′ (NW crs Patuxent) S—1,500′ (NW crs Richmond) S—1,000′ (NW crs Tappahannock) (final) W—2,000′ (SW crs Arcola)	None	S	10 mi.—1,500 E side S crs 15 mi.—1,500 E side S crs 20 mi.—1,500 E side S crs 25 mi.—2,000 E side S crs	1,000	20	4.6	10	R (R) S* A T	800 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	800 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	Turn left and climb to 1,500′ on W crs within 15 mi. *Runway No. 2.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
QUONSET POINT, R. I. Quonset Point NAS 356 kc; NCO SBMRLZ	E—Min. en route altitude SE—Min. en route altitude W—1,500' (SW crs Providence) NW—1,500' (Providence Range)	None	SE	10 mi.—1,500 W side SE crs 15 mi.—1,500 W side SE crs 20 mi.—1,500 W side SE crs 25 mi.—1,500 W side SE crs	800	329	2.2	14	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Make 180° turn to right (East) and climb to 1,500' on SE crs. *Runway No. 34.
RALEIGH, N. C. Raleigh-Durham Arpt. 330 kc; RDU SBRAZ-DTV	NE—1,800' (SE crs Blackstone) SE—Min. en route altitude SE—1,300' (Knightdale FM) (final) S—1,900' (NE crs Florence) NW—2,100' (NE crs Greensboro)	None	SE	10 mi.—1,600 N side SE crs 15 mi.—1,600 N side SE crs 20 mi.—1,600 N side SE crs 25 mi.—1,600 N side SE crs	1,300	296	3.9	435	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000' on NW crs within 25 mi., or alternate procedure (when directed by ATC), climb to 1,800' on NE course. *Runway No. 32.
Raleigh Arpt.	(MAKE LET-DOWN TO RALEIGH-DURHAM ARPT. FLY VFR FROM RALEIGH-DURHAM ARPT. TO RALEIGH ARPT.)													
RANDOLPH (San Antonio), TEX. Randolph Field 359 kc; RND BMRLZ	NE—Min. en route altitude SE—Min. en route altitude SW—2,100' (S crs San Antonio) NW—2,500' (N crs San Antonio)	On SE crs to 2,000' within 8 mi.	SE	8 mi.—2,000' E side SE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,500	325	2.2	761	R A T	500 800 300	2.0 2.0 1.0	500 800 300	3.0 2.0 1.0	If not contact within 3 mi. after passing range, make climbing right turn to 2,500' and proceed out NE crs.
RAPID CITY, S. DAK. Rapid City AFB 221 kc; RAP SBRAZ-DTV	N—Min. en route altitude E—4,400' (Pierre Range) S—5,500' (NE crs Scottsbluff) W—9,000' (SE crs Sheridan)	None	S	10 mi.—5,000' E side S crs 15 mi.—5,500' E side S crs 20 mi.—5,500' E side S crs 25 mi.—5,500' E side S crs	*4,500	345	5.5	3,242	R (R) S# A T	600 600 600 800 300	2.0 1.5 1.5 2.0 1.0	600 600 600 800 300	3.0 2.0 2.0 2.0 1.0	Climb to 5,000' on N crs within 25 mi. *CAUTION: 3,425' terrain 2.5 mi. N of range. #Runway No. 35.
RAWLINS, WYO. Rawlins Arpt.	(SEE SINCLAIR, WYO.)													
RED BLUFF, CALIF. Bidwell Field 338 kc; RBL SBRAZ-DTV	N—11,000' (Fort Jones Range) N—3,000' (Redding FM) E—Min. en route altitude S—3,000' (Williams Range) W—Min. en route altitude	None	S	10 mi.—2,000' E side S crs 15 mi.—2,000' E side S crs 20 mi.—2,000' E side S crs 25 mi.—2,000' E side S crs	1,200	323	2.5	354	R (R) S* A T	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	600 600 600 800 300	1.5 1.0 1.0 2.0 1.0	Make 180° right turn and climb to 2,000' on S crs within 25 mi. *Runway No. 33.
REDMOND, OREG. Roberts Field 368 kc; RDM SBRAZ-DTV	N—7,000' (The Dalles Range) SE—Min. en route altitude S—10,000' (Klamath Falls Range) NW—Min. en route altitude	On N crs to 7,000'	NW	12 mi.—6,000' N side NW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	4,600	161	4.3	3,077	R (R) A T	700 700 800 300	1.5 1.0 2.0 1.0	700 700 800 300	1.5 1.5 2.0 1.0	Make right turn and climb to 7,000' on S crs within 15 mi. CAUTION: 6,400' terrain 5 mi S NW crs, 20 mi. from range.
RENO, NEV. United Air Lines Arpt. 254 kc; RNO SBRAZ-DTV	NE—11,000' (Lovelock Range) NE—9,500' (Wadsworth FM) S—Min. en route altitude SW—12,000' (Donner Summit Rhn) N—Min. en route altitude	N crs to 10,000' within 25 mi.	N	10 mi.—8,500' E side N crs 15 mi.—8,500' E side N crs 20 mi.—8,500' E side N crs 25 mi.—9,000' E side N crs	7,000	161	3.0	4,404	R A T	2,500 2,500 1,000	3.0 3.0 3.0	2,500 2,500 1,000	3.0 3.0 3.0	Make immediate left turn and climb to 9,000 on N crs within 20 mi.
Reno Arpt.	(MAKE LET-DOWN TO UNITED AIRLINES ARPT. FLY VFR FROM UNITED AIRLINES ARPT TO RENO ARPT.)													
RICHMOND, TEX.	(NO ASSOCIATED AIRPORT)													
RICHMOND, VA. Byrd Field 260 kc; RIC SBRAZ-DTV	N—1,500' (SW crs Washington) SE—1,500' (W crs Langley) SW—1,500' (SE crs Blackstone) SW—700' (Chester FM) (final) NW—3,000' (NE crs Lynchburg)	None	SW	10 mi.—1,500' E side SW crs 15 mi.—1,500' E side SW crs 20 mi.—1,500' E side SW crs 25 mi.—1,500' E side SW crs	700	20	2.0	167	R (R) S* A T	500 *500 500 800 300	1.5 2.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on N crs. *Rwy No. 2.
RIVERSIDE, CALIF. March AFB 388 kc; RIV SBMRAZ-DTV	N—12,000' (W crs Daggett) N—5,000' (Fontana FM) E—13,000' (SE crs Daggett) E—10,000' (Banning FM) S—Min. en route altitude W—5,000' (Los Angeles Range)	None	W	10 mi.—4,000' N side W crs 15 mi.—5,000' N side W crs 20 mi.—5,000' N side W crs 25 mi.—5,000' N side W crs	4,000	128	2.9	1,536	R A T	2,000 2,000 1,000	1.5 3.0 1.0	2,000 2,000 1,000	2.0 3.0 2.0	Make right turn and climb to 5,000' on W crs.
Arlington Arpt.	(MAKE LET-DOWN TO MARCH AFB. FLY VFR FROM MARCH AFB TO ARLINGTON AIRPORT.)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
ROANOKE, VA. Woodrum Field 371 kc; ROA SBRAZ-DTV	N—Min. en route altitude NE—6,200' (N crs Lynchburg) S—5,600' (NE crs Greensboro) S—*3,160' (Red Hill FM) (final) W—6,000' (SE crs Pulaski)	None	S	10 mi.—4,600' E side S crs 15 mi.—4,600' E side S crs 20 mi.—4,600' E side S crs 25 mi.—4,600' E side S crs	*4,600	349	1.8	1,174	R A T	2,000 2,500 1,000	2.0 3.0 2.0	2,000 2,500 1,000	2.0 3.0 2.0	If not contact over range, climb to 5,700' on N crs within 25 mi. of range, maintaining at least 500 F/M climb—or alternate procedure (when directed by ATC). If unable to maintain 500 F/M climb, shuttle between range and a point 6 mi. S to the min. en route altitude. *Descent to pass over range at 3,160' authorized after passing Red Hill FM; if Red Hill FM not rec'd, altitude over range is 4,600'. NOTE: Take-off to N and landing to S is not authorized at night.
ROCHESTER, MINN. Rochester Arpt. 256 kc; RST SBMRLZ-DTV	N—2,400' (SE crs Minneapolis) E—2,600' (NW crs La Crosse) S—2,500' (SW crs La Crosse) S—1,900' (Stewartville FM) (final). W—2,400' (Ellendale Int.)	None	S	10 mi.—2,400' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	1,900	330	2.6	1,041	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,400' on N crs. CAUTION: 1,300' msl water tank located 3.3 mi. NW of range. *Runway No. 35.
ROCHESTER, N. Y. Rochester Arpt. 206 kc; ROC SBMRLZ-DTV	N—2,000' (S crs Stirling) E—2,300' (W crs Syracuse) S—2,500' (NW crs Elmira) W—2,000' (NE crs Buffalo)	None	E	10 mi.—2,300' N side E crs 15 mi.—2,300' N side E crs 20 mi.—2,300' N side E crs 25 mi.—2,300' N side E crs	1,500	312	2.1	560	R A T	600 800 300	1.5 2.0 1.0	600 800 300	2.0 2.0 1.0	Climb to 2,000' on W crs.
ROCKFORD, ILL. Machesney Arpt. 338 kc; RFD SBRAZ-DTV	E—2,500' (NW crs Chicago) SE—2,100' (W crs Chicago) W—2,500' (N crs Moline) NW—2,300' (SW crs Madison)	None	W	10 mi.—2,100' S side W crs 15 mi.—2,200' S side W crs 20 mi.—2,200' S side W crs 25 mi.—2,200' S side W crs	1,500	45	1.7	732	R A T	600 800 *300	1.5 2.0 1.0	600 800 800	2.0 2.0 1.0	Climb to 2,500' on E crs. NOTE: Minima apply to acft having stall speeds of 75 mph or less.
ROCK SPRINGS, WYO. Rock Springs Arpt 290 kc; RKS SBRAZ-DTV	N—Min. en route altitude E—10,000' (Sinclair Range) E—8,700' (Point of Rocks FM) (final) E—8,200'* (ILS outer marker) (final) S—Min. en route altitude W—10,000' (Ft. Bridger Range)	None	E	10 mi.—9,200' N side E crs 15 mi.—9,200' N side E crs 20 mi.—9,200' N side E crs 25 mi.—9,200' N side E crs	*8,700	246	2.9	6,752	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 2.0 2.0 1.0	Climb to 10,000' on W crs within 25 mi. *Descent to 8,200' to pass over rng at 8,200' may be started after passing the ILS outer mrkr, provided position over outer mrkr is verified by visual and aural signals. If outer marker not received final approach alt over the range is 8,700'.
RODEO, N. MEX. CAA Int. Field 254 kc; ROD BMRLZ-DTV	NE—Min. en route altitude E—9,000' (Columbus Range) SW—11,000' (NE crs Douglas) W—12,000' (Cochise Range) W—11,000' (Hilltop FM)	On NE crs to 9,500' within 25 mi	NE	10 mi.—8,000' W side NE crs 15 mi.—8,000' W side NE crs 20 mi.—8,000' W side NE crs 25 mi.—8,000' W side NE crs	6,640	13	.0	4,137	R A T	2,500 3,000 2,000	2.0 3.0 1.0	2,500 3,000 2,000	2.0 3.0 2.0	If not contact over range, make immediate right turn climb-to 8,000' on NE crs within 25 mi. CAUTION: 6,922' terrain 5.5 mi. SE of range.
ROME, GA. Russell Field 388 kc; RMG MRLWZ	N—3,900' (Chattanooga Range) E—2,800' (Cartersville Int.) S—Min. en route altitude W—Min. en route altitude	None	S	10 mi.—2,500' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	1,650	3	3.0	655	R S* A T	1,000 #600 1,500 300	2.0 2.0 2.0 1.0	1,000 #600 1,500 300	2.0 2.0 2.0 1.0	Climb to 3,900' on N crs within 25 mi. *Rwy No. 36. #500-1 for acft with stall speeds of 75 mph or less.
ROSWELL, N. MEX. Roswell Arpt. 395 kc; ROW SBMRAZ-DTV Walker AFB	NE—5,500' (W crs Lubbock) SE—5,300' (W crs Hobbs) SW—8,800' (Clint Rdo Ben) NW—12,000' (Albuquerque Range)	SE crs to 5,500' within 30 mi. of range	SE	10 mi.—5,000' N side SE crs 15 mi.—5,000' N side SE crs 20 mi.—5,000' N side SE crs 25 mi.—5,000' N side SE crs	4,500	311	9.1	3,621	R A T	500 800 300	2.0 2.0 1.0	500 800 300	2.0 2.0 1.0	Climb to 12,000' on NW crs.
	NE—Min. en route altitude SE—6,000' (W crs Hobbs) SW—Min. en route altitude NW—Min. en route altitude	None	SE	10 mi.—6,000' N side SE crs 15 mi.—6,000' N side SE crs 20 mi.—6,000' N side SE crs 25 mi.—6,000' N side SE crs	4,500	255	3.5	3,666	R A T	600 1,000 300	1.5 3.0 2.0	600 1,000 300	2.5 3.0 3.0	Make climbing left turn to 6,000' on SE crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
SACRAMENTO, CALIF. Sacramento Arpt. 236 kc; SAC SBMRAZ-DTV	NE—*3,500' (E crs Williams) NE—1,500' (McClellan Rbn) SE—3,000' (Fresno Range) SE—1,200' (N crs Stockton) SW—5,000' (Oakland Range) SW—1,100' (Bay Point FM) SW—600' (Clarksburg FM) (final) NW—3,000' (Williams Range)	None	SW	10 mi.—1,100' E side SW crs 15 mi.—1,100' E side SW crs 20 mi.—1,100' E side SW crs 25 mi.—1,100' E side SW crs	600	22	1.7	21	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500' on NE crs within 20 mi. *Acft may descend to 7,000' after passing Blue Canyon FM to execute standard instrument approach at Sacramento.
ST. JOSEPH, MO. Rosecrans Field 341 kc; STJ SBMRLZ-DTV	N—Min. en route altitude E—2,400' (NE crs Kansas City) S—2,400' (SW crs Kansas City) W—2,200' (N crs Topeka VAR)	None	S	10 mi.—2,100' W side S crs* 15 mi.—2,100' W side S crs* 20 mi.—2,100' W side S crs* 25 mi.—2,100' W side S crs*	1,600	350	3.7	821	R (R) S A T	700 500 500 800 300	1.5 1.0 1.0 2.0 1.0	700 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,400' on N crs within 25 mi., or alternate procedure (when directed by ATC), climb to 2,500' on NW crs Kans. City. *Procedure turn W so as to permit simultaneous approaches at St. Joseph & Kansas City. CAUTION: 400' bluffs W, NW, and E of field. #Rwy No. 35.
ST. LOUIS, MO. Lambert Field 209 kc; STL SBRAZ-DTXV	N—1,800' (SW crs Springfield) E—1,700' (NW crs Scott) E—1,200' (Spanish Lake FM) (final) S—2,000' (SW crs Scott) W—2,000' (Columbia Range) W—1,700' (St. Peters FM)	None	E	10 mi.—1,600' N side E crs 15 mi.—1,800' N side E crs 20 mi.—1,800' N side E crs 25 mi.—1,800' N side E crs	1,300	255	2.0	552	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000' on W crs, or alternate procedure (when directed by ATC), climb to 2,000' on S crs St. Louis.
SALINAS, CALIF. Salinas Arpt. 280 kc; SNS BMRL-DTV	NE—Must be on top. Top not to exceed 3,500' SE—Must be on top. Top not to exceed 3,500' SW—Must be on top. Top not to exceed 3,500' NW—Must be on top. Top not to exceed 3,500'	None	NW	10 mi.—2,000' W side NW crs 15 mi.—On top 20 mi.—On top 25 mi.—On top	780	311	2.4	86	R A T	700 1,000 (BCOB) 300	2.0 2.0 1.0	700 1,000 (BCOB) 300	2.0 2.0 1.0	If not contact over range, climb to on top on SE crs.
SALT FLAT, TEX. CAA Int. Fld. 335 kc; SFL BMRLPZ-DTV	N—Min. en route altitude E—10,000' (Wink Range) S—Min. en route altitude W—8,000' (El Paso Range)	None	N	10 mi.—5,000' W side N crs 15 mi.—5,000' W side N crs 20 mi.—7,000' W side N crs 25 mi.—7,000' W side N crs	4,500	162	2.7	3,727	R A T	500 800 300	1.5 2.0 1.0	500 800 300	2.0 2.0 2.0	Make right climbing turn to 8,000' on W crs within 25 mi.
SALT LAKE CITY, UTAH Salt Lake City Arpt. No. 1 227 kc; SLC SBRAZ-DTXV	N—11,000' (Malad City Range) N—10,000' (Corrine FM) N—6,500' (Ogden Range) N—4,900' (Layton FM) (final) E—15,500' (S crs Ft. Bridger) S—12,000' (E crs Fairfield) S—11,000' (Riverton FM) W—11,000' (Wendover Range) W—11,000' (Timpie FM)	None	N	10 mi.—7,500' W side N crs 15 mi.—7,500' W side N crs 20 mi.—7,500' W side N crs 25 mi.—10,000' W side N crs	4,900	163	2.7	4,222	R A T	800 800 400	2.0 2.0 1.0	800 800 400	2.0 2.0 1.0	Make right turn and climb to 11,000' on W crs. NOTE: High terrain 8 mi. E of N and S crs and W side of S crs, also S of W crs, 12 mi. from range.
Utah Central Arpt.	(MAKE LET-DOWN TO SALT LAKE CITY ARPT #1. FLY VFR FROM SALT LAKE CITY ARPT #1 TO UTAH CENTRAL ARPT.)													
SAN ANGELO, TEX. Mathis Arpt. 227 kc; SJT BMRLZ-DTV	N—3,500' (E crs Big Spring) E—3,500' (S crs Abilene) S—3,500' (NW crs Junction VAR) W—3,500' (SE crs Big Spring)	None	E	10 mi.—3,500' N side E crs 15 mi.—3,500' N side E crs 20 mi.—3,500' N side E crs 25 mi.—3,500' N side E crs	3,500	260	6.5	1,915	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 3,500' on W crs.
Goodfellow Field	N—3,500' (E crs Big Spring) E—3,000' (S crs Abilene) S—3,500' (NW crs Junction Var) W—3,500' (SE crs Big Spring)	None	S	10 mi.—3,100' E side S crs 15 mi.—3,500 E side S crs 20 mi.—3,500' E side crs 25 mi.—3,500' E side S crs	2,500	229	2.5	1,877	R A T	600 800 300	2.0 2.0 2.0	600 800 300	3.0 2.0 3.0	Climb to 3,500' on N crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
SAN ANTONIO, TEX. San Antonio Arpt. 344 kc; SAT SMRLZ-DTV	N—2,500' (SW crs Austin) N—1,720' (Cibolo Creek FM) (final) N—1,500' (Yates Rbn) (final) E—2,200' (NE crs Randolph) S—2,200' (SE crs Kelly) W—2,700' (NW crs Kelly) W—2,200' (NE crs Kelly)	None	N	10 mi.—2,500' W side N crs 15 mi.—2,500' W side N crs 20 mi.—NA 25 mi.—NA	*1,720	174	2.6	800	R (R) #S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 2,500' on S crs within 25 mi. *If Yates Rbn received, alt over range on final approach is 1,500'. #Runway No. 17.
Kelly AFB	(SEE KELLY (SAN ANTONIO), TEX.)													
SAN DIEGO, CALIF. Lindbergh Field 224 kc; SAN SBRAZ-DTV	N—4,000' (SE crs Long Beach) N—2,800' (Oceanside FM) N—1,200' (La Jolla FM) (final) E—9,000' (El Centro Range) E—2,600' (La Mesa FM) SE—Min. en route altitude SE—2,600' (Coronado FM) W—Min. en route altitude	None	N	10 mi.—2,000' W side N crs 15 mi.—2,000' W side N crs 20 mi.—2,000' W side N crs 25 mi.—2,000' W side N crs	*1,800	155	2.2	14	R (R) A T	700 600 800 300	2.0 2.0 2.0 1.0	700 600 800 300	2.0 2.0 2.0 1.0	Climb to 3,000' on SE crs within 15 mi. (Mexican Border). *Descent to 1,200' to pass over range at 1,200' may be started after passing La Jolla FM; if La Jolla FM not received, final approach alt over range is 1,500'. CAUTION: High terrain 9 mi E of N course.
Ream NAAS	(MAKE LET-DOWN TO LINDBERGH FLD. FLY VFR FROM LINDBERGH FLD TO REAM NAAS.)													
SAN FRANCISCO, CALIF. San Francisco Arpt. 227 kc; SFO SBMRLZ-DTV	NE—5,000' (Sacramento Range) NE—3,000' (Bay Point FM) SE—6,000' (W crs Fresno) SE—1,700' (Evergreen FM) SE—700' (Belmont FM) (final)* SW—Min. en route altitude SW—3,000' (Half Moon Bay FM) NW—Min. en route altitude NW—3,000' (SW crs Fairfield-Suisun)	None	SE	10 mi.—1,500' E side SE crs 15 mi.—1,500' E side SE crs 20 mi.—4,000' E side SE crs 25 mi.—4,000' E side SE crs	*1,300	291	3.0	10	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	If not contact within 2 mi after passing range, climb to 3,000' on NW crs within 25 mi. *If Belmont FM is received descent to 700' over range on final approach is authorized.
Belmont Arpt.	(MAKE LET-DOWN TO SAN FRANCISCO ARPT. FLY VFR FROM SAN FRANCISCO ARPT TO BELMONT ARPT.)													
San Mateo Arpt.	(MAKE LET-DOWN TO SAN FRANCISCO ARPT. FLY VFR FROM SAN FRANCISCO ARPT TO SAN MATEO ARPT.)													
SANTA BARBARA, CALIF. Santa Barbara Arpt. 350 kc; SBA SBMRLZ-DTV	N—On top not over 3,500' E—On top not over 3,500' S—On top not over 3,500' W—On top not over 3,500'	None	W	10 mi.—1,500' S side W crs 15 mi.—1,500' S side W crs 20 mi.—On top 25 mi.—On top	700	74	2.2	14	R S* A T	700 #500 800 300	2.0 1.0 2.0 1.0	700 #500 800 300	2.0 1.0 2.0 1.0	Climb to on top on S crs. CAUTION: 4,299' hills 6 mi N of W crs, 10 mi from range. *Runway No. 7. #500-1 authorized straight-in, day and night, for acft with stall speeds of 75 mph or less.
SAULT STE. MARIE, MICH. Sault Ste. Marie Arpt. 356 kc; SSM SBMRAZ-DTV	N—Min. en route altitude SE—Min. en route altitude S—2,000' (E crs Grand Marais) W—2,500' (Grand Marais Range)	None	SE	10 mi.—1,800' E side SE crs 15 mi.—1,800' E side SE crs 20 mi.—1,800' E side SE crs 25 mi.—1,800' E side SE crs	1,300	315	2.2	722	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA	-------	Climb to 2,500' on W crs.
Sault Ste. Marie-Kinross Arpt.	(MAKE LET-DOWN TO SAULT STE. MARIE RANGE. FLY VFR FROM RANGE TO SAULT STE. MARIE-KINROSS ARPT.)					202	14.0	780	R A T	1,000 1,000 (BCOB) 300	2.0 3.0 1.0	1,000 1,000 (BCOB) 300	2.0 3.0 1.0	

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
SAVANNAH, GA. Hunter Field 263 kc; SAV; SBMRLZ-DTV	NE—1,300' (Charleston Range) SE—Min. en route altitude SW—Min. en route altitude SW—600' (Richmond Hill FM) (final) NW—1,300' (NE crs Alma)	None	SW	10 mi.—1,100' S side SW crs 15 mi.—1,100' S side SW crs 20 mi.—1,100' S side SW crs 25 mi.—1,100' S side SW crs	600	51	2.2	40	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,300' on NE crs, or alternate procedure (when directed by ATC), turn left, climb to 1,300' on NW crs. *Runway No. 5.
Chatham AFB	NE—1,300' (Charleston Range) SE—Min. en route altitude SW—Min. en route altitude NW—1,300' (NE crs Alma)	None	SE	10 mi.—1,100' W side SE crs 15 mi.—1,100' W side SE crs 20 mi.—NA 25 mi.—NA	1,000	349	8.5	50	R A T	700 1,500 300	2.0 3.0 2.0	700 1,500 300	2.0 3.0 3.0	Climb to 1,300' on NW crs.
SCOTT (Belleville), ILL. Scott AFB 263 kc; BLV SBMRAPZ-DT	NE—2,000' (NW crs Evansville) SE—Min. en route altitude SW—2,000' (S crs St. Louis) NW—2,000' (E crs St. Louis)	None	SW	10 mi.—1,800' S side SW crs 15 mi.—2,000' S side SW crs 20 mi.—2,000' S side SW crs; 25 mi.—2,000' S side SW crs	1,200	49	2.2	447	R A T	600 800 300	1.5 2.0 2.0	600 800 300	2.0 2.0 3.0	Climb to 2,000' on NE crs.
SCOTTSBLUFF, NEBR. Scottsbluff Arpt. 341 kc; SCT SBMRLZ-DTZ	NE—5,500' (S crs Rapid City) SE—5,500' (W crs North Platte) SW—6,500' (E crs Cheyenne) NW—Min. en route altitude	None	SE	10 mi.—5,500' E side SE crs 15 mi.—5,500' E side SE crs 20 mi.—5,500' E side SE crs 25 mi.—5,500' E side SE crs	5,000	301	3.1	3965	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 6,000' on NW crs. within 25 mi.
SEATTLE, WASH. Boeing Field 269 kc; SEA SBRAZ-DTXPV	NE—Min. en route altitude NE—3,000' (E crs Everett) E—8,000' (Ellensburg Range) E—7,000' (Easton FM) E—5,000' (Hobart FM) S—5,000' (Toledo Range) S—4,000' (SE crs McChord) S—1,500' (Lakeview FM)* NW—5,000' (S crs Patricia Bay) NW—2,000' (W crs Everett)	None	S	10 mi.—2,000' E side S crs 15 mi.—2,000' E side S crs 20 mi.—NA 25 mi.—NA	*1,200	292	2.8	15	R A T	800 800 300	2.0 2.0 1.0	800 800 300	2.0 2.0 1.0	Climb to 1,500' on NW crs within 15 mi. *Descent to cross range at 1,200' may be started after passing LOM (Ident—SE, 9 mi. S of range); if LOM not received, final approach altitude over range is 1,500'.
Seattle-Tacoma Arpt.	NE—Min. en route altitude NE—3,000' (E crs Everett) E—8,000' (Ellensburg Range) E—7,000' (Easton FM) E—5,000' (Hobart FM) S—5,000' (Toledo Range) S—4,000' (SE crs McChord) S—1,500' (Lakeview FM) NW—5,000' (S crs Patricia Bay) NW—2,000' (W crs Everett)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—NA 25 mi.—NA	1,500	193	4.0	405	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 1.0 2.0 1.0	Climb to 1,500' on S crs within 15 mi. of range.
Seattle NAS	(MAKE LET-DOWN TO SEATTLE OR EVERETT RANGE. FLY VFR FROM SEATTLE OR EVERETT RANGE TO SEATTLE NAS)													
SELFRIDGE (Mt. Clemens), MICH. Selfridge AFB 230 kc; MTC SBRAZ	N—2,500' (NW crs Sarnia) SE—2,500' (NE crs Windsor) S—2,500' (NE crs Windsor) NE—2,500' (NW crs Windsor)	None	N	6.5 mi.—1,800' W side N crs 15 mi.—2,500' W side N crs 20 mi.—2,500' W side N crs 25 mi.—2,500' W side N crs	1,500	183	5.6	583	R A T	800 800 300	1.5 2.0 2.0	800 800 300	2.0 2.0 3.0	Climb to 2,500' on S crs.
SEWART (Smyrna) TENN. Sewart AFB 291 kc; SYM MRLW	NE—3,500' (NW crs Smithville) SE—4,000' (S crs Smithville) SW—2,500' (NE crs Muscle Shoals) NW—2,500' (Nashville Range)	SE crs	SE	10 mi.—2,500' E side SE crs 15 mi.—2,500' E side SE crs 20 mi.—3,500' E side SE crs 25 mi.—3,500' E side SE crs	1,020	320	0.8	512	R A T	500 1,500 500	2.0 3.0 1.0	500 1,500 500	3.0 3.0 2.0	Climb to 3,500' on NE crs.
SHAW (Sumter), S. C. Shaw AFB 375 kc; SSC MRAZ	NE—1,500' (NW crs Florence) SE—1,200' (N crs Charleston) SW—1,500' (SE crs Columbia) NW—1,900' (Spartanburg Range)	SW crs within 15 mi. (all turns to E)	SW	10 mi.—1,200' E side SW crs 15 mi.—1,200' E side SW crs 20 mi.—1,200' E side SW crs 25 mi.—1,200' E side SW crs	1,200	35	4.4	236	R A T	600 1,500 A F Minimums	2.0 3.0	600 1,500	3.0 3.0	Climb to 1,500' on NE crs and request further instructions.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
SHERIDAN, WYO. Sheridan Co. Arpt. 239 kc; SHR SBRAZ-DTV	NE—Min. en route altitude SE—7,000′ (N crs Casper) SE—5,500′ (Ucross FM) SE—5,000′ (Sheridan FM) (final) SW—Min. en route altitude NW—3,000′ (Billings Range)	None	SE	10 mi.—6,000′ E side SE crs 15 mi.—6,000′ E side SE crs 20 mi.—6,000′ E side SE crs 25 mi.—6,000′ E side SE crs	*5,500	298	1.6	4,021	R A T	800 800 500	2.0 2.0 1.0	800 800 500	3.0 2.0 2.0	Climb to 8,000′ on NW crs within 25 mi., or alternate procedure (when directed by ATC), climb to 6,500′ on NE crs within 25 mi. *If both visual and aural signals are received over Sheridan FM, min. alt. over rng is 5,000′.
SHREVEPORT, LA. Shreveport Arpt. 230 kc; SHV SBRAZ-DTV	E—1,500′ (Monroe Range) S—1,500′ (NW crs Alexandria) W—1,800′ (NE crs Tyler) NW—1,700′ (W crs Texarkana) NW—800′ (Dixie FM) (final)	None	NW	10 mi.—1,700′ W side NW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	800	152	1.8	179	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500′ on S crs. *Runway No. 13.
SILVER LAKE, CALIF. CAA Int. Fld. 209 kc; SIL SMBRAZ-DTV	NE—9,500′ (Las Vegas Range) SE—9,000′ (E crs Daggett) SW—6,000′ (Daggett Range) NW—Min. en route altitude	None	NW	10 mi.—5,000′ E side NW crs 15 mi.—7,500′ E side NW crs 20 mi.—7,500′ E side NW crs 25 mi.—7,500′ E side NW crs	4,000	132	1.9	919	R A T	3,000 3,000 2,000	3.0 3.0 3.0	3,000 3,000 2,000	3.0 3.0 3.0	If not contact over rng, climb to 9,000′ on SE crs. CAUTION: High terrain immediately W of NW crs.
SINCLAIR, WYO. Rawlins Arpt. 368 kc; SIR SBRAZ-DTV	N—Min. en route altitude E—12,000′ (NW crs Laramie) S—Min. en route altitude W—10,000′ (Rock Springs Range	None	E	10 mi.—*8,600′ N side E crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	8,300	255	5.1	6,789	R A T	1,500 1,500 500	3.0 3.0 1.0	2,000 2,000 (BCOB) #1,500	5.0 5.0 3.0	If not contact over rng, climb to 10,000′ on W crs. *Procedure turn must be accomplished within 10 mi. high terrain to E. CAUTION: High, unlighted terrain surrounding arpt. #Take-off to SW only.
SIOUX CITY, IOWA Sioux City Arpt. 333 kc; SUX SBMRLZ-DTV	N—2,700′ (Sioux Falls Range) E—Min. en route altitude S—2,500′ (Omaha Range) S—1,900′ (Sloan FM) (final) W—Min. en route altitude	None	S	10 mi.—2,400′ E side S crs 15 mi.—2,400′ E side S crs 20 mi.—2,400′ E side S crs 25 mi.—2,400′ E side S crs	1,900	330	1.9	1,097	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,700′ on N crs, or alternate procedure (when directed by ATC), climb to 3,000′ on W crs.
SIOUX FALLS, S. DAK. Sioux Falls Arpt. 245 kc; SUJ SBRAZ-DTV	NE—3,000′ (Minneapolis Range) SE—2,700′ (Sioux City Range) SW—Min. en route altitude NW—2,800′ (Huron Range)	None	NW	10 mi.—2,700′ W side NW crs 15 mi.—2,800′ W side NW crs 20 mi.—2,800′ W side NW crs 25 mi.—2,800′ W side NW crs	2,300	93	2.2	1,423	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 3,000′ on NE crs within 25 mi.
SMITHVILLE, TENN. Smithville Arpt. 347 kc; SMV SBMRAZ-DTV	N—3,000′ (NE crs Nashville) E—4,500′ (NE crs Chattanooga) S—4,000′ (NW crs Chattanooga) NW—3,500′ (NE crs Nashville)	None	N	10 mi.—2,500′ W side N crs 15 mi.—2,500′ W side N crs 20 mi.—2,500′ W side N crs 25 mi.—2,500′ W side N crs	2,000	177	1.4	1,073	R A T	800 1,000 500	2.0 3.0 1.0	800 1,000 500	2.0 3.0 1.0	Climb to 4,000′ on S crs.
SOUTH BEND, IND. St. Joseph Co. Arpt. 215 kc; SBN SBMRAZ-DTV	N—2,000′ (NE crs Chicago) E—2,000′ (N crs Goshen) S—2,000′ (W crs Goshen) W—2,000′ (SE crs Glenview) W—1,500′ (New Carlisle FM) (final)	None	W	10 mi.—2,000′ N side W crs 15 mi.—2,000′ N side W crs 20 mi.—2,000′ N side W crs 25 mi.—2,000′ N side W crs	1,500	75	3.3	776	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,000′ on E crs. *Runways 9 and 4.
SPARTANBURG, S. C. Spartanburg Arpt. 268 kc; SPA SBRAZ-DTV	NE—2,800′ (Mocresville Int.) NE—2,200′ (NW crs Columbia) SE—2,000′ (W crs Columbia) SW—2,500′ (S crs Greenville) NW—Min. en route altitude NW—5,000′ (N crs Greenville)	None	SW	10 mi.—2,000′ S side SW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,500	65	1.5	836	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,800′ on NE crs.
SPOKANE, WASH. Geiger Field 365 kc; GEG SBRAZ-DTV	N—Min. en route altitude E—7,000′ (Coeur D'Alene Range) S—5,500′ (Walla Walla Range) W—5,000′ (Ephrata Range)	None	E	10 mi.—4,700′ N side E crs 15 mi.—4,700′ N side E crs 20 mi.—6,000′ N side E crs 25 mi.—6,000′ N side E crs	3,380	222	10.0	2,372	R A T	1,000 1,000 300	1.0 2.0 1.0	1,000 1,000 300	1.0 2.0 1.0	If not contact over range, climb to 5,000′ on W crs within 18 mi. of station.
Calkins Arpt.	(MAKE LET-DOWN TO SPOKANE RANGE. FLY VFR FROM SPOKANE RANGE TO CALKINS AIRPORT)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)		Ceiling and visibility minimums: Day Ceiling (ft.)	Day Visibility (mi.)	Night Ceiling (ft.)	Night Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished remarks
SPOKANE, WASH.—Con. Spokane AFB	(MAKE LET-DOWN TO SPOKANE RANGE. FLY VFR FROM SPOKANE RANGE TO SPOKANE AFB.)				3,400	226	15.2	2,430	R S A T	1,000 1,000 1,500 500	3.0 3.0 3.0 1.0	1,000 1,000 1,500 500	3.0 3.0 3.0 2.0	
SPRAGUEVILLE, MAINE Presque Isle Arpt. 365 kc; SPR SMLZ-DT	N—2,700′ (Presque Isle Range) E—3,000′ (SE crs Presque Isle) S—2,700′ (W crs Houlton) W—Min. en route altitude	None	S	10 mi.—2,700′ E side S crs 15 mi.—2,700′ E side S crs 20 mi.—2,700′ E side S crs 25 mi.—2,700′ E side S crs	2,000	9	3.6	534	R (R) S A T	500 500 500 1,000 500	1.5 1.0 1.0 3.0 1.0	600 500 500 1,000 500	2.0 1.5 1.5 3.0 1.0	Climb to 2,700′ on N crs.
SPRINGFIELD, ILL. Capital Arpt. 229 kc; SPI SBMRAZ-DTV	NE—2,000′ (E crs Peoria) SE—2,000′ (W crs Effingham) SW—1,900′ (N crs St. Louis) NW—1,800′ (W crs Peoria)	None	SW	10 mi.—1,800′ S side SW crs 15 mi.—1,800′ S side SW crs 20 mi.—1,800′ S side SW crs 25 mi.—1,800′ S side SW crs	1,300	30	5.3	585	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,000′ on NE crs.
Southwest Arpt.	(MAKE LET-DOWN TO CAPITAL ARPT. FLY VFR FROM CAPITAL ARPT. TO SOUTHWEST ARPT.)													
SPRINGFIELD, MO. Springfield Arpt. 254 kc; SGF SBRAZ-DTV	NE—2,600′ (Vichy Range) SE—Min. en route altitude SW—2,400′ (S crs Joplin) NW—2,500′ (N crs Joplin)	None	NW	10 mi.—2,400′ W side NW crs 15 mi.—2,400′ W side NW crs 20 mi.—2,400′ W side NW crs 25 mi.—2,400′ W side NW crs	1,900	134	4.3	1,267	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 3,000′ on SE crs. within 25 mi. *Runway No. 13.
SQUANTUM, MASS. Squantum NAS 233 kc; NZW MRLWZ	NE—1,500′ (E crs Boston) SE—Min. en route altitude SE—1,500′ (Brant Rock FM) SW—1,700′ (Providence Range) NW—1,700′ (W crs Boston)	None	SE	10 mi.—1,500′ E side SE crs 15 mi.—1,500′ E side SE crs 20 mi.—1,500′ E side SE crs 25 mi.—1,500′ E side SE crs	1,000	321	3.7	10	R A T	500 NA 300	1.5 1.0	500 NA 300	1.5 1.0	Make 180° left climbing turn and climb to 1,500′ on SE crs within 25 mi.
STOCKTON, CALIF. Stockton Arpt. 212 kc; SCK BMRL-DTV	N—3,000′ (SE crs Sacramento) E—2,500′ (SE crs Sacramento) S—3,000′ (W crs Fresno) W—5,000′ (Oakland Range) W—3,000′ (S crs Fairfield-Suisun)	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	800	285	3.2	28	R (R) A T	600 500 1,000 (BCOB) 300	1.5 1.0 2.0 1.0	600 500 1,000 (BCOB) 300	1.5 1.0 2.0 1.0	Climb to 2,000′ on N crs within 25 mi.
STUTTGART, ARK. Stuttgart Arpt. 269 kc; SGT MRLWZ	N—1,500′ (NE crs Little Rock) E—1,500′ (SW crs Memphis) S—1,500′ (SE crs Little Rock) W—1,500′ (Little Rock Range)	None	N	10 mi.—1,500′ W side N crs 15 mi.—1,500′ W side N crs 20 mi.—1,500′ W side N crs 25 mi.—1,500′ W side N crs	1,300	173	4.5	223	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500′ on S crs within 25 mi.
SUPERIOR, MONT.	(NO INSTRUMENT LET-DOWNS AUTHORIZED)													
SYRACUSE, N. Y. Hancock Arpt. 250 kc; SYR SBRAZ-DTV	N—2,000′ (NW crs Utica) E—1,900′ (SW crs Utica) S—3,500′ (SW crs Utica) W-2,300′ (E crs Rochester)	None	W	10 mi.—2,100′ S side W crs 15 mi.—2,100′ S side W crs 20 mi.—2,100′ S side W crs 25 mi.—2,100′ S side W crs	1,400	98	7.4	419	R (R) S A T	700 600 600 800 *300	2.0 2.0 2.0 2.0 1.0	700 600 600 800 *300	2.0 2.0 2.0 2.0 1.0	Climb to 1,900′ on E crs. *600-1 required for take-off to SE. CAUTION: 833′ MSL radio mast located 1.3 mi. SE of arpt. Avoid flight over downtown Syracuse—902′ MSL radio mast approx. 4 mi. SW of Hancock Arpt.
TALLAHASSEE, FLA. Dale Mabry Field. 379 kc; TLH SBRAZ-DTV	N—1,400′ (Albany Range) E—1,400′ (NW crs Cross City) S—Min. en route altitude NW-1,400′ (E crs Crestview) NW—700′ (Quincy FM) (final)	None	NW	10 mi.—1,300′ S side NW crs 15 mi.—1,300′ S side NW crs 20 mi.—1,300′ S side NW crs 25 mi.—1,400′ S side NW crs	700	65	2.7	70	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,400′ on E crs, or alternate procedure (when directed by ATC), turn left and climb to 1,400′ on N crs. *Runway No. 9.
TAMPA, FLA. International Arpt. 388 kc; TPA SBRAZ-DTV	NE—1,500′ (Orlando Range) S—Min. en route altitude S-700′ (Ruskin FM) (final) SW—Min. en route altitude N—1,300′ (SE crs Cross City)	None	S	10 mi.—1,200′ E side S crs 15 mi.—1,200′ E side S crs 20 mi.—1,200′ E side S crs 25 mi.—1,200′ E side S crs	700	339	5.7	26	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,300′ on N crs, or alternate procedure (when directed by ATC), climb to 1,500′ on NE crs.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
TAMPA, FLA.—Con.	NE—1,500′ (Orlando Range) S—Min. en route altitude S—1,200′ (Ruskin FM) SW—Min. en route altitude N—1,300′ (SE crs Cross City)	None	SW	10 mi.—1,300′ S side SW crs 15 mi.—1,300′ S side SW crs 20 mi.—1,300′ S side SW crs 25 mi.—1,300′ S side SW crs	700	55	2.7	8	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	1.5 1.5 1.5 3.0 1.0	Climb to 1,500′ on NE crs. Approach control operates from International Arpt. on a 24-hour basis.
MacDill AFB	NE—1,500′ (Orlando Range) S—Min. en route altitude S—1,300′ (Ruskin FM) SW—Min. en route altitude N—1,300′ (SE crs Cross City)	None	NE	10 mi.—1,500′ S side NE crs 15 mi.—1,500′ S side NE crs 20 mi.—1,500′ S side NE crs 25 mi.—1,500′ S side NE crs	700	210	2.3	13	R A T	500 1,000 300	1.5 3.0 2.0	500 1,000 300	1.5 3.0 3.0	Climb to 1,500′ on SW crs, or alternate procedure (when directed by ATC), climb to 1,300′ on S crs.
TAPPAHANNOCK, VA.	(NO PROCEDURE APPROVED)													
TERRE HAUTE, IND. Hulman Fld. 248 kc; HUF BMRLZ-DTV	N—1,900′ (SW crs La Fayette) E—2,000′ (Indianapolis Range) S—1,900′ (Evansville Range) W—2,000′ (Effingham Range)	None	W	10 mi.—1,800′ S side W crs 15 mi.—1,800′ S side W crs 20 mi.—1,800′ S side W crs 25 mi.—1,800′ S side W crs	1,300	72	5.8	585	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	2.0 2.0 2.0 1.0	Climb to 2,000′ on E crs.
Cox Arpt.	N—1,900′ (SW crs La Fayette) E—2,000′ (Indianapolis Range) S—1,900′ (Evansville Range) W—2,000′ (Effingham Range)	None	W	10 mi.—1,800′ S side W crs 15 mi.—1,800′ S side W crs 20 mi.—1,800′ S side W crs 25 mi.—1,800′ S side W crs	1,300	182	.0	485	R A T	800 800 300	1.5 2.0 1.0	NA NA NA	----------	If not contact over range, climb to 2,000′ on E crs.
TEXARKANA, ARK. Texarkana Arpt. 329 kc; TXK SBRAZ-DTV	NE—1,500′ (SW crs Little Rock) S—1,700′ (NW crs Shreveport) W—1,800′ (Sulphur Springs Rbn) N—3,800′ (Fort Smith Rbn)	None	N	10 mi.—1,400′ W side N crs 15 mi.—1,400′ W side N crs 20 mi.—1,700′ W side N crs 25 mi.—1,800′ W side N crs	*900	115	2.1	389	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,700′ on S crs. within 25 mi. *CAUTION: 750′ MSL tower located approx 3 mi SW of rng. #Runway No. 13.
THE DALLES, OREG. The Dalles Arpt. 230 kc; DLS SBRAZ-DTV	NE—8,000′ (Yakima Range) E—4,000′ (W crs Pendleton) S—7,000′ (Redmond Range) W—7,000′ (Portland Range)	None	E	10 mi.—4,000′ *S side E crs 15 mi.—4,000′ *S side E crs 20 mi.—4,000′ *S side E crs 25 mi.—4,000′ *S side E crs	2,740	251	2.8	243	R A T	2,500 2,500 1,500	3.0 3.0 1.0	#2,500 #2,500 NA	3.0 3.0	Make immediate 180° left turn and climb to 4,000′ on E crs within 15 mi. of rng. *Procedure turn on N side of E crs not authorized because of high terrain. #Night operations applicable to acft with stall speeds of 75 mph or less only.
THERMAL, CALIF.	(NO PROCEDURE APPROVED)													
TINKER (Oklahoma City), OKLA. Tinker AFB	(SEE OKLAHOMA CITY, OKLA.)													
TOLEDO, OHIO Toledo Arpt. 239 kc; TOL SBMRAZ-DTV	N—2,100′ (SW crs Windsor) E—2,000′ (SE crs Detroit) S—2,100′ (Findlay Rbn) S—1,400′ (Bowling Green FM) (final) W—2,000′ (NE crs Ft. Wayne)	None	S	10 mi.—1,900′ E side S crs 15 mi.—1,900′ E side S crs 20 mi.—1,900′ E side S crs 25 mi.—1,900′ E side S crs	1,400	17	2.8	622	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.5 2.0 1.0	Climb to 2,100′ on N crs.
TOLEDO, WASH. CAA Int. Fld. 239 kc; TDO BMRLZ-DTV	N—5,000′ (Seattle Range) E—Min. en route altitude S—5,600′ Portland Range) W—Min. en route altitude	None	N	12 mi.—3,500′ W side N crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,880	235	2.9	371	R A T	1,500 1,500 300	2.0 2.0 1.0	1,500 1,500 300	2.0 2.0 1.0	Turn right and climb to 3,500′ on N crs within 12 mi, then continue to climb to 3,000′.
TONOPAH, NEV. Tonopah Mun. Arpt. 221 kc; TPH SRAWZ	N—Min. en route altitude E—Min. en route altitude S—Min. en route altitude W—Min. en route altitude	None	N	10 mi.—9,500′ W side N crs 15 mi.—9,500′ W side N crs 20 mi.—10,500′ W side N crs 25 mi.—10,500′ W side N crs	8,000	168	2.6	5,426	R A T	2,500 2,500 1,000	2.0 3.0 1.0	2,500 2,500 1,000	3.0 3.0 2.0	Climb to 10,000′ on S crs within 25 mi.
TRAVERSE CITY, MICH. Traverse City Arpt. 365 kc; TVC SBMRAZ-DTV	NE—2,300′ (Sault Ste. Marie Range) SE—2,500′ (Saginaw Rbn) SW—Min. en route altitude NW—Min. en route altitude	None	SE	10 mi.—2,100′ E side SE crs 15 mi.—2,400′ E side SE crs 20 mi.—2,400′ E side SE crs 25 mi.—2,500′ E side SE crs	1,600	319	2.2	623	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,300′ on NW crs within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
TRI-CITY (Bristol), TENN. Tri-City Arpt. 221 kc; TRI SBMRAZ—DTV	NE—7,000' (Pulaski Range) NE—6,500' (Chilhowie FM) SE—Min. en route altitude SW—5,000' (Knoxville Range) NW—Min. en route altitude	On SW ers to 4,000' within 12 mi.	NE	10 mi.—4,000' N side NE ers 15 mi.—7,000' N side NE ers 20 mi.—7,000' N side NE ers 25 mi.—7,000' N side NE ers	2,800	252	3.2	1,523	R (R) S* A T	700 800 800 1,000 300	1.0 1.0 1.0 2.0 1.0	700 800 800 1,000 300	1.5 1.0 1.0 2.0 1.0	Climb to 5,000' on SW ers. *Runway No. 27.
TRINIDAD, COLO. Trinidad Arpt. 329 kc; TAD SBMRAZ-DTV	N—7,500' (Pueblo Range) NE—Min. enroute altitude S—11,000' (NE ers Las Vegas) SW—Min. en route altitude	None	N	10 mi.—7,100' E side N ers* 15 mi.—7,100' E side N ers* 20 mi.—7,100' E side N ers* 25 mi.—7,100' E side N ers*	6,700	162	2.5	5,761	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	NA NA NA NA	----------	Make immediate 180° left turn and climb to 7,500' on N ers. *Procedure turn to E account high terrain to W. CAUTION: Abrupt rise in terrain 8 mi. SSW of arpt. 9586' terrain, 12 mi. SSW of arpt.
TUCSON, ARIZ. Davis-Monthan AFB 338 kc; TUS SBRAZ-DTV	NE—Min. en route altitude SE—10,000' (W ers Cochise) SW—Min. en route altitude NW—10,000' (Casa Grande Int.)	None	SW	10 mi.—7,200' S side SW ers 15 mi.—7,200' S side SW ers 20 mi.—7,200'* S side SW ers 25 mi.—10,000'* S side SW ers	3,800	308	3.1	2,626	R A T	900 1,000 300	1.0 3.0 1.0	900 1,000 300	2.0 3.0 1.0	Climb to 7,600' on NW ers within 25 mi, or alternate procedure (when directed by ATC), make left turn and climb to 7,200' on SW ers within 25 mi. *Procedure turn beyond 15 mi authorized for military acft only. CAUTION: 3,703' peak, 7 mi NW of SW ers, 11 mi from range.
Tucson Arpt.	(MAKE LET-DOWN TO DAVIS-MONTHAN AFB. FLY VFR FROM DAVIS-MONTHAN AFB TO TUCSON AIRPORT)													
TUCUMCARI, N. MEX. Tucumcari Arpt. 212 kc; TCC SBMRAZ-DTV	N—Min. en route altitude E—5,200' (Amarillo Range) S—Min. en route altitude W—7,000' (SE ers Las Vegas)	None	W	10 mi.—6,000' S side W ers 15 mi.—6,500' S side W ers 20 mi.—6,500' S side W ers 25 mi.—6,500' S side W ers	5,520	76	2.2	4,063	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 5,200' on E ers. 5,020' terrain, 4.5 mi. SW of range.
TULSA, OKLA. Tulsa Arpt. 245 kc; TUL SBRAZ-DTV	NE—2,200' (S ers Joplin) NE—1,300' (Verdigris R. FM) (final). SE—2,600' (Ft. Smith Rbn) SW—2,200' (E ers Oklahoma City). NW—2,200' (S ers Wichita) NW—2,000' (Skiatook FM)	None	NE	10 mi.—1,800' N side NE ers 15 mi.—2,000' N side NE ers 20 mi.—2,000' N side NE ers 25 mi.—2,000' N side NE ers	1,300	220	1.0	674	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,200' on SW ers within 25 mi., or alternate procedure (when directed by ATC), turn right, climb to 2,300' on NW ers within 25 mi.
TYLER, TEX. Pounds Field 320 kc; TYR BMRLZ-DTV	NE—1,700' (W ers Shreveport) SE—Min. en route altitude SW—Min. en route altitude NW—1,800' (E ers Dallas)	None	NW	10 mi.—1,600' W side NW ers 15 mi.—1,600' W side NW ers 20 mi.—1,600' W side NW ers 25 mi.—1,600' W side NW ers	1,100	120	1.6	544	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 2,000' on SE ers within 25 mi. *Runway No. 12.
TYNDALL (Panama City), FLA. Tyndall AFB 203 kc; PAM BMRAZ	NE—1,400' (NW ers Tallahassee) SE—Min. en route altitude SW—Min. en route altitude NW—1,400' (E ers Eglin)	None	SE	10 mi.—1,100' E side SE ers 15 mi.—1,100' E side SE ers 20 mi.—1,100' E side SE ers 25 mi.—1,100' E side SE ers	600	320	3.0	16	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,400' on NW ers.
UTICA, N. Y. Utica Arpt. 359 kc; UCA BMRLZ-DTV	NE—Min. en route altitude SE—Min. en route altitude SW—4,000' (S ers Syracuse) NW—3,000' (N ers Syracuse)	None	NW	10 mi.—3,000' W side NW ers 15 mi.—3,000' W side NW ers 20 mi.—3,000' W side NW ers 25 mi.—3,000' W side NW ers	2,000	151	3.0	525	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	600 500 800 300	1.5 1.5 2.0 1.0	Climb to 3,000' on SE ers within 25 mi.
Griffiss AFB (Rome)	NE—Min. en route altitude SE—Min. en route altitude SW—4,000' (S ers Syracuse) NW—3,000' (N ers Syracuse)	None	SE	10 mi.—3,000' E side SE ers 15 mi.—3,000' E side SE ers 20 mi.—3,000' S side SE ers 25 mi.—3,000' E side SE ers	2,500	315	2.9	487	R A T	800 1,500 300	2.0 3.0 1.0	800 1,500 300	3.0 3.0 2.0	Climb to 3,000' on NW ers.
VANCE (Enid), OKLA. Vance AFB 209 kc; END BMRLZ-V	NE—3,000' (S ers Wichita) SE—3,000' (N ers Okla. City) SW—Min. en route altitude NW—3,000' (SW ers Wichita)	None	NE	10 mi.—2,500' N side NE ers 15 mi.—2,500' N side NE ers 20 mi.—2,500' N side NE ers 25 mi.—2,500' N side NE ers	2,000	214	3.7	1,290	R A T	800 1,000 300	1.5 3.0 2.0	800 1,000 300	2.0 3.0 3.0	Climb to 2,500' on SW ers within 25 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished remarks
VICHY, MO. CAA Int. Field 344 kc; VIH BMRLZ-DTV	NE—2,200' (W crs St. Louis) SE—3,000' (W crs Advance) SW—2,600' (Springfield Range) NW—2,500' (S crs Columbia)	None	SE	10 mi.—2,300' N side SE crs 15 mi.—2,300' N side SE crs 20 mi.—2,300' N side SE crs 25 mi.—2,300' N side SE crs	1,800	310	2.7	1,148	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 2,500' on NW crs within 25 mi. * Rwy No. 32.
WACO, TEX. Waco Arpt. 385 kc; ACT SBRAZ-DTV	N—1,900' (Dallas Range) SE—2,000' (Bryan Range) S—2,000' (Austin Range) NW—1,700' (S crs Ft. Worth)	None	S	10 mi.—1,600' E side S crs 15 mi.—1,600' E side S crs 20 mi.—1,600' E side S crs 25 mi.—1,600' E side S crs	1,400	9	2.6	513	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.5 2.0 1.0	Climb to 1,900' on N crs. *Runway No. 36.
Connally AFB	N—1,900' (Dallas Range) SE—2,000' (Bryan Range) S—2,000' (Austin Range) NW—1,700' (S crs Ft. Worth)	None	S	10 mi.—1,600' E side S crs 15 mi.—1,600' E side S crs 20 mi.—1,600' E side S crs 25 mi.—1,600' E side S crs	1,400	59	10.7	470	R A T	700 800 300	2.0 2.0 2.0	700 800 300	3.0 3.0 3.0	Climb to 1,900' on N crs.
WALLA WALLA, WASH. City-County Arpt. 212 kc; ALW SBMRLZ-DTV	NE—5,500' (Spokane Range) SE—Min. en route altitude SW—5,000' (E crs Pendleton) NW—Min. en route altitude	None	NE	10 mi.—2,000' W side NE crs 15 mi.—3,500' W side NE crs 20 mi.—3,500' W side NE crs 25 mi.—3,500' W side NE crs	2,500	175	3.6	1,205	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	700 700 800 300	1.5 1.0 2.0 1.0	Climb to 5,000' on SW crs within 20 mi. CAUTION: High terrain 5 mi. E of NE course.
Martin Arpt.	(MAKE LET-DOWN TO CITY-COUNTY ARPT. FLY VFR FROM CITY-COUNTY ARPT TO MARTIN ARPT.)													
WALNUT RIDGE, ARK. Walnut Ridge Arpt. 308 kc; ARG BMRLZ-DTV	NE—1,500' (W crs Advance) E—2,500' (N crs Memphis) SW—1,700' (Little Rock Range) W—Min. en route altitude	None	NE	10 mi.—1,500' W side NE crs 15 mi.—1,500' W side NE crs 20 mi.—1,500' W side NE crs 25 mi.—1,500' W side NE crs	900	173	5.1	275	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		Climb to 1,700' on S W cr within 25 mi. *Runway No. 18.
WASHINGTON, D. C. Washington National Arpt. 232 kc; DCA SBRA-DTXV	NE—1,600' (W crs Baltimore) SE—1,500' (S crs Baltimore) SW—1,500' (N crs Richmond) SW—1,000' (Mt. Vernon FM) (final) NW—1,800' (E crs Front Royal)	None	SW	10 mi.—1,500' E side SW crs 15 mi.—1,500' E side SW crs 20 mi.—1,500' E side SW crs 25 mi.—NA*	1,000	1	5.3	16	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,800' on NW crs. *Danger area adjacent to airway approx. 22 mi. S of range. #Runway No. 36.
Anacostia NAS	NE—1,600' (W crs Baltimore) SE—1,500' (S crs Baltimore) SW—1,500' (N crs Richmond) SW—1,000' (Mt. Vernon FM) (final) NW-1,800' (E crs Front Royal)	None	SW	10 mi.—1,500' E side SW crs 15 mi.—1,500' E side SW crs 20 mi.—1,500' E side SW crs 25 mi.—NA*	1,000	12 (To Bolling AFB)	4.7	10	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	If not contact within 4.7 mi. after passing range, mak climbing left turn to 1,800' on NW crs. *Danger area adjacent to airway approx. 22 mi. S of range CAUTION: Hills with approx. 392' obstructions immed E of fld. #Runway No. 3.
Bolling AFB	NE—1,600' (W crs Baltimore) SE—1,500' (S crs Baltimore) SW—1,500' (N crs Richmond) SW—1,000' (Mt. Vernon FM) (final) NW—1,800' (E crs Front Royal)	None	SW	10 mi.—1,500' E side SW crs 15 mi.—1,500' E side SW crs 20 mi.—1,500' E side SW crs 25 mi.—NA*	1,000	12	4.7	20	R A T	800 800 300	1.0 2.0 2.0	800 800 300	2.0 2.0 3.0	Climb to 1,800' on NW crs. *Danger area adjacent to airway approx. 22 mi. S of range.
WATERTOWN, S. D. Watertown Arpt. 282 kc; ATY BMRLZ-DTV	N—Min. en route altitude E—3,000' (Willmar Range) S—Min. en route altitude W—Min. en route altitude	None	E	10 mi.—3,000' N side E crs 15 mi.—3,000' N side E crs 20 mi.—3,000' N side E crs 25 mi.—3,000' N side E crs	2,500	251	2.9	1,747	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	NA NA NA NA	----------	Climb to 3,000' on W crs.
WEEKSVILLE, N. C. Elizabeth City CGAS 254 kc; NFG SMRLZ	NE—Min. en route altitude NE—1,300' (Aydlette FM) SE—Min. en route altitude SE—1,300' (Manteo FM) SW—Min. en route altitude SW—1,300' (Edenton FM) NW—Min. en route altitude NW—1,300' (South Mills FM)	On SE crs within 8 mi. of range.	SE	10 mi.—1,100' N side SE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	800	319	2.8	10	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	2.0 1.0 1.0 2.0 1.0	Climb to 1,300' on NW crs within 25 mi. CGAS closed to all civil air traffic except in emergency or when given special authorization. *Runway No. 32.
WENDOVER, UTAH Wendover AFB	(NO PROCEDURE APPROVED)													

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
WESTFIELD, MASS. Barnes Arpt. 230 kc; BAF BMRLZ-DTV	N—3,000' (W crs Boston) E—2,500' (NE crs Hartford) S—2,500' (NW crs Hartford) NW—4,500' (Albany Range)	None	S	10 mi.—2,500' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	1,600	27	1.7	270	R A T	1,000 1,500 700	2.0 3.0 1.0	1,000 1,500 700	2.0 3.0 2.0	Climb to 3,000' on N crs.
WESTOVER (Chicopee Falls), MASS. Westover AFB 272 kc; CEE BMRAZ	NE—3,000' (W crs Boston) SE—3,000' (E crs Westfield) SW—3,000' (S crs Westfield) NW—3,000' (N crs Westfield)	To 2,500' on NE crs between LOM and Quabbin marker	NE	10 mi.—2,500' N side NE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,500	239	2.6	244	R A T	600 800 500	2.0 2.0 1.0	600 800 500	3.0 2.0 2.0	Climb to 2,500' to Westfield range, via SW crs Westover and SE crs Westfield and await further clearance.
WEST PALM BEACH FLA. Palm Beach International Arpt. 209 kc; PBI SBMRA-DTV	N—1,500' (SE crs Melbourne) E—Min. en route altitude S—1,400' (Miami Range) S—1,200' (Ft. Lauderdale FM) W—Min. en route altitude	None	W	10 mi.—1,200' S side W crs 15 mi.—1,200' S side W crs 20 mi.—1,200' S side W crs 25 mi.—1,200' S side W crs	600	89	2.4	19	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,200' on E crs.
WHIDBEY ISLAND, WASH. Whidbey Island NAS 253 kc; NUW SBMRLZ	NE—1,500' (S crs Bellingham) SE—1,500' (N crs Everett) SW—1,500' (NW crs Seattle) NW—3,000' (E crs Patricia Bay)	None	SE	6 mi.—1,500' W side SE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,000	303	2.3	17	R A T	850 1,000 500	1.5 3.0 1.0	850 1,000 500	2.0 3.0 1.0	Climb to 3,000' on NW crs.
Oak Harbor NAS	NE—1,500' (S crs Bellingham) SE—1,500' (N crs Everett) SW—1,500' (NW crs Seattle) NW—3,000' (E crs Patricia Bay)	None	NW	10 mi.—1,500' W side NW crs 15 mi.—1,500' W side NW crs 20 mi.—3,000' W side NW crs 25 mi.—3,000' W side NW crs	1,000	123	3.7	S. L.	R (R) S A T	600 600 600 1,000 500	1.5 1.0 1.0 3.0 1.0	600 600 600 1,000 500	1.5 1.0 1.0 3.0 1.0	Climb to 1,500' on SE crs.
WHITEHALL, MONT. CAA Int. Field 284 kc; HIA SBRAZ-DTV	N—10,500' (Helena Range) N—9,000' (Whitetail FM) E—9,000' (Bozeman Range) S—10,500' (Dillon Range) NW—10,000' (S crs Butte)	None	E	10 mi.—8,000' N side E crs 15 mi.—8,000' N side E crs 20 mi.—8,000 N side E crs 25 mi.—8,000' N side E crs	7,000	183	2.3	4,674	R A T	2,000 2,000 1,500	2.0 2.0 1.0	2,000 2,000 1,500	2.0 2.0 2.0	Turn left and return to range, then climb to 9,000' on E crs within 25 mi.
WHITING (Milton), FLA. Whiting NAAS (No.) 524 kc; NSE MRLWZ	NE—1,200' (N crs Crestview) SE—1,300' (NE crs Pensacola) SW—1,200' (N crs Pensacola) NW—1,200' (N crs Pensacola)	None	NW	10 mi.—1,200' S side NW crs 15 mi.—1,200' S side NW crs 20 mi.—1,200' S side NW crs 25 mi.—1,200' S side NW crs	700	130	3.2	177	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	Climb to 1,300' on SE crs.
WHITMORE, CALIF.	(NO ASSOCIATED AIRPORT)													
WICHITA, KANS. Wichita Arpt. 332 kc; ICT SBRAZ-DTV	NE—2,800' (W crs Chanute) S—2,500' (NW crs Tulsa) SW—3,000' (S crs Hutchinson) SW—2,600' (Viola FM) N—2,800' (E crs Hutchinson) N—2,300' (Kechi FM) (final)	None	N	10 mi.—2,800' W side N crs 15 mi.—2,800' W side N crs 20 mi.—2,800' W side N crs 25 mi.—2,800' W side N crs	2,300	167	1.2	1,372	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 *300	1.5 1.0 2.0 1.0	Climb to 2,500' on S crs, or alternate procedure (when directed by ATC), climb to 3,000' on SW crs. *Rwys 6L and 24R not authorized.
WICHITA FALLS, TEX. Kell/Sheppard AFB 229 kc; ICH SBMRAZ-DTV	NE—2,500' (S crs Okla. City) SE—2,200' (N crs Ft. Worth) SE—1,700' (Jolly FM) (final) SW—3,000' (Abilene Range) NW—2,500' (NE crs Childress VAR)	None	SE	10 mi.—2,200' E side SE crs 15 mi.—2,200' E side SE crs 20 mi.—2,200' E side SE crs 25 mi.—2,200' E side SE crs	1,700	305	1.7	1,029	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,500' on NW crs, within 25 mi. or alternate procedure (when directed by ATC), turn right and climb to 2,500' on NE crs. Only rwys 35R and 17L open.
WILKES-BARRE, PA. Wilkes-Barre-Scranton Arpt. 272 kc; AVP SBMRAZ-DTV	E—3,800' (NW crs La Guardia) S—3,500' (W crs Allentown) SW—4,000' (Selinsgrove Rbn) SW—3,500' (E crs Williamsport) N—3,300' (SE crs Elmira)	None	SW	10 mi.—3,500' S side SW crs 15 mi.—3,500' S side SW crs 20 mi.—3,500' S side SW crs 25 mi.—3,500' S side SW crs	2,600	91	4.0	956	R A T	1,000 1,200 600	2.0 2.0 2.0	1,300 1,600 800	2.0 3.0 2.0	Climb to 3,500' on SW crs within 25 mi. High terrain to E, SE, and S of arpt. within 3 mi.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport: Magnetic bearing (deg.)	Station to airport: Distance (mi.)	Field elevation (ft.)	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
WILLIAMS, CALIF. CAA Int. Field 326 kc; ILA SBMRAZ-DTV	N—3,000′ (Red Bluff Range) E—5,000′ (NE crs Sacramento) S—6,000′ (NW crs Oakland) W—Min. en route alt.	N crs within 25 mi.	N	10 mi.—1,200′ W side N crs 15 mi.—1,300′ W side N crs 20 mi.—1,300′ W side N crs 25 mi.—1,300′ W side N crs	740	244	2.2	139	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	600 600 800 300	1.5 1.0 2.0 1.0	If not contact over range, climb to 3,000′ on N course.
WILLIAMSPORT, PA. Williamsport Arpt. 375 kc; IPT BMRLZ-DTV	N—4,000′ (SE crs Elmira) E—8,600′ (SW crs Wilkes-Barre) S—3,500′ (Selinsgrove Rbn) W—3,500′ (SW crs Elmira)	None	E	10 mi.—3,500′ S side E crs 15 mi.—4,000′ S side E crs 20 mi.—4,000′ S side E crs 25 mi.—4,000′ S side E crs	2,500	285	6.9	529	R A T	1,500 1,500 800	1.0 2.0 1.0	1,500 1,500 800	2.0 3.0 1.0	If not contact within 2 mi. after passing range, climb to 3,600′ on E crs.
WILLMAR, MINN. Willmar Arpt. 254 kc; ILL SBMRAZ-DTV	N—2,400′ (SE crs Alexandria) E—2,300′ (NW crs Minneapolis) S—2,300′ (NE crs Sioux Falls) W—3,000′ (Watertown Range)	None	S	10 mi.—2,300′ E side S crs 15 mi.—2,300′ E side S crs 20 mi.—2,300′ E side S crs 25 mi.—2,300′ E side S crs	1,800	337	3.3	1,130	(R) A T	500 800 300	1.0 2.0 1.0	500 800 300	1.0 2.0 1.0	Climb to 2,400′ on N course.
WILLOW GROVE, PA. Willow Grove NAS 391 kc; NXX SMRLZ	NE—1,500′ (NW crs Lakehurst) SE—2,000′ (NE crs Philadelphia) SW—1,900′ (N crs Philadelphia) NW—2,000′ (SW crs Allentown)	None	NE	10 mi.—1,500′ N side NE crs 15 mi.—1,500′ N side NE crs 20 mi.—1,500′ N side NE crs 25 mi.—1,500′ N side NE crs	1,000	241	3.3	369	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,900′ on SW crs. *Runway No. 24.
Johnsville NAS	NE—1,500′ (NW crs Lakehurst) SE—2,000′ (NE crs Philadelphia) SW—1,900′ (N crs Philadelphia) NW—2,000′ (SW crs Allentown)	None	NW	10 mi.—2,000′ N side NW crs 15 mi.—2,000′ N side NW crs 20 mi.—2,000′ N side NW crs 25 mi.—2,000′ N side NW crs	1,000	160	1.6	366	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Make climbing right (West) turn and climb to 1,900′ on SW crs.
WILMINGTON, DEL. New Castle Co. Arpt. 350 kc; ILG MRLWZ	N—1,800′ (W crs Philadelphia) E—1,500′ (S crs Philadelphia) S—1,500′ (SW crs Millville) W—1,800′ (W crs Philadelphia)	None	S	10 mi.—1,500′ E side S crs 15 mi.—1,500′ E side S crs 20 mi.—1,500′ E side S crs 25 mi.—1,500′ E side S crs	1,000	19	2.6	77	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Turn right and climb to 1,500′ on S crs, or alternate procedure (when directed by ATC), climb to 1,500′ on E crs.
WINDSOR, ONT. Detroit City Arpt. 333 kc; QG SBRAZ-T	NE—1,700′ (SE crs Selfridge) SE—1,900′ (Cleveland Range) SW—2,300′ (SE crs Detroit) NW—2,300′ (NW crs Selfridge)	None	SE	10 mi.—1,700′ E side SE crs 15 mi.—1,700′ E side SE crs 20 mi.—1,700′ E side SE crs 25 mi.—1,700′ E side SE crs	1,640	329	9.0	626	R A T*	1,000 1,000 500	2.0 3.0 1.0	1,000 1,000 500	2.0 3.0 1.0	If not contact over range, climb to 2,300′ on NW crs. *300–1 Rwy 33 only. NOTE: Detroit Rbn located 2.6 mi SE of Detroit City Arpt.
WINK, TEX. Wink Arpt. 260 kc; INK SBRAZ-DTV	E—4,500′ (NW crs Midland) S—Min. en route alt. S—3,500′ (SW crs Midland) (final) W—4,500′ (SE crs Carlsbad) N—5,000′ (E crs Hobbs)	None	S	10 mi.—4,000′ E side S crs 15 mi.—4,000′ E side S crs 20 mi.—4,000′ E side S crs 25 mi.—4,000′ E side S crs	3,600	353	1.2	2,824	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 5,000′ on N crs.
WINSLOW, ARIZ. Winslow Arpt. 266 kc; INW SBRAZ-DTV	N—Min. en route altitude E—10,000′ (Zuni Range) E—7,000′ (Joseph City FM) S—Min. en route (altitude) W—10,000′ (Prescott Range) W—7,500′ (Meteor Crater FM)	None	E	10 mi.—7,000′ S side E crs 15 mi.—7,000′ S side E crs 20 mi.—7,000′ S side E crs 25 mi.—7,000′ S side E crs	5,560	160	1.5	4,937	R A T	600 800 400	2.0 2.0 1.0	600 800 400	2.0 2.0 2.0	If not contact over range, climb to 7,000′ on E crs within 25 mi.
WINSTON, SALEM, N.C. Smith-Reynolds Arpt. 382 kc; INT SMRLZ-D	NE—2,400′ (NW crs Greensboro) SE—Min. en route altitude SE—2,000′ (SW crs Greensboro) (final). SE—2,000′ (High Point FM) (final). SW—2,500′ (N crs Charlotte) NW—Min. en route altitude	None	SE	10 mi.—2,200′ W side SE crs 15 mi.—2,500′ W side SE crs 20 mi.—2,500′ W side SE crs 25 mi.—2,500′ W side SE crs	2,000	328	5.1	969	R (R) S* A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	700 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 3,500′ on NW crs within 23 mi., or alternate procedure (when directed by ATC), turn left, climb to 2,500′ on SW course within 25 mi. *Runway No. 33.

LOW FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
WRIGHT-PATTERSON (Dayton), OHIO Patterson AFB 379 kc; FFO SBMRAZ-DTV	NE—2,500' (W crs Columbus) E—2,500' (S crs Columbus) S—2,300' (Cincinnati Range) W—2,500' (S crs Dayton)	None	S	10 mi.—2,500' E side S crs 15 mi.—2,500' E side S crs 20 mi.—2,500' E side S crs 25 mi.—2,500' E side S crs	2,000	22	2.9	819	R A T	500 1,000 500	3.0 3.0 1.0	500 1,000 500	3.0 3.0 1.0	Climb to 2,500' on NE crs of Wright-Patterson and make procedure turn to right, S of E crs of Dayton.
YAKIMA, WASH. Yakima Arpt. 320 kc; YKM BMRLZ-DTV	NE—NA (Danger Area) SE—5,000' (NW crs Pendleton) SW—8,000' (The Dalles Range) NW—Min. en route altitude NW—4,000' (S crs Ellensburg)	None	SE	10 mi.—4,000' *S side SE crs 15 mi.—4,000' *S side SE crs 20 mi.—4,000' *S side SE crs 25 mi.—4,000' *S side SE crs	3,000	265	4.7	1,077	R (R) S A T	800 800 800 800 500	2.0 2.0 2.0 2.0 1.0	1,000 800 800 1,000 800	2.0 2.0 2.0 2.0 2.0	Turn right and climb to 4,000' on NW course within 10 mi of range, make procedure turn on N side NW course. *Procedure turn not authorized on N side of SE course due to high terrain. WARNING: Do not proceed more than 10 mi from range on NW course.
YOUNGSTOWN, OHIO Youngstown Arpt. 212 kc; YNG SBMRLZ-DTV	N—2,200' (SW crs Erie) N—1700' (Gustavus FM) (final) E—2,600' (S crs Erie) S—2,600' (NW crs Pittsburgh) W—2,200' (NE crs Akron)	None	N	10 mi.—2,200' W side N crs 15 mi.—2,200' W side N crs 20 mi.—2,200' W side N crs 25 mi.—2,200' W side N crs	1,700	185	3.4	1,196	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2900' on S course. *Rwy No. 18.
YUMA, ARIZ. Yuma Co. Arpt. 220 kc; YUM SBMRAZ-DTV	N—5,000' (Blythe Range) E—4,000' (Gila Bend Range) S—Min. en route altitude W—3,000' (El Centro Range)	None	N	10 mi.—3,000' E side N crs 15 mi.—3,000' E side N crs 20 mi.—3,000' E side N crs 25 mi.—3,000' E side N crs	2,500	165	5.8	213	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	600 600 800 300	1.5 1.0 2.0 1.0	Climb to 3,500' on S course within 20 mi of range.
ZUNI, N. Mex. El Morro CAA Int Field 353 kc; ZUN BMRLZ-DTV	N—Min. en route altitude E—11,000' (Acomita Range) S—Min. en route altitude W—10,000' (Winslow Range)	None	W	10 mi.—10,000' S side W crs 15 mi.—10,000' S side W crs 20 mi.—10,000' S side W crs 25 mi.—10,000' S side W crs	8,620	78	0.1	7,114	R A T	1,500 2,000 1,000	5.0 5.0 5.0	1,500 2,000 1,000	5.0 5.0 5.0	If not contact over range, make climbing left turn to 10,000' on W crs within 25 mi.

2. The high frequency range procedures prescribed in § 60.46–5 are amended to read:

VERY HIGH FREQUENCY RANGE PROCEDURES

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
ATLANTIC CITY, N. J. Atlantic City NAS 108.7 mc; AIY BVAR-DTV	NE—1,300′ (NE crs Millville) SE—Min. en route altitude SW—1,300′ (SE crs Millville) NW—1,300′ (NE crs Millville)	None	NE (Vis.)	10 mi.—1,300′ W side NE crs 15 mi.—1,300′ W side NC crs 20 mi.—1,300′ W side NE crs 25 mi.—1,300′ W side NE crs	800	140	2.7	70	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 2.0 1.0	Climb to 1,300′ on SW crs.
CHILDRESS, TEX. Childress Arpt. 109.1 mc; CDS BVAR-DTVJ	NE—3,500′ (Hobart VAR) SE—3,500′ (E crs Guthrie VAR) SW—4,500′ (Lubbock Range) NW—3,900′ (SE crs Clarendon)	None	SE (Aur.)	10 mi.—3,000′ E side SE crs 15 mi.—3,500′ E side SE crs 20 mi.—3,500′ E side SE crs 25 mi.—3,500′ E side SE crs	2,460	345	4.0	1,952	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		Climb to 3,900′ on NW crs within 25 mi.
DALHART, TEX. Dalhart Mun Arpt. 108.3 mc; DHT BVAR-DTVJ	NE—Min. en route altitude SE—5,200′ (Amarillo Range) SW—Min. en route altitude NW—Min. en route altitude	None	NW (Vis.)	10 mi.—5,200′ W side NW crs 15 mi.—5,200′ W side NW crs 20 mi.—5,200′ W side NW crs 25 mi.—5,200′ W side NW crs	4,700	179	5.0	3,989	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA		Climb to 5,200′ on SE crs.
DOVER, DEL. Dover AFB 108.3 mc; DOV VARW	NE—1,500′ (SE crs. Millville) SE—1,500′ (NE crs. Salisbury VAR) SW—1,500′ (E crs. Baltimore) NW—1,500′ (S crs. New Castle)	None	NE (Vis.)	10 mi.—1,500′ N side NE crs 15 mi.—1,500′ N side NE crs 20 mi.—1,500′ N side NE crs 25 mi.—1,500′ N side NE crs	1,000	166	7.1	28	R A T	500 1,000 300	2.0 2.0 1.0	500 1,000 300	2.0 2.0 1.0	Climb to 1,500′ on SE crs.
DYERSBURG, TENN Dyersburg Arpt. 109.1 mc; DYR BVAR-DTVJ	N—1,600′ (Paducah VAR) E—1,500′ (N crs Jackson) S—2,000′ (NE crs Memphis) W—1,600′ (S crs Advance)	None	E (Aur.)	10 mi.—1,500′ N side E crs 15 mi.—1,500′ N side E crs 20 mi.—1,500′ N side E crs 25 mi.—1,600′ N side E crs	1,200	268	5.0	337	R A T	600 800 300	1.5 2.0 1.0	NA NA NA		Climb to 1,600′ on W crs.
EL DORADO, ARK. El Dorado-Goodwin Field. 109.1 mc; ELD BVAR-DTVJ	NE—1,500′ (SW crs Pine Bluff VAR) SE—1,500′ (Monroe Range) SW—1,500′ (Shreveport Range) NW—1,500′ (NE crs Texarkana)	None	NE (Vis.)	10 mi.—1,500′ N side NE crs. 15 mi.—1,500′ N side NE crs. 20 mi.—1,500′ N side NE crs. 25 mi.—1,500′ N side NE crs.	1,000	200	3.5	277	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,500′ on SW crs within 25 mi. *Runway No. 22.
ELIZABETH CITY, N. C. Elizabeth City CGAS 108.3; ECG BVAR-DTVJ	N—1,500′ (E crs Norfolk VAR) E—Min. en route altitude S—* W—1,500′ (N crs Williamston VAR)	None	W (Vis.)	10 mi.—1,200′ N side W crs. 15 mi.—1,200′ N side W crs. 20 mi.—1,200′ N side W crs. 25 mi.—1,200′ N side W crs.	800	93	6.4	10	R A T	600 800 300	1.5 2.0 1.0	700 800 300	1.5 2.0 1.0	Turn left and climb to 1,500′ on N course. *This course is through danger areas. CAUTION: 150′ msl radio tower, 0.4 mi SSE of arpt.
GOODLAND, KANS. Goodland Arpt 108.3; GLD BVAR-DTV	N—Min. en route altitude E—5,000′ (Hill City VAR) S—Min. en route altitude W—6,000′ (Thurman VAR)	None	N (Aur.)	10 mi.—4,700′ W side N crs. 15 mi.—4,700′ W side N crs. 20 mi.—4,700′ W side N crs. 25 mi.—4,700′ W side N crs.	4,190	194	1.2	3,655	R A T	*500 1,000 300	1.0 3.0 1.0	NA NA NA		Climb to 5,000′ on S crs within 25 mi. *Not approved for acft having stall speed greater than 75 mph.
GRAND JUNCTION, COLO. Walker Field 108.7; GJT BVAR-DTVJ	N—Min. en route altitude E—14,000′ (Eagle VAR) S—Min. en route altitude W—10,000′ (Hanksville VAR)	None	W (Vis.)	10 mi.—10,000′ N side W crs. 15 mi.—10,000′ N side W crs. 20 mi.—10,000′ N side W crs. 25 mi.—10,000′ N side W crs.	*8,500	50	12.5	4,858	R A T	3,000 3,000 1,000	5.0 5.0 2.0	3,000 3,000 1,000	5.0 5.0 2.0	If not contact within 7 mi afte passing range, turn left and climb to 10,000′ on W crs within 25 mi. *After passing range, continue descent on E crs.
HANKSVILLE, UTAH CAA Int. Field 109.1 mc; HVE BVAR-DPVJ	NE—10,000′ (Grand Junction VAR) SE—Min. en route altitude SW—13,000′ (Bryce Canyon VAR) NW—Min. en route altitude	NW crs to 11,000′ within 20 mi.	SE (Aur.)	10 mi.—8,000′ E side SE crs 15 mi.—8,000′ E side SE crs 20 mi.—8,000′ E side SE crs 25 mi.—8,000′ E side SE crs	5,700	314	2.8	4,480	R A T	1,200 1,200 600	2.0 2.0 2.0	1,200 1,200 600	2.0 2.0 2.0	Climb to 11,000′ on NW crs, within 25 mi.
HOBART, OKLA. Hobart Arpt. 108.7 mc; HBR BVAR-DTVJ	NE—3,000′ (Oklahoma City Range) SE—3,100′ (Wichita Falls Range) SW—3,500′ (Childress VAR) NW—3,500′ (W crs Oklahoma City)	None	SE (Aur.)	10 mi.—3,100′ S side SE crs* 15 mi.—3,100′ S side SE crs* 20 mi.—3,100′ S side SE crs* 25 mi.—3,100′ S side SE crs*	2,600	354	8.9	1,583	R A T	500 800 300	2.0 2.0 1.0	NA NA NA		If not contact over range, climb to 3,500′ on NW crs within 25 mi. *Non-standard account Ft. Sill danger area.

VERY HIGH FREQUENCY RANGE PROCEDURES—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
LEXINGTON, KY. Blue Grass Arpt. 108.3 mc; LEX BVAR	N—2,300′ (E crs Louisville) E—Min. en route altitude S—3,600′ (Corbin VAR) W—2,600′ (S crs Louisville)	None	S (Vis.)	10 mi.—2,500′ E side S crs 15 mi.—3,100′ E side S crs 20 mi.—3,100′ E side S crs 25 mi.—3,100′ E side S crs	2,000	303	8.4	978	R A T	1,000 1,000 (BCOB) 300	1.0 2.0 1.0	1,000 1,000 (BCOB) 300	1.0 2.0 1.0	Climb to 2,300′ on N crs within 20 mi.
MARFA, TEX. Marfa-Alpine Arpt. 108.3 mc; MRF BVAR-DTVJ	NE—Min. en route altitude SE—9,000′ (Big Bend VAR) SW—Min. en route altitude NW—9,000′ (Sierra Blanca VAR)	On NW crs, 4 min. out.	NW (Vis.)	10 mi.—6,000′ W side NW crs 15 mi.—6,000′ W side NW crs 20 mi.—6,000′ W side NW crs 25 mi.—6,000′ W side NW crs	5,500	114	4.0	4,885	R (R) S A T	700 600 600 800 300	1.5 1.0 1.0 2.0 1.0	700 600 600 800 300	2.0 2.0 2.0 2.0 1.0	Make 180° turn to right and climb to 9,000′ on NW crs.
MASSENA, N. Y. Richards Field 108.7 mc; MSS BVAR-DTVJ	NE—2,000′ (SE crs Montreal) SE—Min. en route altitude SW—3,000′ (NE crs Watertown VAR) NW—2,000′ (E crs Ottawa)	None	NE (Vis.)	10 mi.—1,700′ N side NE crs 15 mi.—1,700′ N side NE crs 20 mi.—1,700′ N side NE crs 25 mi.—1,700′ N side NE crs	1,200	299	5.9	200	R (R) A T	600 600 800 300	1.5 1.0 2.0 1.0	600 600 800 300	1.5 1.0 2.0 1.0	Climb to 3,000′ on SW crs.
MYRTLE BEACH, S. C. Myrtle Beach Arpt. 108.3 mc; MYR BVAR-DTVJ	NE—1,200′ (Wilmington VAR) SE—Min. en route altitude SW—1,200′ (Charleston Range) NW—Min en route altitude	None	NE (Vis.)	10 mi.—1,200′ N side NE crs 15 mi.—1,200′ N side NE crs 20 mi.—1,200′ N side NE crs 25 mi.—1,200′ N side NE crs	700	227	3.5	24	R (R) S* A T	600 500 500 800 300	1.0 1.0 1.0 2.0 1.0	600 500 500 800 300	1.0 1.0 1.0 2.0 1.0	Climb to 1,200′ on SW crs. *Runway No. 26. NOTE: Arpt not suitable for acft with weight in excess of 50,000 lbs.
NANTUCKET, MASS. Nantucket Memorial Arpt. 109.1 mc; ACK BVAR-DTVJ (Procedure No. 1)	N—1,500′ (E crs Providence) E—Min. en route altitude S—Min. en route altitude W—Min. en route altitude	None	N (Vis.)	10 mi.—1,300′ W side N crs 15 mi.—1,300′ W side N crs 20 mi.—1,300′ W side N crs 25 mi.—1,300′ W side N crs	800	106	3.2	47	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Turn left and climb to 1,500′ on N crs.
(Procedure No. 2)	N—1,500′ (E crs Providence) E—Min. en route altitude S—Min. en route altitude W—Min. en route altitude	None	W (Aur.)	10 mi.—1,300′ S side W crs 15 mi.—1,300′ S side W crs 20 mi.—1,300′ S side W crs 25 mi.—1,300′ S side W crs	800	106	3.2	47	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 1,500′ on E crs.
NEW BERN, N. C. Simmons-Nott Arpt. 109.9 mc; EWN BVAR-DTVJ	NE—1,200′ (E crs Williamston VAR) SE—Min. en route altitude SW—1,200′ (Wilmington VAR) NW—1,600′ (Raleigh Range)	None	SW (Vis.)	10 mi.—1,100′ N side SW crs 15 mi.—1,100′ N side SW crs 20 mi.—1,100′ N side SW crs 25 mi.—1,100′ N side SW crs	700	108	4.3	18	R A T	*500 800 300	1.0 2.0 1.0	*500 800 300	1.5 2.0 1.0	Climb to 1,200′ on NE crs. *Apply to acft with stall speeds of 75 mph or less only.
OTTUMWA, IOWA Ottumwa Arpt. 108.7 mc; OTM BVAR-DTVBJ	NE—Min. en route altitude SE—2,000′ (NE crs Kirksville) SW—Min. en route altitude NW—2,000′ (E crs Des Moines)	None	SE (Vis.)	10 mi.—2,000′ E side SE crs 15 mi.—2,000′ E side SE crs 20 mi.—2,000′ E side SE crs 25 mi.—2,000′ E side SE crs	1,500	304	7.6	841	R A T	500 800 300	1.5 2.0 1.0	500 800 300	2.0 2.0 1.0	Climb to 2,000′ on NW crs within 25 mi.
PADUCAH, KY. Barkley Field 108.7 mc; PUK VARW-TVJ	N—Min. en route altitude E—Min. en route altitude S—1,600′ (NE crs Blytheville) W—1,700′ (E crs Advance)	None	S (Vis.)	10 mi.—1,600′ E side S crs 15 mi.—1,600′ E side S crs 20 mi.—1,600′ E side S crs 25 mi.—1,600′ E side S crs	1,100	41	4.8	405	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 1,600′ on N crs.
PARKERSBURG, W. VA. Wood Co. Arpt. 109.1 mc; PKB BVAR-DTVJ	NE—2,500′ (SW crs Wheeling VAR) SE—3,700′ (W crs Elkins) SW—2,500′ (N crs Charleston LF) NW—Min. en route altitude	None	SW (Vis.)	10 mi.—2,500′ S side SW crs 15 mi.—2,500′ S side SW crs 20 mi.—2,500′ S side SW crs 25 mi.—2,500′ S side SW crs	2,000	18	3.6	856	R A T	700 800 300	1.5 2.0 1.0	800 800 300	1.5 2.0 1.0	Climb to 2,600′ on NE crs within 25 mi.
PASO ROBLES, CALIF. San Luis Obispo Arpt. 108.7 mc; PRB BVAR-DTVJ	NE—Min. en route altitude SE—7,000′ (NW crs Santa Barbara VAR) SW—Min. en route altitude NW—7,000′ (Salinas VAR) NW—4,000′ (SW crs Fresno)	To 4,000′ on NW crs within 20 mi.	NW (Vis.)	10 mi.—4,000′ W side NW crs 15 mi.—4,000′ W side NW crs 20 mi.—4,000′ W side NW crs 25 mi.—4,000′ W side NW crs	2,000	85	1.8	806	R A T	1,200 1,500 300	1.0 2.0 1.0	NA NA NA		If not contact over range, make right turn and climb to 4,000′ on NW crs within 20 mi.
PINE BLUFF, ARK. Grider Field 108.7 mc; PBF BVAR-DTVJ	N—1,300′ (W crs Stuttgart) N—900′ (SE crs Little Rock) (final) E—1,500′ (SE crs Little Rock) S—1,500′ (NE crs El Dorado VAR) W—1,500′ (SW crs Little Rock)	None	N (Vis.)	10 mi.—1,500′ W side N crs 15 mi.—1,500′ W side N crs 20 mi.—1,500′ W side N crs 25 mi.—1,500′ W side N crs	900	150	5.0	210	R (R) S A T	500 500 500 800 200	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.5 1.5 2.0 1.0	Climb to 1,800′ on S crs within 25 mi.

VERY HIGH FREQUENCY RANGE PROCEDURE—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range—final approach (ft.)	Station to airport		Field elevation (ft.)	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished; remarks
						Magnetic bearing (deg.)	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
ROCKY MOUNT, N. C. Rocky Mount Arpt. 109.1 mc; RMT BVAR-DTVJ	NE—1,500' (NW crs Elizabeth City VAR) SE—Min. en route altitude SW—1,500' (SE crs Raleigh) NW—1,800' (SW crs Richmond)	None	NE (Vis.)	10 mi.—1,200' N side NE crs 15 mi.—1,200' N side NE crs 20 mi.—1,200' N side NE crs 25 mi.—1,200' N side NE crs	800	261	4.3	97	R A T	600 1,000 300	1.5 2.0 1.0	NA NA NA	----------	Climb to 1,800' on SW crs. Arpt rnwys and obstructions not lighted.
SALINA, KANS. Salina Arpt. 108.3 mc; SLN BVAR-DTVJ	N—Min. en route altitude E—Min. en route altitude S—2,800' (E crs Hutchinson) W—3,000' (S crs Waldo VAR)	None	N (Aur.)	10 mi.—3,000' W side N crs 15 mi.—3,000' W side N crs 20 mi.—3,000' W side N crs 25 mi.—3,000' W side N crs	2,200	132	5.0	1,309	R A T	500 800 300	1.5 2.0 1.0	500 800 300	1.5 2.0 1.0	Climb to 2,800' on S crs. NOTE: Use paved runway only.
Smoky Hill AFB	N—Min. en route altitude E—Min. en route altitude S—2,800' (E crs Hutchinson) W—3,000' (S crs Waldo VAR)	None	N (Aur.)	10 mi.—3,000' W side N crs 15 mi.—3,000' W side N crs 20 mi.—3,000' W side N crs 25 mi.—3,000' W side N crs	2,200	180	6.0	1,289	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	NA NA NA NA NA	----------	Climb to 2,800' on S crs within 25 mi. *Runway No. 17.
SALINAS, CALIF. Salinas Arpt. 109.1 mc; SNS BVAR	NE—6,000' (SE crs Oakland) SE—7,000' (Paso Robles VAR) SW—Min. en route altitude NW—6,000' (SW crs San Francisco)	NW crs within 25 mi (min. alt. 3,000')	NW (Vis.)	10 mi.—2,000' W side NW crs 15 mi.—2,000' W side NW crs 20 mi.—2,000' W side NW crs 25 mi.—3,000' W side NW crs	1,100	124	3.4	86	R A T	600 800 300	1.5 2.0 1.0	600 800 300	1.5 1.0 2.0	Climb to 6,000' on SE (Visual) crs within 25 miles. NOTE: Final approach to be monitored by use of ADF on LF range.
SALISBURY, MD. Salisbury Arpt. 109.5 mc; SBY BVAR-DTVJ	NE—1,500' (SW crs Atlantic City VAR) SE—Min. en route altitude SW—1,500' (W crs Chincoteague) NW—Min. en route altitude	None	NE (Vis.)	10 mi.—1,300' N side NE crs 15 mi.—1,300' N side NE crs 20 mi.—1,300' N side NE crs 25 mi.—1,300' N side NE crs	800	291	4.8	52	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	*500 500 *800 *300	2.0 2.0 2.0 1.0	Climb to 1,500' on SW crs. *Night minimums approved for DC-3 or smaller acft only. Prior notice required for night lighting facilities.
SANTA BARBARA, CALIF. Santa Barbara Arpt. 108.3 mc; SBA BVAR	NE—Min. en route altitude SE—Min. en route altitude SE—820' (Int. SE crs Santa Barbara VAR and 83° bearing on Camarillo range) (final) SW—Min. en route altitude NW—Min. en route altitude NW—6,000' (El Capitan FM)	On SE crs. (All turns to S.)	SE (Vis.)	10 mi.—1,500' S side SE crs 15 mi.—1,500' S side SE crs 20 mi.—1,500' S side SE crs 25 mi.—1,500' S side SE crs	820	77	1.6	14	R A T	800 800 300	2.0 2.0 1.0	800 800 300	2.0 2.0 1.0	If not contact over range, turn left 180° and proceed out SE crs, climbing to 1,500' within 20 mi. Final approach to be monitored by use of ADF on LF range. Restricted to qualified personnel having previously flown thru let-down procedure under visual conditions.
TOPEKA, KANS. Phillip Billard Arpt. 109.1 mc; TOP BVAR-DTV	N—2,400' (W crs St. Joseph) E—2,500' (S crs St. Joseph) S—2,400' (NE crs Lebo) W—2,500' (Salina VAR)	None	N (Aur.)	10 mi.—2,300' W side N crs 15 mi.—2,300' W side N crs 20 mi.—2,300' W side N crs 25 mi.—2,300' W side N crs	1,800	211	6.0	879	R A T	700 800 300	1.5 2.0 1.0	700 800 300	1.5 2.0 1.0	Climb to 2,400' on S crs, or alternate procedure (when directed by ATC), climb to 2,500' on W crs.
WATERTOWN, N. Y. Watertown Arpt. 109.1 mc; ART BVAR-DTVJ	NE—3,000' (SW crs Massena VAR SE—Min. en route altitude SW—2,000 (Syracuse Range) SW—1,200' (N crs Syracuse) (final). NW—Min. en route altitude	None	SW (Vis.)	10 mi.—1,500' N side SW crs 15 mi.—1,500' N side SW crs 20 mi.—1,500' N side SW crs 25 mi.—1,500' N side SW crs	1,200	45	3.1	325	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 3,000' on NE crs. *Runway No. 6.
WHEELING, W. VA Ohio Co. Arpt. 108.2 mc; HLG BVAR-DTVJ	NE—2,700' (NW crs Pittsburgh) SE—3,000' (NE crs Morgantown) SW—2,600' (NE crs Parkersburg VAR). NW—2,500' (SE crs Cleveland)	None	NE (Vis.)	10 mi.—2,700' N side NE crs 15 mi.—2,700' N side NE crs 20 mi.—NA 25 mi.—NA	2,200	220	6.7	1,195	R (R) A T	700 600 1,000 300	1.5 1.0 2.0 1.0	700 600 1,000 300	1.5 1.0 2.0 1.0	Climb to 2,600' on SW crs within 25 mi. NOTE: Night operations NA on Rwy 9/27. Take-off on on Rwy 9, landing on Rwy 27, not authorized.
WILMINGTON, N. C. New Hanover Co. Arpt. 108.7 mc; IMN BVAR-DTV	NE—1,200' (New Bern VAR) SE—Min. en route altitude SW—1,200' (SE crs Myrtle Beach VAR). NW—Min. en route altitude.	None	NE (Vis.)	10 mi.—1,200' N side NE crs 15 mi.—1,200' N side NE crs 20 mi.—1,200' N side NE crs 20 mi.—1,200' N side NE crs	700	193	4.3	31	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,200' on SW crs.

3. The automatic direction finding procedures prescribed in § 60.46–7 are amended to read:

AUTOMATIC DIRECTION FINDING PROCEDURES

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
AKRON, OHIO Akron-Canton Arpt. 236 kc; CA LOM	Akron Range	LOM	176	12.8	2,500	S	10 mi.—2,500′ E side S track	2,200	4.31	1,228	R	500	1.5	Climb to 2,500′ on track at 45° to SE crs of Akron. CAUTION: Monitor LOM at all times during instrument approach.
	Int SW crs Akron & 90° bearing on LOM	LOM	90	13.8	2,500	6	15 mi.—2,500′ E side S track				(R)	500	1.0	
	Int. SE crs Akron & 270° bearing on LOM	LOM	270	18.0	2,500	186	20 mi.—2,500′ E side S track.				S	500	1.0	
	Int SE crs Akron & 186° bearing on LOM	LOM	186	11.4	2,500						A	800	2.0	
	Bergholz Rbn	LOM	314	33	2,500		25 mi.—2,500′ E side S track.				T	300	1.0	
ANDERSON, S. C. Anderson Arpt. 236 kc; AND BMHTV	Atlanta Range	Rbn	58	115.0	2,800	SW	10 mi.—2,000′ S side SW track	1,280	.9	783	R	500	2.0	Make 180° left turn and climb to 2,000′ on track of 238° within 25 mi of Anderson Rbn on the SW crs of Spartanburg range.
	Bold Springs Int.	Rbn	55	81.0	2,000	58	15 mi.—NA				A	1,000	2.0	
	Spartanburg Range	Rbn	238	50.0	2,800	238	20 mi.—NA					(BCOB)		
	Int S crs Greenville & SW crs Spartanburg	Rbn	238	23.5	2,200		25 mi.—NA				T	300	1.0	
ARDMORE, OKLA. Ardmore Arpt. 400 kc; ADM BMH–DTV	Oklahoma City Range	Rbn	150	81.0	2,500	SE	10 mi.—2,000′ E side SE track	1,480	.0	875	R	600	1.0	Climb to 2,500′ on track of 333° within 25 mi. *Ngt minimums NOTE: Minimums approved for acft with stall speeds of 75 mph or less only.
	Int. N crs Ft. Worth & N crs Dallas	Rbn	345	26.0	2,500	315	15 mi.—NA				*	600	2.0	
						135	20 mi.—NA				A	800	2.0	
	Tulsa Range	Rbn	209	152	2,700		25 mi.—NA				T	300	1.0	
ATLANTA, GA. Atlanta Arpt. 219 kc; AT LOM	Campbellton Range	LOM	94	8.8	2,300	W	10 mi.—2,300′ S side W track	2,260	4.7	1,024	R	500	1.5	Turn right, climb to 2,000′ on track of 144° within 25 mi and proceed to Walnut Int. Alternate procedure) when directed by ATC), climb to 3,000′ on E crs of Campbellton range, then proceed to Walnut Int. *Runway No. 9R.
	Atlanta NAS Range	LOM	208	18.0	2,600	88	15 mi.—2,300′ S side W track				(R)	500	1.0	
	Atlanta Range	LOM	284	6.9	2,300	268	20 mi.—2,300′ S side W track				S*	500	1.0	
	Int. E crs ILS and NE crs Atlanta	LOM	268	9.5	2,300		25 mi.—2,300′ S side W track				A	1,000	2.0	
	Int. SE crs Atlanta and S crs Atlanta VAR	LOM	311	20.3	2,300							(BCOB)		
	Madras FM	LOM	33	19.0	2,300						T	300	1.0	
	Smyrna FM	LOM	167	18.0	2,300									
ATLANTIC, IOWA Atlantic Arpt. 257 kc; AIC BMH–TV	Des Moines Range	Rbn	260	73.0	2,500	S	10 mi.—2,300′ E side S track	1,860	.0	1,153	R	700	1.5	Climb to 2,400′ on track of 360° within 25 mi. Atlantic Rbn operates on req. Monitor stn aurally while on approach. Approved for smaller than DC-3 acft; heavy acft use in emerg only. *Ngt minimums.
	Omaha Range	Rbn	73	45.0	2,500	360	15 mi.—2,300′ E side S track				(*)	700	2.0	
						180	20 mi.—2,300′ E side S track				A	1,000	3.0	
							25 mi.—2,300′ E side S track				T	500	1.0	
AUGUSTA, GA. Bush Field 385 kc; AGS SBRAZ–DTV (Procedure No. 1)	Columbia Range	Rng	241	73	1,600	W	10 mi.—1,600′ S side W track	1,140	127°, 7.3	140	R	600	2.0	Climb to 2,000′ on E crs of Augusta range within 25 mi.
	Charleston Range	Rng	287	110	1,600	107	15 mi.—1,600′ S side W track				A	800	2.0	
	Int. W crs Augusta and NE crs Atlanta	Rng	107	110	2,800	287	20 mi.—1,600′ S side W track				T	300	1.0	
							25 mi.—1,600′ S side W track							
Bush Field (Procedure No. 2) 233 kc; AG LOM	Augusta Range	LOM	146	12.25	1,400	S	10 mi.—1,300′ E side S track.	1,200	5.0	140	R	600	2.0	Climb to 2,000′ on track of 348° intersect E crs Augusta and proceed to Augusta range. Alternate procedure (when directed by ATC), climb to 3,000′ on track of 348°, intersect NE crs Augusta and proceed to Augusta range. *Rwy 35.
						348	15 mi.—1,300′ E side S track.				(R)	500	1.0	
						168	20 mi.—1,300′ E side S track.				S*	500	1.0	
							25 mi.—1,300′ E side S track.				A	800	2.0	
											T	300	2.0	

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
BALTIMORE, MD. Friendship Intn'l Arpt. 291 kc; BA LOM	Int. NE crs Washington & W crs Baltimore	LOM	222	4.5	1,500	W 101 281	10 mi.—1,500′ N side W track. 15 mi.—1,500′ N side W track. 20 mi.—1,500′ N side W track. 25 mi.—1,500′ N side W track.	1,200	4.3	147	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500′ on track of 80° within 12.6 miles. *Rwy 10.
	Beltsville FM	LOM	33	10.5	1,500									
BEDFORD, MASS. Hanscom Field 260 kc; BE LOM	Int. NE crs Westover & W crs Boston	LOM	116	33	3,000	W 112 292	10 mi.—1,600′ N side W track 15 mi.—1,600′ N side W track 20 mi.—2,000′ N side W track 25 mi.—3,000′ N side W track	1,200	4.5	135	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn left and climb to 1,600′ returning to LOM. *Runway No. 11 only.
	Int. N crs Providence & SW crs Boston	LOM	12	27	1,700									
	Boston Range	LOM	302	21.5	1,700									
BOISE, IDAHO Boise Air Terminal 236 kc; BO LOM	Boise Range	LOM	256	2.1	4,000	NW 96 276	10 mi.—4,000′ S side NW track 15 mi.—4,000′ S side NW track 20 mi.—4,000′ S side NW track 25 mi.—4,000′ S side NW track	3,900	4.36	2,858	R * (R) S# A T	600 700 500 500 800 300	1.5 1.5 1.0 1.0 2.0 1.0	Turn right, climb to 4,000′ on NW crs of Boise range within 10 mi. of range. *Night minimums. #Runway No. 10.
	Eagle FM	LOM	129	10.6	4,000									
BOSTON, MASS. Logan Arpt. 302 kc; BO LOM	Int. SW crs Boston & N crs Providence	LOM	66	20.0	1,700	SW 35 215	10 mi.—1,700′ E side SW track 15 mi.—1,700′ E side SW track 20 mi.—1,700′ E side SW track 25 mi.—1,700′ E side SW track	1,200 (over LOM) 720 (over LMM)	5.67 (LOM) 1.40 (LMM)	19	R (R) A T	600 *500 800 300	1.5 1.0 2.0 1.0	Climb to 1,500′ on track of 31° from Boston range within 15 mi; or alternate procedure (when directed by ATC), make climbing turn to rgt to 1,500′ on E crs of Boston. *600-1 required when circling W of arpt.
	Boston Range	LOM	220	8.4	1,700									
BRADFORD, PA. Bradford-McKean Co. 284 kc; BFD MHW	Phillipsburg Range	Rbn	344	59.0	4,000	SE 321 141	10 mi.—3,400′ N side SE track 15 mi.—3,400′ N side SE track 20 mi.—3,400′ N side SE track 25 mi.—3,400′ N side SE track	2,900	321°, 4.4	2,140	R (R) (*) S# (*) A T	500 500 500 500 500 1,000 (BCOB) 300	1.5 1.0 1.5 1.0 1.5 2.0 1.0	Climb to 3,400′ on track of 321° within 10 mi., make climbing left turn and return to Bradford Rbn climbing to 4,000′. *Night minimums. #Runway No. 32.
	Brookville Rbn	Rbn	40	50.0	4,000									
BRUNSWICK, GA. McKinnon Arpt. 206 kc; SSI BMH-TV	Jacksonville Range	Rbn	13	32.0	1,200	N 192 12	10 mi.—1,200′ W side N track 15 mi.—1,200′ W side N track 20 mi.—1,200′ W side N track 25 mi.—1,200′ W side N track	520	.0	20	R (R) A T	600 500 1,000 (BCOB) 300	1.5 1.0 2.0 1.0	Climb to 1,200′ on track of 192° within 25 mi.
	Savannah Range	Rbn	190	59.0	1,200									
BUTLER, PA. Graham Arpt. 206 kc; BTP HW	Pittsburgh Range	Rbn	10	25.0	3,000	S 10 190	10 mi.—3,000′ E side S track 15 mi.—3,000′ E side S track 20 mi.—3,000′ E side S track 25 mi.—3,000′ E side S track	2,400	3°, 5.25	1,235	R (*) S# (*) A T	600 600 500 600 1,000 (BCOB) 300	1.0 2.0 1.0 2.0 2.0 1.0	Climb to 2,600′ on track of 347° within 10 miles. *Night minimums. #Runway No. 36. NOTE: Minimums are for acft with stall speeds of 75 mph or less.
	Inst E crs Youngstown & S crs Erie	Rbn	167	41.0	2,600									
	Brookville Rbn	Rbn	241	53.0	3,600									
CAMARILLO, CALIF. Ventura Co. (Oxnard) 359 kc; CAV SBAWZ	Int. W crs Los Angeles VAR and SE crs Camarillo	Rng	305	24.0	5,000	W 70 250	10 mi.—1,500′ S side W track 15 mi.—1,500′ S side W track 20 mi.—1,500′ S side W track 25 mi.—1,500′ S side W track	940	250°, 6.0	43	R A T	900 1,000 (BCOB) 300	3.0 2.0 1.0	If not contact over Camarillo range, turn right and climb to 3,000′ on track of 250 within 20 mi. of range. *Descent to 940′ authorized since this is both the initial and final approach.
	*Int. SE crs Santa Barbara VAR and 70° bearing on Camarillo	Rng	70	19.0	940									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station: From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)	Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums:	Ceiling (ft.)	Visibility (mi.)	If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
CAMARILLO, CALIF.—Con. Oxnard Flight Strip	Int. W crs Los Angeles VAR and SE crs Camarillo	Rng	305	24.0	5,000	W 70 250	10 mi.—1,500′ S side W track 15 mi.—1,500′ S side W track 20 mi.—1,500′ S side W track 25 mi.—1,500′ S side W track	980	250′, 0	84	R# A# T#	900 1,000 (BCOB) 300	1.0 2.0 1.0	If not contact over Camarillo range, turn right and climb to 3,000′ on track of 250 within 20 mi of range. *Descent to 980′ authorized since this is both the initial and final approach. #These minima approved for acft with stall speeds of 75 mph or less only. Night landings and take-offs NA.
	*Int. SE crs Santa Barbara VAR and 70° bearing on Camarillo	Rng	70	19.0	980									
CHADRON, NEBR. Chadron Arpt. 206 kc; CDR BMH-TV	Rapid City Range	Rbn	166	84.0	5,500	NE 225 45	10 mi.—4,500′ N side NE track 15 mi.—4,500′ N side NE track 20 mi.—4,500′ N side NE track 25 mi.—4,500′ N side NE track	4,120	.0	3,312	R (*) A T (*)	800 800 1,000 500 500	1.0 2.0 3.0 1.0 2.0	Turn right to a track of 345 climbing to 5,500′ within 25 mi. NOTE: Arpt not satisfactory for DC-3 or larger acft. *Night minimums.
	Int. NE crs Scottsbluff and 346° bearing on Rbn	Rbn	346	41.0	5,500									
CHARLESTON, W. VA. Kanawha Co. Arpt. 227 kc; HW; LMM	Charleston Range	LMM	108	11.0	2,500	NE 230 50	10 mi.—2,300′ N side NE track 15 mi.—2,300′ N side NE track 20 mi.—2,300′ N side NE track 25 mi.—2,300′ N side NE track	1,800	.63	983	R A T	600 800 300	1.5 2.0 1.0	Climb to 2,500′ returning to Charleston range.
	Int. E crs Charleston & 230° bearing on LMM	LMM	230	3.5	2,500									
	Int. S crs Charleston & 50° bearing on LMM	LMM	50	13.5	2,500									
CHICAGO, ILL. Chicago-Midway Arpt. (Procedure No. 1—using ILS Back Course Compass Locator) 248 kc; IH; LOM	Chicago Range	LOM	140	7.0	2,000	SE 312 132	10 mi.—2,000′ E side SE track 15 mi.—2,000′ E side SE track 20 mi.—2,000′ E side SE track 25 mi.—2,000′ E side SE track	1,500	3.82	618	R (R) (*) S# A T	500 500 500 500 800 300	1.5 1.0 1.5 1.0 2.0 1.0	Make right turn and climb to 2,500′ on NE crs of Chicago, or alternate procedure (when directed by ATC), make left turn, climb to 2,500′ on W crs of Chicago, proceeding to Downers Grove Int. ILS Back Course Compass Locator must be monitored continuously during approach. #Runways No. 31L and 31R. *Ngt minimums.
	Lake Shore Int.	LOM	306	10.5	2,300									
	Int. E crs Harvey & 312° bearing on LOM	LOM	312	15.2	2,000 (Final 1,500)									
	Crib Int.	LOM	250	10.0	2,000									
	Howell Int.	LOM	33	5.9	2,000									
	Thornton Int.	LOM	341	11.0	2,000 (Final 1,500)									
219 kc; CH; LOM (Procedure No. 2)	Chicago Range	LOM	305	3.8	2,300	NW 132 312	10 mi.—2,300′ W side NW track 15 mi.—2,300′ W side NW track 20 mi.—2,300′ W side NW track 25 mi.—2,300′ W side NW track	1,800	4.8	618	R (R) (*) S# A T	500 500 500 500 800 300	1.5 1.0 1.5 1.0 2.0 1.0	Climb to 2,500′ on N and S crs of Harvey range, proceeding S to Momence Int., or as directed by ATC. *Night minimums. #Runways No. 13R and 13L.
	Int. SE crs Chicago & 312° bearing on LOM	LOM	312	5.8	2,300									
	Int. NE crs Joliet & 165° bearing on LOM	LOM	165	9.0	2,300									
	Int. NE crs Joliet & 68° bearing on LOM	LOM	68	11.0	2,300									
	Lakeshore Int.	LOM	260	13.0	2,300									
O'Hare Int'l Arpt. 362 kc; OR LOM	Int. S crs Milwaukee & E crs Rockford	LOM	122	13.3	2,500	NW 138 318	10 mi.—2,500′ W side NW track 15 mi.—2,500′ W side NW track 20 mi.—2,500′ W side NW track 25 mi.—2,500′ W side NW track	2,000	4.4	657	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn left and climb to 2,500′ on track of 50° to SE crs Glenview, then proceed NW to Int. NW crs Glenview & E crs Rockford.
	Int. E crs Rockford & 138° bearing on LOM	LOM	138	19.3	2,500									
	Int. NW crs Chicago & 318° bearing on LOM	LOM	318	6.8	2,500									
	Glenview Range	LOM	219	7.8	2,500									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
CINCINNATI, OHIO Greater Cincinnati Arpt. 215 kc; CV LOM	Mt. Healthy Int.	LOM	190	19.0	2,300	S	10 mi.—2,000' E side S track 15 mi.—2,000' E side S track 20 mi.—2,000' E side S track 25 mi.—2,000' E side S track	1,500	4.3	890	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,300' on track of 360° within 15 mi of LOM. *Runway No. 36.
	Cincinnati Range	LOM	235	22.5	2,400	360								
	Int. SW crs Cincinnati & 320° bearing on LOM	LOM	320	2.9	2,000	180								
	Union FM	LOM	360	4.0	2,000									
CLEVELAND, OHIO Cleveland Arpt. 201 kc; CL LOM	Cleveland Range	LOM	224	4.3	2,000	SW	10 mi.—2,000' *N side SW track 15 mi.—2,000' *N side SW track 20 mi.—2,000' *N side SW track 25 mi.—2,000' *N side SW track	2,000	4.5	789	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,600' on track of 100° within 25 mi. *1,168' MSL tower, 4.8 mi E of Grafton, Ohio. #Runway No. 5L. CAUTION: Monitor LOM during approach.
	Elyria FM	LOM	123	9.1	2,000	54								
	Avon Lake FM	LOM	171	9.7	2,000	234								
	N. Royalton FM	LOM	303	10.9	2,200									
	Int. NW crs Akron & 54° bearing on LOM	LOM	54	8.0	2,000									
COLORADO SPRINGS, COLO. Peterson Field 239 kc; CO LOM	Colo. Sprgs Range	LOM	166	9.8	7,500	S	10 mi.—7,000" E side S track 15 mi.—7,000' E side S track 20 mi.—7,000' E side S track 25 mi.—7,000' E side S track	7,000	4.23	6,172	(R) ** (E) ** (S*) A T	600 600 600 600 500 800 300	1.5 2.0 1.0 2.0 1.0 2.0 1.0	Climb to 8,000' on E crs of Colorado Springs range within 25 mi. *Runway No. 35. **Night minimums. CAUTION: (1) 7,190' msl radio twr lctd 5.5 mi N of range. (2) 7,710' msl terrain located 13.5 mi N of range.
	Pueblo Range	LOM	340	36.2	8,000	346								
	Fountain FM	LOM (final)	345	3.6	7,000	166								
COLUMBIA, S. C. Columbia Arpt. 218 kc; CA LOM	Columbia Range	LOM	251	11.8	1,600	SW	10 mi.—1,600' S side SW track 15 mi.—1,600' S side SW track 20 mi.—1,600' S side SW track 25 mi.—1,600' S side SW track	1,000	4.5	244	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Turn right, climb to 1,600' and proceed to the Columbia rang via the W crs of Columbia. *Runway No. 5.
	Int. SE crs Spartanburg & W crs Columbia	LOM	135	4.6	1,600	46 226								
COLUMBUS, OHIO Port Columbus Arpt. 257 kc; CM LOM	Columbus Range	LOM	286	2.7	2,300	W	10 mi.—2,300' S side W track 15 mi.—2,300' S side W track 20 mi.—2,300' S side W track 25 mi.—2,300' S side W track	1,820	4.5	817	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,300' on track of 84° within 25 mi, of Columbus range #Runway No. 9. NOTE: LOM to be monitored continuously during approach. LMM located 6.65 mi from arpt, Freq, 233 kc; Ident, MH.
	Hilliard FM	LOM	87	7.4	2,300	96								
	Hilliard FM (final)	LOM	87	7.4	1,820	275								
DALLAS, TEX. Love Field 371 kc; DA LOM	Dallas Range	LOM	327	6.5	1,700	NW	10 mi.—1,700' N side NW track 15 mi.—1,700' N side NW track 20 mi.—1,700' N side NW track 25 mi.—1,700' N side NW track	1,700	4.35	483	R (R) S (*) A T	500 500 500 500 300 300	1.5 1.0 1.0 1.5 2.0 1.0	Climb to 2,000' on track of 126° and proceed out E crs of Dallas. CAUTION: 1,229' msl tower, 7 mi W of LOM. *Night minimums.
	Int E crs Ft. Worth & N crs Dallas	LOM	193	3.1	1,700	126 306								
DANVILLE, VA. Danville Arpt. 244 kc; DAN BMH-TV	Int. NE crs Greensboro & S crs Lynchburg	Rbn	257	24.0	2,300	NE 241 61	10 mi.—1,600' N side NE track 15 mi.—1,600' N side NE track 20 mi.—1,600' N side NE track 25 mi.—1,600' N side NE track.	*1,050	.0	562	R A ** T	600 1,000 (BCOB) 1,500 (BCOB) 300	1.5 2.0 2.0 1.0	Climb to 2,300' on track of 220° within 25 mi. *Descent must be discontinued at authorized landing minimums. **Night minimums.
	Int. NE crs Greensboro & NW crs Raleigh	Rbn	44	32.0	2,300									
	Int. NE crs Greensboro & 327° bearing on Rbn	Rbn	327	7.5	2,000									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
DEMING, N. MEX. Deming Arpt. 290 kc; DMN HW	Columbus Range	Rbn	325	33.0	9,400	W 77 257	10 mi.—6,500′ S side W track 15 mi.—6,500′ S side W track. 20 mi.—6,500′ S side W track 25 mi.—6,500′ S side W track	5,920	4.5	4,314	R (R) #S T	*500 *500 *500 NA *200	1.5 1.0 1.0 1.0	Climb to 9,400′ on track of 77° within 25 mi., then return to stn. Shuttle: W on track of 257° within 25 mi. of stn. *Landings and take-offs NA at night. #Runway No. 8L.
DETROIT, MICH. Detroit City Arpt. 338 kc; DET MH	Windsor Range (final)	Rbn	329	6.4	1,620	SE 329 149	10 mi.—1,900′ E side SE track 15 mi.—1,900′ E side SE track 20 mi.—1,900′ E side SE track 25 mi.—1,900′ E side SE track	1,620	329°, 2.6	626	R A T	700 1,000 500	1.0 3.0 1.0	Climb to 2,300′ on track of 329° within 25 mi. of Detroit Rbn. NOTE: Minimums apply to acft with stall speeds of 75 mph or less only. CAUTION: Detroit Rbn must be monitored continuously during approach.
	Pontiac Int.	Rbn	149	19.2	2,300									
	Detroit Range	Rbn	67	24.0	2,300									
Willow Run Arpt. 359 kc; YI; LOM (Procedure No. 1)	Detroit Range	LOM	255	12.4	2,000	SW 50 230	10 mi.—2,000′ S side SW track 15 mi.—2,000′ S side SW track 20 mi.—2,000′ S side SW track 25 mi.—2,000′ S side SW track	2,000	5.8	716	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000′ on track of 50° to N crs of Detroit range. CAUTION: Monitor LOM during approach. *Runways 4R & 4L. NOTE: LMM located 0.65 mi. from arpt. Freq. 223 kc, Ident. IP.
	Ann Arbor FM	LOM	92	7.5	2,000									
	Int. N crs Detroit & 230° bearing on LOM	LOM	230	13.6	2,000									
	Int N crs Toledo & 50° bearing on LOM	LOM	50	5.0	2,000									
(Procedure No. 2—Using ILS Back Course Compass Locator) 375 kc; PI; LOM	Int W crs Detroit & 50° bearing on LOM	LOM	50	9.5	2,000	NE 230 50	10 mi.—2,000′ N side NE track 15 mi.—2,200′ N side NE track 20 mi.—2,200′ N side NE track 25 mi.—2,200′ N side NE track	1,400	4.5	716	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000′ on W crs of Detroit range within 15 mi. CAUTION: Monitor ILS back course compass locator during approach. *Runway No. 22.
	Detroit Range	LOM	327	5.0	2,000									
	Int. N crs Toledo & 50° bearing on LOM	LCM	50	16.5	2,000									
	Salem VAR	LOM	143	10.9	2,000									
	Int. N crs Detroit & 230° bearing on LOM													
DODGE CITY, KANS. Dodge City Arpt. 362 kc; DDC BMH-TV	Hutchinson Range	Rbn	252	115.0	3,600	N 169 349	10 mi.—3,600′ W side N track 15 mi.—3,600′ W side N track 20 mi.—3,600′ W side N track 25 mi.—3,600′ W side N track	3,100	.0	2,594	R (*) A T	500 500 800 300	1.5 2.0 2.0 1.0	Climb to 3,600′ on track of 169° within 25 mi. *Night minimums.
	Garden City Range	Rbn	100	50.0	4,000									
DUNKIRK, N. Y. Dunkirk Arpt. 266 kc; DKK BH-TV	Int. E crs Clear Creek & SW crs Buffalo	Rbn	230	18.0	2,000	N 185 5	10 mi.—1,800′ W side N track 15 mi.—1,800′ W side N track 20 mi.—1,800′ W side N track 25 mi.—1,800′ W side N track	1,300	.0	692	R A T	600 1,000 300	1.0 2.0 1.0	Immediately make standard rate climbing turn to the right and climb to 2,500′ on track of 264° within 10 mi. NOTE: Buffalo range and Dunkirk Rbn shall be monitored continuously during approach
	Erie Range	Rbn	64	57.0	2,500									
EL PASO, TEX. El Paso-Anderson Intn'l Arpt. 295 kc; EL; LOM	El Paso Range	LOM	326	4.0	5,000	NE	10 mi.—6,000′ N side NE track	5,000	4.56	3,936	R (R)	500 500	1.5 1.0	Turn left to 125°, climb to 6,000′ on S crs El Paso within 25 mi. *Maintain 8,000′ until 5 mi W of Hueco Mtn FM (high terrain to N). #Rwy 22.
	Newman Rbn	LOM	191	7.5	5,000	287								
	Hueco Mtn FM	LOM	268	20.0	7,000*	37	15 mi.—NA N side NE track 20 mi.—NA N side NE track 25 mi.—NA N side NE track				S# A T	500 800 300	1.0 2.0 1.0	

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
FLINT, MICH. Bishop Arpt. 335 kc; FNT BMH-TV	Lansing Range	Rbn	77	43.0	2,300	N 164 344	10 mi.—2,000′ W side N track 15 mi.—2,000′ W side N track 20 mi.—2,000′ W side N track 25 mi.—2,000′ W side N track	1,380	0.0	781	R A T	600 800 300	1.5 2.0 1.0	Climb to 2,300′ on track of 164° within 25 mi.
	Int. E crs Lansing and N crs Detroit	Rbn	344	22.0	2,200									
	Saginaw Rbn	Rbn	160	43.0	2,000									
FORT SMITH, ARK. Fort Smith Arpt. 323 kc; FSM BH-DTV	Int. NE crs Tulsa and S crs Joplin	Rbn	168	103.0	3,000	N 180 360	10 mi.—2,200′ W side N track 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,600	165°, 1.25	462	R (*) A T	800 800 1,000 (BCOB) 300	1.5 2.0 2.0 1.0	Climb to 3,800′ on track of 163° within 25 mi of Rbn. *Night minimums.
	Little Rock Range	Rbn	283	133.0	3,800									
	Texarkana Range	Rbn	343	130.0	3,800									
	Tulsa Range	Rbn	118	103.0	2,600									
FRESNO, CALIF. Fresno Air Terminal 266 kc; FN LOM	Fresno LF Range	LOM	71	11.1	1,700	SE 288 108	10 mi.—1,700′ **S side SE track 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,000	4.44	331	R (R) S# A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500′ on track of 288°, turn left and climb to 2,000′ on W crs of Fresno within 20 mi. *When approaching from Bowles FM, execute left turn of 265° at LOM to intercept outbound track of 108°. **Procedure turn S side of track due to terrain. #Runway No. 29.
	*Bowles FM	LOM	15	11.5	1,700									
GADSDEN, ALA. Gadsden Arpt. 292 kc; GAD HW	Atlanta Range	Rbn	281	98.0	4,000	NE 240 60	10 mi.—2,500′ N side NE track 15 mi.—2,700′ N side NE track 20 mi.—2,700′ N side NE track 25 mi.—2,700′ N side NE track	1,960	2.5	560	R A T (*)	700 1,000 (BCOB) (BCOB) 300 300	1.5 2.0 1.0 1.5	Climb to 3,500′ on track of 240° within 25 mi. *Night minimums.
	Birmingham Range	Rbn	54	48.0	3,500									
	Chattanooga Range	Rbn	210	90.0	4,000									
	Garden City Int.	Rbn	89	40.0	3,500									
GOODING, IDAHO Gooding Arpt. 215 kc; GNG BH-DTV	Int. SE crs Boise & W crs Burley	Rbn	85	28.0	7,000	W 70 250	10 mi.—5,700′ S side W track 15 mi.—NA 20 mi.—NA 25 mi.—NA	4,320	.0	3,730	R (*) A T	700 700 800 300	1.5 2.0 2.0 1.0	Climb to 7,000′ on track of 120° within 15 mi, intercept W crs Burley and continue climbing to 7,000′ to Burley or 9,000′ to Boise. *Night minimums.
	Burley Range	Rbn	275	38.0	7,000									
	Int. W crs Burley & 70° bearing on Rbn	Rbn	70	9.4	6,000									
	Int. W crs Burley & 300° bearing on Rbn	Rbn	300	7.3	6,000									
GOODLAND, KANS. Goodland Arpt. 353 kc; GLD BH-TV	Int. N. Crs Garden City & 267° bearing on Rbn	Rbn	267	50.0	5,000	E 267 87	10 mi.—4,700′ N side E track 15 mi.—4,700′ N side E track 20 mi.—4,700′ N side E track 25 mi.—4,700′ N side E track	4,290	.0	3,655	R A T (**)	*500 1,000 *300	1.0 3.0 1.0	Climb to 6,000′ on track of 267° within 25 mi. *Not approved for acft having stall speed greater than 75 mph. **Night landings and take-offs not authorized.
	Int. S crs Akron & 87° bearing on Rbn	Rbn	87	62.0	6,000									
HAMILTON (San Rafael), CALIF. Hamilton AFB 328 kc; SRF HZ	Richmond Itn. (NW Oakland & SW Fairfield-Suisun)	Ron	315	13.0	4,000	E 260 80	10 mi.—2,000′ N side E track 15 mi.—2,000′ N side E track 20 mi.—NA 25 mi.—NA	500	183°, 1.8	2	R (*) A T (*)	500 500 800 300 300	2.0 3.0 2.0 1.0 2.0	If contact not estab over Rbn, make left turn and climb to 4,000′ on track of 130° within 10 mi. *Night minimums.
HOUSTON, TEX. Houston Arpt. 219 kc; HO LOM	Houston Range	LOM	241	5.75	1,200	SW 55 215	10 mi.—1,200′ S side SW track 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,200	4.73	50	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,600′ on track of 35° within 25 mi. of LOM. *Runway No. 3. NOTE: Take-offs on Rwy 26 and landings on Rwys 8 & 26 NA.
	Int. E crs Richmond & SW crs	LOM	23	7.5	1,200									
	Houston FM	LOM	168	8.5	1,200									
	Webster FM	LOM	284	12.6	1,200									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
INDIANAPOLIS, IND. Weir Cook Arpt. 341 kc; IN LOM	Clayton FM	LOM	36	10.5	1,900	SW 44 224	10 mi.—1,900' S side SW track 15 mi.—1,900' S side SW track 20 mi.—1,900' S side SW track 25 mi.—1,900' S side SW track	1,400	4.2	796	R (R) S* A T	500 500 500 300 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,300' on E crs of Indianapolis to NW crs of Cincinnati. NOTE: LOM must be monitored continuously during approach. *Runway No. 4.
	Indianapolis Range	LOM	177	2.8	1,900									
	Int. E crs Indianapolis & 224° bearing on LOM	LOM	224	5.0	1,900									
	Int. S crs Indianapolis & 224° bearing on LOM	LOM	224	.3	1,900									
IOWA CITY, IOWA Iowa City Arpt. 248 kc; IOW BHTV	Des Moines Range	Rbn	76	107.0	2,200	E 265 85°	10 mi.—2,000' N side E track 15 mi.—2,000' N side E track 20 mi.—2,000' N side E track 25 mi.—2,000' N side E track	1,460	Adj. E of Arpt	661	R (*) A T	800 800 800 300	1.5 2.0 2.0 1.0	Climb to 2,200' on track of 265° within 25 mi. *Night minimums.
	Moline Range	Rbn	271	47.0	2,000									
JACKSON, MICH. Reynolds Field 281 kc; JXN BMH-TV	Battle Creek Range	Rbn	90	40.0	2,300	NW 135 315	10 mi.—2,200' W side NW track 15 mi.—2,200' W side NW track 20 mi.—2,200' W side NW track 25 mi.—2,200' W side NW track	1,500	.0	990	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,300' on track of 135° within 5 mi of Jackson Rbn. CAUTION: Rbn must be monitored at all times while making an ADF approach. 1,330' msl tower located approx 4.5 mi SE of Rbn.
	Int SE crs Lansing & W crs Detroit	Rbn	300	22.0	2,300									
	Int. SE crs Lansing & 210° bearing on Rbn	Rbn	210	19.0	2,300									
	Int. W crs Detroit & 30° bearing on Rbn.	Rbn	30	19.0	2,300									
JOHNSTOWN, PA. Johnstown Arpt. 368 kc; JST MHW	Int. NE crs Morgantown & SE crs Pittsburgh (Mt. Pleasant Int.)	Rbn	88	36.0	4,500	SW 45 225	10 mi.—3,700' S side SW track 15 mi.—3,700' S side SW track 20 mi.—3,700' S side SW track 25 mi.—3,700' S side SW track	3,000	5.4	2,282	R (R) S A T	600 500 500 1,000 (BCOB) 300	1.5 1.0 1.0 2.0 1.0	Make climbing right turn, return to Johnstown Rbn and hold at 4,500' on track of 225° within 10 mi SW of Rbn. NOTE: Runway lights at arp are available by prior request only.
	Altoona Range	Rbn	266	31.0	4,500									
	Int. NW crs W Washington & S crs Altoona (Flintstone Int.)	Rbn	335	38.0	4,500									
LAFAYETTE, LA. Lafayette Arpt. 266 kc; LFT BMH-DTV	New Orleans Range	Rbn	271	109.1	1,400	N 180 360	10 mi.—1,200' W side N track 15 mi.—1,200' W side N track 20 mi.—1,200' W side N track 25 mi.—1,200' W side N track	640	.0	40	R (*) (R) (*) A T	600 600 600 600 800 300	1.5 2.0 1.0 2.0 2.0 1.0	Climb to 1,400' on track of 180° within 25 mi. *Night minimums.
	Lake Charles Range	Rbn	81	69.0	1,400									
	Baton Rouge Range	Rbn	235	54.0	1,500									
LEXINGTON, KY. Blue Grass Arpt. 229 kc; LEX BMH-DTV	Louisville Range	Rbn	104	55.5	2,400	SW 45 225	10 mi.—2,100' S side SW track 15 mi.—2,100' S side SW track 20 mi.—2,100' S side SW track 25 mi.—2,100' S side SW track	1,600	45°, 2.2	978	R A T	500 800 300	1.5 2.0 1.0	Climb to 2,300' on track of 45° within 15 mi. Rbn must be monitored at all times while making an ADF approach.
	Lexington VAR Range	Rbn	287	9.2	2,100									
	Int. S crs Louisville & E crs Godman	Rbn	87	54.3	2,100									
	Int. E crs Louisville & N crs Lexington VAR	Rbn	190	18.0	2,300									
LOUISVILLE, KY. Standiford Field 224 kc; SD LOM	Int. N crs Godman & W crs Louisville	LOM	113	16.0	2,000	S 10 190	10 mi.—2,000' W side S track 15 mi.—2,000' W side S track 20 mi.—2,000' W side S track 25 mi.—2,000' W side S track	1,900	4.22	497	R (*) (R) (*) S# A T	600 600 500 500 500 800 300	1.5 2.0 1.0 2.0 1.0 2.0 1.0	Climb to 2,000' on track of 10° until intersecting W crs of Louisville, then turn left and proceed out W crs. CAUTION: Danger area 10 mi. S of LOM. Start procedure turn within 2 mi of LOM. Monitor LOM at all times while making an ADF Approach. *Night minimums. #Runway No. 1.
	Eastwood FM	LOM	240	18.0	2,000									
	Int. E crs Godman & S crs Louisville	LOM	322	10.9	2,000									
	Louisville Range	LOM	218	10.2	2,000									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track: degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
LUBBOCK, TEX. Lubbock Arpt. 219 kc; LB LOM	Lubbock Range	LOM	52	13.0	4,500	N 170 350	10 mi.—4,500′ E side N track 15 mi.—4,500′ E side N track 20 mi.—NA 25 mi.—NA	4,000	4.76	3,256	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 4,500′ on track of 170° within 25 mi of LOM.
LUMBERTON, N. C. Lumberton Arpt. 359 kc; LBT BMH-TV	Florence Range	Rbn	52	46.5	1,300	SW 52 232	10 mi.—1,200′ S side SW track 15 mi.—1,200′ S side SW track 20 mi.—1,200′ S side SW track 25 mi.—1,200′ S side SW track	726	.0	126	R A (BCOB) T	600 1,000 300	2.0 2.0 1.0	Climb to 1,900′ on track of 15° within 25 mi.
	Raleigh Range	Rbn	195	83.0	1,900									
MANSFIELD, OHIO Mansfield Arpt. 329 kc; MFD BMH-TV	Dayton Range	Rbn	59	113	2,500	NW 130 310	10 mi.—2,500′ W side NW track 15 mi.—2,500′ W side NW track 20 mi.—2,500′ W side NW track 25-mi.—2,500′ W side NW track	2,000	.0	1,296	R (R) A T	700 700 800 300	1.5 1.0 2.0 1.0	Climb to 2,500′ on track of 130° within 25 mi. CAUTION: Rbn must be monitored at all times while making an ADF approac..
	Elyria FM	Rbn	210	47	2,600									
	Columbus Range	Rbn	23	61	2,500									
	Int. NE crs Columbus & 330° track to Rbn	Rbn	310	3.4	2,500									
	Findlay Rbn	Rbn	105	61	2,500									
	Int. SW crs Akron & 285° bearing on Rbn	Rbn	285	33	2,500									
MASON CITY, IOWA Mason City Arpt. 227 kc; MCW BHTV	LeRoy Int.	Rbn	236	54.0	2,500	S 360 180	10 mi.—2,200′ E side S track 15 mi.—2,200′ E side S track 20 mi.—2,200′ E side S track 25 mi.—2,200′ E side S track	1,7.0	.0	1,215	R (*) A T	500 500 800 300	1.5 2.0 2.0 1.0	Climb to 2,500′ on track of 360° within 19 mi. *Night minimums.
	Des Moines Rng	Rbn	91	113.0	2,500									
	Minneapolis Rng	Rbn	178	117.0	2,500									
	Waterloo Rbn	Rbn	306	63.0	2,500									
MATHER (Sacramento), CALIF. Mather AFB 476 kc; MHR HW	Int. NE crs Sacramento & E crs Williams (Auburn Int.)	Rbn	189	24.0	3,500	NE 220 40	10 mi.—2,500′ N side NE track 15 mi.—NA 20 mi.—NA 25 mi.—NA	600	216°, 0.8	91	R *A T (*)	500 500 1,500 300	2.0 3.0 3.0 2.0	Climb to 3,000′ on track of 160° SCHUTTLE: SW crs Sacramento—holding as directed by Sacramento Radio. NOTE: Complete procedure turn within 10 mi of station. CAUTION: 2,100′ terrain, 15 mi out on approach crs. *Night minimums.
	Galt Int.	Rbn	351	20.0	3,000							300	3.0	
	Sacramento Rng	Rbn	48	14.0	3,000									
	Williams Rng	Rbn	112	57.0	3,000									
MEMPHIS, TENN. Memphis Arpt. 215 kc; ME; LOM 201 kc; EM; LMM	Memphis Rng	LOM	202	5.3	1,700	W 86 266	10 mi.—1,700′ S side W track 15 mi.—1,700′ S side W track 20 mi.—1,700′ S side W track 25 mi.—1,700′ S side W track (From LOM)	1,700 (LOM)	4.88 (LOM) 0.69 (LMM)	269	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	If not contact at LMM, climb to 2,000′ on track of 86° to NE course of Memphis.
	Cuba FM	LOM	192	19.0	2,000									
	Bruins FM	LOM	48	22.0	1,700									
MILWAUKEE, WIS. Gen. Mitchell Arpt. 219 kc; MK LOM	Milwaukee Rng	LOM	194	2.5	2,000	S 5 185	10 mi.—2,000′ E side S track 15 mi.—2,000′ E side S track 20 mi.—2,000′ E side S track 25 mi.—2,000′ E side S track	1,400	4.6	698	R (R) (*) S# A T	500 500 500 500 800 300	1.5 1.0 1.5 1.0 2.0 1.0	Climb to 2,300′ on track of 5° within 25 mi of LOM. *Night minimums. #Runways No. 1 and 36. NOTES: (1) LOM to be monitored continuously during approach. (2) LMM located 0.67 mi. from end of Rwy No. 1; Freq. 201 kc; Ident. KE.
	Genesee FM	LOM	105	25.0	2,000									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
MINERAL WELLS, TEX Mineral Wells Arpt 284 kc; MWL BMH-TV	Abilene Range	Rbn	67	96.5	3,000	SE 300 120	10 mi.—2,000' E side SE track 15 mi.—2,000' E side SE track 20 mi.—2,000' E side SE track 25 mi.—2,000' E side SE track	1,540	6.0	964	R (*) (R) (*) S (*) A T	700 700 600 600 600 600 800 300	1.5 2.0 1.0 2.0 1.0 2.0 2.0 1.0	Climb to 2,500' and return to station. *Night minimums.
	Ft. Worth Range	Rbn	253	41.5	2,500									
MINNEAPOLIS, MINN. Minneapolis-St. Paul Intn'l Arpt. 219 kc; MS; LOM	Minneapolis Range	LOM	219	4.9	2,300	S 354 174	10 mi.—2,300' W side S track 15 mi.—2,300' W side S track 20 mi.—2,300' W side S track 25 mi.—2,300' W side S track	2,000	4.7	838	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500' on track of 354° to NW ers of Minneapolis to Hamel FM. CAUTION: Monitor LOM during approach. NOTE: LMM on 201 kc, Ident. SP, located 0.7 mi. from airport. *Runway No. 35.
	Stanton Rbn	LOM	336	24.0	2,300									
	Int. SW ers Minneapolis & 65° bearing on LOM	LOM	65	13.5	2,300									
	Int. NW ers Minneapolis & 130° bearing on LOM	LOM	130	23.0	2,300									
MOBILE, ALA. Bates Field 371 kc; MO LOM	Mobile Range	LOM	290	12.0	1,400	NW 139 319	10 mi.—1,400' W side NW track 15 mi.—1,400' W side NW track 20 mi.—1,400' W side NW track	1,400	4.5	217	R (R) S* A T	500 500 500 1,000 (BCOB) 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on W ers o Mobile. *Runway No. 13.
	Int. SE ers ILS & W ers Mobile	LOM	319	8.5	1,400									
MONTAGUE, CALIF. Siskiyou Co. Arpt. 288 kc; SIY BH-DTV	Ft. Jones Range (approaches from any other direction NOT AUTHORIZED)	Rbn	34	24.0	9,000	S 340 160	10 mi.—6,500' * E side S track 15 mi.—NA 20 mi.—NA 25 mi.—NA	5,140	351°, 3.2	2,047	R (#) A T (#)	2,500 2,500 2,500 1,500 1,500	2.0 3.0 3.0 2.0 3.0	Turn left, climb to 9,000' on track of 160° outbound, 340° inbound, within 10 mi. *In no case will procedure turn be accomplished S of the E ers of Ft. Jones. PROCEDURE NOT AUTHORIZED DURING SEVERE STORM CONDITIONS. #Night minimums.
NEW BEDFORD, MASS. New Bedford Arpt. 208 kc; WB; LMM	Providence Range	LMM	107	25.8	1,500	SW 53 233	10 mi.—1,400' S side SW track 15 mi.—1,400' S side SW track 20 mi.—NA 25 mi.—NA (Procedure turn not authorized beyond 15 mi.)	700	.60	79	R (R) A T	600 500 800 300	1.5 1.0 2.0 1.0	Make a standard rate climbing turn to the right, return to LMM and hold at 1,400' on S side of 233° track within 10 mi. of LMM.
	Int. E ers Providence & 336° bearing on LMM	LMM	336	2.7	1,400									
NEW ORLEANS, LA. Moisant Intn'l Arpt. 219 kc; MS LOM	New Orleans Range	LOM	274	8.2	1,400	W 99 279	10 mi.—1,400' S side W track 15 mi.—NA 20 mi.—NA 25 mi.—NA	940	3.92	3	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 1,500' on E ers of New Orleans range, or alternate procedure (when directed by ATC), climb to 1,300' on S ers of New Orleans. *Runway No. 10.
	LaPlace FM	LOM	87	8.4	1,400									
Alvin-Callender Arpt. 254 kc; BCS HW	New Orleans Range	Rbn	126	15.0	1,600	S 350 170	10 mi.—1,200' E side S track 15 mi.—1,200' E side S track 20 mi.—1,200' E side S track 25 mi.—1,200' E side S track	600	.0	5	R (*) A T	600 600 800 (BCOB) 300	1.5 2.0 2.0 1.0	Make 180° right turn and climb to 1,200' on track of 170° within 25 mi. *Night minimums.
NIAGARA FALLS, N. Y. Niagara Falls Arpt. 251 kc; GF LOM	Buffalo Range	LOM	338	12.0	2,000	E 278 98	10 mi.—2,000' N side E track 15 mi.—2,000' N side E track 20 mi.—2,000' N side E track 25 mi.—2,000' N side of track	1,800	4.6	590	R (R) S A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,000' on track of 278° within 25 mi.
	Wolcottsville FM	LOM	286	14.0	2,000									
	Int. SE ers Toronto & 223° bearing on LOM	LOM	223	4.0	2,000									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
OKLAHOMA CITY, OKLA. Will Rogers Field (Procedure No. 1 using Tulakes Rbn) 311 kc; TWO MHW	Int. N crs Okla. City & W crs Tinker	Rbn	152	2.8	2,300	N 170 350	10 mi.—2,500′ W side N track 15 mi.—2,500′ W side N track 20 mi.—2,500′ W side N track 25 mi.—2,500′ W side N track	1,960	4.54	1,283	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500′ on a track of 170° within 25 miles of Tulakes Rbn. *Runway No. 17.
	Int. E crs Okla. City & S crs Tinker	Rbn	279	13.5	2,700									
	Okla. City Range	Rbn	15	5.2	2,400									
	Mustang FM	Rbn	52	11.6	2,400									
	ILS LOM	Rbn	350	10.0	2,500									
(Procedure No. 2 using ILS Outer Compass Locator) 219 kc; OK LOM	Okla. City Range	LOM	148	5.6	2,500	S 350 170	10 mi.—2,500′ E side S track 15 mi.—NA 25 mi.—NA 25 mi.—NA	2,440	4.6	1,283	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 3,000′ on track of 350° within 25 mi, or alternate procedure (when directed by ATC), turn left, climb to 2,500′ on W crs Okla. City. *Runway No. 35.
	Int. SW crs Tulsa & E crs Okla. City	LOM	245	24.5	2,700									
	Washington FM	LOM	337	18.5	2,500									
	Mustang FM	LOM	104	11.3	2,500									
OLD TOWN, MAINE Old Town Arpt. 260 kc; OLD MHW	Bangor Range	Rbn	72	9.0	2,500	SW 72 252	10 mi.—1,600′ S side SW track 15 mi.—1,600′ S side SW track 20 mi.—2,200′ S side SW track 25 mi.—2,300′ S side SW track	900	57°, 2.3	126	R A T	600 800 300	1.5 2.0 1.0	Climb to 2,500′ on NE crs of Bangor within 15 mi. NOTE: Bangor radio will be monitored during approach
	Bangor Range (final)	Rbn	72	9.0	900									
	Int. NE crs Bangor & S crs Houlton	Rbn	252	60.0	2,500									
OTIS (Falmouth), MASS. Otis AFB 269 kc; AUA MH	Providence Range	Rbn	106	44.0	2,000	NE 230 50	10 mi.—1,200′ NW side NE track 15 mi.—NA 20 mi.—NA 25 mi.—NA (Turn to be made 1 minute from Rbn)	615	.0	126	R (*) A T (*)	500 500 800 300 300	2.0 3.0 2.0 1.0 2.0	Climb to 2,000′ on track of 270°. SHUTTLE: to 3,000′ on track of 230° outbound, 140° inbound, within 10 mi. of Rbn. *Night minimums. NOTE: Rbn operates on request only. Otis control tower must be called as soon as communications can be established, when operation of Rbn is requested.
OXNARD, CALIF. Ventura Co. Arpt.	(SEE CAMARILLO, CALIF.)													
PELLSTON, MICH. Emmet, Co. Arpt. 322 kc; PLN BMH-TV	Sault Ste. Marie Range	Rbn	204	65.0	2,500	SE 315 135	10 mi.—2,200′ S side SE track 15 mi.—2,200′ S side SE track 20 mi.—2,200′ S side SE track 25 mi.—2,200′ S side SE track	1,520	.0	720	R (R) A T	800 800 1,500 (BCOB) 500	1.5 1.0 2.0 1.0	Climb to 2,500′ on track of 315° within 25 mi.
	Traverse City Range	Rbn	34	69.0	2,500									
PITTSBURGH, PA. Allegheny Co. Arpt. (Using McKeesport Rbn) 287 kc; MKP MHW	Pittsburgh Range	Rbn	92	9.6	2,600	E 275 95	10 mi.—2,600′ N side E track 15 mi.—2,600′ N side E track 20 mi.—2,600′ N side E track 25 mi.—2,600′ N side E track	2,500	6.3	1,252	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500′ on track of 267° within 20 mi and proceed to Cecil holding pattern; or alternate procedure (when directed by ATC), climb to 2,600′ on track of 308° from Pittsburgh range and proceed to Clinton holding pattern. *Runway No. 27.
	Int. E crs Pittsburgh & NE crs Morgantown	Rbn	263	23.0	2,600									
	Int. SE crs Pittsburgh & NE crs Morgantown	Rbn	322	16.8	2,600									
PONCA CITY, OKLA. Ponca City Arpt. 368 kc; PNC BMH-DTV	Wichita Range	Rbn	162	64.0	2,500	N 170 350	10 mi.—2,300′ W side N track 15 mi.—2,300′ W side N track 20 mi.—2,300′ W side N track 25 mi.—2,300′ W side N track	1,600	.0	1,014	R (*) A T	600 600 800 300	1.0 2.0 2.0 1.0	Turn right and climb to 2,300′ on a track of 350° within 25 mi. *Night minimums.
	Oklahoma City Range	Rbn	09	90.0	3,100									
	Tulsa Range	Rbn	287	77.5	2,400									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
PORTLAND, OREG. Portland Arpt. 291 kc; DX LMM	Woodland FM to Sauvies Is.	Rbn	176	19	2,000	NW 113 293	10 mi.—2,500' *N side NW track 15 mi.—2,500' *N side NW track 20 mi.—NA 25 mi.—NA (Make procedure turn within 5 mi. NW of Sauvies Is. Rbn)	1,300 (over OM**)	4.81 (from OM)	23	R ## (E) S# A T	600 700 600 600 800 300	1.5 1.5 1.0 1.0 2.0 1.0	If not contact over LMM, execute a climbing right turn and proceed out S crs of Portland climbing to 3,000'. *Procedure turn on S side of track NA due to high terrain. **If OM not received both aurally and visually, maintain 1,300' to LMM, then climb to 3,000' on S crs of Portland. #Runway No. 11. ##Night minimums.
	Sauvies Is. Rbn (FINAL)	LMM	113	10.9	1,300									
	Willamette FM to Sauvies Is.	Rbn	333	25	3,000									
	Portland Range to Sauvies Is.	Rbn	278	10	2,300									
QUINCY, ILL. Baldwin-Quincy Arpt. 269 kc; UIN BH-DTV	Ottumwa Rbn	Rbn	138	104	2,000	SW 32 212	10 mi.—1,800' S side SW track 15 mi.—1,800' S side SW track 20 mi.—1,800' S side SW track 25 mi.—1,800' S side SW track	1,300	4.5	769	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,500' on track 25° of within 25 mi of Rbn *Runway No. 3. NOTE: Quincy Rbn must be monitored continuously while making an approach.
	Jerseyville Int.	Rbn	321	91	2,000									
	Kirksville Range	Rbn	96	69	2,500									
	Peoria Range	Rbn	229	100	1,900									
SAGINAW, MICH. Tri-City Arpt. 212 kc; SGW BMH-TV	Flint Rbn	Rbn	340	43	2,000	NW 170 350	10 mi.—2,000' W side NW track 15 mi.—2,000 W side NW track 20 mi.—2,000' W side NW track 25 mi.—2,000' W side NW track	1,160	.0	667	R (E) A T	600 500 800 300	1.5 1.0 2.0 1.0	Climb to 2,000' on track of 170° within 25 mi of Saginaw Rbn. CAUTION: Rbn must be monitored continuously during an ADF approach. 900' msl radio tower 3.5 mi, 170° from arpt. 780' msl elevator 2.5 mi W of arpt.
	Gladwin Rbn	Rbn	147	36	2,500									
	Lansing Range	Rbn	25	57	2,000									
SALEM, OREG. McNary Field 266 kc; SL LOM	Portland Range	LOM	176	57.0	3,000	SE 310 130	10 mi.—2,500' *W side SE track 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,600	4.6	207	R (**) (R) S# A T	800 1,000 500 500 1,000 300	1.5 1.5 1.0 1.0 2.0 1.0	Turn right and climb to 2,500' on track of 130° within 10 mi. *Procedure turn to E not authorized due to terrain. **Night minimums. #Rwy No. 31.
	Eugene Range	LOM	356	47.0	3,000									
SAN FRANCISCO, CALIF. San Francisco Arpt. 350 kc; SF LOM	Moffett Range	LOM	298	18.2	1,600	SE 278 98	10 mi.—2,000' S side SE track 15 mi.—2,000' S side SE track 20 mi.—2,000' S side SE track 25 mi.—2,000' S side SE track	1,600	6.6	10	R (R) S A T	600 500 500 800 300	2.0 1.0 1.0 2.0 1.0	Climb to 3,000' on track of 285° within 25 mi of LOM.
	San Francisco Range	LOM	89	3.6	2,000									
	Oakland Range	LOM	174	12.0	3,000									
SANTA FE, N. MEX. Santa Fe Arpt. (New) 206 kc; SAF BMH-DTV	Las Vegas Range	Rbn	258	51.5	12,500	SW 56 236	10 mi.—8,200' S side SW track 15 mi.—8,200' S side SW track 20 mi.—NA 25 mi.—NA	7,700	.0	6,344	R (*) A T (*)	1,500 1,500 2,000 (BCOB) 1,000 1,000	1.5 2.0 3.0 1.0 #2.0	Immediately turn right and climb to 8,200' on track of 236° within 15 mi. SHUTTLE: On track of 236° outbound 56°, inbound to 8,200' within 15 mi. *Night minimums. #Night take-off on Rwy 29 not authorized.
	Otto Range	Rbn	340	38.0	10,600									
SAULT STE. MARIE, MICH. Sault Ste. Marie-Kinross Arpt. 219 kc; KN LOM	Sault Ste. Marie Range	LOM	256	13.0	2,100	NW 154 334	10 mi.—2,100' W side NW track 15 mi.—2,100' W side NW track 20 mi.—2,100' W side NW track 25 mi.—2,100' W side NW track	1,600	6.65	780	R (R) S* A T	500 500 500 800 300	1.5 1.0 1.0 2.0 1.0	Climb to 2,100' on track of 154° within 25 mi of LOM. *Runway No. 15. NOTE: ILS LOM must be monitored continuously during approach. ILS LMM located 0.67 mi from rwy—freq. 261 kc, ident. NR.
	Int. W crs Sault Ste. Marie & 154° bearing on LOM	LOM	154	12.0	2,100									
	Int. S crs Sault Ste. Marie & 334° bearing on LOM	LOM	334	10.0	2,100									

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
SOUTH BEND, IND. St. Joseph Co. Arpt. 341 kc; SB OMK	South Bend Range	LOM	82	8.6	2,000	E 268 88	10 mi.—2,000′ N side E track 15 mi.—2,000′ N side E track 20 mi.—2,000′ N side E track 25 mi.—2,000′ N side E track	2,000	4.39	776	R (R) (#) S* A T	500 500 500 500 800 300	1.5 1.0 1.5 1.0 2.0 1.0	Climb to 2,000′ on track of 278° within 25 mi of LOM. CAUTION: Monitor LOM during approach. LMM located 0.71 mi from arpt on 352 kc—ident. BN. *Runway No. 27. #Night minimums.
	Int. W crs S. Bend & 88° bearing on LOM	LOM	88	14.8	2,000									
	Int. N crs S. Bend & 88° bearing on LOM	LOM	88	4.5	2,000									
	Int. N crs Goshen & 268° bearing on LOM	LOM	268	20.0	2,200									
	Int. N crs Goshen & E crs S. Bend	LOM	252	22.0	2,000									
SPOKANE, WASH. Geiger Field 236 kc; GE LOM	Int. W crs Spokane & 203° bearing on LOM	LOM	203	7.1	4,000	SW 23 203	5 mi.—4,000′ S side SW track 15 mi.—NA 20 mi.—NA 25 mi.—NA	3,500	4.42	2,372	R (*) (R) S# A T	500 600 300 500 1,000 300	1.5 1.5 1.0 1.0 2.0 1.0	Climb to 5,000′ on W crs Spokane within 18 mi of range. *Night minimums. #Runway No. 2.
	Spangle FM	LOM	312	13.6	4,100									
	Harrington FM	LOM	60	32.0	4,000									
TINKER (Oklahoma City), OKLA. Tinker AFB 263 kc; TIK MRAWZ	Oklahoma City Range	Rng	54	16.6	2,700	N 170 350	10 mi.—2,700′ E side N track 15 mi.—2,700′ E side N track 20 mi.—2,700′ E side N track 25 mi.—2,700′ E side N track	2,400	170°, 5.1	1,272	R (*) A T (*)	600 600 1,500 300 300	2.0 3.0 3.0 2.0 3.0	Climb to 3,500′ on track of 170° and contact Okla. City approach control. *Night minimums.
VALDOSTA, GA. Valdosta Arpt. 224 kc; VLD BMH-TV	Int. E crs Tallahassee & 17° bearing on Rbn	Rbn	17	24.0	1,400	NE 240 60	10 mi.—1,300′ N side NE track 15 mi.—1,300′ N side NE track 20 mi.—1,300′ N side NE track 25 mi.—1,300′ N side NE track	720	.0	203	R A T	500 1,000 (BCOB) 300	1.5 2.0 1.0	Climb to 1,500′ on track of 240° within 25 mi.
	Int. W crs Jacksonville & 311° bearing on Rbn	Rbn	311	33.0	1,400									
VERO BEACH, FLA. Vero Beach Arpt. 359 kc; VRB; BMH-TV	Int. S crs Melbourne & N crs W. Palm Beach	Rbn	338	49.0	1,500	W 110 290	10 mi.—1,200′ S side W track 15 mi.—1,200′ S side W track 20 mi.—1,200′ S side W track 25 mi.—1,200′ S side W track	520	.0	25	R (R) A T	500 500 1,000 (BCOB) 300	1.5 1.0 2.0 1.0	Climb to 1,200′ on track of 110° within 25 mi.
	Melbourne Range	Rbn	158	37.0	1,200									
WASHINGTON, D. C. Washington National Arpt. (Using Riverdale Rbn) 344 kc; RVD MHW	Washington Range	Rbn	25	11.3	1,600	NE 221 41	10 mi.—1,800′ W side NE track 15 mi.—1,800′ W side NE track 20 mi.—NA 25 mi.—NA	1,000	226°, 6.2	16	R A T	800 1,000 300	2.0 3.0 1.0	If not contact within 2 mi after passing Riverdale Rbn, make climbing left turn and climb to 1,600′ returning to Riverdale Rbn.
	Int. NE crs Washington & W crs Baltimore	Rbn	218	24.5	1,500									
	Beltsville FM (final)	Rbn	221	10.0	1,000									
	Andrews Range	Rbn	348	16.3	1,600									
	Largo VAR	Rbn	252	9.5	1,600									
Anacostia NAS	(Procedure identical to above)								216°, 4.2 naut. mi.		R A T	800 1,000 300	2.0 3.0 1.0	

AUTOMATIC DIRECTION FINDING PROCEDURES—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
WHITE PLAINS, N. Y. Westchester Co. Arpt. 281 kc; HP; LOM 260 kc; PN; LMM	Int. S crs Poughkeepsie & 161° bearing on LOM	LOM	161	17.0	2,000	NW 161 341	10 mi.—2,000′ W side NW track 15 mi.—2,000′ W side NW track 20 mi.—2,500′ W side NW track 25 mi.—2,500′ W side NW track	1,500 (LOM)	5.8 (LOM) .66 (LMM)	435	R (R) S A T	600 500 500 800 300	1.5 1.0 1.0 2.0 1.0	If not contact over LMM, turn left and climb to 2,000′ on track of 341° to LOM.
	Rye Rbn	LOM	360	12.1	2,000									
	Int. NE crs Newark & 341° bearing on LMM	LMM	341	8.5	2,000									
WILMINGTON, N. C. New Hanover Co. Arpt. 368 kc; IMN BMH–TV	New Bern VAR	Rbn	222	73.0	1,200	NW 160 340	10 mi.—1,200′ W side NW track. 15 mi.—1,200′ W side NW track. 20 mi.—1,200′ W side NW track. 25 mi.—1,200′ W side NW track.	530	.0	31	R (R) A T	500 500 800 300	1.5 1.0 2.0 1.0	Climb to 1,200′ on track of 160° within 15 mi.
	Wilmington VAR	Rbn	197	5.0	1,200									
	Myrtle Beach VAR	Rbn	59	68.0	1,200									
	Bald Head Rbn	Rbn	13	28.0	1,200									
	Lumberton Rbn	Rbn	112	70.0	1,500									
WORCESTER, MASS. Worcester Arpt. 254 kc; OR LOM *	From Int. NE crs Hartford & SE crs Westover to Int NE crs Hartford & LOM		64	25.0	2,400	SE 332 152	10 mi.—2,000′ N side SE track. 15 mi.—2,000′ N side SE track. 20 mi.—2,000′ N side SE track. 25 mi.—2,000′ N side SE track.	*1,500	4.6	1,009	R (R) A T	600 500 1,000 300	1.5 1.0 2.0 1.0	Climb to 2,400′ in standard rate left climbing turn to NE crs Hartford. *Descent must be discontinued at alt of authorized landing minimums.
	From Int. W crs Boston & NW crs Squantum to Int NE crs Hartford & LOM		244	28.0	2,400									
ZANESVILLE, OHIO Zanesville Arpt. 263 kc; ZZV BMH–TV	Int. E crs Columbus & 180° bearing on Rbn	Rbn	180	9.0	2,600	S 360 180	10 mi.—2,300′ E side S track. 15 mi.—2,300′ E side S track. 20 mi.—2,300′ E side S track. 25 mi.—2,300′ E side S track.	1,700	.0	901	R (R) A T	800 800 1,000 300	1.5 1.0 3.0 1.0	Make left turn, climb to 2,600′ on track of 180° within 10 mi. CAUTION: Rbn must be monitored at all times while making an ADF approach.

4. The instrument landing system procedures prescribed in § 60.46–9 are amended to read:

INSTRUMENT LANDING SYSTEM PROCEDURES

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
AKRON, OHIO Akron-Canton Arpt. Freq. 109.5 mc Ident. CAK	Bergholz Rbn.	Outer Mrkr	313	33.0	2,500	S 6 186	2,500'—E side S crs	2,400	2,340	1,430	4.31	0.78	1,228	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on track of 45° to SE crs of Akron. *Runway No. 36.
	Akron Range	Outer Mrkr	176	12.8	2,500												
	Int. SE crs Akron & 270° bearing on OM.	Outer Mrkr	270	18.0	2,500												
	Int. SW crs Akron & 90° on OM.	Outer Mrkr	90	13.8	2,500												
	Int. SE crs Akron & N crs ILS	Outer Mrkr	186	11.4	2,500												
ALBUQUERQUE, N. M. Kirtland Field Freq. 110.3 mc Ident. ABQ	Albuquerque Range	Peralta FM	174	12.0	8,000	S 349 169	8,000'—W side S crs	6,400	6,400	5,530	4.32	.66	5,330	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn left and climb to 10,000' on W crs of Albuquerque. *Runway No. 35.
	Peralta FM	Outer Mrkr	349	10.5	6,400												
AMARILLO, TEX. Amarillo Air Terminal Freq. 110.3 mc Ident. AMA	Amarillo Range	NE crs ILS	77	2.5	4,700	NE 209 29	4,700'—N side NE crs	*4,700	5,200	3,800	5.92	.62	3,604	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 5,000' on S crs of Amarillo, or alternate procedure (when directed by ATC), turn right, climb to 5,200' on W crs Amarillo within 25 mi. *Glide path will be intercepted between OM and MM. #Runway No. 21.
	Int. S crs Amarillo & SW crs ILS	Outer Mrkr	29	10.4	4,700												
	Int. E crs Amarillo & NW crs Clarendon	Outer Mrkr	279	12.5	4,700												
ATLANTA, GA. Atlanta Arpt. Freq. 109.9 mc Ident. ATL	Atlanta Range	Outer Mrkr	284	6.9	2,300	W 88 268	2,260'—S side W crs	2,260	2,260	1,225	4.70	.62	1,024	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn right, climb to 2,000' on SE crs of Atlanta and proceed to Walnut Int.; or alternate procedure (when directed by ATC), climb to 3,000' on E crs of Campbellton and proceed to Walnut Int. *Runway No. 9R.
	Campbellton Range	Outer Mrkr	94	8.8	2,300												
	Int. E crs ILS & NE crs Atlanta	Outer Mrkr	268	9.5	2,300												
	Int. SE crs Atlanta & S crs Atlanta VAR (Walnut)	Outer Mrkr	311	20.3	2,300												
	Atlanta NAS Range	Outer Mrkr	208	18.0	2,600												
	Madras FM	Outer Mrkr	33	19.0	2,300												
	Smyrna FM	Outer Mrkr	167	18.0	2,300												
AUSTIN, TEX. Mueller Arpt. Freq. 109.5 mc Ident. AUS	Austin Range	NW crs ILS	117	1.7	1,900	SE 305 125	1,900'—E side SE crs	*1,900	2,120	820	6.12	.69	621	R S# A T	500 400 800 300	1.5 1.0 2.0 1.0	Climb to 2,500' on NW crs of Austin within 25 mi., or alternate procedure (when directed by ATC), turn right and climb to 2,000' on NE crs of Austin within 25 mi. *Glide path intercepted after crossing outer marker inbound. #Runway No. 30L.
	Int. SE crs Austin & W crs Richmond	SE crs ILS	287	22.0	1,900												
BAKERSFIELD, CALIF. Bakersfield-Kern Co. Arpt. #1 Freq. 109.5 mc Ident. BFL	Bakersfield Range	SE crs ILS	141	1.5	2,000	SE 300 120	1,820'S—side SE crs	1,820	1,820	690	4.71	.62	515	R S# A T*	700 400 800 300	2.0 1.0 2.0 1.0	Climb to 3,000' on NW crs of Bakersfield, or alternate procedure (when directed by ATC), climb to 3,000' on SW crs of Bakersfield within 15 mi. *Take-offs or landings not authorized on Runway 7–25. #Runway No. 30.

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: eastbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
BALTIMORE, MD. Friendship International Arpt. Freq. 110.3 mc Ident. BAL	Int. NE crs Washington & W crs Baltimore.	LOM	222	4.5	1,500	W 101 281	1,500'—N side W crs	1,500	1,200	340	4.30	0.71	147	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Proceed to the Baltimore range, climbing to 1,500', or as directed by ATC. *Runway No. 10.
	Int. W crs ILS & NE crs Arcola	LOM	102	23.0	1,600												
	Beltsville FM	LOM	33	10.5	1,500												
BARKSDALE (Shreveport), LA. Barksdale AFB Freq. 109.9 mc	Dixie FM	NW crs ILS	110	3.0	1,500	NW 143 323	1,500'—N side NW crs	1,500	1,230	465	4.31	.89	167	R (*) S# A T	500 500 400 1,000 300	2.0 3.0 3/4 3.0 1.0	Climb to 1,500' on SE crs of ILS. *Night minimums. #Runway No. 14.
	Shreveport Range	Outer Mrkr	89	3.5	1,500												
BEDFORD, MASS. Hanscom Field Freq. 108.3 mc Ident. BED.	Boston Range	W crs ILS	300	26.5	1,600	W 112 292	1,600'—N side W crs	1,600	1,520	390	4.50	.69	135	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn left and climb to 1,600' on W crs of the ILS within 20 mi. *Runway No. 11.
	Int. W crs Boston & NW crs Squantum	W crs ILS	321	3.7	1,600												
BILLINGS, MONT. Billings Arpt. Freq. 110.3 mc Ident. BIL	Billings Range	E crs ILS	73	(*)	5,000	E 253 73	5,000'—N side E crs	5,000	4,659	3,722	4.30	.60	3,612	R S# A T	500 400 800 300	1.5 1.0 2.0 1.0	Climb to 5,500' on W crs of Billings range within 25 mi. *Billings range is adjacent to N side of E crs of ILS. #Runway No. 25.
	Nibbe FM	E crs ILS	215	17.0	5,000												
	Park City FM	W crs ILS	62	5.0	5,500												
BIRMINGHAM, ALA. Birmingham Arpt. Freq. 110.3 mc Ident. BHM	Birmingham Range	Outer Mrkr	215	8.7	2,500	SW 52 232	2,000'—W side SW crs	1,820	1,820	845	5.40	.72	643	R S* A T	900 500 1,000 300	1.5 3/4 2.0 1.0	Climb to 2,500' on N crs of Birmingham range, or alternate procedure (when authorized by ATC), climb to 2,700' on NE crs of ILS. Upon reaching 2,700' make 180° right turn and proceed to S crs of Birmingham, hold on S crs of Birmingham. *Runway No. 6.
	Int. SW crs Birmingham & SW crs ILS.	Outer Mrkr	52	26.0	2,000												
	Int. S crs Birmingham & SW crs ILS.	Outer Mrkr	232	6.7	2,500												
	Int. E crs Birmingham & NE crs ILS.	Outer Mrkr	232	11.3	2,500												
BISMARCK, N. D. Bismarck Arpt. Freq. 110.3 mc Ident. BIS	Int. W crs Bismarck & SE crs ILS.	Outer Mrkr	124	5.0	3,400	SE 304 124	3,000'—E side SE crs	2,700	2,660	1,855	4.30	.70	1,653	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 3,300' on W crs of Bismarck range to Judson FM. #Runway No. 30.
	Bismarck Range	Outer Mrkr	153	3.8	3,400												
	Bismarck Range	SE crs ILS	223	1.8	3,400												
	Driscoll FM	Outer Mrkr	243	30.0	3,400												
BOISE, IDAHO Boise Air Terminal Freq. 109.9 mc Ident. BOI	Boise Range	Outer Mrkr	256	2.1	4,000	NW 96 276	4,000'—S side NW crs	3,900	3,900	3,055	4.36	.69	2,858	R (*) S# A T	600 700 400 800 300	1.5 1.5 3/4 2.0 1.0	Turn right and climb to 4,000' on NW crs of Boise within 10 mi. *Night minimums. #Runway No. 10L.
	Int. SE crs Boise & NW crs ILS.	Outer Mrkr	276	4.6	4,000												
	Eagle FM	NW crs ILS (Inbound)	114	8.2	4,000												
BOSTON, MASS. Logan Arpt. Freq. 110.3 mc Ident. BOS	Boston Range	LOM	220	8.4	1,800	SW 35 215	1,800'—E side SW crs	1,800	1,770	460	5.67	1.40	19	R S# A T	600 400 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on NE crs ILS, or alternate procedure (when directed by ATC), make right turn, climb to 1,500' on E crs Boston. #Runway No. 4 L.
	Int. SW crs Boston & N crs Providence	SW crs ILS	85	14.0	1,800												
	Int. W crs Boston & N crs Providence	LOM	150	16.0	1,800												
BROWNSVILLE, TEX. International Arpt. Freq. 110.3 mc Ident. BRO	Brownsville Range	N crs ILS	35	1.5	1,200	N 173 353	1,200'—W side N crs	1,150	1,150	240	4.34	.66	22	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,200' on S crs of Brownsville, or alternate procedure (when directed by ATC), turn left and climb to 1,200' on E crs of Brownsville. *Runway No. 17L.
	Los Fresnos FM	N crs ILS	126	5.5	1,200												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)—	Visibility (mi.)	
BUFFALO, N. Y. Buffalo Arpt. Freq. 109.9 mc Ident. BUF	SW crs Buffalo	SW crs ILS	52	0.0	2,000	NE 232 52	1,800'—N side NE crs	1,800	1,750	930	4.20	0.66	711	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on SW crs. *Runway No. 23.
	Wolcottsville FM	Outer Mrkr	224	8.4	1,800												
CHARLESTON, S. C. Charleston Arpt. Freq. 110.3 mc Indent. CHS	Charleston Range	Outer Mrkr	5	3.4	1,200	NW 148 328	1,200'—W side NW crs	1,200	1,100	345	4.60	.64	45	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,500' on E crs of Charleston, or alternate procedure (when directed by ATC), turn right and climb to 1,300' on SW crs of Charleston range. *Runway No. 15.
	Int. E crs Charleston & ILS localiser	Outer Mrkr	328	5.6	1,200												
	Int. N crs Charleston & NW crs ILS	Outer Mrkr	328	.8	1,200												
CHARLOTTE, N. C. Douglas Field Freq. 109.5 mc Indent. CLT	Charlotte Range	Outer Mrkr	251	5.3	2,200	SW 50 230	2,150'—S side SW crs	2,150	2,150	922	5.90	.62	748	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn left and climb to 2,500' on N crs of Charlotte within 25 mi, or alternate procedure (when directed by ATC), turn right, climb to 2,200' on E crs of Charlotte within 25 mi. NOTE: 1,116' MSL radio tower located 4 mi. SE of range and 2.3 mi. E of the S course. *Runway No. 4.
	Int. W crs Charlotte & SW crs ILS	Outer Mrkr	230	2.5	2,200												
	Int. N crs Charlotte & SW crs ILS	Outer Mrkr	230	6.75	2,200												
	Fort Mill FM	SW crs ILS	293	8.75	2,200												
CHEYENNE, WYO. Cheyenne Arpt. Freq. 110.3 mc Indent. CYS	Cheyenne Range	Outer Mrk	81	8.4	7,300	E 262 82	7,500'—N side E crs	7,500	7,500	6,300	6.11	.66	6,156	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 7,500' on S crs of Cheyenne, or alternate procedure (when directed by ATC), climb to 7,500' on crs Cheyenne. *Runway No .26.
	Hillsdale FM	E crs ILS	230	5.0	7,300												
	Silver Crown FM	Outer Mrkr	92	30.0	8,000												
CHICAGO, ILL. Chicago-Midway Arpt. Freq. 109.9 mc Ident. CHI	Chicago Range	NW crs ILS	290	2.2	2,300	NW 132 312	2,300'—W side NW crs	2,300	2,260	875	5.80	.79	618	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on N & S course of Harvey proceeding S to Momence Int.; or alternate procedure (when directed by ATC), make left climbing turn to 2,500' on SE crs of Chicago proceeding to Thornton Int. *Runway No. 13 R.
	Int. SE crs Chicago & NW crs ILS	Outer Mrkr	312	5.8	2,300												
	Int. NE crs Joliet & NW crs Chicago	NW crs ILS	180	6.7	2,300												
	Int. NE crs Joliet & NW crs ILS	Outer Mrkr	132	6.6	2,300												
	Lakeshore Int.	Outer Mrkr	260	13.0	2,300												
O'Hare-International Arpt. Freq. 109.9 mc Ident. ORD	Int. S crs Milwaukee & E crs Rockford	NW crs ILS	230	10.9	2,500	NW 138 318	2,500'—W side NW crs	1,900	1,820	875	4.40	.70	657	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn left after passing NE crs Joliet and climb to 2,500' on track of 50° to SE crs Glenview, proceed to Mundelein Int. (NW Glenview & E Rockford) via Glenview range sta. #Runway No. 14.
	Int. E crs Rockford & NW crs ILS	Outer Mrkr	138	19.3	2,500												
	Int. NE crs Joliet & SE crs ILS	Outer Mrkr	318	6.3	2,500												
	Glenview Range	Outer Mrkr	219	7.8	2,500												
	Int. NW crs Chicago & SE crs ILS	Outer Mrkr	318	7.6	2,500												
CINCINNATI, OHIO Greater Cincinnati Arpt. Freq. 109.9 mc Ident. CVG	Int. NW crs Cincinnati & N crs ILS	Outer Mrkr	180	23.1	2,300	S 360 180	2,000'—E side S crs	2,000	2,000	1,050	4.30	.60	890	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on N crs of ILS to NW crs of Cincinnati. *Runway No. 36.
	Union FM	Outer Mrkr	360	4.0	2,000												
	Cincinnati Range	Outer Mrkr	235	21.4	2,400												
	Int. SW crs Cincinnati to 320° bearing on OM	Outer Mrkr	320	2.9	2,000												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
CLEVELAND, OHIO Cleveland Arpt. Freq. 109.9 mc Ident. CLE	Cleveland Range	Outer Mrkr	224	4.3	2,100	SW 54 234	2,100'—*N side SW crs	2,100	2,020	1,020	4.50	0.71	789	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,600' on E crs of Cleveland Range. *1168 MSL tower, 4.0 mi E of Grafton, Ohio. #Runway No. 5L.
	Elyria FM	SW crs ILS	135	8.4	2,100												
	Avon Lake FM	Outer Mrkr	171	9.7	2,100												
	No. Royalton FM	Outer Mrkr	303	10.9	2,600												
	Int. NW crs Akron & SW crs ILS	Outer Mrkr	54	8.0	2,200												
COLUMBUS, OHIO Port Columbus Arpt. Freq. 109.5 mc Ident. CMH	Columbus Range	Outer Mrkr	286	2.8	2,300	W 95 275	2,300'—S side W crs	2,000	2,000	1,020	4.50	.66	817	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on E crs of Columbus within 25 mi. *Runway No. 9.
	Hilliard FM	Outer Mrkr	87	7.3	2,300												
	Int. E crs Columbus & W crs ILS	Outer Mrkr	275	5.8	2,300												
	Int. NE crs Columbus & W crs ILS	Outer Mrkr	275	2.9	2,300												
DALLAS, TEX. Love Field Freq. 110.3 mc Ident. DAL	Dallas Range	NW crs ILS	341	4.0	1,700	NW* 126 306	1,700'—N side NW crs	1,700	1,500	700	4.35	.70	483	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on SE crs of ILS & proceed out E crs of Dallas, or alternate procedure (when directed by ATC), turn left, climb to 2,000' on track of 45°, intersect and proceed E on E crs of Ft. Worth. *Runway No. 13.
	Int. E crs Dallas & SE crs ILS	Outer Mrkr	306	9.0	1,700												
	Int. E crs Ft. Worth & N crs Dallas	Outer Mrkr	196	3.0	1,700												
	Int. E crs Ft. Worth & NW crs of ILS	Outer Mrkr	126	3.0	1,700												
DAYTON, OHIO Dayton Arpt. Freq. 110.3 mc Ident. DAY	Dayton Range	SW crs ILS	200	3.8	2,300	SW 55 235	2,300'—S side SW crs	2,300	2,270	1,235	4.40	.66	1,007	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on N crs of Dayton to W crs Columbus (Troy Int.). *Runway No. 6.
	Verona FM	SW crs ILS	125	8.3	2,300												
	Int. S crs Dayton & SW crs ILS	Outer Mrkr	55	1.3	2,300												
	Int. W crs Columbus to NE crs ILS	Outer Mrkr	235	18.0	2,300												
DENVER, COLO. Stapleton Field Freq. 110.3 mc Ident. DEN	Denver (LF) Range	NE crs ILS	59	4.3	6,300	NE 206 26	6,300'—S side NE crs	6,300	6,300	5,480	4.19	.65	5,325	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 8,900' on S crs Denver, LF range, or alternate procedure (when directed by ATC)-climb to 8,500', intercepting an ADF track of 120° outbound from Denver LF range within 25 mi. *CAUTION: 6,715' MSL hill, 6 mi NW Franktown. #Runway No. 2L.
	Henderson FM	NE crs ILS	93	8.1	6,300												
	Franktown FM (Northbound only)	NE crs ILS	341	34.0	*7,000												
	Int. E crs Denver VAR & NE crs Denver LF	NE crs ILS	261	5.9	6,300												
	Int. W crs Denver VAR & N crs Denver LF	Outer Mrkr	128	5.3	6,300												
DES MOINES, IOWA Des Moines Arpt. Freq. 110.3 mc Ident. DSM	Des Moines Range	Outer Mrkr	99	3.3	2,300	SE 305 125	2,300'—E side SE crs	2,300	2,300	1,170	4.77	.61	957	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on N crs of Des Moines within 25 mi., or alternate procedure (when directed by ATC), climb to 2,500' on NW crs ILS within 25 mi. *Runway No. 31.
	Indianola FM	SE crs ILS	31	9.3	2,300												
	Int SE crs ILS & N crs Des Moines	Outer Mrkr	125	4.2	2,300												
	Int. E crs Des Moines & SE crs ILS	Outer Mrkr	125	1.5	2,300												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS apprs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
DETROIT, MICH. Willow Run Arpt. Freq. 109.5 mc Ident. YIP	Ann Arbor FM	Outer Mrkr	92	7.5	2,000	SW 50 230	2,000'—S side SW crs	2,000	2,250	930	5.80	0.65	716	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on NE crs of ILS to N crs of Detroit. *Runway No. 4R.
	Int. N crs Detroit & NE crs ILS	Outer Mrkr	230	13.6	2,000												
	Detroit Range	Outer Mrkr	255	12.4	2,000												
	Int. N crs Toledo & SW crs ILS	Outer Mrkr	80	5.0	2,000												
EL PASO, TEX. El Paso International Arpt. Freq. 109.5 mc Ident. ELP	El Paso Range	LOM	326	4.0	5,000	NE 217 37	5,000'—N side NE crs	5,000	5,090	4,140	4.56	.67	3,936	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Make left turn to 125° and climb to 6,000' on S crs of El Paso. *Maintain 8,000' until 5 mi W of Hueco Mtn. FM (high terrain N). #Runway No. 22.
	Int. W. crs El Paso & NE crs ILS	LOM	37	5.9	5,000												
	Hueco Mountain FM*	LOM	268	20.0	7,000												
	Newman Rbn	NE crs ILS	177	3.0	5,000												
ERIE, PA. Port Erie Arpt. Freq. 110.3 mc Ident. EBI	Erie Range	Outer Mrkr	236	2.4	2,000	SW 60 240	2,000'—S side SW crs	2,000	1,970	935	4.35	.62	732	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on NE crs ILS. *Runway No. 6.
	North Springfield FM	Outer Mrkr	74	9.1	2,000												
EUGENE, OREG. Mahlon Sweet Arpt. Freq. 109.9 mc Ident. EUG	Eugene Range	N crs ILS (Outbound)	339	*	2,000	N 159 339	2,000'—**E side N crs	1,300	1,310	555	4.40	.63	365	R S# A T	600 400 800 300	1.5 1.0 2.0 1.0	Turn right and climb to 3,000' on N crs of Eugene within 15 mi. *Eugene range is adjacent to W side of N crs of ILS. **Procedure turn to W not authorized due to high terrain. #Runway No. 16.
	Cottage Grove FM	S crs ILS (Inbound)	320	12.0	3,100												
FAIRFIELD-SUISUN, CALIF. Fairfield-Suisun AFB Freq. 110.3 mc Ident. SUU	Fairfield-Suisun Range	NE crs ILS	*	*	3,000	NE 210 30	3,000'—N side —NE** crs	3,000	1,245	305 (MM) 108 (IM)	4.40 .05	.66 (IM)	58	R # S A T #	500 500 400 800 300 300	1.5 2.0 1.0 2.0 2.0 3.0	Climb to 4,000' on SW course on Fairfield-Suisun range. *ILS localizer course passes over range; descend to 3,000' on NE course ILS within 15 mi. **Procedure turn to be made not less than 6 mi NE of OM. #Night minimums. NOTE: Straight-in-approach to Runway No. 21L.
FARGO, N. DAK. Fargo Arpt. Freq. 110.3 mc Ident. FAR	Fargo Range	Outer Mrkr	188	5.2	2,300	S 351 171	2,300'—W side S crs	2,100	2,090	1,125	4.55	.71	900	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on N crs of Fargo range. *Runway No. 35.
	West Fargo FM	Outer Mrkr	132	6.6	2,300												
	Barnesville FM	S crs ILS	285	20.0	2,300												
	Int. N crs Fargo & N crs ILS	Outer Mrkr	171	12.0	2,300												
FORT WAYNE, IND. Baer Field Freq. 109.9 mc Ident. FWA	Int. NW crs Fort Wayne & NW crs ILS	Middle Mrkr	135	23.0	2,100	SE 315 135	2,100'—E side SE crs	2,100	2,080	1,085	4.44	.82	861	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,100' on NW crs ILS, make left turn within 20 mi. and proceed to middle marker. *Runway No. 31. NOTE: LMM located 0.82 mi. from end of runway, freq. 201 kc, ident-WA.
	Int. NE crs Fort Wayne & ILS crs	Middle Mrkr	135	1.4	2,100												
	Fort Wayne Range	Middle Mrkr	70	4.5	2,100												
FORT WORTH, TEX. Meacham Field Freq. 109.9 mc Ident. FTW	Ft. Worth Range	Outer Mrkr	353	1.7	2,000	N 173 353	2,000'—W side N crs	2,000	2,000	950	4.10	.71	692	R S* A T	600 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on S crs of Ft. Worth. CAUTION: Do not descend below glide path; bldg 283' below glide path, 2.4 mi. S of outer marker. *Runway No. 17.
	Haslet Rbn	N crs ILS	215	2.0	2,000												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
FRESNO, CALIF. Fresno Air Terminal Freq. 110.3 mc Ident. FNO	Fresno LF Range	SE crs ILS (LOM)	70	11.0	1,700	SE 288 108	1,700'—*S side SE crs (within 10 mi. of outer marker.	1,700	1,400	535	4.44	0.70	331	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb on a heading of 288° to 1,500', then turn left and climb to 2,000' on W crs of Fresno LF rng within 20 mi. *Procedure turn S side of course due to more suitable terrain. #Runway No. 29.
	Bowles FM	SE crs ILS (LOM)	13	11.5	1,700												
GRAND JUNCTION, COLO. Walker Field Freq. 110.3 mc Ident. GJT	Grand Jct. VAR Range.	Int. N crs Grand Jct. & NW crs ILS.	329	17.5	8,100	NW 110 290	8,000'—W side NW crs (At int. N crs Grand Jct & NW crs of ILS)	6,200	6,060	5,050	4.32	.69	4,858	R S# A T*	1,000 500 1,500 500	2.0 2.0 2.0 2.0	Make 180° turn to right and climb to 8,000' on NW crs of the ILS to the int. of the N crs of Grand Jct. *300-1 authorized for 4-engine acft only. #Runway No. 12.
	Int. E crs Grand Jct. & SE crs ILS	Int. N crs Grand Jct. & NW crs ILS	290	21.5	8,000												
HOUSTON, TEX. Houston Arpt. Freq. 109.9 mc Ident. HOU	Houston Range	SW crs ILS	309	2.5	1,200	SW 35 215	1,200'—S side SW crs	1,200	1,260	250	4.78	.66	50	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,600' on NE crs of ILS, or alternate procedure (when directed by ATC), turn left, climb to 1,600' on NW crs of Houston within 25 mi. *Runway No. 3. NOTE: Take-offs on Runway 26 and landings on Runways 8 and 26 not authorized.
	Houston FM	SW crs ILS	129	6.3	1,200												
	Arcola FM	SW crs ILS	15	4.7	1,200												
	Houston FM	Outer Mrkr	168	8.5	1,200												
	Webster FM	Outer Mrkr	284	12.6	1,200												
INDIANAPOLIS, IND. Weir Cook Arpt. Freq. 109.9 mc Idnt. IND	Clayton FM	SW crs ILS	105	7.8	1,900	SW 44 224	1,900'—S side SW crs (to be made not less than 4 mi. SW outer marker)	1,900	1,720	1,015	4.30	.89	796	R S* A T	500 400 800 300	1.5 3/4 2.0 3/4	Climb to 2,400' on E crs of Indianapolis to NW crs of Cincinnati. *Runway No. 4.
	Indianapolis Range	SW crs ILS	193	2.6	1,900												
	Int. E crs Indianapolis & SW crs ILS	Outer Mrkr	224	5.9	1,900												
	Int. S crs Indianapolis & SW crs ILS	Outer Mrkr	224	1.1	1,900												
ISLIP, N. Y MacArthur Field Freq. 108.7 mc Ident. ISL	Int. NW crs Islip VAR & NE crs ILS	Middle Mrkr	237	8.6	1,500	SW 57 237	1,500'—S side SW crs	1,500	1,340	300	6.10	.71	98	R S* A T	600 400 800 300	1.5 1.0 2.0 1.0	Climb to 1,000' on NE crs ILS, make a standard rate climbing turn to the right, returning to ILS middle marker at 1,500'. *Runway No. 6.
	Int. E crs Mitchel & SW crs ILS	Outer Mrkr	57	8.3	1,500												
	Islip VAR	Middle Mrkr	254	9.3	1,500												
JACKSON, MISS. Hawkins Field Freq. 109.9 mc Ident. JAN	Jackson Range	Outer Mrkr	256	4.8	1,500	NW 108 288	1,500'—S side NW crs	1,500	1,490	375	4.43	.70	343	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,600' on S crs of Jackson, or alternate procedure (when directed by ATC), climb to 1,800' on E crs of Jackson range. *Runway No. 11.
	Int. S crs Jackson & ILS localizer	Outer Mrkr	288	5.3	1,500												
	Int. W crs Jackson & NW crs ILS	Outer Mrkr	108	2.5	1,500												
JACKSONVILLE, FLA. Imeson Arpt Freq. 116.3 mc Ident. JAX	Int. N crs Jacksonville & NE crs ILS	Outer Mrkr	224	9.9	1,200	SW 44 224	1,200'—N side SW crs	1,200	1,150	270	4.60	.66	52	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,200' on N crs of Jacksonville, or alternate procedure (when directed by (ATC), turn right and climb to 1,200' on E crs of Jacksonville. *Runway No. 5.
	Jacksonville Range	Outer Mrkr	237	7.3	1,200												
	Bryceville FM	Outer Mrkr	104	14.0	1,200												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
KANSAS CITY, MO. Kansas City Arpt. Freq. 109.9 mc Ident. MKC	Int. E crs Topeka VAR & NE crs Kansas City	N crs ILS	252	13.0	2,500	N 173 353	2,500'—E side N crs	2,500	2,500	1,505 (790 boundary mrkr)	6.35	2.60	744	R S# A A T*	700 400 800 300	1.5 1.0 2.0 1.0	Immediately execute right climbing turn to 2,500', intercept ADF track of 270° msg. outbound from Kansas City range, proceed to and hold at Fairmount Int., or alternate procedure (when authorized by ATC), immediately climb to 2,200' on SW crs of Kansas City. CAUTION: 902' MSL cracking plant and 869' MSL stack 0.6 mi NW of approach end of runway 17, 1,670' MSL tower 2.9 mi S of arpt and directly in line with N/S runway. *Take-offs to S and SW, when weather is below 1000-3, will assume heading of at least 210° mag as soon as practicable after take-off and maintain heading until reaching 2,500' prior to making left turn. #Runway No. 17.
	Int. E crs Topeka VAR & NW crs Kansas City	N crs ILS	72	4.0	2,500												
	Kansas City Range	Outer Mrkr	8	4.0	2,500												
	Linkville FM	N crs ILS	95	6.0	2,500												
	Excelsior Springs FM	N crs ILS	256	23.0	2,500												
Fairfax Field (Kansas City, Kans.)	Int. E crs Topeka VAR & NE crs Kansas City	N crs ILS	232	13.0	2,500	N 173 353	2,500'—E side N crs	2,500	2,500	1,505	187°, 0.9 (from MM to nearest edge of field)		746	R A T*	700 800 300	1.5 2.0 1.0	Immediately execute right climbing turn to 2,500' intercept ADF track of 270° mag outbound from Kansas City rng, proceed to and hold at Fairmount Int., or alternate procedure (when authorized by ATC), immediately climb to 2,300' on SW crs of Kansas City. CAUTION: 902' MSL cracking plant and 869' MSL stack-0.5 ESE of approach end of Rnwy 35. 1,670' MSL tower-4.7 mi SSE of arpt. 1,430' MSL obstructions located 2.5 mi SE of arpt, adhere strictly to pull-up procedure. *Take-offs to S and SW, when weather is below 1,000-3, will assume a heading of at least 210° mag. as soon as practicable after take-off and maintain heading until reaching 2,500' prior to making left turn.
	Int. E crs Topeka VAR & NW crs Kansas City	N crs ILS	72	4.0	2,500												
	Kansas City Range.	Outer Mrkr	8	4.0	2,500												
	Linkville FM	N crs ILS	95	6.0	2,500												
	Excelsior Springs FM	N crs ILS	256	23.0	2,500												
KNOXVILLE, TENN. McGhee-Tyson Arpt. Freq. 110.3 mc Ident. TYS	Int. NE crs Knoxville & NE crs ILS	Outer Mrkr	224	12.5	2,500	SW 44 224	2,500'—W side SW crs	2,500	2,040	1,190	4.34	.61	989	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn left and climb to 3,000' on N course Knoxville within 25 mi, or alternate procedure (when directed by ATC), climb to 4,000' on NE course of Knoxville within 25 mi. *Runway No. 4.
	Knoxville Range	Outer Mrkr	214	8.5	2,500												
LONG BEACH, CALIF. Long Beach Arpt. Freq. 110.3 mc Ident. LGB	Long Beach Range	Outer Mrkr	139	1.5	1,500	SE 302 122	1,500'—W side SE crs	1,320	1,320	245	5.42	.77	56	R S* A T	600 400 800 300	2.0 1.0 2.0 1.0	Climb to 800' on NW crs of Long Beach. After reaching 800', make climbing left turn to 250°, intersect S crs Los Angeles LF range and proceed to San Pedro Int. at minimum alt of 2,000'. *Runway No. 30.
	Huntington Beach FM	Outer Mrkr	295	9.2	1,300												
	San Pedro Int.	Outer Mrkr	39	7.3	1,500												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimums on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
LOS ANGELES, CALIF. Los Angeles International Arpt. Freq. 109.9 mc Ident. LAX	Los Angeles Range	Outer Mrkr	67	2.2	2,000	E 247 67	2,000'—S side E crs*	1,650	1,650	300	6.04	0.64	109	R S# A T	600 400 800 300	1.5 1.0 2.0 1.0	Climb to 2,000' on W crs of Los Angeles. *In no case farther E than Downey FM. #Runway No. 25L.
	Downey FM	Outer Mrkr	252	8.8	1,650												
LOUISVILLE, KY. Standiford Field Freq. 110.3 mc Ident. SDF	Int W crs Louisville & N crs ILS	Outer Mrkr	189	7.8	2,000	S 9 189	2,000'—W side S crs	2,000	1,560	690	4.22	.68	497	R * (R) * S# A T	600 600 500 500 400 800 300	1.5 2.0 1.0 2.0 3/4 2.0 1.0	Climb to 2,000' on ouisviers ILS to W crs of Ld out W then turn left, procee crs of Louisville. *Night minimums. #Runway No. 1. CAUTION: Do not cross E crs of Godman while on S crs of ILS. Danger area 10 mi S of outer marker.
	Louisville Range	Outer Mrkr	218	10.2	2,000												
	Eastwood FM	Outer Mrkr	240	18.0	2,000												
	Int. S crs Louisville & E crs Godman	Outer Mrkr	322	10.9	2,000												
	Int. W crs Louisville & N crs Godman	Outer Mrkr	113	16.0	2,000												
MEMPHIS, TENN. Memphis Arpt. Freq. 109.9 mc Ident. MEM	Memphis Range	W crs ILS	356	2.4	1,700	W 86 266	1,700'—S side W crs	1,700	1,420	470	4.88	.69	269	R S* A T	500 400 800 300	1.5 4/3 2.0 1.0	Climb to 2,000' on NE crs of Memphis range, or alternate procedure (when directed by ATC), turn right and climb to 1,300' on S crs of Memphis range. *Runway No. 9.
	Cuba FM	Outer Mrkr	188	18.3	2,000												
	Int. NE crs Memphis & E crs ILS	Outer Mrkr	266	9.8	2,000												
	Int. SW crs Memphis & 45° bearing on LOM	Outer Mrkr	48	21.0	1,700												
MERIDIAN, MISS. Key Field Freq. 110.3 cm Ident. MEI	Meridian Range	Outer Mrkr	169	8.5	1,700	S 355 175	1,700'—E side S crs	1,380	1,380	495	4.89	.71	297	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on NE crs of Meridian rng within 25 mi. *Runway No. 36.
	Int. S crs Meridian & ILS localiser	Outer Mrkr	175	6.0	1,700												
MIAMI, FLA. Miami International Arpt. Freq. 109.5 mc Ident. MIA	*Miami Range	Int. N crs Miami & NW crs ILS	351	4.7	1,100	NW 117 297	1,100'—S side NW crs	1,100	1,020	200	4.31	.63	9	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,400' on SE crs ILS, or alternate procedure (when directed by ATC), turn left, climb to 1,400' on E crs of Miami. *Right turn to localiser not authorized. *Runway No. 12.
	Int. N crs Miami & NW crs ILS	Outer Mrkr	117	3.2	1,100												
	Int. E crs Miami & NW crs ILS	Outer Mrkr	297	2.5	2,000												
	Krome FM	NW crs ILS	59	8.9	1,100												
MIDLAND, TEX. Midland Air Term. Freq. 110.3 mc Ident. MAF	Midland Range	SW crs ILS	188	1.0	4,200	SW 42 222	4,200'—S side SW crs	4,010	4,010	3,070	4.69	.70	2,867	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 4,000' on SE crs of Midland within 10 mi. *Runway No. 4.
MILWAUKEE, WIS. Gen. Mitchell Arpt. Freq. 110.3 mc Ident. MKE	Milwaukee Range	Outer Mrkr	194	2.5	2,000	S 5 185	2,000'—E side S crs	2,000	2,010	925	4.60	.67	698	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on N crs of Milwaukee within 25 mi. *Runway No. 1.
	Int. N crs Milwaukee & S crs ILS	Outer Mrkr	185	4.3	2,000												
	Franksville FM	S crs ILS	340	3.8	2,000												
	Int. W crs Milwaukee & S crs ILS	Outer Mrkr	185	2.5	2,000												
MINNEAPOLIS, MINN. Minneapolis-St. Paul International Arpt Freq. 109.9 mc Ident. MSP	Stanton Rbn	S crs ILS	324	15.0	2,300	S 354 174	2,300'—W side S crs	2,000	1,960	1,040	4.70	.70	838	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on NW crs of Minneapolis to Hamel FM. *Runway No. 35
	Minneapolis Range	Outer Mrkr	219	4.9	2,300												
	Jordan FM	S crs ILS	82	20.0	2,300												
	Hamel FM	Outer Mrkr	130	22.0	2,300												
	Hastings FM	S crs ILS	270	12.9	2,300												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
MOLINE, ILL. Quad-City Arpt Freq. 110.3 mc Ident. MLI	Moline Range	W crs ILS	174	8.5	1,900	W 86 266	1,900'—S side W crs	1,900	1,820	785	5.10	0.53	590	R (*) S# A T	600 600 600 800 300	1.5 2.0 3/4 2.0 1.0	Climb to 2,100' on E crs of ILS within 25 mi. of arpt. *Night minimums. #Runway No. 9.
	Int. W crs Moline & N crs Burlington	W crs ILS	130	12.0	1,900												
	Int. S crs Moline & E crs Burlington	W crs ILS	354	29.0	2,000												
NASHVILLE, TENN. Berry Field Freq. 109.9 mc Ident. BNA	Nashville Range	Outer Mrkr	213	8.3	2,500	S 14 194	2,500'—E side S crs	2,500	1,905	800	5.70	.81	606	R S* A T	500 400 800 300	1.5 1.0 2.0 1.0	Climb to 2,000' on NW crs of Nashville, or alternate procedure (when directed by ATC), turn right and climb to 2,500' on NE crs of Nashville range within 25 mi. *Runway No. 21.
	Walterhill FM	Outer Mrkr	280	19.5	2,500												
	Fairview FM	Outer Mrkr	80	20.0	2,500												
	Int. NW crs Nashville & N crs ILS		194	10.7	2,500												
NEWARK, N. J. Newark Arpt. Freq. 110.3 mc Ident. EWR	Newark Range	LOM	238	3.5	1,500	SW 58 238	1,400'—S side SW crs	1,400	1,280	230	4.64	.61	18	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,000' on NE scr of Newark, then turn left, climbing to 2,700' on heading of 290° mag before proceeding out W crs of Newark. Alternate procedure (when directed by ATC): Climb to 1,000' on NE crs Newark, turn left and proceed to Paterson, climbing to 2,000 *Runway No. 6.
	Int. E crs Allentown & SW crs Newark	LOM	58	19.8	1,500												
NEW BEDFORD, MASS. New Bedford Arpt. Freq. 109.5 mc Ident. EWB	Int. E crs Providence & SW crs ILS	Outer Mrkr	233	1.5	1,400	SW 53 233	1,400'—S side SW crs (not authorized beyond 10 mi.)	1,400	1,240	265	4.40	.66	79	R S* A T	600 400 800 300	1.5 3/4 2.0 1.0	Make standard rate climbing turn to the right, climb to 1,400' on SW crs ILS. Hold on S side SW crs ILS between LMM and 10 mi SW of sta. *Runway No. 5.
NEW ORLEANS, La. Moisant International Arpt. Freq. 109.9 mc Ident. MSY	New Orleans Range	W crs ILS	271	4.5	1,400	W 99 279	1,400'—S side W crs	1,040	1,040	220	3.92	.67	3	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,500' on E crs of New Orleans, or alternate procedure (when directed by ATC), climb to 1,300' on S crs of New Orleans. *Runway No. 10.
	Int. S crs New Orleans & E crs ILS	Outer Mrkr	279	8.5	1,400												
	LaPlace FM	Outer Mrkr (final)	87	8.4	1,040												
NEW YORK, N. Y. LaGuardia Field Freq. 109.9 mc Ident. LGA	LaGuardia Range	LOM	224	8.6	1,500	SW 44 224	1,500'—S side SW crs	1,500	1,350	340	4.49	.81	20	R S* A T	600 400 800 300	1.5 1.0 2.0 1.0	Climb to 1,500' on NE crs of ILS. *Runway No. 4.
	Int. SE crs Newark & SW crs LaGuardia	LOM	44	12.3	1,500												
New York International Arpt. Freq. 109.5 mc Ident. IDL	Int. SW crs ILS & SE crs Newark	Outer Mrkr	42	11.2	1,000	SW 42 222	1,200'—S side SW crs	1,000	700	200	2.90	.63	12	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Turn right to a heading of 130° intersecting the SW crs of Mitchel, climb to 1,500' on SW crs of Mitchel, or alternate procedure (when directed by ATC), turn right climb to 1,200' on SW course of ILS within 25 mi. *Runway No. 4.
	Idlewild Range	Outer Mrkr	222	.0	1,200												
NIAGARA FALLS, N. Y. Niagara Falls Arpt. Freq. 109.5 mc Ident. GFS	Wolcottsville FM	Outer Mrkr	286	14.0	2,000	E 278 98	2,000'—N side E crs	2,000	1,830	810	4.60	.70	590	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on W crs of the ILS. *Runway No. 27.
	Int. E crs ILS & NE crs Buffalo	Outer Mrkr	278	15.0	2,000												
	Buffalo Range	Outer Mrkr	338	12.0	2,000												
NORFOLK, VA. Norfolk Arpt. Freq. 109.9 mc Ident. ORK	Norfolk Range	Outer Mrkr	223	.9	1,400	SW 43 223	1,400'—S side SW crs	1,400	1,050	220	4.30	.63	25	R S* A T	500 400 800 300	1.5 1.0 2.0 1.0	Turn right and climb to 1,500' on SW crs of Norfolk *Runway No. 4.
	Int. SW crs Norfolk & S crs Langley	Outer Mrkr	43	7.0	1,400												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
OAKLAND, CALIF. Oakland Arpt. Freq. 109.9 mc Ident. OAK	Newark FM	Hayward Rbn	340	7.2	2,590	E 274 94	3,500'*—S side E crs**	1,320	1,320	230	4.75	0.66	*5	R S# A T	600 400 800 300	1.5 3/4 2.0 1.0	Climb to 3,600' on NW crs of Oakland within 25 mi. *Upon completion of procedure turn and transition to localizer course inbound, descent is authorized to cross Hayward Rbn at 2,590'. **Oakland range transition only. Higher terrain to N. Newark FM transition, left turn from 340° to localizer course of 274°. #Runway No. 27R.
	Oakland range	Hayward Rbn	98	11.0	3,500												
OKLAHOMA CITY, OKLA. Will Rogers Field Freq. 109.9 mc Ident. OKC	Okla. City range	S crs ILS	78	2.5	2,500	S 350 170	2,500'—E side S crs	2,450	2,450	1,500	4.60	.66	1,283	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 3,000' on N crs of Oklahoma City, or alternate procedure (when directed by ATC), turn left, climb to 2,500' on W crs of Oklahoma City within 25 mi. *Runway No. 35.
	Washington FM	S crs ILS	335	16.0	2,500												
	Int. N crs Okla. City & N crs ILS	Outer Mrkr	170	15.0	2,500												
	Okla. City FM	S crs ILS	258	3.2	2,500												
	Mustang FM	Outer Mrkr	104	11.3	2,500												
PHILADELPHIA, PA. Philadelphia International Arpt. Freq 109.9 mc Ident. PHL	Boothwyn FM	W crs ILS	85	0	1,700	W 85 265	1,600'*—S side W crs	1,600	1,625	225	5.66	.66	10	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,800' on NE crs of Philadelphia, or alternate procedure (when directed by ATC), make right turn, climb to 1,500' on S crs of Philadelphia. *1,800' if made more than 10 mi. from LOM. #Runway No. 9.
	Philadelphia range	Outer Mrkr	253	2.8	1,700												
PHOENIX, ARIZ. Sky Harbor Arpt. Freq. 110.3 mc Ident. PHX	Phoenix (LF) Range	E crs ILS	90	.5	2,600	E 256 76	2,600'*—S side E crs	2,300	2,290	1,330	4.40	.63	1,120	R S# A T	600 400 800 300	1.5 3/4 2.0 1.0	Climb to 6,000' on W crs Phoenix (LF) range. *Procedure turn NA beyond 15 mi. (Mess FM). #Runway No. 26R.
PITTSBURGH, PA. Allegheny Co. Arpt. Freq. 109.5 mc Ident. PIT	Pittsburgh Range	Outer Mrkr	91	7.5	2,600	E 275 95	2,600'—N side E crs	2,600	2,510	1,460	4.70	.65	1,252	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on W crs of Pittsburgh, proceed to Cecil holding pattern, or alternate procedure (when directed by ATC), climb to 2,600' on NW crs of Pittsburgh, proceed to Clinton holding pattern. *Runway No. 27.
	McKeesport Rbn	Outer Mrkr	275	1.6	2,600												
RALEIGH, N. C. Raleigh-Durham Arpt. Freq. 110.3 mc Ident. RDU	Int. NE crs Raleigh & NE crs ILS	Outer Mrkr	228	30.0	1,500	SW 48 228	1,500'—S side SW crs	1,500	1,410	635	4.43	.62	435	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,600' on NE crs of Raleigh, or alternate procedure (when directed by ATC), turn left and climb to 2,000' on NW crs of Raleigh within 25 mi. *Runway No. 5.
	Int. NW crs Raleigh & SW crs ILS	Outer Mrkr	228	4.8	1,500												
	Raleigh Range	Outer Mrkr	262	7.7	1,500												
READING, PA. Reading Arpt. Freq. 109.5 mc Ident. RDG	Int. SW crs Allentown & S crs ILS	Outer Mrkr	181	3.9	2,500	S 1 181	2,100'—E side S crs	2,100	1,560	535	4.50	.59	343	R S# A (**) T*	800 500 1,000 (BCOB) 1,500 300	2.0 1.0 2.0 2.0 1.0	Climb to 2,500' on N crs of ILS within 10 mi. *Take-offs on Rnwys 18 and 13, make right turn as soon as practicable to avoid 1,270' mountain and tower located 3.6 mi SE of arpt. **Night minimums. #Runway No. 36.

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
RICHMOND, VA. Byrd Field Freq. 109.9 mc Ident. RIC	Richmond Rng	Outer Mrkr	264	3.0	1,500	SW 63 243	1,500'—S side SW crs	1,500	1,370	370	4.36	0.68	167	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 1,500' on NE crs ILS, or alternate procedure (when directed by ATC), make left climbing turn, climb to 1,500' on N crs of Richmond. *Runway No. 6.
	Int. N crs Richmond & SW crs ILS	Outer Mrkr	243	3.5	1,500												
	Chester FM	SW crs ILS	10	6.0	1,500												
	Manakin Rbn	Outer Mrkr	129	19.5	1,500												
ROCK SPRINGS, WYO. Rock Springs Arpt. Freq. 110.3 mc Ident. RKS	Rock Springs Range	Outer Mrkr	93	1.5	9,200	E 254 74	9,200'—N side E crs	8,700	7,850	6,950	4.50	.70	6,752	R (*) (R) (*) S# A T	500 500 500 500 400 800 300	1.5 2.0 1.0 2.0 3/4 2.0 1.0	Climb to 10,000' on W crs of Rock Springs. *Night minimums. #Runway No. 25.
	Int. E crs ILS & W crs Rock Springs	Outer Mrkr	74	6.0	9,200												
	Point of Rocks FM	E crs ILS	235	9.0	9,200												
ST. JOSEPH, MO. Rosecrans Field Freq. 110.3 mc Ident. STJ	St. Joseph Range	Outer Mrkr.	125	.8	2,100	S 352 172	2,100'—W side S crs*	2,100	1,800	1,050	4.18	.86	823	R S# A T	700 400 800 300	1.5 1.0 2.0 1.0	Climb to 2,400' on N crs St. Joseph within 25 mi, or alternate procedure (when directed by ATC), climb to 2,500' on NW crs of Kansas City. *Procedure turn W to permit simultaneous approaches to St. Joseph and Kansas City. #Runway No. 35.
ST. LOUIS, MO. Lambert Field Freq. 110.3 mc Ident. STL	St. Louis Range	NE crs ILS	35	2.0	1,700	NE 238 58	1,600'—N side NE crs	1,600	1,600	760	3.98	.64	552	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,000' on W crs of St. Louis range, or alternate procedure (when directed by ATC), climb to 2,000' on SW crs of ILS, within 25 mi. *Runway No. 24.
	Spanish Lake FM	NE crs ILS	290	2.5	1,600												
	Woodriver Int.	NE crs ILS	285	3.8	1,700												
	Alton Int.	Outer Mrkr	238	9.0	1,700												
	Int. SW crs ILS & NE crs Vichy	Outer Mrkr	58	51.0	3,000												
SALT LAKE CITY, UTAH Salt Lake City Arpt. #1. Freq. 110.3 mc Ident. SLC (Procedure No. 1)	Salt Lake City Range	N crs ILS	162	.0	5,600	S 343 163	5,600'—W side S crs	5,600	5,560	4,440	4.69	.71	4,222	R S* A T	800 400 800 300	2.0 1.0 2.0 1.0	Climb to cross Salt Lake City range at 6,500', then climb to 11,000' on W crs of Salt Lake City. *Runway No. 34.
(Procedure No. 2)	Riverton FM	S crs ILS	343	.0	7,500	S 343 163	None	7,500	5,560	4,440	4.69	.71	4,222	R S# A T	800 400 800 300	2.0 1.0 2.0 1.0	Climb to cross Salt Lake City range at 6,500', then climb to 11,000' on W crs of Salt Lake City. *ILS Procedure No. 2 for status conditions only; acft must be on top, inbound on glide path, before starting final approach; tops between Riverton FM and Salt Lake City not to exceed 7,500' MSL (otherwise ILS Procedure #1 will be used). NOTE: Night landings and take-offs not authorized. #Runway No. 34.

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
SAN ANTONIO, TEX. San Antonio Arpt. Freq. 109.9 mc Ident. SAT	San Antonio Range	SW crs ILS	174	3.6	2,200	SW 31 211	2,200'—W side SW crs	2,050	2,050	1,000	4.38	0.68	800	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on NE crs of ILS. *Runway No. 3.
	Kelly Range	Outer Mrkr	39	8.2	2,200												
	Int. S crs San Antonio & SE crs Kelly	SW crs ILS	338	13.0	2,200												
	Int. E crs San Antonio & NE crs ILS	Outer Mrkr	211	8.8	2,200												
	Int. SW crs Kelly & SW crs ILS	Outer Mrkr	31	12.0	2,200												
SAN FRANCISCO, CALIF. San Francisco Arpt. Freq. 109.5 mc Ident. SFO	San Francisco Range	*Belmont FM (via SE crs San Francisco)	107	5.8	1,660	SE 278 98	None	1,660	1,660	230	6.60	.66	10	R S# A T	600 400 800 300	2.0 3/4 2.0 1.0	Climb to 3,000' on NW crs San Francisco range within 25 mi. *After passing thru Belmont FM, make left turn of 180° to intercept localizer crs SE of LOM inbound (278°). **After take-off from Oakland Arpt., home on San Francisco LOM (Doolin marker) making good a track of 175° while climbing to 2,000'. Cross LOM at 2,000' and execute standard rate left turn to intercept localizer crs SE of LOM, inbound. #Runway No. 28.
	Moffett Range	SE crs ILS (LOM)	298	18.2	1,660												
	Oakland Arpt.	SE crs ILS (LOM)	175	10.8	**												
SAULT STE. MARIE, MICH. Sault Ste. Marie-Kinross Arpt. Freq. 110.3 mc Ident. KNR	Sault Ste. Marie Range	LOM	296	13.0	2,300	NW 154 334	2,300'—W side NW crs	2,300	*2,220	*980	6.05	.67	780	R S# A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,300' on SE crs of the ILS within 25 mi. *Outer and middle markers and middle marker compass locator not monitored-Outer marker compass locator only monitored. #Runway No. 15.
	Int. W crs Sault Ste. Marie and NW crs ILS	LOM	154	12.0	2,300												
	Int. S crs Sault Ste. Marie and SE crs ILS	LOM	334	10.0	2,300												
SEATTLE, WASH. Seattle-Tacoma International Arpt. Freq. 110.3 mc Ident. SEA (Procedure No. 1)	Seattle Range	S crs ILS (outbound)	175	8.6	2,000	S 338 158	2,000'—E side S crs	1,600	1,590	585	4.60	.65	405	R (**) S# A T	500 500 400 800 300	1.5 2.0 3/4 2.0 1.0	Climb to 1,500' on NW crs of Seattle within 15 mi of range. *Acft shall maintain 3,000' to intxn NW crs Seattle & N crs ILS. **Night minimums. #Runway No. 34.
	Lakeview FM	S crs ILS (inbound)	26	7.0	1,600												
	Everett Range	N crs ILS (inbound)	180	3.0	*3,000												
(Procedure No. 2*** Using Back Course ILS Localizer)	Port Gamble Int.	Int. NW crs Seattle & N crs ILS	116	27.0	2,000	N 158 338	2,000'—W side N crs	1,500 (over Int. NW crs Seattle & N crs ILS)			5.5 (from Int. NW crs Seattle & N crs ILS)		405	R (**) S# A T	500 500 400 800 300	1.5 2.0 1.0 2.0 1.0	Climb to 1,500' on S crs ILS within 10 mi of LOM. *2,000' authorized from Everett as final approach altitude over Int. NW crs Seattle & N crs ILS. **Night minimums. ***Procedure utilizing Int. NW crs Seattle & N crs ILS for approaches to Runway No. 16. #Runway No. 16.
	Everett Range	N crs ILS (final)*	180	3.0	2,000												
	Seattle Range	Int. NW crs Seattle & N crs ILS	296	3.8	2,000												
	Int. S crs Seattle & S Seattle & S crs ILs	Int. NW crs Seattle & N crs ILS	338	10.5	2,000												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
SOUTH BEND, IND. St. Joseph Co. Arpt. Freq. 110.3 mc Ident. SBN	Goshen Range	E crs ILS	311	21.0	2,200	E 268 88	2,000'—N side E crs	2,000	1,930	985	4.39	0.71	776	R S* A T	500 400 800 300	1.5 ¾ 2.0 1.0	Climb to 2,000' on W crs of South Bend. *Runway No. 27.
	So. Bend Range	Outer Mrkr	82	8.6	2,000												
	Int. W crs So. Bend & W crs ILS.	Outer Mrkr	88	14.8	2,000												
	Int. N crs So. Bend & W crs ILS.	Outer Mrkr	88	8.5	2,000												
	Int. N crs Goshen & & E crs ILS.	Glide path Intercept'n	268	18.0	2,200												
	Int. N crs Goshen & E crs South Bend.	E crs ILS	250	19.0	2,000												
SPOKANE, WASH. Geiger Field Freq. 109.9 mc Ident. GEG	Int. W crs Spokane & NE crs ILS.	Outer Mrkr	203	7.1	4,000	SW 23 203	4,000'—S side SW crs (within 10 mi of OM). NOTE: Acft approaching from Spangle FM will make procedure turn on W side SW crs at 4,000'.	*3,560	3,560	2,595	4.42	.62	2,372	R ** S# A T	500 600 400 1,000 300	1.5 1.5 2.0 2.0 1.0	Climb to 5,000' on W crs of Spokane within 18 mi of rng. *Acft approaching from Harrington FM will intercept glide path at 4,000'. **Night minimums. #Runway No. 2.
	Harrington FM	SW crs ILS	76	25.0	4,000												
	Spangle FM	Outer Mrkr	312	13.5	4,100												
SYRACUSE, N. Y. Hancock Arpt. Freq. 109.9 mc Ident. SYR	Syracuse Range	Outer Mrkr	99	13.2	1,900	E 278 98	1,600'—N side E crs	1,600	1,535	636	4.40	.71	419	R S# A T	700 400 800 *300	1.5 ¾ 2.0 1.0	Climb to 2,300' on W crs of Syracuse. *600-1 required for take-off to SE. CAUTION: 833' MSL radio tower, 1.3 mi SE of arpt. #Runway No. 28.
	Syracuse Omniring	Outer Mrkr	Radial 120	10.2	1,900												
TALLAHASSEE, FLA. Dale Mabry Field Freq. 110.3 mc Ident. TLH	Int. N crs Tallahassee & NW crs ILS	Outer Mrkr	123	8.5	1,200	SE 303 123	1,200'—N side SE crs	1,200	1,100	270	4.69	.67	70	R S* A T	500 400 800 300	1.5 ¾ 2.0 1.0	Climb to 1,400' on N crs of Tallahassee, or alternate procedure (when directed by ATC), turn left and climb to 1,400' on S crs of Tallahassee. *Runway No. 31.
	Tallahassee Range	Outer Mrkr	109	7.8	1,200												
	Int. E crs Tallahassee & SE crs ILS	Outer Mrkr	123	5.3	1,200												
TAMPA, FLA. International Arpt. Freq. 110.3 mc Ident. TPA	Int. N crs Tampa & N crs ILS	Outer Mrkr	180	16.3	1,000	N 180 360	1,200'—W side N crs	1,030	1,030	225	4.70	.61	26	R S* A T	500 400 800 300	1.5 ¾ 2.0 1.0	Climb to 1,300' on S crs of Tampa, or alternate procedure (when directed by ATC), turn left and climb to 1,500' on NE crs of Tampa. *Runway No. 18.
	Tampa Range	ILS Localiser	336	5.8	1,200												
TETERBORO, N. J. Teterboro Air Terminal Freq. 108.5 mc Ident. TEB	Linden Rbn	SW crs ILS	30	8.0	1,800	SW 60 240	1,800'—W side SW crs	1,400	1,360	230	4.70	.64	5	R S* A T	1,000 600 1,000 300	1.5 1.0 2.0 1.0	Climb on a heading of 60° to 1,000', turn left, climb to 2,000' on ADF track to Paterson Rbn, or alternate procedure (when directed by ATC) climb on heading of 60° to 1,000', turn left and climb to 1,800' on SW crs of ILS. *Runway No. 6.
	Int. SW crs ILS & W crs Newark	SW crs ILS	60	0	1,800												
	Paterson Rbn	Middle Mrkr	164	8.0	2,000												
TOPEKA, KANS. Phillip Billard Arpt. Freq. 109.5 mc Ident. TOP	Topeka VAR	Outer Mrkr	252	7.3	2,100	NW 125 305	2,100'—W side NW crs	2,000	1,880	1,080	4.67	.83	879	R S# (*) A T	600 400 400 800 300	1.5 ¾ 1.0 2.0 1.0	Climb to 2,400' on S crs of Topeka VAR (Aural). Alternate procedure (when directed by ATC): Climb to 2,400' on SE crs of ILS within 25 mi. *Night minimums. #Runway No. 12.
	Int. SE crs ILS & S crs Topeka VAR	Outer Mrkr	305	12.5	2,100												

INSTRUMENT LANDING SYSTEM PROCEDURES—Continued

ILS location and range from which initial approach to ILS shall be made	Transition to ILS					Final ILS appr. crs: degrees inbound: outbound:	Procedure turn minimum on ILS	Minimum altitude at glide path interception (ft.)	Glide path altitude over markers (ft.)		Distance from markers to approach end of runway (mi.)		Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (deg.)	Distance (mi.)	Minimum altitude (ft.)				Outer	Middle	Outer	Middle			Ceiling (ft.)	Visibility (mi.)	
TRI-CITY, TENN. Tri-City Arpt. Freq. 109.9 mc Ident. TRI	Tri-City Range	Emmett Rbn	96	11.5	5,500	E 270 90	5,500'—N side E crs	5,500	2,740	1,750	4.30	6.65	1,823	R (*) S# A T	800 800 600 1,000 300	1.5 2.0 3/4 2.0 1.0	Climb to 5,000' on SW crs of Tri-City range. NOTE: Descent inbound on glide path not to be started until definitely past Emmett Rbn. (Emmett MHW facility is located 14.5 mi E of runway; Freq. 308 kc, Ident. EME). Glide path altitude is 5,500' MSL at Emmett Rbn. *Night minimums. #Runway No. 27.
	Int. SW crs Tri-City & E crs ILS	Emmett Rbn	90	13.5	5,500												
	Int. SE crs Tri-City & E crs ILS	Emmett Rbn	90	11.0	5,500												
TULSA, OKLA. Tulsa Arpt. Freq. 110.3 mc Ident. TUL	Tulsa Range	S crs ILS	225	1.3	2,200	S 354 174	2,200'—E side S crs	2,020	2,020	850	4.82	.65	674	R S* A T	500 400 500 300	1.5 3/4 2.0 1.0	Turn left and climb to 2,300' on NW crs of Tulsa, or alternate procedure (when directed by ATC), turn right, climb to 2,200' on NE crs of Tulsa. *Runway No. 35R.
	Skiatook FM	Outer Mrkr	138	22.0	2,200												
	Red Fork FM	Outer Mrkr	70	9.7	2,200												
	Verdigris River FM	Outer Mrkr	211	15.6	2,200												
WASHINGTON, D. C. Washington National Arpt. Freq. 109.9 mc Ident. DCA	Washington Range	Outer Mrkr	360	0	1,400	S 360 180	1,400'—W side S crs	1,400	1,360	205	5.30	.62	16	R S* A T	500 400 500 300	1.5 3/4 2.0 1.0	If contact not established at LMM, make climbing turn to the left as soon as practicable and climb to 1,500' on NW crs of Washington range. *Runway No. 36. CAUTION: 596' monument 2 miles N of arpt.
	Int. SW crs Andrews & S crs ILS	Outer Mrkr	360	9.7	1,500												
WHITE PLAINS, N. Y. Westchester Co Arpt. Freq. 109.5 mc Ident. HPN	Int S crs Poughkeepsie & NW crs ILS	Outer Mrkr	161	17.0	2,000	NW 161 341	2,000'—W side NW crs	2,000	1,980	645	5.80	.66	433	R S* A T	600 400 800 300	1.5 3/4 2.0 1.0	Make climbing left turn to 2,000' on NW crs of ILS. *Runway No. 16.
	Rye Rbn	Outer Mrkr	360	12.1	2,000												
	Paterson Rbn	Outer Mrkr	68	24.0	2,000												
WICHITA, KANS. Wichita Arpt. Freq. 110.3 mc Ident. ICT	Wichita Range	S crs ILS (Outbound)	167	2.2	2,800	S 3 183	2,800'—E side S crs	2,750	2,750	1,565	5.37	.60	1,372	R S# A T*	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,800' on NE crs of Wichita, or alternate procedure (when directed by ATC), climb to 2,800' on N crs of ILS within 25 mi. Flight path to be conducted in the Blue sector. *Runways 6L and 24R not authorized at night. #Runway No. 36L.
	Viola FM	S crs ILS (Inbound)	73	18.5	2,800												
	Oxford FM	S crs ILS (Inbound)	337	16.2	2,800												
	Int. N crs ILS & NE crs Wichita	Outer Mrkr	183	7.8	2,800												
	Towanda FM	Outer Mrkr	214	24.2	2,800												
WINDSOR LOCKS, CONN. Bradley Field Freq. 110.3 mc Ident. BDL	Int. NW crs Hartford & SW crs ILS	Outer Mrkr	58	3.0	2,500	SW 58 238	2,500'—S side SW crs	2,500	1,740	400	5.50	.66	171	R S* A T	500 400 800 300	1.5 3/4 2.0 1.0	Climb to 2,500' on NE crs of ILS. *Runway No. 6.
	Int. E crs Westfield & NE crs ILS	Outer Mrkr	238	19.0	2,500												
	Int. S crs Westfield & SW crs ILS	Outer Mrkr	58	2.5	2,500												
WINSLOW, ARIZ. Winslow Arpt. Freq. 110.3 mc Ident. INW	Winslow Range	E crs ILS (MM)	127	3.2	6,500	E 286 106	6,500'—S side E crs	5,920	5,920	5,070	4.21	.68	4,937	R S* A T	600 400 800 300	2.0 3/4 2.0 1.0	Climb to 7,000' on E crs of Winslow within 25 mi. *Runway No. 28.
WORCESTER, MASS. Worcester Arpt. Freq. 109.5 mc Ident. ORH	Worcester LOM	SE crs ILS	332	0	2,400	SE 332 152	2,100'—E side SE crs	2,100	2,150	1,305	4.60	.66	1,009	R S* A T	600 400 1,000 300	1.5 3/4 3.0 1.0	Make climbing left turn to 2,400' on SE crs of ILS, but not to exceed 6 mi. SE of LOM. *Runway No. 33.

(Sec. 205, 52 Stat. 984, as amended; 49 U. S. C. 425. Interpret or apply sec. 601, 52 Stat. 1007, as amended; 49 U. S. C. and Sup. 551)

These procedures shall become effective upon publication in the FEDERAL REGISTER.

[SEAL] F. B. LEE,
Acting Administrator of Civil Aeronautics.

[F. R. Doc. 50-8925; Filed, Nov. 9, 1950; 8:45 a. m.]

[Supp. 3, Amdt. 5]

PART 60—AIR TRAFFIC RULES

STANDARD INSTRUMENT APPROACH PROCEDURES

Under section 205 (a) of the Civil Aeronautics Act of 1938, as amended, the Administrator of Civil Aeronautics is authorized to make and amend such rules, regulations, and procedures as are necessary to carry out the provisions of, and to perform and exercise his powers and duties under, the act. Under section 601 of the Civil Aeronautics Act of 1938, as amended, the Civil Aeronautics Board is empowered to delegate to the Administrator of Civil Aeronautics the authority to prescribe rules, regulations, and standards which promote safety of flight in air commerce. Under §§ 42.55 (b), 42.56, and 60.46 of the Civil Air Regulations, the Civil Aeronautics Board has authorized the Administrator of Civil Aeronautics to prescribe standard instrument approach procedures, including ceiling and visibility minimums.

Acting pursuant to the foregoing statutes and regulations, standard instrument approach procedures were prescribed. Those procedures are hereby amended. This amendment is made effective without delay, in order to promote safety of the flying public. Compliance with the notice, procedures, and effective date provisions of section 4 of the Administrative Procedure Act would be impracticable and contrary to the public interest, and therefore is not required.

1. The low frequency range procedures prescribed in § 60.46–4 are amended by adding:

LOW FREQUENCY RANGE PROCEDURE—ALASKA

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport: Magnetic bearing	Station to airport: Distance (mi.)	Field elevation	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
ADAK Davis AFB 347 kc; ADK SBRAZ-P-D	NE—Min. en route altitude SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	SW crs to 4,900' within 25 mi.	NE (hold close "A" twilight returning to range)	10 mi.—3,100' N side NE crs 15 mi.—6,800' N side NE crs 20 mi.—6,800' N side NE crs 25 mi.—6,800' N side NE crs	1,000	228	3.2	17	R (R) S A T	1,000 900 900 1,500 *1,000	2.5 2.0 2.0 3.0 2.0	1,000 900 900 1,500 *1,000	3.0 3.0 3.0 3.0 3.0	If visual contact not estab within 2 mi after passing range, make left turn and climb to 7,800' on "A" side of NE crs within 10 mi of range. CAUTION: Cape Adagdak 2,100' high 3 mi NW of NE (approach) crs, 5 mi N of rng. *300-1 day and night using Rwy 5. NOTE: If at any time VFR cannot be maintained from rng to arpt, immediately execute missed approach procedure. Course alignment must be checked with ADF prior to descent. Runway 23.
ANCHORAGE Elmendorf AFB 338 kc; ANC SBRAZ-P-DTXV	NE—6,800' (E crs Skwentna) SE—9,000' (W crs Cordova, or 500' on top, day only) SW—1,500' (SE crs Kenai) NW—1,500' (NE crs Kenai)	SW crs Anchorage to 1,500' (all turns to W).	SW	10 mi.—1,500' *W side SW crs 15 mi.—1,500' *W side SW crs 20 mi.—1,500' *W side SW crs 25 mi.—1,500' *W side SW crs	800	11	3.9	212	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Turn left, proceed out NW crs climbing to 1,500' to NE crs Kenai. *W procedure turn due to mountain radge E of SW crs.
Merrill Field (Procedure No. 1—Using Anchorage Range for let-down)	NE—6,800' (E crs Skwentna) SE—9,000' (W crs Cordova, or 500' on top, day only) SW—1,500' (SE crs Kenai) NW—1,500' (NE crs Kenai)	SW crs Anchorage to 1,500' (all turns to W)	SW	10 mi.—1,500' *W side SW crs 15 mi.—1,500' *W side SW crs 20 mi.—1,500' *W side SW crs 25 mi.—1,500' *W side SW crs	800	359	.9	139	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 2.0	Turn left, proceed out NW crs climbing to 1,500' to NE crs Kenai. *W procedure turn due to mountain range E of SW crs.
(Procedure No. 2—Using Merrill Range for let-down) 239 kc; ALZ MRLW	NE—6,800' (E crs Skwentna) SE—9,000' (W crs Cordova, or 500' on top, day only) SW—1,500' (SE crs Kenai) NW—1,500' (NE crs Kenai)	SW crs Anchorage to 1,500' (all turns to W)	W (Merrill Range)	10 mi.—1,500' S side W crs 15 mi.—1,500' S side W crs 20 mi.—1,500' S side W crs 25 mi.—1,500' S side W crs	800 (Over Int. W Merrill & NW Anchorage)	63 (From Int. W Merrill & NW Anchorage)	1.3	139	R S A T	500 NA 1,000 300	1.5 3.0 1.0	500 NA 1,000 300	2.0 3.0 2.0	Proceed out NW crs Anchorage, climbing to 1,500' to Intsn NW Anchorage and NE Kenai. NOTE: Straight-in approach authorized from Delta Int. (NE crs Kenai & W crs Merrill).
ANIAK Aniak Airport 359 kc; ANI BMRLZ-P-D	NE—5,800' (McGrath Range) SE—Min. en route altitude SW—2,300' (Bethel Range) NW—3,000' (S crs Unalakleet)	None	SW	10 mi.—1,600' S side SW crs 15 mi.—1,800' S side SW crs 20 mi.—2,300' S side SW crs 25 mi.—2,300' S side SW crs	1,080	85	1.6	86	R A T	1,000 1,000 500	2.0 3.0 1.0	1,000 1,000 1,000	3.0 3.0 2.0	If visual contact not estab over range, turn right and climb to 2,300' on SW crs to Bethel range. NOTE: Hold "N" twilight, procedure turn to range.

LOW FREQUENCY RANGE PROCEDURE—ALASKA—Continued

Station: frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport: Magnetic bearing	Station to airport: Distance (mi.)	Field elevation	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
ANNETTE IS. Annette Is. Airport 266 kc; ANN SBRAZ-P-DTXV	NE—Min. en route altitude S—2,800' (NE crs Sandspit) SW—4,700' (NW crs Sandspit) NW—5,700' (Petersburg Range).	None	S	10 mi.—1,500' W side S crs 15 mi.—2,800' W side S crs 20 mi.—2,800' W side S crs 25 mi.—2,800' W side S crs	920	122	1.5	119	R A T	800 1,000 *500	1.5 3.0 1.0	800 1,000 *500	2.0 3.0 2.0	If visual contact not estab over range, turn left and proceed out S crs climbing to 2,800' to NE crs Sandspit. *Take-off to NE is 600', 1.0 mi. day, 600', 2 mi night. NOTE: W procedure turn only due to mountains E of S crs.
ATTU Casco Cove NAS 285 kc; ATU SRAW	E—3,200' (Shemya Range) S—Min. en route altitude SW—Min. en route altitude N—Min. en route altitude	None	S	10 mi.—1,500' W side S crs 15 mi.—1,500' W side S crs 20 mi.—1,500' W side S crs 25 mi.—1,500' W side S crs	See Remarks	353	1.6	40	R A T	1,000 1,000 *300	2.5 3.0 1.0	1,000 1,000 *300	3.0 3.0 1.0	If visual contact not estab at authorized minimums 10 mi S of range, turn right and climb to 3,200' on E crs to Shemya rng. Maneuvering W and N of arpt not authorized due to precipitous terrain in close proximity to arpt. NOTE: If, after establishing contact 10 mi S of range, visual contact cannot be maintained, execute missed approach procedure at once. *Runways 10 or 20 only.
BETHEL Bethel Airport 388 kc; BET SBRAZ-P-DV	NE—2,300' (Aniak Range) SE—7,500' (King Salmon Rng) SW—Min. en route altitude NW—Min. en route altitude	SE crs to 1,200' within 25 mi.	SE	10 mi.—1,200' S side SE crs 15 mi.—1,200' S side SE crs 20 mi.—1,200' S side SE crs 25 mi.—1,200' S side SE crs	300	298	2.4	9	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	2.0 2.0 2.0 3.0 1.0	Make climbing right turn and climb to 2,300' on NE crs to Aniak range.
BETTLES Bettles Airport 391 kc; BTT SBRAZ-P-DV	E—Min. en route altitude SE—4,500' (NE crs Tanana) W—Min. en route altitude N—9,800' (SE crs Umiat)	SE crs to 2,500' within 10 mi.	SE	10 mi.—2,500' W side SE crs 15 mi.—2,500' W side SE crs 20 mi.—3,100' W side SE crs 25 mi.—3,500' W side SE crs	1,750	5	1.4	750	R A T	1,000 1,500 *500	2.5 3.0 2.0	1,000 1,500 *500	3.0 3.0 3.0	If contact not estab over range, turn right, climb to 3,500' on SE crs within 25 mi of rng. *Take-off N: 900-2 day, 900-3 night; turn left, climb on SE crs. CAUTION: (a) 200' hill, 3 mi NNE Rwy 36. (b) 866' msl range towers, 1.4 mi S of Rwy 18. (c) 2,000' hills, 8 mi W and 6 mi E of arpt.
BIG DELTA Big Delta Airport 347 kc; BIG SBRAZ-P-DT	NE—Min. en route altitude SE—7,000' (Tanacross Range) S—13,000' (Gulkana Range) NW—6,000' (E crs Fairbanks)	NW crs to 4,500' within 10 mi.	NW	10 mi.—2,800' W side NW crs 15 mi.—2,800' W side NW crs 20 mi.—4,000' W side NW crs 25 mi.—4,400' W side NW crs	1,900	178	1.5	1,256	R A T	700 1,000 500	1.5 3.0 1.0	700 1,000 500	2.0 3.0 2.0	If visual contact not estab over range, proceed out NW crs climbing to 6,000' to E crs Fairbanks.
COLD BAY Thornbrough AFB 368 kc; CDB SBRAZ-P-D	NE—9,900' (Port Heiden Range, or 500' on top. day only) S—Min. en route altitude W—Min. en route altitude N—Min. en route altitude	N crs to 2,800' within 25 mi.	N	10 mi.—1,400' E side N crs 15 mi.—2,800' E side N crs 20 mi.—2,800' E side N crs 25 mi.—2,800' E side N crs	860	141	2.4	89	R A T	*500 1,500 #300	2.0 3.0 1.0	*500 1,500 #300	3.0 3.0 1.0	Turn left and proceed out N crs climbing to 2,800' within 25 mi. #Take-off on Rwy 26 is 800-2 day, 1,000-3 night—right turn after take-off. *Landing on Rwy 8 is 1,200-2 day, 1,200-3 night. NOTE: Perform all maneuvering prior to landing to N and E of arpt. (Mt. Simeon, 900' high, 2 mi W of arpt). CAUTION: Amak Is., 1,800' high, 18½ mi. NW of rng within 10 mi of N (approach) crs.
CORDOVA (Procedure No. 1) Mile 13 Airport 394 kc; CDV MLZ-D	(To be made on CXD Range) NE—9,500' (S crs Gulkana) E—5,000' (S crs Yakataga) SW—Min. en route altitude W—8,800' (SE crs Anchorage)	E crs CXD rng between CXD rng & SE crs CDV localizer to 2,500'	SE crs CDV localizer	(CDV localizer) 10 mi.—1,000' S side SE crs 15 mi.—1,000' S side SE crs 20 mi.—1,000' S side SE crs 25 mi.—1,000' S side SE crs	940 (CDV localizer)	265 (From CDV localizer)	2.1	38	R A T	900 1,500 *500	2.5 3.0 1.0	900 1,500 *500	3.0 3.0 2.0	If visual contact not estab over CDV localizer, turn left and proceed out SE crs CDV localizer climbing to 1,000' to E crs CXD range. *Turn S after take-off.

LOW FREQUENCY RANGE PROCEDURE—ALASKA—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport		Field elevation	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
						Magnetic bearing	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
CORDVOA (Procedure No. 2) 362 kc; CXD SBRAZ-P-DTV	(To be made on CXD Range) NE—9,500' (S crs Gulkana) E—5,000' (S crs Yakataga) SW—Min. en route altitude W—8,000' (SE crs Anchorage)	E crs CXD range between CXD range & SE crs CDV localizer to 2,800'	SW crs CDV localizer	(CDV localizer) 10 mi.—1,000' S side SW crs 15 mi.—1,000' S side SW crs 20 mi.—1,000' S side SW crs 25 mi.—1,000' S side SW crs	(See Remarks)	265 (From CDV localizer)	2.1	28	R A T	900 1,500 *500	2.5 3.0 1.0	900 1,500 *500	3.0 3.0 2.0	If visual contact not estab at authorized minimums 6 mi after passing intxn of SW crs CDV and E crs CXD, turn right and climb to 1,000' on SW crs CDV. (NOTE: If at any time after establishing contact, VFR cannot be maintained, immediately execute the missed approach procedure. *Turn S after take-off.
FAIRBANKS Ladd AFB 260 kc; FAI SBRAZ-P-DTX	N—Min. enroute altitude E—4,000' (NW crs Big Delta) S—9,500' SE crs Nenana) W—3,900' (NW crs Nenana)	E or W crs to 4,000'	E	10 mi.—2,200 S side E crs 15 mi.—2,200' S side E crs 20 mi.—2,500' S side E crs 25 mi.—3,700' S side E crs	1,400	239	2.7	448	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Turn left, climb to 3,900' on NE crs of Nenana to Nenana range.
Weeks Field 260 kc; FAI SBRAZ-P-DTX	N—Min. en route altitude E—4,000' (NW crs Big Delta) S—9,500' (SE crs Nenana) W—3,900' (NW crs Nenana)	E or W crs to 4,000'	E	10 mi.—2,200' S side E crs 15 mi.—2,200' S side E crs 20 mi.—2,500' S side E crs 25 mi.—3,700' S side E crs	1,400	239	6.9	435	R A T	700 1,000 500	3.0 3.0 2.0	1,000 1,000 500	3.0 3.0 3.0	If visual contact not estab within 2.7 mi. after passing range, turn left, climb to 3,900' on NE crs of Nenana to Nenana range. NOTE: If, after establishing reference with the ground, VFR cannot be maintained, execute the missed approach procedure at once.
FAREWELL Farewell Airport 206 kc; FWL BMRLZ-P-DV	NE—7,000' (Minchumina Range) SE—11,700' (Skwentna Range, or 500' on top, day only) SW—Min. en route altitude NW—5,500' (McGrath Range)	NW crs to 5,500' (all turns to N)	NW	10 mi.—*2,500' N side NW crs 15 mi.—*2,500' N side NW crs 20 mi.—*2,500' N side NW crs 25 mi.—*2,500' N side NW crs	(See Remarks)	80	1.5	1,538	R A T	1,000 3,000 500	2.5 3.0 1.0	1,000 3,000 900	3.0 3.0 2.0	If visual contact not estab 10 mi. NW of range, turn left, climb to 5,500' on NW crs to McGrath rng. *Altitude in procedure turn shall not be less than the authorized minimums for the applicable flight. (CAUTION: High hills 1.5 mi. SE of arpt. Turns to SE not authorized.
GALENA Galena Airport 371 kc; GAL SBRAZ-P-DTV	N—Min. en route altitude E—5,800' (SW crs Tanana) S—4,900' (NW crs McGrath) W—6,000' (N crs Unalakleet)	W crs to 2,500' within 10 mi.	E	10 mi.—2,500' S side E crs 15 mi.—3,000' S side E crs 20 mi.—3,300' S side E crs 25 mi.—4,000' S side E crs	1,020	245	3.3	120	R (R) S A T	900 800 500 1,000 300	1.5 1.0 1.0 3.0 1.0	900 800 500 1,000 500	2.5 2.0 2.5 3.0 2.0	Climb on W crs to 6,000' to N crs Unalakleet. NOTE: Dike at either end of rwy, 12-15 ft high.
GULKANA Gulkana Airport 265 kc; GKN SBRAZ-P-DV	NE—10,500' (SW crs Northway) S—9,500' (N crs Cordova) W—10,500' (NE crs Anchorage) N—13,000' (Big Delta Range)	N crs to 3,200' within 10 mi.	N	10 mi.—3,200' W side N crs 15 mi.—3,200' W side N crs 20 mi.—3,200' W side N crs 25 mi.—3,200' W side N crs	2,300	142	2.3	1,571	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 400	2.0 3.0 1.0	Climb to 9,500' on S crs to N crs of Cordova.
GUSTAVUS (Procedure No. 1) Gustavus Airport 236 kc; GST SBRAZ-P-DTV	NE—Min. en route altitude SE—7,000' (NE crs Sitka) SW—5,500' (NW crs Sitka) NW—Min. en route altitude	SE & NW courses to 4,500' within 10 mi.	SE (within 10 mi. of FM)	10 mi.—3,200' S side SE crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,500 (over Pleasant Isle FM)	285 (From Pleasant Isle FM)	3.2	28	R A T	800 1,000 *500	1.5 3.0 1.0	800 1,000 *500	2.5 3.0 1.0	Turn left, climb to 7,000' on SE crs to NE crs Sitka. NOTES: If Pleasant Isle FM not received on final approach, maintain 1,500' to range. If visual contact not estab over range, turn left, climb to 7,000' on SE crs. Maneuvering N of arpt not authorized due to high terrain. *Take-off to N is 700-2, day and night.

LOW FREQUENCY RANGE PROCEDURE—ALASKA—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport		Field elevation	Ceiling and visibility minimums					If visual contact not established over airport at authorised landing minimums, or if landing not accomplished, remarks
						Magnetic bearing	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
GUSTAVUS (Procedure No. 2) Gustavus Airport 236 kc; GST SBRAZ-P-DTV	NE—Min. en route altitude SE—7,000' (NE crs Sitka) SW—5,500' (NW crs Sitka) NW—Min. en route altitude	SE & NW courses to 4,500' within 10 mi.	*NW	10 mi.—2,700' W side NW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	1,400	105	2.7	28	R A T	800 1,000 #500	1.5 3.0 1.0	800 1,000 #500	2.5 3.0 1.0	Proceed out SE crs climbing to 7,000' to NE crs of Sitka. *Prior to proceeding out NW crs for final approach, a positive check of wind velocity and direction must be made on SE crs. #Take-off to N is 700-2, day and night. NOTE: Maneuvering to N of arpt not authorized due to high terrain.
HOMER Homer Airport 220 kc; HOM SBRAZ-P-DTV	N—4,000' (SE crs Kenai) E—Min. en route altitude S—6,000' (Kodiak Range) W—3,300' (SW crs Kenai)	None	*W	10 mi.—2,600' S side W crs 15 mi.—2,600' S side W crs 20 mi.—2,600' S side W crs 25 mi.—2,600' S side W crs	1,300	75	1.7	72	R A T	1,000 1,000 #500	2.5 3.0 1.0	1,000 1,000 #500	3.0 3.0 2.0	Turn right and climb to 3,300' on W crs. *Hold "A" twilight returning to range on final. #S turn only after take-off. NOTE: Let down from range to minimum authorized altitude on heading of 90° Straight-in approach authorized from Anchor Point Int. (APO).
ILIAMNA Iliamna Airport 233 kc; ILI BMRLZ-P-D	N—Min. en route altitude SE—6,000' (W crs Homer) S—4,100' (NE crs King Salmon) NW—Min. en route altitude	None	S	10 mi.—1,200' E side S crs 15 mi.—1,200' E side S crs 20 mi.—3,100' E side S crs 25 mi.—3,100' E side S crs	980	250	2.6	190	R A T	800 1,000 500	2.5 3.0 1.0	1,000 1,000 500	3.0 3.0 2.0	If visual contact not estab over range, turn left and climb to 4,100' on S crs to NE crs of King Salmon.
KENAI Kenai Airport 279 kc; ENA BMRLZ-P-DT	NE—1,500' (NW crs Anchorage) SE—1,500' (N crs Homer) SW—1,400' (W crs Homer) NW—Min. en route altitude	None	NE	10 mi.—1,400' N side NE crs 15 mi.—1,400' N side NE crs 20 mi.—1,400' N side NE crs 25 mi.—1,400' N side NE crs	800	184	1.4	91	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Proceed out SW crs climbing to 1,400' to W crs Homer.
KING SALMON King Salmon Airport 400 kc; AKN SBRAZ-P-D	NE—4,500' (S crs Iliamna) SE—10,300' (W crs Kodiak) SW—1-500' (Fort Helden Range) NW—7,500' (Bethel Range or 500' on top, day only)	NW crs to 3,000' within 25 mi.	NW	10 mi.—1,300' W side NW crs 15 mi.—1,300' W side NW crs 20 mi.—1,300' W side NW crs 25 mi.—1,300' W side NW crs	500	111	1.3	52	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Turn right, climb to 1,500' on SW crs.
KODIAK Kodiak NAS 242 kc; NHB SBRAZ-P-DTX	N—6,000' (Homer Range) E—Min. en route altitude S—Min. en route altitude W—6,000' (SE crs King Salmon)	E crs to 4,000' within 25 mi.	E	10 mi.—1,500' N side E crs 15 mi.—1,500' N side E crs 20 mi.—1,500' N side E crs 25 mi.—1,500' N side E crs	1,000	232	6.5	60	R (R) S# A T	1,000 800 800 1,500 *500	4.0 4.0 4.0 4.0 1.0	1,000 800 800 1,500 *500	4.0 4.0 4.0 4.0 2.0	If visual contact not estab within 3.5 mi after passing range, turn left (So.) and climb to 4,000' on E crs within 25 mi of range. *E turns only after take-off. CAUTION: 400' tower adjacent to range towers. High terrain within 1.5 mi N, W and S of arpt. #Runway 25.
McGRATH McGrath Airport 350 kc; MCG SBRAZ-P-DV	NE—6,800' (Minchumina Range) SE—5,500' (Farewell Range) SW—5,800' (Aniak Range) NW—6,800' (Unalakleet Range)	None	SE	10 mi.—2,000' S side SE crs 15 mi.—2,000' S side SE crs 20 mi.—2,000' S side SE crs 25 mi.—2,000' S side SE crs	1,340	275	1.6	367	R A T	1,000 1,500 500	2.5 3.0 1.0	1,000 1,500 500	2.5 3.0 2.0	Turn right and climb to 5,500' on SE crs to Farewell range. Take-off to N or S, turn left, climb on SE crs. EXTREME CAUTION: 1,300' terrain 3 mi S of Rwy 15.
MINCHUMINA Minchumina Airport 227 kc; MHM BMRL-P-DV	NE—4,800' (Nenana Range) SE—Min. en route altitude SW—6,800' (McGrath Range) NW—5,100' (SW crs Tanana)	SE crs within 25 mi.	SE	10 mi.—2,100' E side SE crs 15 mi.—2,100' E side SE crs 20 mi.—2,100' E side SE crs 25 mi.—2,100' E side SE crs	1,600	114	0.7	707	R A T	900 1,500 700	3.0 3.0 2.0	1,000 1,500 1,000	3.0 3.0 2.0	If visual contact not estab over range, turn left and climb to 6,800' on SW crs to McGrath range. NOTE: All turns must be made to the left when returning to arpt and while circling the arpt.

LOW FREQUENCY RANGE PROCEDURE—ALASKA—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport		Field elevation	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
						Magnetic bearing	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
MOSES POINT Moses Point Airport 263 kc; MOS BMRLZ-P-DTV	N—5,200' (NE crs Nome) E—4,000' (N crs Unalakleet) S—4,100' (W crs Unalakleet) W—5,000' (Nome Range)	None	E	10 mi.—1,000' S side E crs 15 mi.—1,000' S side E crs 20 mi.—1,500' S side E crs 25 mi.—1,500' S side E crs	(See remarks)	56	3.0	14	R A T	1,000 1,500 500	2.5 3.0 1.0	1,000 1,500 500	3.0 3.0 2.0	If contact not estab at authorized minimums 10 mi E of range, make left turn and climb to 4,100' on S crs to W crs Unalakleet. CAUTION: 1,200' terrain 3 mi NW of range.
NENANA Nenana Airport 226 kc; ENN BMRLZ-P-DT	NE—3,900' (Fairbanks Range). SE—9,500' (S crs Fairbanks) SW—4,800' (Minchumina Range). NW—2,600' (W crs Fairbanks)	NE & SW courses to 3,900' within 10 mi.	SW	10 mi.—1,500' W side SW crs 15 mi.—1,500' W side SW crs 20 mi.—1,500' W side SW crs 25 mi.—2,400' W side SW crs	1,360	302	2.4	360	R A T	900 1,000 500	3.0 3.0 1.0	1,000 1,000 500	3.0 3.0 2.0	If visual contact not estab over range, turn right, climb to 2,400' on SW crs withing 25 mi. CAUTION: 1,600' hill 2.5 mi N of arpt.
NOME Marks AFB 739 kc; MKF SBRAZ-P-DTX	NE—6,000' (Kotzebue Eddo-Ben). E—5,000' (S crs Moses Point) SW—Min. en route altitude W—Min. en route altitude	None	E	10 mi.—2,500' S side E crs 15 mi.—2,500' S side E crs 20 mi.—2,500' S side E crs 25 mi.—2,500' S side E crs	800	264	3.5	37	R (R) S A T	500 500 500 1,000 *300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 *300	2.0 2.0 2.0 3.0 2.0	Climb to 2,800' on W crs within 25 mi of range. *Right turn after take-off on Rwy 2.
NORTHWAY Northway Airport 400 kc; ORT SBRAZ-P-DTV	NE—Min. en route altitude SE—6,400' (Snag Range) SW—10,500' (S crs Tanacross) NW—8,000' (Tanacross Range)	NW crs to 4,200' within 10 mi.	NW	10 mi.—2,900' W side NW crs 15 mi.—3,000' W side NW crs 20 mi.—3,100' W side NW crs 25 mi.—3,700' W side NW crs	2,320	138	.95	1,716	R A T	600 1,000 400	1.5 3.0 1.0	600 1,000 400	2.5 3.0 2.0	If visual contact not estab over range, climb to 6,400' on SE crs to Snag Range. CAUTION: 1,900' MSL radio tower between range and arpt.
POINT BARROW Point Barrow NAS 236 kc; PBA BMRLZ-P-D	NE—Min. en route altitude SE—2,500' (Umiat Range) SW—Min. en route altitude NW—Min. en route altitude	None	SW	10 mi.—1,100' S side SW crs 15 mi.—1,100' S side SW crs 20 mi.—1,100' S side SW crs 25 mi.—1,100' S side SW crs	520	20	2.6	9	R (R) S A T	500 500 500 1,000 300	1.5 1.0 1.0 3.0 1.0	500 500 500 1,000 300	2.0 1.0 1.0 3.0 1.0	Climb to 2,500' on SE crs to Umiat Range.
PORT HEIDEN Port Heiden Airport 203 kc; PTH SBRAZ-P-D	NE—1,500' (King Salmon Range) SE—Min. en route altitude SW—2,900' (Cold Bay Range or 500' on top, day only) NW—Min. en route altitud	None	NW	10 mi—1,000' W side NW crs 15 mi—1,000' W side NW crs 20 mi—1,000' W side NW crs 25 mi—1,000' W side NW crs	589	308	2.5	86	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	If visual contact not established over range, turn right, climb to 1,500' on NE crs to King Salmon Range. CAUTION: 225' MSL radio tower between range and arpt.
SHEMYA Shemya AFB 226 kc; SYA MRLWZ-P-D	N—Min. en route altitude E—Min. en route altitude S—Min. en route altitude W—3,200' (Attu Range)	None	W	10 mi—1,700' S side W crs 15 mi—1,700' S side W crs 20 mi—1,700' S side W crs 25 mi—1,700' S side W crs	1,200	100	1.4	95	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Climb to 1,700' on E crs within 25 mi. NOTE: Course alignment must be checked with ADF prior to descent. CAUTION: 680' terrain 8 mi W of range, 1 mi N of W crs.
SITKA Seaplane Landing Area 223 kc; STK SBRAZ-P-D	NE—Min. en route altitude SE—5,600' (SW crs Petersburg) SW—Min. en route altitude NW—5,300' (SW crs Gustavus)	SW crs	SW	10 mi—1,500' S side SW crs 15 mi—1,500' S side SW crs 20 mi—1,500' S side SW crs 25 mi—1,500' S side SW crs	1,000	360	15.3	S. L.	R A T	1,000 1,500 (BCOB) 1,000	2.0 3.0 1.0	NA NA NA	----------	If visual contact not established within 7 mi after passing Sitka range, turn left, climb to 1,500' on SW crs. NOTE: After passing range sta, a track of 360° mag. shall be maintained. If visual contact is not estab within 7 mi (a beam Povorotni Point), turn left to a track of 180° on Sitka range and climb to 1,500' on SW crs. If at any time after establishing contact, VFR cannot be maintained, execute the missed approach procedure. CAUTION: High terrain N side of NW crs.

LOW FREQUENCY RANGE PROCEDURE—ALASKA—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport		Field elevation	Ceiling and visibility minimums					If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
						Magnetic bearing	Distance (mi.)			Day		Night		
										Ceiling (ft.)	Visibility (mi.)	Ceiling (ft.)	Visibility (mi.)	
SKWENTNA Skwentna Airport 269 kc; SKW SBRAZ-P-DT	NE—10,000' (Summit Range or 500' on top, day only) E—5,900' (N crs Merrill) SW—Min. en route altitude W—11,700' (Farewell Range or 500' on top, day only)	None	E	10 mi.—1,500' N side E crs 15 mi.—1,500' N side E crs 20 mi.—1,500' N side E crs 25 mi.—1,500' N side E crs	960	90	1.6	158	R A T	800 1,000 500	1.5 3.0 1.0	800 1,000 500	2.5 3.0 2.0	If visual contact not estab over range, turn left, proceed out E crs climbing to 5,000' to N crs of Merrill.
SUMMIT Summit Airport 326 kc; SIT SBRAZ-P-DV	NE—9,500' (SE crs Nenana) E—Min. en route altitude SW—10,000' (E crs Skwentna) W—Min. en route altitude	SW crs within 6 mi of sta.	SW	6 mi.—7,600' W side SW crs 15 mi.—NA 20 mi.—NA 25 mi.—NA	7,100	30	1.8	2,418	R A T	*4,300 (BCOB) *4,300 (BCOB) *4,300 (BCOB)	3.0 3.0 3.0	*4,300 (BCOB) *4,300 (BCOB) *4,300 (BCOB)	5.0 5.0 5.0	Proceed out NE crs climbing to 9,500' to SE crs Nenana. *Shuttle on SW crs within 6 mi of sta to 8,000' before climbing on course. CAUTION: 6,600' terrain 1 mi E of SW crs at point of completion of procedure turn. Mountainous terrain all sides of arpt.
TANACROSS Tanacross Airport 317 kc; TSG MRLWZ	N—Min. en route altitude E—5,200' (NW crs Northway) S—10,500' (SW crs Northway) W—11,700' (S crs Big Delta)	E crs to 4,500' within 10 mi.	E	10 mi.—4,200' N side E crs 15 mi.—4,200' N side E crs 20 mi.—4,200' N side E crs 25 mi.—4,500' N side E crs	2,440	297	1.6	1,549	R A T	900 1,500 500	2.5 3.0 1.0	1,000 1,500 1,000	3.0 3.0 2.0	If visual contact not estab over range, turn right, proceed out E crs climbing to 5,200' to NW crs of Northway.
TANANA Tanana Airport 212 kc; TAL BMRLZ-P-DV	NE—Min. en route altitude SE—4,000' (W crs Fairbanks) SW—3,800' (E crs Galena) NW—Min. en route altitude	None	SW	10 mi.—1,700' S side SW crs 15 mi.—1,700' S side SW crs 20 mi.—2,400' S side SW crs 25 mi.—2,600' S side SW crs	1,120	62	2.3	225	R A T	900 1,500 500	3.0 3.0 1.0	1,000 1,500 500	3.0 3.0 2.0	If visual contact not estab over range, turn right, proceed out SE crs climbing to 4,000' to W crs Fairbanks.
UMIAT Umiat NAS 360 kc; UMT BMRLZ-P-D	E—Min. en route altitude SE—Min. en route altitude W—Min. en route altitude NW—2,500' (Pt. Barrow Range	W crs	W	10 mi.—2,100' S side W crs 15 mi.—2,500' S side W crs 20 mi.—2,500' S side W crs 25 mi.—2,500' S side W crs	1,500	52	2.6	352	R A T	1,000 1,000 *500	2.5 3.0 2.0	1,000 1,000 *700	3.0 3.0 3.0	Turn right, climb to 2,500' on W crs within 25 mi. *S turn only after take-off. CAUTION: 900' bluff 2.5 mi ENE of arpt.
UMNAK Cape AFB 370 kc; UMK SMRLZ-P-D	NE—7,700' (W crs Cold Bay) SE—Min. en route altitude SW—Min. en route altitude NW—Min. en route altitude	NE crs to 3,000' within 25 mi.	SW (hold "N" twilight)	10 mi.—1,500' S side SW crs 15 mi.—3,700' S side SW crs 20 mi.—3,700' S side SW crs 25 mi.—3,700' S side SW crs	820	30	2.2	127	R S A T	800 600 1,500 *700	3.0 2.0 3.0 2.0	800 600 1,500 *700	3.0 2.0 3.0 2.0	Climb on mag track of 40° on S side of NE crs to 3,000' within 20 mi of range. *Using Rwy 24, make S turn after take-off. NOTE: Course alignment must be checked with ADF prior to beginning descent for instrument approach. CAUTION: 4,100' volcano 4 mi W of range. 2,700' terrain within in 10 mi W of SW crs and 15-25 mi from range.
UNALAKLEET Unalakleet Airport 382 kc; UNK SBRAZ-P-DTV	N—4,300' (E crs Moses Point) E—6,000' (McGrath Range) S—5,400' (NW crs Aniak) W—2,800' (S crs Moses Point)	None	W	10 mi.—1,000' S side W crs 15 mi.—1,000' S side W crs 20 mi.—1,000' S side W crs 25 mi.—1,000' S side W crs	(See Remarks)	269	2.5	21	R A T	1,000 1,000 500	2.5 3.0 1.0	1,000 1,000 500	3.0 3.0 2.0	If visual contact not estab at authorized minimums 5 mi W of range, turn right and climb to 2,800' on W crs to S crs of Moses Point. CAUTION: 700' terrain, 2.5 mi N of arpt. All turns to NE not authorized.
YAKATAGA Yakataga Airport 385 kc; YAA BMRLZ-P-D	N—Min. en route altitude E—Min. en route altitude S—Min. en route altitude W—5,000' (Cordova Range)	S crs to 4,000'	S	10 mi.—1,000' W side S crs 15 mi.—1,000' W side S crs 20 mi.—1,000' W side S crs 25 mi.—1,000' W side S crs	(See Remarks)	194	0.5	12	R A T	1,000 2,000 *500	2.5 3.0 1.0	1,000 2,000 *500	3.0 3.0 1.0	If visual contact not estab at authorized minimums 10 mi S of range, turn left and climb to 5,000' on E crs of Cordova. *Take-off to NE is 1,000-2 day or night. NOTE: If, after establishing contact 10 mi S of range, VFR cannot be maintained, execute the missed approach procedure at once. Restrict any maneuvering E of fld due to high terrain E of arpt.

LOW FREQUENCY RANGE PROCEDURES—ALASKA—Continued

Station; frequency; identification; class	Minimum initial approach altitude from the direction and radio fix indicated	Shuttle	Final approach range course	Procedure turn minimum at distances from radio range station	Minimum altitude over range, final approach	Station to airport: Magnetic bearing	Station to airport: Distance (mi.)	Field elevation	Ceiling and visibility minimums	Day: Ceiling (ft.)	Day: Visibility (mi.)	Night: Ceiling (ft.)	Night: Visibility (mi.)	If visual contact not established over airport at authorized landing minimums, or if landing not accomplished, remarks
YAKUTAT Yakutat Airport 332 kc; YAK SBRAZ-P-DV	NE—Min. en route altitude SE—1,200' (SW ers Gustavus) SW—Min. en route altitude W—1,500' (S ers Yakataga)	W ers	W	10 mi.—1,200' S side W ers 15 mi.—1,200' S side W ers 20 mi.—1,200' S side W ers 25 mi.—1,200' S side W ers	700	105	3.1	32	R A T	500 1,000 300	1.5 3.0 1.0	500 1,000 300	2.0 3.0 1.0	Proceed out SE ers climbing to 1,200' to SW ers Gustavus.

2. The automatic direction finding procedures prescribed in § 60.46–7 are amended by adding:

AUTOMATIC DIRECTION FINDING PROCEDURES—ALASKA

Station; frequency; identification; class	Initial approach to station: From—	To—	Magnetic course (degs.)	Distance (mi.)	Minimum altitude (ft.)	Final approach track; degrees inbound: outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums	Ceiling (ft.)	Visibility (mi.)	If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
ANCHORAGE Elmendorf AFB 287 kc; EDF LMM	Anchorage Range	LMM	343	3.6	1,300	W 53 233	10 mi.—1,500' N side W track 15 mi.—NA 20 mi.—NA 25 mi.—NA	700	53°, 0.48	212	R * A T	500 500 1,000 300	1.5 2.0 3.0 1.0	If visual contact not estab over LMM, turn left and climb to 1,500' on NW ers Anchorage to NE ers Kenai. SHUTTLE: SW ers Anchorage to 1,500' (All turns to W—mountain range to E). *Night minimums.
ANNETTE ISLAND Annette Island Arpt 266 kc; ANN SBRAZ-P-DTXV	Petersburg Range	Range	122	131	5,700	S 220 140	10 mi.—1,500' W side S track 15 mi.—2,800' W side S track 20 mi.—2,800' W side S track 25 mi.—2,800' W side S track	820	122°, 1.5	119	R * A T *	800 800 1,000 #500 #500	1.5 2.0 3.0 1.0 2.0	If visual contact not estab over range, turn left and climb to 2,800' on track of 140° from Annette Is. to NE ers Sandspit. * Night minimums. #Take-off to NE is 600-1 day, 600-1.5 night. NOTE: W procedure turn due to mountains E of S ers. CAUTION: High terrain within 2.5 mi of range.
	Ketchikan Range	Range	99	196	6,000									
	Dixon Int.	Range	27	70	4,700									
	Banks Int.	Range	320	93	2,800									
FAIRBANKS Weeks Field 260 kc; FAI SBRAZ-P-DTX	Nenana Range	Range	35	50	3,900	E 239 59	10 mi.—2,200' S side E track 15 mi.—2,200' S side E track 20 mi.—2,500' S side E track 25 mi.—3,700' S side E track	1,400	239°, 6.9	435	R * A T *	700 1,000 1,000 500 500	3.0 3.0 3.0 2.0 3.0	If visual contact not estab within 2.7 mi of range, climb to 3,900' on track of 215° to Nenana rng. *Night minimums. NOTE: If, after establishing reference with the ground, VFR cannot be maintained, execute the missed approach procedure at once. SHUTTLE: To 4,000' on track of 59° outbound, 239° inbound within 25 mi of range.
	Big Delta Range	Range	286	79	6,000									
	Int. E ers Fairbanks & NW ers Big Delta	Range	239	26	4,000									
KOTZEBUE Kotzebue Airport 356 kc; OTZ BHP-DTV	Nome Range	Rbn	6	182	6,000	NE 209 29	10 mi.—1,200' N side NE track 15 mi.—1,200' N side NE track 20 mi.—2,000' N side NE track 25 mi.—2,700' N side NE track	700	209°, 1.8	9	R * A T *	500 500 1,000 300 300	2.5 3.0 3.0 1.0 2.0	Climb to 6,000' on track of 180° to Nome rng. *Night minimums. SHUTTLE: 29° outbound, 209° inbound to 3,500' within 25 mi.
	Galena Range	Rbn	290	221	7,000									
NOME Marks AFB 239 kc; MKF SBRAZ-P-DTX	Moses Point Range	Range	244	95	5,000	E 249 69	10 mi.—2,500' S side E track 15 mi.—2,500' S side E track 20 mi.—2,500' S side E track 25 mi.—2,500' S side E track	800	264°, 2.5	37	R * S * A T	500 500 500 500 1,000 300	1.5 2.0 1.0 2.0 3.0 1.0	Climb to 2,800' on track of 264° within 25 mi of range. NOTE: Right turn after take-off on Rwy 2. *Night minimums.
	Unalakleet Range	Range	268	145	2,800									
	Kotzebue Rbn.	Range	187	182	6,000									
PORT HEIDEN Port Heiden Arpt 203 kc; PTH SBRAZ-P-D	King Salmon Range	Range	189	140	1,500	NW 128 308	10 mi.—1,000' W side NW track 15 mi.—1,000' W side NW track 20 mi.—1,000' W side NW track 25 mi.—1,000' W side NW track	580	308°, 2.5	86	R * A T	500 500 1,000 300	1.5 2.0 3.0 1.0	If not contact over range, turn right and climb to 1,500' on track of 9° within 25 mi. *Night minimums. CAUTION: 235' msl radio tower between range and airport.
	Cold Bay Range	Range	35	200	9,900 or 500 on top day only									

AUTOMATIC DIRECTION FINDING PROCEDURES—ALASKA—Continued

Station; frequency; identification; class	Initial approach to station					Final approach track; degrees inbound; outbound	Procedure turn minimum at distances from station	Minimum altitude over station on final approach (ft.)	Distance from station to approach end of runway (mi.)	Field elevation (ft.)	Minimums			If visual contact not established at authorized landing minimums, or if landing not accomplished; remarks
	From—	To—	Magnetic course (degs.)	Distance (mi.)	Minimum altitude (ft.)							Ceiling (ft.)	Visibility (mi.)	
ST. PAUL ISLAND St. Paul Airport 324 kc; K; HW	Cold Bay Range	Rbn	274	313	2,000	SW	10 mi.—1,500' S side SW track 15 mi.—1,500' S side SW track 20 mi.—1,500' S side SW track 25 mi.—1,500' S side SW track	840	23°, 2.7	42	R • A T •	900 NA NA #500 NA	4.0 2.0	If visual contact not estab over rbn, turn right and climb to 2,000' on track of 100° within 25 mi. #200-1 on Rwy 17 with E turn after take-off. •Night minimums. NOTE: If at any time contact cannot be maintained between rbn and arpt, immediately execute missed approach procedure.
	King Salmon Range	Rbn	234	305	2,000	45								
	Bethel Range	Rbn	230	405	2,000	225								
SKWENTNA Skwentna Airport 269 kc; SKW SBRAZ-P-DT	Anchorage Range	Range	295	70	5,300	E	10 mi.—1,500' N side E track 15 mi.—1,500' N side E track 20 mi.—1,500' N side E track 25 mi.—1,500' N side E track	860	90°, 1.6	158	R • A T •	700 700 800 400 400	1.0 2.0 2.0 1.0 2.0	If visual contact not estab over range, turn left and climb to 5,000' on track of 85° within 25 mi. •Night minimums.
	Farewell Range	Range	88	98	11,700	265								
	Talkeetna Rbn	Range	210	45	5,000	85								
TALKEETNA Talkeetna Airport 317 kc; TKA BHP-DTV	Summit Range	Rbn	173	73	10,000	S	10 mi.—1,700' W side S tract 15 mi.—1,700' W side S trset 20 mi.—1,700' W side S track 25 mi.—1,700' W side S track	1,360	90°, 0.5	358	R # A # T # •	1,000 1,500 1,000 1,500 •1,000 •1,000	3.0 5.0 3.0 5.0 2.0 3.0	If visual contact not estab over rbn, turn left and climb to 4,800' on track of 147° to intxn N crs Merrill & E crs Skwentna. SHUTTLE: Outbound track of 180°, inbound track of 360° to 4,800' within 25 mi (All turns to W**). # Night minimums. • Left turn only after take-off N. ** Rbn located 0.5 mi W of the N end of the rwy. High mtns to E and N within 3-5 mi. of arpt. 300' hill 0.5 mi S and E of arpt. NOTE: All maneuvering preparatory to landing should be accomplished W of arpt.
	Merrill Range	Rbn	327	79	4,800	360 180								

These procedures shall become effective upon publication in the FEDERAL REGISTER.

(Sec. 205, 52 Stat. 984, as amended; 49 U. S. C. 425. Interpret or apply sec. 601, 52 Stat. 1007, as amended; 49 U. S. C. and Sup., 551)

[SEAL] LEONARD W. JURDEN,
Acting Administrator of Civil Aeronautics.

[F. R. Doc. 50-10050; Filed, Nov. 9, 1950; 8:48 a. m.]

TITLE 32A—NATIONAL DEFENSE, APPENDIX

Chapter I—National Production Authority, Department of Commerce

[NPA Reg. 3]

PART 12—OPERATIONS OF THE PRIORITIES SYSTEMS BETWEEN CANADA AND THE UNITED STATES

This regulation is found necessary and appropriate to promote the national defense and is issued pursuant to the authority granted by the Defense Production Act of 1950 and to implement the provisions of the Statement of Principles for Economic Cooperation issued by the governments of the United States and Canada dated October 26, 1950.

Sec.
12.1 What this part does.
12.2 Rated orders from the United States under the United States Defense Program, placed with Canadian suppliers.
12.3 Rated orders from Canada under the Canadian Defense Program placed with United States suppliers.
12.4 Certifications by Canadian purchasers to suppliers in the United States.
12.5 Specific authorizations required.
12.6 Applicability of Part 11 of this chapter (NPA Reg. 2).

AUTHORITY: §§ 12.1 to 12.6 issued under sec. 704, Pub. Law 774, 81st Cong. Interpret or apply sec. 101, Pub. Law 774, 81st Cong., sec. 101, E. O. 10161, Sept. 9, 1950, 15 F. R. 6105.

§ 12.1 *What this part does.* This part describes how and to what extent contracts and orders required to support either the United States Defense Program or the Canadian Defense Program may have the benefits of the priorities system of the United States. Instructions are being issued by the Canadian Government under appropriate Canadian legislation to provide for priority assistance in obtaining essential Canadian materials and services for the United States defense program.

§ 12.2 *Rated orders from the United States under the United States Defense Program, placed with Canadian suppliers.* (a) If a rated order from a person in the United States, under Part 11 of this chapter (NPA Reg. 2), is placed with a supplier located in Canada, either directly or through extension, the Canadian supplier may extend the rating received by him in order to acquire materials in the United States, to the extent permitted by the provisions of § 11.5, and subject to the restrictions of § 11.6, of NPA Reg. 2.

(b) A person located in Canada extending the rating to a supplier located in the United States, pursuant to this section, shall mark on his purchase order the prefix DO and the two digit designation appearing on the rated order received by him, and shall affix thereto the additional statement to read as follows: "Certified under NPA Reg. 3". Such certification shall be signed by him in the manner prescribed in § 11.8 (NPA Reg. 2).

(c) If a supplier located in Canada requires priority assistance to obtain materials of Canadian origin to fill a rated order under the United States Defense Program, received by him from the United States, application should be made by him to the Projects Division, Department of Trade and Commerce, Ottawa, Canada, for appropriate action.

§ 12.3 *Rated orders under the Canadian Defense Program, placed with United States suppliers.* (a) The Canadian Commercial Corporation, Canadian Arsenals Ltd., and National Research Council, on behalf of the Atomic Energy Control Board, charged with the responsibility for procurement under the Canadian Defense Program, or any person located in Canada who has an order in connection with the Canadian Defense Program calling for delivery of materials to such persons located in Canada by persons located in the United States, may make application through the Projects Division, Department of Trade and Commerce, Ottawa, Canada, to the National Production Authority, Washington 25, D. C., Ref: Reg. 3, for authority to apply to such orders the DO rat-

ings authorized by Part 11 of this chapter (NPA Reg. 2). All such orders to which such DO ratings shall be so applied shall have equal standing to the ratings applied to United States orders under Part 11 of this chapter (NPA Reg. 2).

(b) All such rated Canadian orders referred to in paragraph (a) of this section which are placed in the United States pursuant to this section shall bear the prefix DO and the two digits which may be assigned by NPA as identifying Canadian orders, together with a statement to read as follows: "Certified under NPA Reg. 3". Such certification shall be signed in the manner prescribed in § 11.8 of NPA Reg. 2.

§ 12.4 *Certifications by Canadian purchasers to suppliers in the United States.* Certifications under §§ 12.2 (b) or 12.3 (b) constitute a representation to the supplier in the United States and to the National Production Authority that the person signing such certification is either extending a rating received by him from a purchaser in the United States under the provisions of Part 11 of this chapter (NPA Reg. 2) or he has been specifically authorized by the National Production Authority under this part to apply the rating for the purposes authorized in connection with the Canadian Defense Program.

§ 12.5 *Specific authorizations required.* Under arrangements made with the Canadian Department of Trade and Commerce, if a person located in Canada receives either a DO rated order from a person located in the United States or a Canadian defense order and cannot fill such order without assistance in obtaining materials from the United States for plant improvement, expansion or construction or obtaining machine tools or other items which he will carry as capital equipment or obtaining maintenance, repair or operating supplies, such Canadian supplier may apply to the Projects Division, Department of Trade and Commerce, Ottawa, Canada, for priorities assistance. Such request for assistance will be examined by the Projects Division, and if approved by it, may be forwarded to National Production Authority, Washington 25, D. C., Ref: NPA Reg. 3 for action. Ratings for items included in this category may be applied only when so approved by the National Production Authority.

§ 12.6 *Applicability of NPA Reg. 2.* All of the provisions of Part 11 of this chapter (NPA Reg. 2) shall apply to all persons affected by the provisions of this part except to the extent where such provisions are inconsistent, in which event the provisions of this part will prevail.

This part shall take effect on November 8, 1950.

NATIONAL PRODUCTION AUTHORITY,
[SEAL] W. H. HARRISON,
Administrator.

[F. R. Doc. 50-10113; Filed, Nov. 8, 1950; 12:28 p. m.]

[NPA Order M-6]

PART 22—STEEL DISTRIBUTORS

This order is found necessary and appropriate to promote the national defense and is issued pursuant to the authority granted by section 101 of the Defense Production Act of 1950. In the formulation of this order, there has been consultation with industry representatives, including trade association representatives, and consideration has been given to their recommendations.

Sec.
22.1 What this part does.
22.2 Definitions.
22.3 Shipments during the current calendar quarter for 1950.
22.4 Allotments for steel distributors.
22.5 Rejection of orders issued pursuant to this part.
22.6 Tonnage limitation.
22.7 Item limitation.
22.8 Extension of DO ratings.
22.9 Application for adjustment or exception.
22.10 Communications.
22.11 Reports.
22.12 Violations.
22.13 List A—Industrial steel products.
22.14 List B—Merchant trade steel products.

AUTHORITY: §§ 22.01 to 22.14 issued under sec. 704, Pub. Law 774, 81st Cong. Interpret or apply secs. 101, 701, Pub. Law 774, 81st Cong., sec. 101, E. O. 10161, Sept. 9, 1950, 15 F. R. 6105.

§ 22.1 *What this part does.* This part applies particularly to steel distributors, and provides rules to assist them in obtaining supplies of steel for the carrying out of their normal functions. It requires steel producers to establish regular allotments of steel for purchase by steel distributors based upon their average monthly purchases over an established base period. It provides special rules for the extension of DO rated orders by steel distributors and specifies a tonnage limitation and item limitation for required acceptance of rated orders by them.

§ 22.2 *Definitions.* As used in this part:

(a) "Steel distributor" means a person engaged in the business of maintaining facilities and equipment for the stocking and distribution of rolled or drawn steel products for sale or resale in the form as received or after performing such operations as cutting to length, shearing to size or shape, pipe threading or sorting and grading. A person who, in connection with any sale of such steel products from his stock, bends, punches or performs any fabricating or processing operation designed to prepare steel for final use or assembly is not a steel distributor with respect to such sale. The term "steel distributor" excludes any person who purchases steel products for resale but does not take physical delivery of the material into his own stock at a location regularly maintained for such purpose.

(b) "Industrial steel products" means all products shown on § 22.13 (List A).

(c) "Merchant steel products" means all products shown on § 22.14 (List B).

(d) "Item" shall mean any steel or iron product which is different from all other steel or iron products by reason of one or more of its specifications, such as width, thickness, temper, alloy, finish or method of manufacture.

§ 22.3 *Shipments during the current calendar quarter for 1950.* Producers of steel who have accepted orders from steel distributors, including the affiliates and subsidiaries of such producers, for steel products for shipment prior to or during the remainder of the current calendar quarter ending December 31, 1950, shall endeavor to make shipment of all such unfilled orders not later than forty-five days immediately following the end of said quarter.

§ 22.4 *Allotments for steel distributors.* In order to determine the monthly allotment for each steel distributor, each steel producer shall first determine the amount of each steel product which will be available for the particular month, after making provision for production under DO rated orders and other orders which such steel producer is required to accept by specific direction of NPA. Each steel producer shall then allot to each of his steel distributor customers the same percentage of each product as may thus remain, on a proportion based on the average monthly percentage of the same product as was delivered by him to each of his steel distributor customers during the period from January 1, 1950 through September 30, 1950. A steel producer must accept orders placed by a steel distributor up to the limit of his allotment: *Provided however*, That such orders are placed not less than 45 days prior to the first day of the month for which the allotment is made. Deliveries under such allotments shall be made in addition to shipments to the same steel distributor pursuant to authorized extension of DO ratings. Orders placed under the provisions hereof must be for substantially the same product as was supplied to each such steel distributor during such period, except for minor variations in size and design. In determining the amount of the monthly allotments, adjustments may be made by a steel producer with the consent of the steel distributors involved, to provide for any abnormal situations which affect any steel products.

§ 22.5 *Rejection of orders issued pursuant to this part.* Unless otherwise specifically directed by NPA, producers of steel need not accept an order issued pursuant to this part which is received less than 45 days prior to the first day of the month in which shipment is requested.

§ 22.6 *Tonnage limitation.* Unless specifically directed by the National Production Authority, no steel distributor shall be required to accept rated orders for shipment from inventory from any one location where he maintains a stock, in any one calendar quarter for a total tonnage of each product or grade in excess of thirty percent of his average quarterly shipments thereof from such locations during the period from January 1, 1950 through September 30, 1950.

§ 22.7 *Item limitation for acceptance of rated orders.* Unless otherwise spe-

cifically directed by the National Production Authority, no steel distributor shall be required to make delivery on a rated order from warehouse stock to any one customer to any one destination, at any one time, steel products in quantities in excess of the following:

Any item of carbon steel more than 8,000 lbs.
Any item of alloy steel (except stainless) more than 5,000 lbs.
Any item of stainless steel sheet more than 2,000 lbs.
Any item of stainless bars and plates more than 1,000 lbs.
Any item of stainless tubing or pipe more than 1,000 lbs. or feet whichever is less.

In no case shall a distributor be required to make such deliveries aggregating 40,000 pounds or more at one time unless the deliveries include ten or more different items with no item to weigh more than shown in the table above.

§ 22.8 *Extension of DO ratings for industrial and merchant trade steel products.* No person may extend a DO rating to secure industrial and/or merchant trade steel products for sale or resale in the form in which such steel is received, except to replace inventory carried regularly in stock which has been delivered by him from stock under DO rated orders. However, this provision shall not prevent the extension of a DO rating to effect delivery of those steel products which are normally sold by steel producers through their own authorized steel distributors for direct shipment from the mill of such steel producers to steel consumers.

§ 22.9 *Adjustments and exceptions.* Any person affected by any provision of this order may file a request for adjustment or exception upon the ground that such provision works an unreasonable hardship upon him not suffered generally by others in the same trade or industry or that its enforcement against him would not be in the interest of National Defense. Each request shall be in writing and shall set forth all pertinent facts and the nature of the relief sought, and shall state the reasons why denial of the request could result in undue and exceptional hardship. All such requests should be addressed to National Production Authority, Washington 25, D. C. Ref: M–6.

§ 22.10 *Communications.* All communications concerning this part shall be addressed to National Production Authority, Washington 25, D. C. Ref: M–6.

§ 22.11 *Reports.* Persons subject to this part shall make such records and submit such reports to the NPA as it shall require, subject to the terms of the Federal Reports Act.

§ 22.12 *Violations.* Any person who wilfully violates any provisions of this part or any other order or regulation of NPA or wilfully conceals a material fact or furnishes false information in the course of operation under this part is guilty of a crime and upon conviction may be punished by fine or imprisonment or both. In addition, administrative action may be taken against any such person to suspend his privilege of making or receiving further deliveries of materials or using facilities under priority or allocation control and to deprive him of further priorities assistance.

§ 22.13 *List A—Industrial steel products.*

Product groups	Types		
	Carbon	Stainless	Other alloy
1. Blooms, billets, slabs, tube rounds, die blocks, sheet and tin bars	X	X	X
2. Structural shapes and piling	X	.....	X
3. Plates (universal and sheared) including skelp, prime scheduled rolling only	X	X	X
4. Rails and track accessories	X	.....	X
5. Hot rolled bars—except concrete reinforcing bars but including forged, galvanized and wrought iron bars	X	X	X
6. Concrete reinforcing bars (unfabricated)	X	.....	.....
7. Cold finished bars	X	X	X
8. Sheets and strip, hot rolled	X	X	X
9. Sheet and strip, cold reduced	X	X	X
10. Tin mill black plate, tin plate and terne plate	X	.....	.....
11. Sheets and strip, all other but not including items on schedule II	X	X	X
12. Welded tubing	X	X	X
13. Seamless tubing	X	X	X
14. Tool steel, including drill rod	X	.....	X
15. Wire rope and strand	X	.....	X

§ 22.14 *List B—Merchant trade steel products.*

CARBON AND LOW ALLOY ONLY

No. *Products*

20. Standard and line pipe, water well tubular products, and couplings[1] (includes steel and wrought iron pipe).
21. Oil country casing, tubing, drill pipe and couplings.[1]
22. Galvanized, lead coated, or painted sheet and strip (including galvanized flat sheets purchased for the manufacture of roofing and siding) formed roofing and siding (painted, black, galvanized or lead coated) valley, ridge roll and flashing, price scheduled rolling only.
23. Nails (cut and wire), fence and netting staples.
24. Wire, drawn.
25. Wire bale ties.
26. Wire (barbed and twisted) and wire fence (woven or welded).
27. Wire netting.
28. Fence posts.
29. Welded wire concrete reinforcing mesh.

This part shall take effect on November 8, 1950.

NATIONAL PRODUCTION AUTHORITY,
[SEAL] W. H. HARRISON,
Administrator.

[F. R. Doc. 50–10112; Filed, Nov. 8, 1950; 12:28 p. m.]

[1] Threaded couplings of the type normally supplied on threaded pipe by pipe producers.

TITLE 8—ALIENS AND NATIONALITY

Chapter I—Immigration and Naturalization Service, Department of Justice

Subchapter A—General Provisions

PART 90—BOARD OF IMMIGRATION APPEALS

Subchapter B—Immigration Regulations

PART 150—DEPORTATION PROCEEDINGS: INVESTIGATION AND ARREST

PART 151—DEPORTATION PROCEEDINGS: HEARING AND ADJUDICATION

PART 152—DEPORTATION PROCEEDINGS: ACTION SUBSEQUENT TO ADJUDICATION

ARREST AND DEPORTATION PROCEDURE

OCTOBER 10, 1950.

The following amendments to Chapter I, Title 8 of the Code of Federal Regulations, are hereby prescribed:

1. Subparagraph (2) of § 90.3 (a) is amended so that when taken with the introductory sentence it will read as follows:

§ 90.3 *Cases appealable to the Board of Immigration Appeals; powers of Board; finality of Board decisions.* (a) When the Commissioner, or officers designated by him, exercise the power and authority of the Attorney General delegated to them by provisions of this chapter by entering orders in proceedings under the immigration, nationality, or other laws administered by the Service, such orders shall be final except that appeals shall lie to the Board of Immigration Appeals where the orders, whatever their nature (excluding mitigation of fines or penalties), arise in:

* * * * *

(2) Deportation proceedings, except as provided in § 90.10 (b) and those cases mentioned in § 151.5 (e) of this chapter in which the decision has become final because the alien has not filed exceptions to the decision of the hearing officer;

* * * * *

2a. The headnote and paragraph (a) of § 150.3 are amended and new paragraphs (c) and (d) are added to that section to read as follows:

§ 150.3 *Voluntary departure; special procedure.* (a) Notwithstanding any other provisions of this part, (1) the officer in charge of a district or suboffice, prior to the issuance or service of a warrant of arrest or after service of such warrant, and before hearing is commenced; or (2) the hearing officer, at any time during the hearing, and before the record of hearing is submitted to the Commissioner, may exercise the authority conferred upon the Attorney General by section 19 (c) (1) of the Immigration Act of 1917, as amended (39 Stat. 889, 54 Stat. 671, 56 Stat. 1044, 62 Stat. 1206; 8 U. S. C. 155 (c) (1)), to permit an alien to depart from the United States to any country of his choice at his own expense if (1) the alien is charged with being

subject to deportation upon any ground other than those set forth in section 19 (d) of the Immigration Act of February 5, 1917, as amended, (2) the alien is willing and able to depart promptly from the United States, (3) he will apparently be admitted to the country of his destination, and (4) the prompt departure of the alien will be advantageous to the Government: *Provided*, That the Commissioner may at any time require certain classes of cases or individual cases where voluntary departure under this section has been authorized to be submitted to him for review and such action by the Commissioner shall defer the operation of any disposition previously made. Upon such review, the Commissioner may adhere to the original decision or may make any other determination which in his discretion he deems proper. In any case where the authority to permit an alien to depart from the United States is exercised under this section, the officer in charge of the district or suboffice shall determine the period of time within which the alien shall depart and the conditions under which he shall depart. The granting of voluntary departure shall serve to defer further hearing in the case, but such hearing shall be resumed if the Commissioner so directs or if the alien does not avail himself of the privilege of the voluntary departure within such period of time or authorized extension thereof and under such conditions as the officer in charge of the district or suboffice determines.

* * * * *

(c) In any case in which voluntary departure is granted pursuant to paragraph (a) of this section, the officer exercising such authority shall have the additional authority, subject to § 142.1 (a) or (b) and § 142.2 of this chapter, to authorize preexamination. Cases in which the application for preexamination is based on § 142.1 (c) of this chapter shall be referred to the Commisioner for decision.

(d) When an alien has been granted voluntary departure, with or without preexamination, in the manner provided for in paragraph (a) of this section, the alien's departure from the United States verified to the satisfaction of the officer in charge of the district or suboffice shall serve to terminate the outstanding proceeding and to cancel any outstanding delivery bond.

b. Paragraphs (b) and (d) of § 150.7 are amended to read as follows:

§ 150.7 *Application, prior to hearing, for suspension of deportation.* * * *

(b) *Prima facie eligibility not established.* If Form I–256 does not establish prima facie eligibility for suspension of deportation the case shall be referred to a hearing officer for hearing as to deportability.

* * * * *

(d) *Submission of record for hearing.* When the necessary entries have been verified and the documents required of the alien and the report of the investigating officer have been submitted, the case shall be referred to a hearing officer for hearing under the warrant of arrest.

3a. Sections 151.1, 151.2, and 151.3 are amended to read as follows:

§ 151.1 *Hearing; when to be accorded.* After the alien has been taken into custody under a warrant of arrest and has been given a reasonable period of time to arrange for the presentation of his case, including if desired, representation by counsel, the case of the alien shall be referred to an appropriate officer for hearing to determine whether the alien is subject to deportation. The alien shall be timely informed of the time and place of hearing.

§ 151.2 *Conduct of hearing*—(a) *Hearing officer; general powers.* The officer assigned to conduct the hearing under a warrant of arrest shall be referred to as the "hearing officer." He shall have authority to (1) administer oaths and affirmations, (2) issue, during the course of the hearing, any subpenas authorized by law, (3) conduct the hearing and rule upon all objections to the introduction of evidence or motions made during the course of the hearing, (4) take or cause depositions to be taken, (5) make decisions in accordance with § 151.5, (6) grant voluntary departure or preexamination pursuant to § 150.3 of this chapter, or both, and (7) take any other action consistent with applicable provisions of law and regulation.

(b) *Hearing officer; general duties.* The hearing officer shall conduct a fair and impartial hearing. He shall use his independent judgment in rendering his decision and shall not perform any duties inconsistent with the duties and responsibilities of an adjudicating officer, except as provided in this part. He shall certify that to the best of his knowledge and belief the record is a true report of everything that was stated during the course of the hearing, including oaths administered and the rulings on objections, except statements made off the record. The hearing officer shall exclude from the record any evidence that is irrelevant, immaterial, or unduly repetitious. He may, within his discretion, exclude from the record any arguments in support of objections, but, in such event, the alien or his counsel or representative may submit argument in the form of a brief to accompany the record.

(c) *Hearing officer; specific duties.* At the commencement of the hearing under the warrant of arrest, the hearing officer shall (1) place the alien under oath or affirmation, (2) enter of record a copy of the warrant of arrest and explain to the alien in simple, understandable language the nature of the charges contained therein, and (3) advise the alien, if not represented by counsel or other qualified representative, that he may be so represented if he desires and require him to state then and there for the record whether he desires such representation. Except as provided in paragraph (f) of this section, the hearing officer shall thereafter present all available evidence, including the interrogation of the alien and all witnesses presented, concerning (1) alienage, (2) date, place and manner of entry into the United States, (3) grounds for deportation, (4) factors bearing upon statutory eligibility for discretionary relief, and (5) such other information as may be pertinent to the issue of the case: *Provided*, That the alien need not be interrogated in any case in which the facts as to such matters are stipulated pursuant to § 151.3 (d). The hearing officer shall, at such time during the hearing under the warrant of arrest as he deems appropriate, advise the alien concerning application for the privilege of departure in lieu of deportation or for suspension of deportation under the provisions of section 19 (c) of the Immigration Act of 1917, as amended, in all cases except (1) those in which the alien is charged with being subject to deportation upon one of the grounds mentioned in section 19 (d) of the said act and (2) those in which the alien was admitted to the United States as a nonimmigrant visitor under section 201 of the United States Information and Education Exchange Act of 1948 (62 Stat. 7; 22 U. S. C. 1446; Pub. Law 402, 80th Cong.).

(d) *Hearing officer; additional charges.* If, during the hearing, it develops that there exists grounds in addition to those stated in the warrant of arrest why the alien is subject to deportation, the hearing officer may lodge additional charges against the alien and shall develop evidence upon such charges in like manner as on the charges contained in the warrant of arrest. When additional charges are lodged against the alien, the hearing officer shall explain these charges to the alien in simple, understandable language and shall advise him, if he is not represented by counsel or other qualified representative, that he may be so represented if he desires, and require the alien to state then and there for the record whether he desires such representation. The hearing officer shall also inform the alien that he may have a reasonable period of time within which to meet the additional charges if the alien so desires and require the alien to state then and there whether he desires such additional time.

(e) *Withdrawal and substitution of hearing officer.* No person shall serve as a hearing officer in any case if he has engaged in the performance of investigative or other enforcement functions in the same case. The hearing officer assigned to conduct the hearing in any case may at any time withdraw if he deems himself disqualified. If a hearing officer becomes unavailable to complete his duties within a reasonable time in connection with any case, another hearing officer shall be assigned to complete the case. In such event the new hearing officer shall first familiarize himself with the case and shall state for the record that he has done so.

(f) *Assignment of examining officer in addition to hearing officer; duties of examining officer.* An officer other than a hearing officer shall be assigned by the officer in charge of the district in which the hearing is to be held to act at the hearing as the "examining officer" in any case in which the warrant of arrest contains a charge, or an additional charge is lodged during the course of the hearing, that the alien is subject to deportation under (1) the act of October 16, 1918, as amended (40 Stat. 1012, 41 Stat. 1008, 54 Stat. 673, Public Law 831, 81st Congress; 8 U. S. C. 137); (2) the act of

February 9, 1909, as amended (35 Stat. 614, 42 Stat. 596; 21 U. S. C. 171, 174–175); (3) the act of February 18, 1931, as amended (46 Stat. 1171, 54 Stat. 673; 8 U. S. C. 156a); or (4) so much of section 19 of the Immigration Act of 1917, as amended (39 Stat. 889–890; 54 Stat. 671–673, 56 Stat. 1044; 8 U. S. C. 155) as relates to criminals, prostitutes, procurers or other immoral persons, anarchists, subversives and mental defectives; or in any other case, in the discretion of the officer in charge of the district. The examining officer shall in such cases conduct the interrogation of the alien and of the witnesses on behalf of the Government and the examination or cross-examination of the alien's witnesses, present all available evidence bearing upon the alien's liability to deportation, lodge such additional charges as he may find to be applicable, and, if the alien has applied for relief from deportation, inquire thoroughly into the alien's statutory eligibility for such relief. The hearing officer in such cases shall exercise all his regular functions not herein assigned to the examining officer, and he may, in addition, take such part in the interrogation of the alien and witnesses as he may deem necessary to assure that a full and proper hearing is held.

(g) *Representation by counsel; waiver.* The alien shall be permitted to obtain counsel or other representative who has qualified under Part 95 of this chapter. If such counsel or representative be selected, he shall be permitted to be present during the entire hearing, to offer evidence to meet any evidence adduced against the alien and to cross-examine witnesses. The alien's attorney or representative shall be permitted to state his objections succinctly, and they shall be entered on the record. If representation be waived the alien shall be permitted to offer evidence to meet any evidence adduced against him, to cross-examine witnesses, and to make objections which shall be entered on the record.

(h) *Interpreters.* If the services of an interpreter are found necessary in the conduct of the hearing, the hearing officer, if qualified, may act as interpreter, or he may request the officer in charge of the district or suboffice to furnish an interpreter, who shall be sworn to interpret and translate accurately, unless the interpreter is an employee of the Service.

(i) *Continuances.* The hearing officer may grant continuances of the hearing as may be reasonable at his own instance or upon motion of the alien or his counsel or representative. A continuance of the hearing for the purpose of allowing the alien to obtain representation shall not be granted more than once, unless sufficient cause for the granting of more time is shown.

(j) *Consolidation.* When two or more cases pending in the same district involve a common question of law or fact, the officer in charge of the district or suboffice may order a joint hearing of any or all matters and issues common to the cases; he may order all such hearings consolidated; and he may make such orders concerning proceedings therein as may tend to avoid unnecessary costs or delay.

§ 151.3 *Contents of record: evidence*—(a) *Record.* The transcript of testimony and exhibits, together with all written motions and other papers and requests filed in the proceeding, shall constitute the record for the decision in the case.

(b) *Use of statements made during course of investigation.* The hearing officer may enter of record any written or recorded statement or satisfactory evidence of any admission made by the alien or any other person during the investigation. If objection thereto is made by the alien or his counsel or representative, the reasons for the objection, as well as the ruling thereon by the hearing officer, shall be made a part of the record.

(c) *Affidavit.* In cases in which an affidavit in narrative form has been made by an alien prior to the issuance of a warrant of arrest as provided in § 150.1 (c) of this chapter and such affidavit satisfactorily establishes the facts necessary for determination as to deportability, the hearing officer may enter the affidavit as an exhibit of record and it may be used as the basis for the decision in the case.

(d) *Stipulation.* When any facts or other matters at issue in the proceeding have been stipulated in writing prior to hearing, upon agreement between an officer of the Service other than a hearing officer and the alien or his counsel or representative, and the stipulation has been signed by all of these persons, the hearing officer may enter the stipulation as an exhibit of record. Nothing in this paragraph shall preclude the concession of any fact by either side during a proceeding under this part.

(e) *Application for discretionary relief.* At any time during the hearing under the warrant of arrest the alien may apply for suspension of deportation, or for the privilege of departing from the United States at his own expense in lieu of deportation, or for the privilege of departing from the United States at his own expense in lieu of deportation with the additional privilege of preexamination. He may apply in the alternative for more than one of these three forms of relief. The burden of establishing that the alien meets the statutory requirements precedent to the exercise of discretionary relief shall be upon the alien. In addition, he may submit any written or oral material which he believes should be considered in the exercise of discretion by the Commissioner.

b. Section 151.4, *Proposals and arguments upon conclusion of hearing,* is revoked.

c. Sections 151.5, 151.6, and 151.7 are amended to read as follows:

§ 151.5 *Decision*—(a) *Preparation by hearing officer of written decision.* Except as provided in paragraph (d) of this section, the hearing officer shall, as soon as practicable after the conclusion of the hearing, prepare in writing a decision, signed by him, which shall set forth a summary of the evidence adduced and his findings of fact and conclusions of law as to deportability. If the alien has applied for relief from deportation, the decision shall also contain a separate determination as to whether or not the hearing officer is satisfied, on the basis of the evidence presented, as to the alien's statutory eligibility for the relief requested. The decision shall be concluded with a statement of the hearing officer's disposition of the case which shall be (1) that the alien be deported, or (2) that the warrant of arrest be cancelled, or (3) that the alien be found to have established statutory eligibility for suspension of deportation, or (4) that the alien be granted voluntary departure at his own expense in lieu of deportation, or (5) that the alien be granted voluntary departure at his own expense in lieu of deportation with the additional privilege of preexamination, or (6) that such other action be taken in the proceedings as may be required for the appropriate determination of the same. The hearing officer shall have no authority to exercise the Attorney General's powers under section 19 (c) (2) of the Immigration Act of February 5, 1917, as amended, or under the 7th proviso to section 3 of that act.

(b) *Notice of decision.* The decision shall be made part of the record of the case. The hearing officer shall cause a signed copy of such decision to be served either in person or by registered mail on the alien or his counsel or representative, together with a notice advising him of his right to file exceptions as provided in paragraph (c) of this section.

(c) *Exceptions to written decision.* The alien or his counsel or representative shall be allowed five business days (which period may be extended by the hearing officer upon a showing of good cause that more time is necessary) in which to submit to the officer in charge of the district or suboffice, for consideration by the Commissioner, exceptions to the hearing officer's decision, which may be accompanied by supporting argument or brief. The alien or his counsel or representative may enter a written waiver of the right herein specified to file exceptions to the decision of the hearing officer.

(d) *Oral decision and exceptions thereto.* In any case which he deems appropriate, the hearing officer may, at the conclusion of the hearing, state for the record in the presence of the alien or his counsel or representative, a brief summary of the evidence, findings of fact and conclusions of law, and his disposition of the case in the manner provided in paragraph (a) of this section. When the hearing officer enters such oral decision, he shall then require the alien or his counsel or representative to state for the record whether the alien desires to take exceptions to any part thereof. If it is desired to file exceptions, the alien or his counsel or representative shall be afforded the same opportunities accorded him by paragraph (c) of this section, except that the time provided for shall commence to run from the date on which the hearing officer orally states his decision.

(e) *Finality of decision.* The hearing officer's decision described in paragraphs

(a) and (d) of this section shall be final except:

(1) When any charge in the warrant of arrest or any additional lodged charge is predicated upon any ground of deportability set forth in section 19 (d) of the Immigration Act of February 5, 1917, as amended; or

(2) When the hearing officer determines that the alien be found to have established statutory eligibility for suspension of deportation and the hearing officer provides for no other disposition of the case; or

(3) When the hearing officer determines that the warrant of arrest shall be cancelled; or

(4) When the alien or his counsel or representative files exceptions to the hearing officer's decision in conformity with paragraph (c) or (d) of this section.

Every case within subparagraphs (1) to (4), inclusive, shall be referred promptly to the Commissioner and the hearing officer's decision shall be final only upon approval by the Commissioner, subject, however, to appeal to the Board of Immigration Appeals as provided in § 90.3 of this chapter. In considering such cases, the Commissioner may enter any order which he deems appropriate for the disposition of the case. Notwithstanding the provisions of this section, the Commissioner may at any time require certain classes of cases or individual cases to be submitted to him for review, and such action by the Commissioner shall defer the operation of any disposition previously made. Upon such review, the Commissioner may adhere to the original decision or may make any other determination which in his discretion he deems appropriate. In any case required by the Commissioner to be submitted to him for review, the officer in charge of the district or suboffice shall cause written notice to be served promptly upon the alien informing him of the forwarding of the record to the Commissioner, and of his right to make such representations to the Commissioner as he may desire, including oral argument as provided in § 151.8, and/or the filing of exceptions and brief in support thereof as provided in paragraph (c) of this section.

§ 151.6 *Reopening of hearing* — (a) *Execution.* At any time prior to the forwarding of the record of hearing to the Commissioner, the hearing officer may at his own instance direct that the case be reopened for proper cause, or he may direct a reopening upon request of the alien or his counsel or representative. Such request shall be in writing, shall state the new facts to be proved, and shall be supported by affidavits or other evidentiary material. A request for reopening shall be filed with the officer in charge of the district or suboffice who shall refer it to the hearing officer.

(b) *Denial.* The denial of an alien's motion to reopen any case in which the decision of the hearing officer has become final as provided in § 151.5 (e) shall be in writing, and a copy thereof shall be served on the alien or his counsel or representative. Upon the denial of such a motion the entire record, including the motion papers and the order denying the motion, shall be forwarded promptly to the Commissioner for review and the Commissioner may enter any order in the case which he deems appropriate.

§ 151.7 *Submission of record to the Commissioner.* In cases in which the decision of the hearing officer requires the approval of the Commissioner as provided in § 151.5 (e), the entire record of hearing, upon completion, shall be forwarded promptly to the Commissioner by the officer in charge of the district or suboffice (a) upon receipt of exceptions as provided in § 151.5, or (b) upon expiration of the time allowed therefor, or (c) upon receipt of a waiver from the alien or his counsel or representative of the right to file exceptions to the decision of the hearing officer.

d. Paragraph (b) of § 151.9, *Decision by the Commissioner*, is amended to read as follows:

(b) *Reopening.* The Comissioner may for proper cause direct a reopening of any case which is before him for consideration. Hearings ordered reopened by the Commissioner shall be conducted by a hearing officer, who need not be the hearing officer originally presiding in the case.

e. Section 151.10, *Applicability of regulations*, is revoked.

4. Paragraph (a) of § 152.2, *Issuance and execution of warrant of deportation*, is amended to read as follows:

(a) *Issuance.* The officer in charge of the appropriate district shall issue a warrant of deportation in any case in which an order of deportation has become final.

The regulations in effect immediately prior to the date of the publication of this order in the FEDERAL REGISTER shall so far as pertinent be applicable to all cases in which proceedings were instituted on or after September 11, 1946, and were concluded prior to September 27, 1950. In any case in which hearings have been commenced but not concluded prior to the date of the publication of this order in the FEDERAL REGISTER, the proceedings shall continue in conformity with the regulations as amended by this order; and in such proceedings the hearing examiner or other officer previously designated and appointed under § 151.2, Chapter I, Title 8, Code of Federal Regulations, or under other applicable provisions of law and regulation, may continue to hear the case until it is concluded, and exercise the duties of hearing officer, as described in the said § 151.2.

This order shall become effective on the date of its publication in the FEDERAL REGISTER. The Supplemental Appropriation Act, 1951, approved September 27, 1950 (Public Law 843, 81st Congress), provides that thereafter proceedings under law relating to the exclusion or expulsion of aliens shall be without regard to the provisions of sections 5, 7, and 8 of the Administrative Procedure Act. The general purpose of the amendments made by this order is to conform the pertinent immigration regulations to the requirements of the said Supplemental Appropriation Act, 1951. Compliance with the provisions of section 4 of the Administrative Procedure Act (60 Stat. 238; 5 U. S. C. 1003) relating to notice of proposed rule making and delayed effective date is impracticable and contrary to the public interest in this instance for the reason that the changes in immigration proceedings made necessary by the said appropriation act should be made effective as speedily as possible.

(Sec. 23, 39 Stat. 892, as amended, sec. 24, 43 Stat. 166, sec. 37, 54 Stat. 675; 8 U. S. C. and Sup., 102, 222, 458)

A. R. MACKEY,
Acting Commissioner,
Immigration and Naturalization.

Approved: November 8, 1950.

J. HOWARD MCGRATH,
Attorney General.

[F. R. Doc. 50-10115; Filed, Nov. 9, 1950; 8:54 a. m.]

TITLE 9—ANIMALS AND ANIMAL PRODUCTS

Chapter I—Bureau of Animal Industry, Department of Agriculture

Subchapter D—Exportation and Importation of Animals and Animal Products

[BAI Order 379, Amdt. 8]

PART 92—IMPORTATION OF CERTAIN ANIMALS AND POULTRY AND CERTAIN ANIMAL AND POULTRY PRODUCTS (EXCEPT FROM MEXICO)

MISCELLANEOUS AMENDMENTS

On July 29, 1950 there was published in the FEDERAL REGISTER (15 F. R. 4896) a notice of the proposed amendment of the regulations relating to importation of certain animals and poultry into the United States (except from Mexico) (9 CFR Part 92, as amended) under sections 6, 7, 8, and 10 of the act of August 30, 1890, as amended (21 U. S. C. 102–105) and section 2 of the act of February 2, 1903, as amended (21 U. S. C. 111). After due consideration of all relevant material submitted pursuant to the notice and under the authority of said statutory provisions, said regulations are hereby amended as follows:

1. The title of Part 92 is changed to read "Part 92—Importation of Certain Animals and Poultry and Certain Animal and Poultry Products (Except from Mexico)."

2. Section 92.4 is amended by changing its heading to read *"Permits for ruminants, swine, and poultry and for animal semen"*; by redesignating the present provisions of the section as paragraph (a), headed *"Ruminants, swine, and poultry"*; and by adding to the section a new paragraph (b) to read as follows:

(b) *Animal semen.* (1) No animal semen may be imported from any part of the world unless the importer first obtains a permit from the Bureau. However, the Chief of Bureau, when he finds that such action may be taken without endangering the livestock industry of the United States, may authorize the importation of animal semen from Canada without such permit. No permit will be issued for the importation of semen derived from domestic ruminants

or swine in any country where foot-and-mouth disease or rinderpest has been determined to exist.

(2) The permit will be in two sections, one for presentation to the American consulate in the district which includes the port of shipment and the other for presentation to the collector of customs at the port of entry specified therein. The semen will be received at the specified port on the date prescribed for its arrival or at any time during three weeks immediately following, after which time the permit shall be void.

3. Section 92.8 is amended to read as follows:

§ 92.8 *Inspection at port of entry.* Inspection shall be made at the port of entry of all horses, ruminants, swine and poultry offered for importation from any part of the world (except Mexico), except as provided in §§ 92.24, 92.25, and 92.29. However, the Chief of Bureau, when he finds that such action may be taken without endangering the poultry industry of the United States, may waive inspection at the port of entry or provide for inspection at some other point with respect to importations from Canada of eggs for hatching, newly hatched poultry and poultry consigned for immediate slaughter. All animals found to be free from communicable disease and not to have been exposed thereto shall be admitted subject to the other provisions in this part. Animals found to be affected with a communicable disease or to have been exposed thereto, shall be refused entry. Ruminants and swine refused entry shall be handled thereafter in accordance with the provisions of section 8 of the act of August 30, 1890, or quarantined or otherwise disposed of as the Chief of Bureau may direct. Horses and poultry refused entry, unless exported within a time fixed in each case by the Chief of Bureau, shall be disposed of as said Chief may direct. Such portions of the transporting vessel, and of its cargo, as have been exposed to any such animals or their emanations shall be disinfected in such manner as may be considered necessary by the inspector in charge at the port of entry, before the cargo is allowed to land.

4. Section 92.19 is amended by deleting the phrase "cattle, sheep, goats, and swine" wherever it appears therein and substituting therefor the phrase "cattle, sheep, goats, swine, and poultry."

5. Paragraph (c) of § 92.20 is amended to read as follows:

(c) *Brucellosis test certificates.* Cattle 6 months or older, offered for importation from Canada—except steers, spayed heifers, and all cattle for immediate slaughter—shall be accompanied by a certificate issued or endorsed by a salaried veterinarian of the Canadian Government showing them to have been tested for brucellosis (Bang's disease), with negative results, within 30 days preceding their being offered for entry. However, such cattle need not have been so tested if they are accompanied by a certificate, similarly issued or endorsed, showing that they were officially vaccinated as calves 6 to 8 months of age, in accordance with Canadian regulations, within 22 months prior to their being offered for entry. The certificate accompanying such vaccinated cattle shall also show the date of vaccination of each animal.

6. Section 92.25 is amended by deleting therefrom the phrase "cattle, sheep, and swine" and substituting therefor the phrase "cattle, sheep, swine, and poultry"; by deleting the word "or" preceding the figures "§ 92.22 (a)"; and by adding after such figures the phrase "or § 92.25a".

7. Section 92.25a is amended to read as follows:

§ 92.25a *Poultry from Canada.* All poultry offered for importation from Canada shall be accompanied by a certificate issued or endorsed by a salaried veterinarian of the Canadian Government showing that such poultry have been inspected on the premises of origin and that, as far as it has been possible to determine, such poultry are free of evidence of any communicable disease or exposure thereto. However, the Chief of Bureau, when he finds that such action may be taken without endangering the poultry industry of the United States, may authorize the importation from Canada, without such certification, of eggs for hatching, newly hatched poultry, and poultry consigned for immediate slaughter.

(Secs. 6, 7, 8, 10, 26 Stat. 416, 417, as amended, sec. 2, 32 Stat. 792, as amended; 21 U. S. C. 102–105, 111)

The foregoing amendments shall be effective 30 days after their publication in the FEDERAL REGISTER.

Done at Washington, D. C., this 6th day of November 1950.

[SEAL] CHARLES F. BRANNAN,
Secretary of Agriculture.

[F. R. Doc. 50–10955; Filed, Nov. 9, 1950; 8:48 a. m.]

TITLE 15—COMMERCE AND FOREIGN TRADE

Chapter III—Bureau of Foreign and Domestic Commerce, Department of Commerce

Subchapter C—Office of International Trade

[5th Gen. Rev. of Export Regs., Amdt. P. L. 23] [1]

PART 399—POSITIVE LIST OF COMMODITIES AND RELATED MATTERS

MISCELLANEOUS AMENDMENTS

Section 399.1 *Appendix A Positive List of Commodities* is amended in the following particulars:

1. The following commodities are added to the Positive List:

Dept. of Commerce Schedule B No.	Commodity	Unit	Processing code and related commodity group	GLV dollar value limits	Validated license required
707918	Electrical machinery, and apparatus: Radio apparatus: Radio receiving-set components: Television picture tubes, having short-image life phosphor, types P-1 and P-4.[1] [2]	.....	RARA 51	None	R

[1] These commodities prior to October 10, 1950, required export authorization from the Department of State. All outstanding licenses for the exportation of these commodities issued by the Department of State remain valid until they expire or are revoked.

[2] Other television picture tubes classified under Schedule B No. 707918 require export authorization from the Department of State.

2. Certain entries on the Positive List are amended by changing commodity descriptions, as set forth below:

Dept. of Commerce Schedule B No.	Commodity	Unit	Processing code and related commodity group	GLV dollar value limits	Validated license required
722700	Construction, excavating, and conveying machinery: Tractor attachments as follows: bulldozers, angledozers, trail builders, tree dozers, logging arches and sulkies, and brush cutters or rakes (report snowplow and ripper or rooter attachments in 722810; crane, pipelayer, excavating and loading, and dragline and clamshell attachments, in 723100; winches and controls for contractors' wheel-type tractors in 722100; winches and controls for track-laying tractors in 788901; and winches and controls for wheel-type tractors in 788905).[3]	No....	CONS	None	R
722810	Other road and airport machines: bituminous paving plants; bituminous distributors; bituminous mixers; bituminous spreaders; snowplow, and ripper or rooter attachments for tractors.[4]	No....	CONS	None	R
722890	Parts and accessories for the following road and airport machines: concrete mixers and pavers; self-propelled road graders; self-propelled scrapers, self-loading; bituminous paving plants; bituminous distributors; bituminous mixers; bituminous spreaders; self-propelled road rollers, steel-tired types; and parts and accessories for the following tractor attachments: bulldozers; angledozers; trail builders; brush cutters or rakes; tree dozers; logging arches and sulkies; snowplows; and rippers or rooters.[5]	.....	CONS	100	R

[3] This amendment does not change the commodity coverage or the license requirements for tractor attachments but classifies specific attachments in lieu of the former general classification "and other similar tractor equipment."

[4] The effect of this amendment is to add to the Positive List, as R commodities, snowplow attachments for tractors under 65 horsepower, and all ripper or rooter attachments.

[5] The effect of this amendment is to add to the Positive List, as R commodities, parts and accessories for snowplow attachments for tractors under 65 horsepower and parts and accessories for all ripper or rooter attachments.

[1] This amendment was published in Current Export Bulletin No. 501 dated November 2, 1950.

Dept. of Commerce Schedule B No.	Commodity	Unit	Processing code and related commodity group	GLV dollar value limits	Validated license required
	Construction, excavating, and conveying machinery—Continued				
722100	Batcher outfits; concrete block machines, high-speed; contractors' off-the-road trucks; contractors' off-the-road wagons and trailers; pile drivers; pile hammers; crane, pipe layer, excavating and loading, dragline and clamshell attachments for tractors and parts therefor.[6]	.....	CONS	100	R
722300	Parts and accessories for contractors' wheel-type tractors (including control and winch attachments, and parts therefor).[7]	.....	CONS	100	R
	Agricultural machinery and implements:				
788501	Parts and accessories for track-laying tractors (including control and winch attachments, and parts therefor).[8]	.....	FARM	100	R
788505	Parts and accessories for wheel-type tractors, except garden tractors (including control and winch attachments, and parts therefor).[8]	.....	FARM	500	R

[6] The effect of this amendment is to add to the Positive List, as R commodities, crane, pipe-layer, excavating and loading, dragline and clamshell attachments for tractors, and parts therefor.
[7] The effect of this amendment is to add to the Positive List, as R commodities, accessories for contractors' wheel-type tractors (including control and winch attachments and parts therefor).
[8] This amendment does not change the commodity coverage or the license requirements. The above revised entry indicates that control and winch attachments and parts therefor are included on the Positive List under this Schedule B number.

3. Certain Schedule B numbers of entries on the Positive List are changed as set forth below. The changes conform with Census Bulletin P. B. 169 B–I and II issued by the Bureau of the Census September 21, 1950.

Dept. of Commerce Schedule B No.	Commodity	Unit	Processing code and related commodity group	GLV dollar value limits	Validated license required
	Machine tools:				
	Power-driven metalworking machine tools (nonportable), and parts:				
[9] 740308	Gun-boring lathes; cartridge-case trimming lathes; cartridge-head finishing lathes; and shell lathes.	No....	TOOL	None	RO
[9] 740311	Boring lathes, except gun-boring; crankshaft turning lathes; and relieving lathes.	No....	TOOL	None	RO
[9] 740604	Shell tappers	No....	TOOL	500	RO
[9] 740607	Pipe threading and cutting-off machines, 8-inch and over; and pipe and nipple-threading machines, 8-inch and over.	No....	TOOL	None	R
[9] 740607	Tapping machines, vertical multiple spindle, adjustable joint.	No....	TOOL	500	RO
[10] 744318	Rifle-working machines	No....	TOOL	750	RO
	Industrial chemicals (exclusive of medicinal chemicals U. S. P. & N. F.):				
	Cases as specified: [11]				
[12] 839520	Genetrons	.....	SALT	25	RO
[12] 839750	Hydrogen peroxide or dioxide (over 50% strength)	Lb....	SALT 64	None	RO

[9] Formerly 740309.
[10] Formerly 740605.
[11] Formerly in 744319. Note that other commodities under Schedule B No. 744319 are on the Positive List.
[12] Many of the gases classified under Schedule B No. 839520 require export authorization from the Department of State. See § 370.5.
[13] Formerly 839500.
[14] Formerly in 839000. Note that other commodities under Schedule B No. 839000 are on the Positive List.

4. The following commodities are changed from R to RO commodities. Accordingly, the entry therefor on the Positive List is amended to read as follows:

Dept. of Commerce Schedule B No.	Commodity	Unit	Processing code and related commodity group	GLV dollar value limits	Validated license required
	Mining, well, and pumping machinery:				
	Mining and quarrying machinery:				
731200	Churn drills, including drill rigs	.....	CONS	None	RO

Shipments of any commodities removed from general license to Country Group R or Country Group O destinations as a result of the changes set forth in Parts 2 and 4 of this amendment, which were on dock, on lighter, laden aboard an exporting carrier, or in transit to a port of exit pursuant to actual orders for export prior to the effective date of this amendment may be exported under the previous general license provisions.

(Sec. 3, 63 Stat. 7; 50 U. S. C. App. Sup., 2023. E. O. 9630, Sept. 27, 1945, 10 F. R. 12245, 3 CFR, 1945 Supp.; E. O. 9919, Jan. 3, 1948, 13 F. R. 59, 3 CFR, 1948 Supp.)

This amendment shall become effective as of November 2, 1950, except that Part 1 thereof shall become effective as of October 10, 1950.

LORING K. MACY,
Deputy Director,
Office of International Trade.

[P. R. Doc. 50-10062; Filed, Nov. 9, 1950; 8:50 a. m.]

TITLE 26—INTERNAL REVENUE

Chapter II—The Tax Court of the United States

PART 701—RULES OF PRACTICE

CHIEF JUDGE

Pursuant to the provisions of section 32, act of June 25, 1948 (Public Law 773), as amended by section 127, Public Law 72, approved May 24, 1949, applicable to The Tax Court of the United States, references to Chairman in § 701.30 and to Presiding Judge in § 701.48 should be changed to read Chief Judge.

Effective: October 30, 1950.

(53 Stat. 160, as amended; 26 U. S. C. 1111)

Dated: November 3, 1950.

By the Court.

[SEAL] JOHN W. KERN,
Chief Judge, The Tax Court of the United States.

[F. R. Doc. 50-10054; Filed, Nov. 9, 1950; 8:48 a. m.]

TITLE 38—PENSIONS, BONUSES, AND VETERANS' RELIEF

Chapter I—Veterans' Administration

PART 1—GENERAL PROVISIONS

RECORDS AND PAPERS

1. In § 1.525, paragraphs (a), (b), and (c) are amended to read as follows:

§ 1.525 *Inspection of records by or disclosure of information to recognized representatives of organizations.* (a) (1) The accredited representatives of any of the organizations recognized under section 200, Public No. 844, 74th Congress (act of June 29, 1936), holding appropriate power of attorney may inspect the Veterans' Administration file of any claimant upon the condition that only such information contained therein as may be properly disclosed under §§ 1.500 through 1.526 will be disclosed by him to the claimant or, if the claimant is incompetent, to his legally constituted fiduciary. All other information in the file shall be treated as confidential and will be used only in determining the status of the cases inspected or in connection with the presentation to officials of the Veterans' Administration of the claim of the claimant. The managers of field stations and the directors of the services concerned in central office will each designate a responsible officer to whom requests for all files must be made, except that managers of district offices and centers with district office activities will designate two responsible officials, recommended by the service directors concerned, one responsible for claims and allied folders and the other for insurance files.

(2) When power of attorney does not obtain, the accredited representative will explain to the designated officer of the Veterans' Administration the reason for requesting information from the file, and the information will be made available only when in the opinion of the designated officer it is justified; in no circumstances will such representatives be allowed to inspect the file; in such cases a contact report will be made out and attached to the case, outlining the reasons which justify the verbal or written release of the information to the accredited representative. In any case where there is an unrevoked power of attorney, no persons or organizations

other than the one named in the power of attorney shall be afforded information from the file; and when any claimant has filed notice with the Veterans' Administration that he does not want his file inspected, such file will not be made available for inspection.

(b) (1) Inspection of folders by accredited representatives shall be in space assigned for such inspection; however, where space or other local problems make it impractical to assign space for this purpose, the assistant administrator or comparable official concerned in central office or the manager at the particular office affected may permit inspection of folders at the desks of the accredited representatives in the office(s) which they regularly occupy. Managers and directors of the services concerned in central office will designate and assign, in or near the space occupied by accredited representatives, one or more employees to whom the folders will be charged and who will at all times know and have a record of the location of such folders during the time they are made available for inspection by the accredited representatives.

(2) When it is necessary that folders which are being inspected by accredited representatives be carried elsewhere within the installation for purposes of official discussion or presentation of claims, such folders shall be carried by Veterans' Administration employees; however, where it is not feasible to use Veterans' Administration employees for these purposes, the manager or the director of the service concerned in central office may permit the folders to be carried by the accredited representatives, provided the designated Veterans' Administration employee to whom the folders are charged is at all times kept informed of their location.

(3) No person other than a Veterans' Administration employee in the performance of his official duties may inspect an insurance file or receive information therefrom except:

(i) An authorized representative of the insured, or, after maturity of the insurance by death of the insured, of the beneficiary, shall, if holding valid power of attorney, be permitted to inspect the claims file containing the basic papers concerning the insurance for the purpose of assisting the insured or beneficiary in perfecting a claim for any benefit under the policy.

(ii) In the absence of a claim for benefits under the insurance policy, the insured or, after the maturity of the insurance by death of the insured, the beneficiary may authorize the release to a third person of such insurance information as the insured or the beneficiary would be entitled to receive, provided there is submitted to the Veterans' Administration a specific authorization in writing for this purpose.

(iii) Unless otherwise authorized by the insured or the beneficiary, as the case may be, such authorized representative shall not release information as to designated beneficiary to anyone other than the insured or to the beneficiary after death of the insured. Otherwise, information in the insurance file shall be subject to the provisions of §§ 1.500 through 1.526.

(4) Clinical records and medical files, including files for outpatient treatment, may be inspected by accredited representatives only to the extent such records or parts thereof are incorporated in the claims folder, or are made available to Veterans' Administration personnel in the adjudication of the claim. Records or data in clinical or medical files which are not incorporated in the claims folder or which are not made available to Veterans' Administration personnel for adjudication purposes will not be inspected by anyone other than those employees of the Veterans' Administration whose duties require same for the purpose of clinical diagnosis or medical treatment.

(5) Under no circumstances shall any paper be removed from a file, except by a Veterans' Administration employee, for purpose of having an authorized photostat made. Copying of material in a file shall not be permitted except in connection with the performance of authorized functions under the power of attorney.

(6) In any case involving litigation against the Government, whether contemplated or initiated, inspection, subject to the foregoing, shall be within the discretion of the solicitor or chief attorney, except that in insurance suits under the World War Veterans Act, 1924, as amended, or the National Service Life Insurance Act, 1940, as amended, inspection shall be within the discretion of the official having jurisdiction of the claim. Files in such cases may be released to the Department of Justice, but close liaison will be maintained to insure their return intact upon termination of the litigation.

(c) Managers and the directors of the services concerned in central office will be responsible for the administrative compliance with and accomplishment of the foregoing within their jurisdiction, and any violations of the prescribed conditions for inspection of files or release of information therefrom will be brought to the immediate attention of the Administrator.

* * * * *

2. In § 1.526, paragraph (c) is amended to read as follows:

§ 1.526 *Copies of records and papers.* * * *

(c) Where only one or a relatively few copies of papers from a claimant's file are desired by the veteran or a dependent, for use in securing one of the governmental benefits or in connection with a claim for commercial insurance, the workmen's compensation, or similar benefit, the necessary copies may, in the discretion of the manager or staff member in central office having jurisdiction in the matter, be supplied without charge.

* * * * *

(Sec. 5, 43 Stat. 608, as amended, sec. 2, 46 Stat. 1016; 38 U. S. C. and Sup., 11a, 426. Interpret or apply sec. 30, 43 Stat. 615, as amended; 38 U. S. C. 456)

This regulation is effective November 10, 1950.

[SEAL] O. W. CLARK,
Deputy Administrator.

[F. R. Doc. 50-9984; Filed, Nov. 9, 1950; 8:45 a. m.]

PART 3—VETERANS' CLAIMS

SPECIALLY ADAPTED HOUSING WHICH CERTAIN VETERANS REQUIRE BY REASON OF SERVICE CONNECTED DISABILITY; CERTIFICATE OF MEDICAL FEASIBILITY

In § 3.1501 (b) (5), subdivisions (ii) and (iii) are amended to read as follows:

§ 3.1501 *Assistance to certain veterans in acquiring specially adapted housing which they require by reason of their service-connected disabilities.* * * *

(b) *Processing of applications.* * * *

(5) *Conditions precedent to execution of certificate of medical feasibility.* * * *

(ii) When the applicant is hospitalized in a Veterans' Administration hospital or center and is suffering from a service-connected disability causing loss or loss of use of both lower extremities, the manager will appoint a board, which shall in all instances include the chief of professional services or the chief of the service responsible for the patient, and at least two other members, for the purpose of examining the applicant and for the preparation of VA Form 4555b. In all cases, membership of the board will include a neurologist and a physiatrist. In addition, a urologist will be included on the board in all paraplegic cases and in other cases where indicated. Any other specialists deemed necessary may also be included.

(iii) When the applicant with the condition set forth in subdivision (ii) of this subparagraph is not currently undergoing hospitalization, the examination and the report referred to above, will be accomplished at the nearest Veterans' Administration hospital, center, or regional office having the required specialistic professional personnel available as specified in subdivision (ii) of this subparagraph. In all instances, the board must consist of a minimum of three physicians. (Instruction 1-C, Public Law 702, 80th Congress)

* * * * *

(Sec. 5, 43 Stat. 608, as amended, sec. 2, 46 Stat. 1016, sec. 7, 48 Stat. 9; 38 U. S. C. and Sup. 11a, 426, 707. Interprets or applies sec. 1, 48 Stat. 8, as amended; 38 U. S. C. and Sup. 701)

This regulation is effective November 10, 1950.

[SEAL] O. W. CLARK,
Deputy Administrator.

[F. R. Doc. 50-9983; Filed, Nov. 9, 1950; 8:45 a. m.]

PART 3—VETERANS' CLAIMS

PART 4—DEPENDENTS' AND BENEFICIARIES' CLAIMS

MISCELLANEOUS AMENDMENTS

1. In Part 3, § 3.228, paragraphs (h) and (i) are amended, and a new paragraph (i-1) has been added, as follows:

§ 3.228 *Computation of annual income for the purposes of Veterans Regulation 1 (a), Part III (38 U. S. C. ch. 12), or section 1 (c) of Public No. 198, 76th Congress (act of July 19, 1939), as amended by section 11, Public Law 144, 78th Congress.* * * *

(h) *Deferred determinations.* Where there is doubt as to whether the antici-

pated income will exceed the statutory limitation, payment of pension will not be made before the end of the calendar year, when the total income received during such year may be determined. Where a determination as to entitlement is deferred in accordance with this paragraph, pension may be payable from the first of that calendar year if notice (constituting an informal claim) that the claimant's income did not exceed the statutory limitation is received at any time within the succeeding calendar year. Any necessary evidence must be received in the Veterans' Administration within 1 year after the date of request. If notice is not received within the period prescribed, payments may not be made for any period prior to the date of receipt of a new claim (formal or informal).

(i) *Reduction of income.* Where a claim has been disallowed or payments discontinued because the claimant's annual income is in excess of the statutory limitation, pension may be payable from the first of the succeeding calendar year if notice (constituting an informal claim) is received during that year that the claimant's actual or anticipated income will not exceed $1,000 (or $2,500, whichever is applicable), and the necessary evidence is furnished within 1 year after the date of request; otherwise, pension may not be paid for any period prior to the date of receipt of a new claim (formal or informal).

(i–1) *Failure to return annual income questionnaire.* When payments have been discontinued as required by §§ 3.292 or 4.86 (g) (2) of this chapter, because of failure to return the annual income questionnaire, pension may be payable, if otherwise in order, from the date of last payment, provided the questionnaire or other evidence that the claimant's income is not in excess of the statutory limitation is received within 1 year from the date of issuance of the questionnaire; otherwise, pension may not be paid for any period prior to the date of receipt of the questionnaire or a new claim (formal or informal).

(Interprets or applies sec. 1, 48 Stat. 1281, as amended, Vet. Reg. 1 (a), as amended; 38 U. S. C. and Sup., 503c, ch. 12 note)

2. In Part 3, § 3.293 is canceled.

§ 3.293 *Restoration of award upon receipt of questionnaire.* [Canceled.]

3. In Part 4, § 4.86 (g) (2) is amended as follows:

§ 4.86 *Public No. 2, 73d Congress (act of March 20, 1933), as amended; sections 28 and 31, Title III, Public No. 141, 73d Congress (act of March 28, 1934), as amended; Public No. 484, 73d Congress (act of June 28, 1934), as amended; and Public Law 301, 79th Congress (act of February 18, 1946).* Where death pension or compensation has been awarded under the provisions of Public No. 2, 73d Congress, or section 28 or 31, Title III, Public No. 141, 73d Congress, or Public No. 484, 73d Congress, as amended, the effective date of reduction or discontinuance of such death pension or compensation shall be in accordance with the facts found, except that: * * *

(g) * * *

(2) A follow-up will be maintained with respect to questionnaires forwarded to payees for the purpose of determining the amount of annual income, as follows: If at the expiration of 30 days, or 60 days if payee resides without the continental limits of the United States, the questionnaire is not returned, another will be forwarded; if the questionnaire is not returned at the expiration of the second 30 days, or 60 days if the payee resides without the continental limits of the United States, the award will be discontinued as of the date of last payment. (See § 3.228.)

* * * * *

(Interprets or applies secs. 1, 4, 9, 20, 48 Stat. 1281, 9, 10, 309, as amended, sec. 5, 57 Stat. 555, sec. 5, 58 Stat. 804, Part III, Vet. Reg. 1 (a), as amended; 38 U. S. C. 503 (c), 704, 707, 709, 729, 734, ch. 12 note)

4. In Part 4, § 4.88 (d) is amended to read as follows:

§ 4.88 *Recommencement of death pension or compensation.* Death pension or compensation previously discontinued will be recommenced as follows: * * *

(d) *Recommencement after discontinuance because of failure to file annual income questionnaire.* See § 3.228 of this chapter.

* * * * *

(Interprets or applies R. S. 4706, as amended, sec. 9, 48 Stat. 10, as amended, 60 Stat. 574; 38 U. S. C. 200, 709, 729a)

(Sec. 5, 43 Stat. 608, as amended, sec. 2, 46 Stat. 1016, sec. 7, 48 Stat. 9; 38 U. S. C. and Sup., 11a, 426, 707)

This regulation effective November 10, 1950.

[SEAL] O. W. CLARK,
Deputy Administrator.

[F. R. Doc. 50-9935; Filed, Nov. 9, 1950; 8:45 a. m.]

TITLE 39—POSTAL SERVICE

Chapter I—Post Office Department

PART 25—GENERAL PROVISIONS RELATING TO POST OFFICES

PENSION VOUCHERS

In § 25.13 *Pension vouchers* (39 CFR 25.13) amend paragraph (i) to read as follows:

(i) *Penalty for false authentication.* To authenticate a voucher prior to the fourth day of the month in which the pension is payable and post-date the jurat to make it appear that the voucher was executed on any other than the actual date of execution; or to authenticate a voucher without the pensioner being present and duly sworn, or in the absence of witnesses where witnesses are required; or to willfully aid or assist in the making or in any wise procure the making or presentation of any false or fraudulent affidavit, declaration, certificate, voucher, or paper, or writing concerning any claim for pension, renders the offender liable to a fine not exceeding $10,000 or to imprisonment for a term of not more than five years.

NOTE: See 18 U. S. C. 289.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10035; Filed, Nov. 9, 1950; 8:45 a. m.]

PART 34—CLASSIFICATION AND RATES OF POSTAGE

MISCELLANEOUS AMENDMENTS

a. In Part 34, Classification and Rates of Postage (39 CFR Part 34), make the following changes:

Amend § 34.80 *Fourth-class rates to Alaska, Hawaiian Islands, and Canal Zone* to read as follows:

§ 34.80 *Fourth-class rates to or from Canal Zone, Canton Island, Guam, Hawaii and American Samoa.* The eighth-zone rates of postage shall apply to fourth-class mail transported: (a) To or from Canton Island, Guam and American Samoa; (b) between the Canal Zone and the United States or its territories or other possessions; (c) between Hawaii and the United States (including Alaska) or its possessions.

NOTE: See § 34.2 for definition of possessions. The parcel-post zoning system has been established for all post offices within the United States, Alaska, Hawaii, Puerto Rico, and Virgin Islands of the United States.

The foregoing amendment shall become effective November 15, 1950.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369)

b. In § 34.88 *Postage rates for air parcel post* (39 CFR 34.88; 15 F. R. 5930) amend subdivision (i) of paragraph (b) (5) to read as follows:

(i) Between any point in continental United States and any point in its territories and possessions falling in the eighth delivery zone, namely, Hawaii, Guam, etc.

The foregoing amendment shall become effective November 15, 1950.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25, sec. 1 (11), 62 Stat. 1097; 5 U. S. C. 22, 369, 39 U. S. C 475)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10036; Filed, Nov. 9, 1950; 8:46 a. m.]

PART 43—TREATMENT OF DOMESTIC MAIL MATTER AT RECEIVING POST OFFICES

PART 137—FIELD SERVICE

DELIVERY OF CERTAIN NONMAILABLE MATTER REACHING OFFICE OF DESTINATION; APPOINTMENT AND REMOVAL OF POSTMASTERS

a. In § 43.40 *Delivery of certain nonmailable matter reaching office of destination* (39 CFR 43.40) amend paragraph (a) by striking "1717" and substituting in lieu thereof "334", and by inserting a note at the end of the paragraph to read as follows:

NOTE: Section 334 of title 18, United States Code, is now codified as section 1461 of title 18, United States Code.

b. In § 137.2 *Appointment and removal of postmasters* (39 CFR 137.2) amend

paragraph (c) by adding a note at the end thereof to read as follows:

NOTE: Sections 208-212 of title 18, United States Code are now codified as sections 602, 603, 606, and 607 of title 18, United States Code.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10037; Filed, Nov. 9, 1950; 8:46 a. m.]

PART 43—TREATMENT OF DOMESTIC MAIL MATTER AT RECEIVING POST OFFICES

PART 50—CITY DELIVERY

PART 71—ISSUE OF DOMESTIC MONEY ORDERS

MISCELLANEOUS AMENDMENTS

a. In § 43.45 *Undeliverable postal and post cards* (39 CFR 43.45) make the following changes:

1. Amend paragraph (a) to read as follows:

(a) *Those which may be returned to sender.* Unpaid, misdirected, unmailable, and unclaimed domestic postal cards and post cards, whether single or double and whether addressed for local delivery or otherwise, shall be returned to the sender only when they bear his return card in the upper left corner of the address side, together with a pledge to pay return postage. Single post and postal cards and double post cards bearing the sender's return card and pledge to pay return postage shall be returned charged with postage due at the rate of 1 cent each, to be collected on delivery, as provided in § 43.48. Double postal cards and double post cards with postage affixed to the return portion bearing the sender's return card and pledge to pay return postage shall be (1) endorsed to show reason for nondelivery and marked on the original portion with the usual inscription for return to sender and (2) reverse folded so that the address side of the reply portion is faced out. Such undeliverable double cards shall be postmarked on the face of the reply portion and returned to the sender without collection of additional postage.

2. Insert a new paragraph (b-1) to read as follows:

(b-1) *Double cards not bearing sender's return card and pledge to pay return postage.* Undeliverable double cards that do not bear the sender's return card and pledge to pay return postage shall be destroyed or disposed of as waste paper. Any uncanceled stamps on such cards shall be canceled.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369)

b. In Part 50, City delivery (39 CFR Part 50), make the following changes:

Amend § 50.11 *Delivery where vicious dogs are kept,* to read as follows:

§ 50.11 *Delivery where dogs are kept.* Carriers are not required to deliver mail where dogs are permitted to interfere with deliveries.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369)

c. In Part 71, Issue of domestic money orders (39 CFR Part 71) make the following changes:

§ 71.14 *Record of money orders* is hereby rescinded.

(R. S. 161, 369, 4027, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369; 39 U. S. C. 711)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10038; Filed, Nov. 9, 1950; 8:46 a. m.]

PART 127—INTERNATIONAL POSTAL SERVICE: POSTAGE RATES, SERVICE AVAILABLE, AND INSTRUCTIONS FOR MAILING

MISCELLANEOUS AMENDMENTS

a. In § 127.3 *Letters and letter packages* (39 CFR 127.3) make the following changes:

1. Amend paragraph (e) to read as follows:

(e) *Dutiable articles.* (1) Articles liable to customs duty (merchandise) may be forwarded in letters or letter packages (surface or air) provided such importation is not prohibited by the country of destination (see subparagraph (2) of this paragraph). The sender should enclose in the article a completed paper form of customs declaration, Form 2976-A, or an invoice, and should affix a green label, Form 2976, to the address side of the article. If the sender knows that the contents of his package are not dutiable he may, if he prefers, omit the above-mentioned forms, in which case acceptance for mailing will be at his risk, and the Department will assume no responsibility with respect to the treatment that may be accorded the article by the foreign customs authorities. Omission of these forms, particularly the green label, may, however, result in delayed delivery and possible penalties against the addressee, regardless of whether the contents are dutiable or not. Special restrictions exist with respect to certain countries. Consult the individual country items. Facsimiles of Forms 2976 (C-1) and 2976-A (C-2) appear in § 127.17.

(2) The following countries will not accept dutiable articles sent in letters or letter packages:

Azores (ordinary).
Bolivia.
Brazil.
British Honduras.
China.
Colombia.
Dominican Republic.
Ecuador (ordinary).
Estonia.
Greece.
Guatemala (ordinary).
Japan.
Korea.
Latvia.
Lithuania.
Madeira Islands (ordinary).
Panama (ordinary).
Peru (ordinary).
Poland.
Rio de Oro.
Ryukyu Islands.
Salvador (El).
Spanish Guinea.
Union of Soviet Socialist Republics.
Uruguay.
Venezuela.

If a letter or package containing merchandise is offered for mailing to one of the countries named in this paragraph it may be accepted only on condition that the sender knows that the contents are not dutiable in the country of destination, in which case Forms 2976 and 2976-A may be omitted. The Department will assume no responsibility with respect to the treatment of these articles by the foreign customs authorities.

(3) The above instructions are applicable only to letters and letter packages (surface or air). Air mail packages prepared as "small packets" or any other Postal Union (regular) mail classification except letters or letter packages, even though such packages are prepaid at the air mail letter rate of postage, must be unsealed, even if registered, and must comply in other respects with §§ 127.5 to 127.13.

2. Delete paragraphs (f), (g) and (h) and insert a new paragraph (f) to read as follows:

(f) It is recommended that senders add the words "Letter (lettre)" on the address side of letters and letter packages which, by reason of their volume or packing, may be mistaken for other articles.

b. Amend § 127.20 *Air-mail service* (39 CFR 127.20) by deleting paragraphs (g), (h) and (i) and by inserting new paragraphs (g) and (h) to read as follows:

(g) Post cards (single) for Canada and Mexico are conveyed by air at the rate of 4 cents each. Single post cards for other countries and double post cards for all countries must be prepaid at the regular air-mail letter rate to the country concerned.

(h) See § 127.3 (e) concerning the acceptance of air mail letters and letter packages containing articles liable to customs duty.

Articles in the Postal Union (regular) mails other than letters, letter packages and post cards (that is, commercial papers, printed matter, samples of merchandise, and small packets, as well as 8-ounce merchandise packages, combination packages and articles grouped together) are transmissible by air provided they are prepared for mailing as prescribed for the class of article involved, as set forth in §§ 127.5 to 127.13, inclusive, and are prepaid at the appropriate air mail rate. This rate will be the letter rate unless a special "other articles" rate is in effect to the country concerned in which case the latter rate will apply. In either case the rate is shown in the subcaption "Air mail service" under the country heading concerned.

c. In § 127.219 *Brazil* (39 CFR 127.219) amend paragraph (a) (5) to read as follows:

(5) *Air mail service.* Postage rates: Letters and post cards, 10 cents per half ounce. Air-letter sheets, 10 cents each. Other regular-mail articles, 55 cents for the first 2 ounces and 35 cents for each additional 2 ounces. (See § 127.20.)

d. In § 127.226 *Burma* (39 CFR 127.226) amend paragraph (b) (4) to read as follows:

(4) *Observations.* (i) Addressees in Burma are required to obtain import licenses to take delivery of parcels, including those sent as gifts. It is recommended that senders inform the addressees of the contents of parcels

prior to mailing, so that they may apply for the necessary import license.

(ii) Parcel-post packages addressed to box numbers and not bearing the actual address of the addressee will be returned to origin.

e. In § 127.232 *Colombia* (39 CFR 127.232) amend paragraph (a) (7) to read as follows:

(7) *Air mail service.* Postage rates: Letters and post cards, 10 cents per half ounce. Air-letter sheets, 10 cents each. Other regular-mail articles, 39 cents for the first 2 ounces and 19 cents for each additional 2 ounces. (See § 127.20.)

f. In § 127.237 *Cyprus* (39 CFR 127.237) amend paragraph (b) (4) to read as follows:

(4) *Observations.* The addressees of all commercial parcels, and of gift parcels exceeding $2.80 in value, are required to obtain import licenses.

g. In § 127.267 *Gold Coast Colony* (39 CFR 127.267) amend subdivision (ii) of paragraph (b) (4) to read as follows:

(ii) The addressees of all parcels except those sent as bona fide gifts are required to obtain import licenses in order to take delivery. All gift parcels must be clearly marked as gifts. Gift parcels should not exceed £10 ($28) in value, and an addressee may not receive more than one gift parcel from any one sender during a month.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369; and the terms of postal conventions and agreements entered into pursuant to R. S. 398, 48 Stat. 943; 5 U. S. C. 372)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10040; Filed, Nov. 9, 1950; 8:46 a. m.]

PART 127—INTERNATIONAL POSTAL SERVICE: POSTAGE RATES, SERVICE AVAILABLE, AND INSTRUCTIONS FOR MAILING

MISCELLANEOUS AMENDMENTS

a. In § 127.10 *Small packets* (39 CFR 127.10) amend paragraph (e) by deleting "Netherlands Indies" and by inserting "Indonesia" between "Hungary" and "Iran" in the alphabetical list of countries shown therein.

b. In § 127.76 *Group shipments* (39 CFR 127.76) amend paragraph (b) by deleting "Netherlands Indies (limited to 3 parcels)" and by inserting "Indonesia (limited to 3 parcels)" between "Iceland" and "Iran (limited to 3 parcels)" in the list of countries shown therein.

c. In § 127.93 *Storage (demurrage) charge* (39 CFR 127.93) amend paragraph (c) by deleting "Netherlands Indies" and by inserting "Indonesia" between "Iceland" and "Italy".

d. In § 127.102 *Special provisions applicable to international insurance service* (39 CFR 127.102) amend paragraph (a) by deleting "Netherlands Indies" and inserting "Indonesia" between "Iceland" and "Ireland" in the list of countries shown therein.

e. In § 127.199 *Alphabetical index to Subpart D* (39 CFR 127.199) delete "Netherlands Indies, 127.310" and insert "Indonesia 127.278a" between "Indo-China, French, 127.257 *French Indo-China*" and "Iran 127.279" in the list of countries shown therein.

f. In § 127.310 *Netherlands Indies* (Bali, Banka, Billiton, Netherlands Borneo, Celebes, Java, Little Soenda (Sunda), Madoera, Moluccas (Moloques), Rhio (Riouw), Sumatra, Netherlands Timor, and Netherlands New Guinea) (39 CFR 127.310) make the following changes:

1. Amend § 127.310 by deleting "§ 127.310" and substituting "§ 127.278a" in lieu thereof; and by inserting the entire § 127.278a between § 127.278 and § 127.279.

2. Amend the section headnote by deleting "Netherlands Indies" and inserting "Indonesia" in lieu thereof.

3. Amend paragraph (b) as follows:

a. Delete "(Netherlands Indies.)" and substitute in lieu thereof "(Indonesia)".

b. Add a new subdivision (iii) to paragraph (b) (5) to read as follows:

(iii) Addressees in Indonesia are required to obtain import licenses for all gift parcels exceeding 25 guilders ($6.50) in value.

(R. S. 161, 396, secs. 301, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369; and the terms of postal conventions and agreements entered into pursuant to R. S. 398, 48 Stat. 943; 5 U. S. C. 372)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10039; Filed, Nov. 9, 1950; 8:46 a. m.]

PART 127—INTERNATIONAL POSTAL SERVICE: POSTAGE RATES, SERVICE AVAILABLE, AND INSTRUCTIONS FOR MAILING

ISRAEL (STATE OF)

In § 127.282 *Israel (State of)* (39 CFR 127.282; 15 F. R. 200, 3151, 4221) amend subdivisions (i) and (ii) of paragraph (b) (4) to read as follows:

(i) Addressees in Israel are required to possess import licenses in order to take delivery of all parcels except the following:

(*a*) Bona fide gift parcels containing food (see below).

(*b*) Parcels containing bona fide gifts other than food or postage stamps, provided the contents do not exceed £40 (about $112) in value and are for the addressee's personal use, and not more than one parcel or group of parcels is received by one family within three months.

(*c*) Parcels containing the property of persons who have settled in Israel or the property of tourists, if duty-free.

(ii) Parcels which do not fall within one of those categories require import licenses to be obtained by the addressees in Israel. It is the responsibility of the mailer in each case to ascertain that the addressee has obtained or can obtain the import license if required.

(R. S. 161, 396, secs. 304, 309, 42 Stat. 24, 25; 5 U. S. C. 22, 369; and the terms of Postal conventions and agreements entered into pursuant to R. S. 398, 48 Stat. 943; 5 U. S. C. 372)

[SEAL] V. C. BURKE,
Acting Postmaster General.

[F. R. Doc. 50-10041; Filed, Nov. 9, 1950; 8:46 a. m.]

TITLE 43—PUBLIC LANDS: INTERIOR

Chapter I—Bureau of Land Management, Department of the Interior

Appendix—Public Land Orders

[Public Land Order 682]

NEW MEXICO

REVOKING IN PART EXECUTIVE ORDER NO. 8874 OF AUGUST 28, 1941

By virtue of the authority vested in the President and pursuant to Executive Order No. 9337 of April 24, 1943, it is ordered as follows:

Executive Order No. 8874 of August 28, 1941, as amended by Executive Order No. 9526 of February 28, 1945, withdrawing public lands for the use of the War Department as a practice bombing range, is hereby revoked so far as it affects the following-described lands:

NEW MEXICO PRINCIPAL MERIDIAN

T. 10 N., R. 1 E.,
Sec. 4, lots 1 to 8 inclusive.

The areas described aggregate 257.76 acres.

The lands are chiefly valuable for grazing purposes.

No application for these lands may be allowed under the homestead, small tract, desert land, or any other nonmineral public-land laws, unless the lands have already been classified as valuable or suitable for such types of applications, or shall be so classified upon consideration of an application.

This order shall not otherwise become effective to change the status of such lands until 10:00 a. m. on the 35th day after the date of this order. At that time the said lands shall, subject to valid existing rights and the provisions of existing withdrawals, become subject to application, petition, location, and selection as follows:

(a) *Ninety-one day period for preference-right filings.* For a period of 91 days, commencing at the hour and on the day specified above, the public lands affected by this order shall be subject only to (1) application under the homestead or the desert-land laws or the Small Tract Act of June 1, 1938, 52 Stat. 609 (43 U. S. C. 682a), as amended, by qualified veterans of World War II and other qualified persons entitled to preference under the act of September 27, 1944, 58 Stat. 747 (43 U. S. C. 279–284), as amended, subject to the requirements of applicable law, and (2) application under any applicable public-land law, based on prior existing valid settlement rights and preference rights conferred by existing laws or equitable claims subject to allowance and confirmation. Applications under subdivision (1) of this paragraph shall be subject to applications and claims of the classes described in subdivision (2) of this paragraph. All applications filed under this paragraph either at or before 10:00 a. m. on the 35th day after the date of this order shall be treated as though filed simultaneously at that time. All applications filed under this paragraph after 10:00 a. m. on the said 35th day shall be considered in the order of filing.

(b) *Date for non-preference-right filings.* Commencing at 10:00 a. m. on the 126th day after the date of this order, any lands remaining unappropriated shall become subject to such application, petition, location, selection, or other appropriation by the public generally as may be authorized by the public-land laws. All such applications filed either at or before 10:00 a. m. on the 126th day after the date of this order, shall be treated as though filed simultaneously at the hour specified on such 126th day. All applications filed thereafter shall be considered in the order of filing.

A veteran shall accompany his application with a complete photostatic, or other copy (both sides), of his certificate of honorable discharge, or of an official document of his branch of the service which shows clearly his honorable discharge as defined in § 181.36 of Title 43 of the Code of Federal Regulations, or constitutes evidence of other facts upon which the claim for preference is based and which shows clearly the period of service. Other persons claiming credit for service of veterans must furnish like proof in support of their claims. Persons asserting preference rights, through settlement or otherwise, and those having equitable claims, shall accompany their applications by duly corroborated statements in support thereof, setting forth in detail all facts relevant to their claims.

Applications for these lands, which shall be filed in the Land and Survey Office, Santa Fe, New Mexico, shall be acted upon in accordance with the regulations contained in § 295.8 of Title 43 of the Code of Federal Regulations and Part 296 of that title, to the extent that such regulations are applicable. Applications under the homestead laws shall be governed by the regulations contained in Parts 166 to 170, inclusive, of Title 43 of the Code of Federal Regulations, and applications under the desert-land laws and the said Small Tract Act of June 1, 1938, shall be governed by the regulations contained in Parts 232 and 257, respectively, of that title.

Inquiries concerning these lands shall be addressed to the Manager, Land and Survey Office, Santa Fe, New Mexico.

OSCAR L. CHAPMAN,
Secretary of the Interior.

NOVEMBER 3, 1950.

[F. R. Doc. 50-10032; Filed, Nov. 9, 1950; 8:45 a. m.]

TITLE 49—TRANSPORTATION

Chapter I—Interstate Commerce Commission

PART 127—CLASSIFICATION OF STEAM RAILROAD EMPLOYEES; REPORTS OF SERVICE AND COMPENSATION

At a session of the Interstate Commerce Commission, Division 1, held at its office in Washington, D. C., on the 19th day of October A. D. 1950.

The matter of the revision of rules governing the classification of railroad employees and reports of their service and compensation being under consideration:

It is ordered, That the order dated July 17, 1940, prescribing certain "Rules Governing the Classification of Steam Railway Employees and Reports of Their Service and Compensation" (49 CFR 127.1–127.5) be, and it is hereby vacated and set aside, effective January 1, 1951, and in lieu thereof the following order is prescribed:

It is further ordered, That: (a) A copy of this order and the annexed rules shall be served on each and every railroad (except electric railways) subject to Part I of the Interstate Commerce Act and upon every receiver, trustee, executor, administrator, or assignee of any such railroad and notice of this order shall be given to the general public, by depositing a copy thereof in the office of the Secretary of this Commission at Washington, D. C., and by filing it with the Director of the Division of the Federal Register.

(b) *Objections may be filed.* Any interested party may on or before December 8, 1950, file with the Commission a written statement of reasons why the said order should not become effective as provided above. Unless otherwise ordered after consideration of such objections, the said order shall become effective as herein ordered.

Sec.
127.1 Rules governing classification of employees, service, and compensation.
127.2 Reports; period covered, forms, date of filing.
127.3 Employees; definition, service hours, and compensation.
127.4 Forms required to be used.
127.5 Explanatory instructions pertaining to Form A.
127.6 Explanatory instructions pertaining to Form B.
127.7 Appendix—Forms A and B.

AUTHORITY: §§ 127.1 to 127.7 issued under sec. 12, 24 Stat. 383, as amended; 49 U. S. C. 12. Interpret or apply sec. 20, 24 Stat. 386, as amended; 49 U. S. C. 20.

§ 127.1 *Rules governing classification of employees, service and compensation.* The annexed revised rules governing the classification of railroad employees and reports of their service and compensation outlined in §§ 127.3 and 127.4 are hereby approved, to become effective on January 1, 1951, and each and every railroad (except electric railways) within the scope of section 20, Part 1, of the Interstate Commerce Act shall be governed by such rules in the preparation and submission of their annual and other periodical reports to the Interstate Commerce Commission in accordance with the forms adopted for such returns (§ 127.2); and also with respect to any other matters covered by these rules.

§ 127.2 *Reports; period covered, forms, date of filing.* Commencing with the month of January 1951 and monthly thereafter until further order, each and every class I railroad (§ 126.1 of this chapter), including class I switching and terminal companies, is hereby required to file monthly reports of employees, service and compensation in accordance with the forms outlined in §§ 127.4 and 127.7: *Provided,* That such carriers shall also file summary reports covering the period January 1 to December 31, inclusive, of each year beginning with returns for the year 1951 in accordance with the said forms. Such monthly reports shall be filed in the Bureau of Transport Economics and Statistics, Interstate Commerce Commission, Washington, D. C., within thirty days after the close of the month to which they relate; and, the annual summaries shall be filed within forty-five days following the close of the period for which they are compiled. Railroads of class I, II, and III (§ 126.1 of this chapter), including switching and terminal companies and lessor companies, shall include a statement of employees, service and compensation in their annual reports to the Interstate Commerce Commission in accordance with the requirements of the annual report forms prescribed for such carriers (49 CFR Part 120).

§ 127.3 *Employees; definition, service hours, and compensation*—(a) *Definition of "employees".* The word "employees," as used in this part, is intended to include every person in the service of the reporting carrier subject to its continuing authority to supervise and direct the manner of rendition of his service. Persons engaged to render only specifically defined service and not subject to the continuing authority of the carrier to supervise and control their acts, such as customhouse brokers, lawyers retained only for specific cases and not under general or continuing retainer, etc., are not employees in the meaning of the term used in this part.

(b) *Counting employees.* Since the number of employees fluctuates, carriers are required to classify and count all of their employees at twelve different times each year.

(c) *Joint employees.* Each person jointly employed shall, if carried on the payrolls of the several joint employers, be counted by each employer and represented in its return of number of employees by a fraction based on the number of employers reporting him; if a person, for example, is reportable by three employers, each should include him in its number of employees as one-third of an employee. When the entire compensation of a joint employee is shown on the payroll of a single joint employer and is paid to the employee by that employer such employee should, for the purpose of returns, be treated as if employed solely by such employer.

(d) *Service hours.* (1) The number of hours on duty, or held for duty, and the number of hours paid for are to be ascertained and recorded for every class of employee. For enginemen and trainmen, the actual number of miles run and miles paid for but not run are to be recorded, as well as the number of hours on duty and the number of hours paid for. (The service time of all classes of employees shall be recorded in hours instead of days or hours as heretofore.)

(2) Whenever an employee works at more than one occupation, or in more than one class of service, both the number of hours worked and the compensation paid, should be separated and reported under the proper Reporting Divisions.

(3) If an employee is paid a day's wage for a smaller number of hours than con-

stitutes a day's work, the number of hours paid for as well as the actual number of hours the employee is on duty should be ascertained and recorded. Time allowed for meals, part holidays, holidays, absences on leave, vacations, etc., should be excluded from time actually worked, but if such time is paid for it should be appropriately reported as "Time paid for but not worked" on Form A or as a "constructive allowance" on Form B. These requirements apply to enginemen and trainmen paid on the basis of trips or of miles run, and to employees paid at piece rates, as well as to employees paid on hourly, daily, weekly, monthly, or other time basis. Service hours for officers and employees who do not receive payment for overtime should be reported as the number of hours in each month at 8 hours per day contemplated for the position.

(e) *Compensation.* The compensation of employees as defined in the rules in this part is to be stated in such manner and detail as the forms adopted for periodical returns require.

§ 127.4 *Forms required to be used.* (a) There are presented outlines of two forms for the report of information to the Interstate Commerce Commission on railroad employees, their service and compensation, designated as Form A (§ 127.5), which relates to employees other than train and engine service employees, and Form B (§ 127.6), which relates to train and engine service employees. These forms are explained by instructions, in order that the intent of the forms may be correctly understood.

(b) With general reference to the statement of the compensation of employees in Forms A and B, it should be understood that the total compensation received by the employees in each Reporting Division, as well as the amount of work they perform, should be shown properly distributed under the prescribed column heads as indicated by the forms. It should be noted that if the work of an employee varies during a report period, his time and compensation should be apportioned accordingly. Amounts reported should be gross compensation paid.

Illustrations. (a) In the course of a month, an employee works as a Machinist, and as such earns $200 and he works also as a Gang Foreman and earns $130. The pay received by him in the occupation of Machinist with corresponding time should be included in returns for Division 61, "Machinists", and his other pay and time as indicated should be included in returns for Division 53, "Gang Foremen and Gang Leaders". (b) When an employee works at two or more of the occupations of engineer, fireman, conductor, and brakeman, his time and compensation should be assigned in accordance with the facts. For example, in a certain month an employee earns as a Through Freight Engineer $110, as a Local or Way Freight Engineer $140, as a Yard Engineer $100, and as a Local or Way Freight Fireman $75. The time and the compensation of this employee should be correspondingly distributed among Reporting Divisions 122, 123, 124, and 127 as they are respectively applicable, without regard to the predominance of the time worked or the amount earned by him in one occupation.

(c) As elsewhere indicated, the statement of the number of employees in the service of a company with respect to Reporting Divisions depends upon the allocation of the individual employees as of the day of count.

(d) The Reporting Divisions shown in Forms A and B following will also be used in connection with the employees' schedule in the annual reports of railroads of Classes I and II to the Commission. Railroads of Class III and lessor companies shall report only the information required in the annual report forms prescribed for such companies.

§ 127.5 *Explanatory instructions pertaining to Form A—Column 1.* Employees should be classified for reporting purposes in accordance with the reporting divisions shown in this column.

Column 2. Enter the total number of employees in service as of the 15th day of the month, except when it falls on a Saturday, Sunday, or a holiday, in which case the count should be made as of the last preceding business day.

Employees whose duties are such as to make them includible in two or more Reporting Divisions should be reported in that Division indicated by the occupation in which engaged on the day of the middle of month count.

The count should not be restricted to employees actually on duty as of the day of the count, but should include all other regular assigned employees, employees under pay on vacation or sick leave and relief employees. Employees who are not subject to call for duty, such as employees absent on definite leave or under suspension; employees on furlough; and pensioners not bound to render service, should be excluded.

Column 3. Enter number of employees who made time during the month, no matter for how short a period, classified by reporting divisions. Employees who worked in more than one occupation during the month should be assigned according to the preponderance of their duties.

Column 4. There should be stated for each of the Reporting Divisions the total number of hours actually worked and paid for at straight time rates by the employees whose service is includible in the particular division. (Time previously reported as overtime at pro rata rates should be included.)

Column 5. Enter the total number of overtime hours paid for at punitive rates.

Column 6. Enter the totals of time paid for and not worked, such as payment for part holidays, holidays, absence on definite leave, vacations, miscellaneous time paid for but not worked, such as pay for attending court, suspensions, sickness, time allowed for meals, and other time that can properly be considered constructive, such as allowance to complete a minimum day when less than a minimum day is worked.

Where vacation allowances are based on compensation earned in previous year, the hours should be computed as in note to Column 7, Form B.

Column 7. Enter the total of columns 4 to 6.

Column 8. Enter the compensation for the time shown in column 4.

Column 9. Enter the compensation for the time shown in column 5.

Column 10. Enter the compensation for the time shown in column 6.

Column 11. Enter the total of columns 8 to 10.

§ 127.6 *Explanatory instructions pertaining to Form B—Column 1.* Employees should be classified for reporting purposes in accordance with the reporting divisions shown in this column.

Column 2. Enter the total number of employees in service as of the 15th day of the month, except when it falls on a Saturday, Sunday, or a holiday, in which case the count should be made as of the last preceding business day.

Employees whose duties are such as to make them includible in two or more Reporting Divisions should be reported in that Division indicated by the occupation in which engaged on the day of middle of month count.

The count should not be restricted to employees actually on duty as of the day of the count, but should include all other regular assigned employees, employees under pay on vacation or sick leave, as well as those listed on the "extra board" in train and engine service who are subject to call for duty. Employees who are not subject to call for duty, such as employees absent on definite leave or under suspension; employees on furlough; and pensioners not bound to render service, should be excluded.

Column 3. Enter number of employees who made time during the month, no matter for how short a period, classified by reporting divisions. Employees who worked in more than one occupation during the month should be assigned according to the preponderance of their duties.

Column 4. Enter the number of straight-time hours actually worked.

Column 5. Enter the total straight-time hours paid for.

Column 6. Enter the number of hours of overtime paid for.

Column 7. Enter the number of constructive hours allowed which does not represent actual train service and for which mileage is not allowed, such as vacation time, pay for "Held away from home terminal" rule, called and not used, runaround, deadheading, attending court, suspensions, investigations, and claim and safety meetings.

NOTE: The total dollar amount of vacation allowance in a particular month, for each reporting division, should be divided by the average straight time hourly rate of such reporting division, to determine the constructive vacation hours paid for. The straight time hourly rate for each such reporting division should be determined by dividing the "Compensation—Straight Time Paid For" by "Service Hours—Straight Time Paid For" for the latest available month.

Column 8. Enter the total of amounts shown in columns 5, 6, and 7.

Column 9. Enter the straight-time compensation for the time shown in column 5.

Column 10. Enter the overtime compensation for the time shown in column 6.

Column 11. Enter the compensation for the time shown in column 7.

Column 12. Enter the sum of the amounts shown in columns 9, 10, and 11.

Column 13. Enter the number of miles run in performing actual train service.

Column 14. Enter the number of miles paid for but not run in connection with actual train service.

Column 15. Include in this column the number of trips made for which not less than a minimum day is paid.

§ 127.7 *Appendix—Forms A and B.*

I. C. C. Wage Statistics

MONTHLY REPORT OF EMPLOYEES SERVICE AND COMPENSATION

FORM A

Full name of reporting company .....

Miles of line covered by this report ..... Month of ....., 19....

Calendar year 19....

(Strike out one)

Division No.	Reporting division (1)	Column headings
	I. EXECUTIVES, OFFICIALS, AND STAFF ASSISTANTS	
1	Executives, general officers, and assistants	
2	Division officers, assistants, and staff assistants	
(001)	Total (executives, officials, and staff assistants)	
	II. PROFESSIONAL, CLERICAL, AND GENERAL	
3	Professional and subprofessional assistants	(2) Number of employees middle of month.
4	Supervisory or chief clerks (major departments)	(3) Number of employees who received pay during month.
5	Chief clerks (minor departments) and assistant chief clerks and supervising cashiers	SERVICE HOURS
6	Clerks and clerical specialists (A)	(4) Time actually worked and paid for at straight time rates.
7	Clerks (B and C)	(5) Overtime paid for at punitive rates.
8	Mechanical device operators (office)	(6) Time paid for but not worked.
9	Stenographers and secretaries (A)	(7) Total time paid for.
10	Stenographers and typists (B)	
11	Storekeepers, sales agents, and buyers	COMPENSATION
12	Ticket agents and assistant ticket agents	(8) Time actually worked and paid for at straight time rates.
13	Traveling auditors or accountants	(9) Overtime paid for at punitive rates.
14	Telephone switchboard operators and office assistants	(10) Time paid for but not worked.
15	Messengers and office boys	(11) Total.
16	Elevator operators and other office attendants	
17	Lieutenants and sergeants of police	
18	Patrolmen and watchmen	
19	Traffic and various other agents, inspectors, and investigators	
20	Claim agents or investigators	
21	Freight claim agents or investigators	
22	Chief claim agents or investigators	
23	Miscellaneous trades workers (other than plumbers)	
24	Motor vehicle and motor car operators	
25	Teamsters and stablemen	
26	Janitors and cleaners	
(002)	Total (professional, clerical, and general)	
	III. MAINTENANCE OF WAY AND STRUCTURES	
27	Roadmasters, general foremen and assistants	(2) Number of employees middle of month.
28	Maintenance of way and scale inspectors	(3) Number of employees who received pay during month.
29	Bridge and building gang foremen (skilled labor)	SERVICE HOURS
30	Bridge and building carpenters	(4) Time actually worked and paid for at straight time rates.
31	Bridge and building ironworkers	(5) Overtime paid for at punitive rates.
32	Bridge and building painters	(6) Time paid for but not worked.
33	Masons, bricklayers, plasterers, and plumbers	(7) Total time paid for.
34	Maintenance of way and structures helpers and apprentices	
35	Portable equipment operators	COMPENSATION
36	Portable equipment operators' helpers	(8) Time actually worked and paid for at straight time rates.
37	Pumping equipment operators	(9) Overtime paid for at punitive rates.
38	Gang foremen (extra gang and work train laborers)	(10) Time paid for but not worked.
39	Gang foremen (bridge and building, signal and telegraph laborers)	(11) Total.
40	Gang or section foremen	
41	Extra gang men	
42	Section men	
43	Maintenance of way laborers (other than track and roadway) and gardeners and farmers	

FORM A—Continued

Division No.	Reporting division (1)	Column headings
	III. MAINTENANCE OF WAY AND STRUCTURES—continued	
44	General and assistant general foremen and inspectors (signal, telegraph, and electrical transmission)	
45	Gang foremen (signal and telegraph skilled trades labor)	
46	Signalmen and signal maintainers	
47	Linemen and groundmen	
48	Assistant signalmen and assistant signal maintainers	
49	Signalmen and signal maintainer helpers	
(003)	Total (maintenance of way and structures)	
	IV. MAINTENANCE OF EQUIPMENT AND STORES	
50	General, assistant general, and department foremen	(2) Number of employees middle of month.
51	General and assistant general foremen (stores)	(3) Number of employees who received pay during month.
52	Equipment, shop, electrical, material and supplies inspectors	SERVICE HOURS
53	Gang foremen and gang leaders (skilled labor)	(4) Time actually worked and paid for at straight time rates.
54	Blacksmiths	(5) Overtime paid for at punitive rates.
55	Boilermakers	(6) Time paid for but not worked.
56	Carmen (A and B)	(7) Total time paid for.
57	Carmen (C and D)	
58	Electrical workers (A)	COMPENSATION
59	Electrical workers (B)	(8) Time actually worked and paid for at straight time rates.
60	Electrical workers (C)	(9) Overtime paid for at punitive rates.
61	Machinists	(10) Time paid for but not worked.
62	Molders	(11) Total.
63	Sheet-metal workers	
64	Skilled trades helpers (M. of E. and Stores)	
65	Helper apprentices (M. of E. and Stores)	
66	Regular apprentices (M. of E. and Stores)	
67	Coach cleaners	
68	Gang foremen (shops, engine houses, and power plants)	
69	Gang foremen (stores and ice, reclamation, and timber-treating plants)	
70	Classified laborers (shops, engine houses, and power plants)	
71	General laborers (shops, engine houses, and power plants)	
72	General laborers (stores and ice, reclamation, and timber-treating plants)	
73	Stationary engineers (steam)	
74	Stationary firemen, oilers, coal passers, and water tenders	
(004)	Total (maintenance of equipment and stores)	
	V. TRANSPORTATION; (OTHER THAN TRAIN, ENGINE, AND YARD)	
75	Chief train dispatchers	(2) Number of employees middle of month.
76	Train dispatchers	(3) Number of employees who received pay during month.
77	Train directors	SERVICE HOURS
78	Station agents (supervisory—major stations—nontelegraphers)	(4) Time actually worked and paid for at straight time rates.
79	Station agents (smaller stations—nontelegraphers)	(5) Overtime paid for at punitive rates.
80	Station agents (telegraphers and telephoners)	(6) Time paid for but not worked.
81	Chief telegraphers and telephoners or wire chiefs	(7) Total time paid for.
82	Clerk-telegraphers and clerk-telephoners	
83	Telegraphers, telephoners, and towermen	COMPENSATION
84	Station masters and assistants	(8) Time actually worked and paid for at straight time rates.
85	Supervising baggage agents	(9) Overtime paid for at punitive rates.
86	Baggage agents and assistants	(10) Time paid for but not worked.
87	Baggage, parcel room, and station attendants	(11) Total.
88	General foremen (freight stations, warehouses, grain elevators, and docks)	
89	Assistant general foremen (freight stations, warehouses, grain elevators, and docks)	
90	Gang foremen (freight stations, warehouse, grain elevator, and dock labor)	
91	Callers, loaders, scalers, sealers, and perishable-freight inspectors	
92	Truckers (stations, warehouses, and platforms)	
93	Laborers (coal and ore docks and grain elevators)	
94	Common laborers (stations, warehouses, platforms, and grain elevators)	
95	Stewards, restaurant and lodging-house managers, and dining-car supervisors	
96	Chefs and cooks (restaurants or dining cars)	
97	Waiters, camp cooks, kitchen helpers, etc.	

FORM A—Continued

Division No.	Reporting division (1)	Column headings
	V. TRANSPORTATION, ETC.—continued	
98	Officers, workers, and attendants on barges, launches, ferry boats, towing vessels, steamers, and shore workers	
99	Transportation and dining-service inspectors	
100	Parlor and sleeping car conductors	
101	Train attendants	
102	Bridge operators and helpers	
103	Crossing and bridge flagmen and gatemen	
104	Foremen (laundry and laundry workers)	
(105)	Total (transportation—other than train, engine, and yard)	
	VI (a). TRANSPORTATION (YARD MASTERS, SWITCHTENDERS, AND HOSTLERS)	
105	Yardmasters	(2) Number of employees middle of month. (3) Number of employees who received pay during month.
106	Assistant yardmasters	
107	Switch tenders	
108	Outside hostlers	
109	Inside hostlers	SERVICE HOURS
110	Outside hostler helpers	(4) Time actually worked and paid for at straight time rates. (5) Overtime paid for at punitive rates. (6) Time paid for but not worked. (7) Total time paid for.
(906)	Total (transportation—yard masters, switch tenders, and hostlers)	
(907)	Total, all groups (except train and engine)	COMPENSATION
		(8) Time actually worked and paid for at straight time rates. (9) Overtime paid for at punitive rates. (10) Time paid for but not worked. (11) Total.

FORM B

Division No.	Reporting division (1)	Column headings
	VI (b). TRANSPORTATION (TRAIN AND ENGINE)	
111	Road passenger conductors	(2) Number of employees middle of month. (3) Number of employees who received pay during month.
112	Assistant road passenger conductors and ticket collectors	
113	Road freight conductors (through freight)	
114	Road freight conductors (local and way freight)	
115	Road passenger baggagemen	SERVICE HOURS
116	Road passenger brakemen and flagmen	(4) Straight time actually worked. (5) Straight time paid for. (6) Overtime paid for. (7) Constructive allowances. (8) Total.
117	Road freight brakemen and flagmen (through freight)	
118	Road freight brakemen and flagmen (local and way freight)	
119	Yard conductors and yard foremen	
120	Yard brakemen and yard helpers	
121	Road passenger engineers and motormen	COMPENSATION
122	Road freight engineers and motormen (through freight)	(9) Straight time paid for. (10) Overtime paid for. (11) Constructive allowances. (12) Total.
123	Road freight engineers and motormen (local and way freight)	
124	Yard engineers and motormen	
125	Road passenger firemen and helpers	MILES
126	Road freight firemen and helpers (through freight)	(13) Actually run. (14) Paid for but not run. (15) Total number of trips for which not less than a minimum day was paid.
127	Road freight firemen and helpers (local and way freight)	
128	Yard firemen and helpers	
(908)	Total (transportation—train and engine)	

NOTE: The reporting requirements of this order have been approved by the Bureau of the Budget in accordance with the Federal Reports Act of 1942.

By the Commission, Division 1.

[SEAL] W. P. BARTEL, *Secretary.*

[F. R. Doc. 50-10053; Filed, Nov. 9, 1950; 8:48 a. m.]

PROPOSED RULE MAKING

DEPARTMENT OF AGRICULTURE

Bureau of Animal Industry

[9 CFR, Part 151]

HORSES

RECOGNITION OF BREEDS AND BOOKS OF RECORD OF PUREBRED ANIMALS

Notice is hereby given that the Secretary of Agriculture, pursuant to the authority vested in him by section 201, paragraph 1606 of the Tariff Act of 1930, as amended (19 U. S. C. 1201, par. 1606), proposes to withdraw recognition of the book of record for Thoroughbred and Arabian horses entitled "Registro-Matricula de Caballos de Pura Sangre" published by the Minister of Agriculture, General Livestock Administration, Madrid, Spain, and to amend the regulations governing the recognition of breeds and books of record of purebred animals by removing the name of the stud book from the list of books of record named in 9 CFR 151.10 (a) (14 F. R. 159), as amended, under the subheading "Horses".

Any person who wishes to submit written data or arguments concerning the proposed amendment may do so by filing them with the Chief of the Bureau of Animal Industry, Agricultural Research Administration, United States Department of Agriculture, Washington 25, D. C., within thirty days after the date of publication of this notice in the FEDERAL REGISTER.

(Sec. 201, par. 1606, 46 Stat. 673; 19 U. S. C. and Supp., 1201, par. 1606)

Done at Washington, D. C., this 6th day of November 1950.

[SEAL] CHARLES F. BRANNAN, *Secretary of Agriculture.*

[F. R. Doc. 50-10056; Filed, Nov. 9, 1950; 8:48 a. m.]

Production and Marketing Administration

[P. & S. Docket No. 425]

SIOUX CITY STOCK YARDS CO.

NOTICE OF PETITION FOR MODIFICATION

On March 19, 1947 (6 A. D. 179), February 14, 1948 (7 A. D. 132), September 9, 1948 (7 A. D. 864), and February 9, 1950 (9 A. D. 203), respectively, by orders of the Judicial Officer of the Department of Agriculture, the respondent was permitted to make effective certain temporary modifications of its charges for the yarding and feeding of livestock. On November 3, 1950, the respondent filed a petition requesting further temporary modifications of its charges as follows:

YARDAGE CHARGES

	Present rate	Proposed rate
Cattle:	*Cents per head*	*Cents per head*
Salable receipts	65	70
Direct to packers	33	35
Resales in commission division	65	70
Calves:		
Salable receipts	38	41
Direct to packers	19	21
Resales in commission division	38	41
Hogs:		
Salable receipts	22	24
Direct to packers	11	12
Resales in commission division	22	24
Sheep:		
Salable receipts	14	15
Direct to packers	7	8
Resales in commission division	14	15

MARGINS ON SALES OF FEED

	Per ton	*Per ton*
Hay—Prairie	$11.00	$12.00
Hay—Alfalfa	11.00	12.00
	Cents per bu.	*Cents per bu.*
Corn	40	50
Oats	25	35

If authorized the modifications will produce additional revenues for the respondent and increase the cost of marketing to the shippers. Accordingly, it appears that public notice should be given of the filing of the petition in order that all interested persons may have an

opportunity to be heard in the matter.

Now, therefore, notice is hereby given to the public and to all interested persons of the filing of such petition.

All interested persons who desire to be heard upon the matter requested in said petition shall notify the Hearing Clerk, United States Department of Agriculture, Washington 25, D. C., within 15 days from the date of publication of this notice.

Done at Washington, D. C., this 6th day of November 1950.

[SEAL] KATHERINE L. MASON,
Hearing Clerk.

[F. R. Doc. 50-10095; Filed, Nov. 9, 1950; 8:53 a. m.]

NOTICES

DEPARTMENT OF THE INTERIOR

Office of the Secretary

[Order No. 2509, Amdt. 10]

DELEGATION OF AUTHORITY WITH RESPECT TO CLAIMS RELATING TO IRRIGATION WORKS

Paragraphs (d) and (e) of section 22 of Order No. 2509 are respectively revised to read as follows:

SEC. 22. *Claims relating to irrigation works.* * * *

(d) The Solicitor of the Department of the Interior is authorized to determine whether claims for damage to or loss of property, personal injury, or death arising out of activities of the Bureau of Reclamation shall be allowed in whole or in part or shall be disallowed.

(e) Subject to the direction and supervision of the Solicitor, the Regional Counsels of the Bureau of Reclamation are severally authorized to determine whether claims not exceeding $1,000 for damage to or loss of property, personal injury, or death arising out of activities of the Bureau of Reclamation shall be allowed in whole or in part or shall be disallowed.

* * * * *

OSCAR L. CHAPMAN,
Secretary of the Interior.

NOVEMBER 3, 1950.

[F. R. Doc. 50-10049; Filed, Nov. 9, 1950; 8:47 a. m.]

DEPARTMENT OF AGRICULTURE

Forest Service

SITGREAVES NATIONAL FOREST

REMOVAL OF TRESPASSING HORSES

Whereas a number of horses are trespassing and grazing on land in the Bear Canyon Allotment and the South Portion of the Pinedale Community Allotment, Pinedale Ranger District, Sitgreaves National Forest, State of Arizona; and

Whereas these horses are consuming forage needed for permitted livestock, are causing extra expense to established permittees, and are injuring national-forest lands;

Now, therefore, by virtue of the authority vested in the Secretary of Agriculture by the act of June 4, 1897 (30 Stat., 35; 16 U. S. C. 551), and the act of February 1, 1905 (33 Stat., 628, 16 U. S. C. 472), the following order is issued for the occupancy, use, protection, and administration of the Bear Canyon Allotment and the south portion of the Pinedale Community Allotment, Pinedale Ranger District, Sitgreaves National Forest:

Temporary closure from livestock grazing. (a) The following-described area within the Sitgreaves National Forest is hereby closed from November 15, 1950 to April 30, 1951 to the grazing of horses, excepting those that are lawfully grazing on or crossing land in such area pursuant to the regulations of the Secretary of Agriculture, or which are used in connection with operations authorized by such regulations, or used as riding, pack, or draft animals by persons traveling over such land:

All of the Bear Canyon Allotment and the South Portion of the Pinedale Community Allotment, Pinedale Ranger District, Sitgreaves National Forest. Beginning at Pinedale Ranger Station approximately in the center of Section 32, T. 11 N., R. 20 E., running south 45 degrees, east 2 miles to the fence at the northeast corner of the Juniper Ridge allotment in the center of Sec. 4, T. 10 N., R. 20 E., along this fence in a southeasterly direction 2 miles to a point ¼ mile east of the southwest corner of Sec. 9, same township and range; thence east 2 miles along the fence between Juniper Ridge and Bear Canyon allotments to ¼ mile east of the southwest corner of Sec. 11, T. 10 N., R. 20 E.; thence southwesterly along fence on the east and south sides of Bear Canyon allotment approximately 6 miles to a point ¼ mile north of the southwest corner of Sec. 31, T. 10 N., R. 20 E.; thence northwesterly along the fence between the Pinedale allotment and Fort Apache Indian Reservation approximately 9 miles to the ¼ corner on the west side of Sec. 7, T. 10 N., R. 19 E.; thence northeasterly along the fence on the west side of Pinedale Community allotment approximately 2½ miles to the southwest corner of Sec. 28, T. 11 N., R. 19 E.; thence easterly along the Aripine, Clay Springs, Pinedale, Linden County road to the place of beginning.

(b) Officers of the United States Forest Service are hereby authorized to dispose of, in the most humane manner, all horses found trespassing or grazing in violation of this order.

(c) Public notice of intention to dispose of such horses shall be given by posting notices in public places or advertising in a newspaper of general circulation in the locality in which the Sitgreaves National Forest is located.

Done at Washington, D. C., this 6th day of November 1950.

Witness my hand and the seal of the Department of Agriculture.

[SEAL] CHARLES F. BRANNAN,
Secretary of Agriculture.

[F. R. Doc. 50-10057; Filed, Nov. 9, 1950; 8:49 a. m.]

Production and Marketing Administration

HAWAII

NOTICE OF ADDITIONAL HEARING ON FAIR PRICES FOR SUGARCANE

Pursuant to the authority contained in section 301 (c) (2) of the Sugar Act of 1948 (61 Stat. 929; U. S. C. Sup. 1131), notice is hereby given that an additional public hearing on the matter of fair and reasonable prices for the 1951 crop of sugarcane will be held on November 21, 1950, in Hilo, on the Island of Hawaii, at 9:30 a. m. The hearing will be held in the Post Office Building unless a change in the location is announced by the Director, Hawaiian Area Office, Honolulu, T. H.

This additional hearing is being called at the request of the Hawaiian sugar industry for the purpose of receiving supplementary evidence likely to be of assistance to the Secretary of Agriculture in establishing fair and reasonable prices for the 1951 crop of sugarcane to be paid under either purchase or toll agreements by processors, who as producers, apply for payments under the act. The original hearing on the same matter was held in Hilo on September 11, 1950.

Thomas H. Allen, A. A. Greenwood, and Will N. King are hereby designated as presiding officers to conduct either jointly or severally the foregoing hearing.

Issued this 7th day of November 1950.

[SEAL] RALPH S. TRIGG,
Administrator.

[F. R. Doc. 50-10094; Filed, Nov. 9, 1950; 8:53 a. m.]

DEPARTMENT OF COMMERCE

National Production Authority

[NPA Delegation 3]

DELEGATION OF AUTHORITY TO NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS

Pursuant to the authority of the Defense Production Act of 1950 (Public Law 774, 81st Cong.) and Executive Order 10161 (15 F. R. 6105) there is hereby delegated to the National Advisory Committee for Aeronautics the authority to apply ratings to direct contracts and purchase orders to meet authorized procurement and construction programs of the National Advisory Committee for Aeronautics.

The National Advisory Committee for Aeronautics is also authorized to assign the right to apply ratings to persons placing orders for materials to be delivered to or for the account of the National Advisory Committee for Aeronautics to meet authorized programs.

This authority may be redelegated by the National Advisory Committee for Aeronautics to appropriate agencies of the National Advisory Committee for Aeronautics or to its authorized agents.

The exercise of this authority shall conform to the terms of the regulations and orders of the National Production Authority and also to priorities and allocations policy directives issued by the

National Advisory Committee for Aeronautics and subject to approval by the National Production Authority.

In applying ratings on direct contracts and purchase orders, the certification and procedure stated in NPA Reg. 2 (15 F. R. 6632) shall be used. In assigning the right to apply ratings on contracts and orders, the following certification shall be used: "By authority of the National Production Authority, rating DO (2-digit program code) is assigned to the deliveries on this purchase order or contract." This certification shall be authenticated with the signature of an authorized official of the National Advisory Committee for Aeronautics or its authorized agents.

The use of this authority is limited to such quantitative allocations as may be assigned by the National Production Authority to the National Advisory Committee for Aeronautics, and to such conditions as may be imposed by the National Production Authority on use, records, and reports.

This authority shall not be used to rate direct procurement or contractors' purchase of construction equipment for use on construction in the Zone of Interior; purchases from exclusively retail establishments, except in emergency situations and only for small amounts to prevent imminent stoppage; or procurement of any of the following items: Commercial office equipment and supplies; flags, bunting, flagstaffs, pennants, insignia and medals; vending machines; portable household fans; commercial type luggage; barber chairs; card tables; books, maps, and periodicals; brooms and mops for household use; and domestic type dishwashing machinery.

This directive shall take effect on November 8, 1950.

NATIONAL PRODUCTION AUTHORITY,
W. H. HARRISON,
Administrator.

Approved:

PHILIP B. FLEMING,
Acting Secretary of Commerce.

[F. R. Doc. 50-10144; Filed, Nov. 8, 1950; 12:28 p. m.]

[NPA Delegation 4]

DELEGATION OF AUTHORITY TO THE UNITED STATES COAST GUARD

Pursuant to the authority of the Defense Production Act of 1950 (Public Law 774, 81st Cong.) and Executive Order 10161 (15 F. R. 6105) there is hereby delegated to the United States Coast Guard the authority to apply ratings to direct contracts and purchase orders to meet authorized procurement and construction programs of the United States Coast Guard.

The United States Coast Guard is also authorized to assign the right to apply ratings to persons placing orders for materials to be delivered to or for the account of the United States Coast Guard to meet authorized programs.

This authority may be redelegated by the United States Coast Guard to appropriate agencies of the United States Coast Guard or to its authorized agents.

The exercise of this authority shall conform to the terms of the regulations and orders of the National Production Authority and also to priorities and allocations policy directives issued by the United States Coast Guard and subject to approval by the National Production Authority.

In applying ratings on direct contracts and purchase orders, the certification and procedure stated in NPA Reg. 2 (15 F. R. 6632) shall be used. In assigning the right to apply ratings on contracts and orders, the following certification shall be used: "By authority of the National Production Authority, rating DO (2-digit program code) is assigned to the deliveries on this purchase order or contract." This certification shall be authenticated with the signature of an authorized official of the United States Coast Guard or its authorized agents.

The use of this authority is limited to such quantitative allocations as may be assigned by the National Production Authority to the United States Coast Guard and to such conditions as may be imposed by the National Production Authority on use, records, and reports.

This authority shall not be used to rate direct procurement or contractors' purchase of construction equipment for use on construction in the Zone of Interior; civilian type items for resale in Post Exchanges and Ship Stores; purchases from exclusively retail establishments, except in emergency situations and only for small amounts to prevent imminent stoppage; or procurement of any of the following items: Commercial office equipment and supplies; flags, bunting, flagstaffs, pennants, insignia and medals; vending machines; portable household fans; commercial type luggage; barber chairs; card tables; books, maps, and periodicals; brooms and mops for household use; and domestic type dishwashing machinery.

This directive shall take effect on November 8, 1950.

NATIONAL PRODUCTION AUTHORITY,
W. H. HARRISON,
Administrator.

Approved:

PHILIP B. FLEMING,
Acting Secretary of Commerce.

[F. R. Doc. 50-10145; Filed, Nov. 8, 1950; 12:28 p. m.]

FEDERAL POWER COMMISSION

[Docket No. G-1521]

GREENFIELD GAS CO., INC.

NOTICE OF APPLICATION

NOVEMBER 6, 1950.

Take notice that Greenfield Gas Company, Inc. (Applicant), an Indiana corporation of 114 North State Street, Greenfield, Indiana, filed on October 31, 1950, an application for a certificate of public convenience and necessity, pursuant to section 7 of the Natural Gas Act, authorizing the acquisition and operation of the gas utility plant, property and equipment presently in use in rendering natural-gas service to the public in and adjacent to the Towns of Fortville, Ingalls, McCordsville, and Mohawk, all in the State of Indiana, which facilities, among others, are being acquired by Indiana Gas & Water Company, Inc., from Eastern Indiana Gas Company in the proceedings in Docket No. G-1441 and in accordance with temporary authorization granted Indiana Gas & Water Company, Inc., on October 4, 1950, all as an alternative to a request for an order disclaiming jurisdiction over the proposed acquisition and operation by Applicant of said facilities.

The application recites that Applicant desires to acquire the aforesaid facilities from Indiana Gas & Water Company, Inc., concurrently with the acquisition thereof by Indiana Gas from Eastern Indiana Gas Company so that there will be no intermediate operation of such properties by Indiana Gas prior to the sale thereof to Applicant under an agreement entered into August 4, 1950, between Applicant and Indiana Gas.

The total cost of the facilities to be acquired by Applicant from Indiana Gas is $45,000, which will be paid in cash from funds on hand and the proceeds of the sale of securities authorized by the Public Service Commission of Indiana.

Protests or petitions to intervene may be filed with the Federal Power Commission, Washington 25, D. C., in accordance with the rules of practice and procedure (18 CFR 1.8 or 1.10) on or before the 24th day of November 1950. The application is on file with the Commission for public inspection.

[SEAL] LEON M. FUQUAY,
Secretary.

[F. R. Doc. 50-10060; Filed, Nov. 9, 1950; 8:50 a. m.]

[Docket No. G-1467]

CITY OF CARLISLE, KY.

NOTICE OF ORDER FOR SALE OF NATURAL GAS

NOVEMBER 6, 1950.

Notice is hereby given that, on November 3, 1950, the Federal Power Commission issued its order entered November 2, 1950, directing Central Kentucky Natural Gas Company to establish physical connection of its facilities, for the sale of natural gas, with those of the City of Carlisle, Kentucky.

[SEAL] LEON M. FUQUAY,
Secretary.

[F. R. Doc. 50-10033; Filed, Nov. 9, 1950; 8:45 a. m.]

[Docket Nos. G-1399, G-1400,

VIRGINIA GAS TRANSMISSION CORP. AND LYNCHBURG PIPE LINE CO.

NOTICE OF CONTINUANCE OF HEARING

NOVEMBER 3, 1950.

Upon consideration of the Petition by Lynchburg Pipe Line Company for continuance of hearing in the above-

designated matters, notice is hereby given that the hearing in the above-designated matters now scheduled for November 9, 1950, be and it is hereby continued to December 8, 1950, at 10:00 a. m., in the Commission's Hearing Room, at 1800 Pennsylvania Avenue NW., Washington, D. C.

[SEAL] LEON M. FUQUAY,
Secretary.

[F. R. Doc. 50-10034; Filed, Nov. 9, 1950; 8:45 a. m.]

GENERAL SERVICES ADMINISTRATION

COMMISSIONER OF PUBLIC BUILDINGS SERVICE, COMMISSIONER OF FEDERAL SUPPLY SERVICE, AND ARCHIVIST OF UNITED STATES

DELEGATION OF AUTHORITY WITH RESPECT TO UNINTERRUPTED PERFORMANCE OF FUNCTIONS

General Regulation No. 2 of September 27, 1950, 15 F. R. 6611, is hereby amended by revoking paragraph 2 thereof, effective 10 days after the date hereof.

Dated: November 3, 1950.

JESS LARSON,
Administrator.

[F. R. Doc. 50-10061; Filed, Nov. 9, 1950; 8:50 a. m]

SECURITIES AND EXCHANGE COMMISSION

[File No. 30-37]

WASHINGTON GAS AND ELECTRIC CO.

NOTICE OF FILING OF APPLICATION

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 6th day of November A. D. 1950.

Notice is hereby given that an application has been filed with this Commission, pursuant to section 5 (d) of the Public Utility Holding Company Act of 1935 (the "act"), by Washington Gas and Electric Company ("Washington"), a registered holding company and a public utility company, for an order under said act declaring that the applicant has ceased to be a holding company.

All interested persons are referred to said application which is on file in the offices of the Commission for a complete statement relating to the requested finding and order, which is summarized as follows:

The application states that pursuant to the plan of reorganization submitted under section 11 (f) of the act by Nathan A. Smyth, as Trustee in Reorganization under Chapter X of the Bankruptcy Act of Washington, which plan was approved by the Commission on March 9, 1949, accepted by the bondholders and general creditors of Washington on April 30, 1949, confirmed by order of the District Court of the United States for the Southern District of New York on October 5, 1949, and the consummation of which was ordered by said court by orders dated April 14, 1950 and July 27, 1950, distribution of the common stock of Washington and of its former sole subsidiary, Southern Utah Power Company, a public utility company, has commenced and Washington has been divested of its interest in the common stock of Southern Utah Power Company.

The application indicates that Washington is now solely a public utility company and that it does not now, directly or indirectly, own, control or hold with power to vote any outstanding voting securities of a public utility company or of a company which is a holding company within the provisions of the act.

Notice is further given that any interested person may, not later than November 20, 1950, at 5:30 p. m., e. s. t., request the Commission in writing that a hearing be held on such matter stating the reasons for such request, the nature of his interest, and the issues of fact or law raised by said application which he desires to controvert, or may request that he be notified if the Commission should order a hearing thereon. At any time thereafter said application, as filed or as amended, may be granted. Any such request should be addressed: Secretary, Securities and Exchange Commission, 425 Second Street NW., Washington 25, D. C.

By the Commission.

[SEAL] NELLYE A. THORSEN,
Assistant Secretary.

[F. R. Doc. 50-10042; Filed, Nov. 9, 1950; 8:47 a. m.]

DUNCAN ROSS CO.

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 3d day of November 1950.

In the matter of Duncan Ross, dba Duncan Ross Company, 342 Madison Avenue, New York, N. Y.

I. The Commission's public official files disclose that Duncan Ross, doing business as Duncan Ross Company, hereinafter referred to as registrant, is registered as a broker-dealer pursuant to section 15 (b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948 or 1949 as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

[1] Filed as part of the original document.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has wilfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950 at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a Hearing Examiner to be designated by the Commission. On such date the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20th, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27th, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN,
Assistant Secretary.

[F. R. Doc. 50-10043; Filed, Nov. 9, 1950; 8:47 a. m.]

FREDERICK J. ROSE

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 3d day of November 1950.

In the matter of Frederick J. Rose, 50 Broad Street, New York City, N. Y.

I. The Commission's public official files disclose that Frederick J. Rose, hereinafter referred to as registrant, is registered as a broker-dealer pursuant to section 15 (b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948 or 1949 as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has willfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950 at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a Hearing Examiner to be designated by the Commission. On such date the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20th, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

[1] Filed as part of the original document.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN, *Assistant Secretary.*

[F. R. Doc. 50-10044; Filed, Nov. 9, 1950; 8:47 a. m.]

EDWARD E. REED

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 3d day of November 1950.

In the matter of Edward E. Reed, 225 Broadway, New York, N. Y.

I. The Commission's public official files disclose that Edward E. Reed, hereinafter referred to as registrant, is registered as a broker-dealer pursuant to section 15 (b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948 or 1949 as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has wilfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950 at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a Hearing Examiner to be designated by the Commission. On such date the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20th, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN, *Assistant Secretary.*

[F. R. Doc. 50-10045; Filed, Nov. 9, 1950; 8:47 a. m.]

GEORGE HARRIS

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its

office in the city of Washington, D. C., on the 3d day of November 1950.

In the matter of George Harris, Windsor Hotel, Cumberland, Maryland.

I. The Commission's public official files disclose that George Harris, hereinafter referred to as registrant, is registered as a broker-dealer pursuant to section 15 (b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948 or 1949 as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has wilfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950 at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a hearing Examiner to be designated by the Commission. On such date the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN,
Assistant Secretary.

[F. R. Doc. 50-10046; Filed, Nov. 9, 1950; 8:47 a. m.]

FRANK B. PRYOR

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 3d day of November 1950.

In the matter of Frank B. Pryor, 1517 Buchanan Street NW., Washington, D. C.

I. The Commission's public official files disclose that Frank B. Pryor, hereinafter referred to as registrant is registered as a broker-dealer pursuant to section 15 (b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948, or 1949, as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has wilfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950, at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a Hearing Examiner to be designated by the Commission. On such date, the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN,
Assistant Secretary.

[F. R. Doc. 50-10047; Filed, Nov. 9, 1950; 8:47 a. m.]

M. SIMPSON CO.

ORDER FOR PROCEEDINGS AND NOTICE OF HEARING

At a regular session of the Securities and Exchange Commission held at its office in the city of Washington, D. C., on the 3d day of November 1950.

[1] Filed as part of the original document.

In the matter of Marcus Simpson dba M. Simpson Company, Delaware Trust Building, Wilmington, Delaware.

I. The Commission's public official files disclose that Marcus Simpson doing business as M. Simpson Company, hereinafter referred to as registrant, is registered as a broker-dealer pursuant to section 15–(b) of the Securities Exchange Act of 1934.

II. The Records Officer of the Commission has filed with the Commission a statement, a copy of which is attached hereto and made a part hereof,[1] stating that registrant did not file with the Commission reports of his financial condition during the calendar years 1943, 1944, 1945, 1946, 1947, 1948, or 1949 as required by section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted thereunder.

III. The information reported to the Commission by its Records Officer as set forth in paragraph II hereof tends, if true, to show that registrant violated Section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section.

IV. The Commission, having considered the aforesaid information, deems it necessary and appropriate in the public interest and for the protection of investors that proceedings be instituted to determine:

(a) Whether the statements set forth in paragraph II hereof are true;

(b) Whether registrant has wilfully violated section 17 (a) of the Securities Exchange Act of 1934 and Rule X-17A-5 adopted under said section;

(c) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, it is in the public interest to revoke registration of registrant; and

(d) Whether, pursuant to section 15 (b) of the Securities Exchange Act of 1934, pending final determination, it is necessary or appropriate in the public interest or for the protection of investors to suspend the registration of registrant.

V. *It is ordered*, That registrant be given an opportunity for hearing as set forth in paragraph IV hereof on the 27th day of November 1950 at the main office of the Securities and Exchange Commission, located at 425 Second Street NW., Washington 25, D. C., before a Hearing Examiner to be designated by the Commission. On such date the Hearing Room Clerk in Room 101, North Building, will advise the parties and the Hearing Examiner as to the room in which such hearing will be held. The Commission will consider any motion with respect to a change of place of said hearing if said motion is filed with the Secretary of the Commission on or before November 20, 1950. Upon completion of any such hearing in this matter the Hearing Examiner shall prepare a recommended decision pursuant to Rule IX of the rules of practice unless such decision is waived.

It is further ordered, That in the event registrant does not appear personally or through a representative at the time and place herein set or as otherwise ordered, the Hearing Room Clerk shall file with the Records Officer of the Commission a written statement to that effect and thereupon the Commission will take the record under advisement for decision.

This order and notice shall be served on registrant personally or by registered mail forthwith, and published in the FEDERAL REGISTER not later than fifteen (15) days prior to November 27, 1950.

In the absence of an appropriate waiver, no officer or employee of the Commission engaged in the performance of investigative or prosecuting functions in this or any factually related proceeding will be permitted to participate or advise in the decision upon the matter except as witness or counsel in proceedings held pursuant to notice. Since this proceeding is not "rule making" within the meaning of section 4 (c) of the Administrative Procedure Act, it is not deemed to be subject to the provisions of the section delaying the effective date of any final Commission action.

By the Commission.

[SEAL] NELLYE A. THORSEN, *Assistant Secretary.*

[F. R. Doc. 50-10048; Filed, Nov. 9, 1950; 8:47 a. m.]

[1] Filed as part of the original document.

TAX COURT OF THE UNITED STATES

USE OF TITLE "CHIEF JUDGE"

Pursuant to the provisions of section 32, act of June 25, 1948 (Public Law 773), as amended by section 127, Public Law 72, approved May 24, 1949, applicable to The Tax Court of the United States, the title "Chief Judge," The Tax Court of the United States, will be used instead of "Presiding Judge," The Tax Court of the United States, effective from October 30, 1950.[1]

Dated: November 3, 1950.

JOHN W. KERN, *Chief Judge, The Tax Court of the United States.*

[F. R. Doc. 50-10146; Filed, Nov. 9, 1950; 8:55 a. m.]

[1] See Title 26, Chapter II, Part 701, *supra.*

DEPARTMENT OF JUSTICE

Office of Alien Property

AUTHORITY: 40 Stat. 411, 55 Stat. 839, Pub. Laws 322, 671, 79th Cong., 60 Stat. 50, 925; 50 U. S. C. and Supp. App. 1, 616; E. O. 9193, July 6, 1942, 3 CFR, Cum. Supp., E. O. 9567, June 8, 1945, 3 CFR, 1945 Supp., E. O. 9788, Oct. 14, 1946, 11 F. R. 11981.

[Vesting Order 15365]

ANNA ELIZABETH LANGE

In re: Stock owned by and debt owing to Anna Elizabeth Lange also known as Anna Elisabeth Lange, as Anna Lange as Anna E. Lange and as Frau Anna Lange. F-28-22917-D-2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Anna Elizabeth Lange, also known as Anna Elisabeth Lange, as Anna Lange as Anna E. Lange and as Frau Anna Lange, whose last known address is 2 Parkstrasse, Oldenburg, I. O., Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the property described as follows:

a. Those certain shares of stock described in Exhibit A, attached hereto and by reference made a part hereof, registered in the names of the persons set forth in Exhibit A, together with all declared and unpaid dividends thereon,

b. Ten (10) shares of $1.00 par value common stock of The Adams Express Company, 40 Wall Street, New York, New York, a Joint Stock Association, organized under the common law of the State of New York, evidenced by a certificate numbered No–133134, registered in the name of A. C. Lange & Son Aktiebolag, Malmo, Sweden, together with all declared and unpaid dividends thereon,

c. Ten and 3,000/10,000 (10.3000) shares of common stock class A (New) of General Gas & Electric Corporation (in dissolution), 67 Broad Street, New York, New York, a corporation organized under the laws of the State of Delaware, evidenced by three (3) certificates numbered GO 72164 for ten (10) shares, IGS 43999 for fifteen hundred ten thousandths (0.1500) of a share and IGS 53551 for fifteen hundred ten thousandths (0.1500) of a share, registered in the name of Anna E. Lange, Oldenburg I Oldbg. Steinweg 53, Germany, together with all declared and unpaid dividends thereon, and any and all rights to exchange the aforesaid certificates for two (2) shares of $7.50 par value common stock of Florida Power Corporation, together with all declared and unpaid dividends thereon and $20.65 in cash under the terms of the Plan of Divestment of Assets of General Gas & Electric Corporation ordered by the District Court of the United States for the Southern District of New York in October 1945 and effective as of December 1, 1945, and

d. That certain debt or other obligation owing to Anna Elizabeth Lange also known as Anna Elisabeth Lange as Anna E. Lange as Anna Lange and as Frau Anna Lange by H. F. Hintzen, Box 976, Tucson, Arizona, in the amount of $41.64, as of August 9, 1950, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, Anna Elizabeth Lange also known as Anna Elisabeth Lange, as Anna Lange as Anna E. Lange and as Frau Anna Lange, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States re-

quires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 25, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

EXHIBIT A

Name and address of issuing corporation	State of incorporation	Certificate Nos.	Number of shares	Par value	Type of stock	Form of registration	File No.
North American Aviation, Inc., 5701 Imperial Highway, Inglewood, Calif.	Delaware	A2929	15	$1.00	Common	Miss Anna E. Lange	F-28-22917-D-2.
Radio Corporation of America, 30 Rockefeller Plaza, New York, N. Y.	do	N0183203	5	No par	do	Miss Anna Lange	F-28-22917-D-3.
		N0001731	10	No par			
Kennecott Copper Corp., 120 Broadway, New York, N. Y.	New York	6907249	5	No par	do	Anna E. Lange	F-28-22917-D-4.
		6207425	5	No par			
International Telephone & Telegraph Corp., 67 Broad St., New York, N. Y.		NNF 254469	9	No par	do	Miss Anna Lange	F-28-22917-D-5.
		NNF 220733	1	No par		do	
		NNF 636561	10	No par		Miss Anna E. Lange	
Trans World Airlines, Inc. (formerly Transcontinental & Western Air, Inc.), 101 West 11th St., Kansas City, Mo.	Delaware	A18-923	1	$5.00	do	do	F-28-22917-D-7.
Houdaille Hershey Corp., 1500 Fisher Bldg., Detroit, Mich.	Michigan	NC08501	10	No par	do	A. G. Lange & Son	F-28-22917-D-8.
Servel, Inc., 51 East 42d St., New York, N. Y.	Delaware	19648	10	$1.00	do	do	F-28-22917-D-10.
Bank of America National Trust & Savings Assn., 550 Montgomery St., San Francisco, Calif.		J 44839	1	$6.25	do	do	F-28-22917-D-11.
		P 64824	1				
Transamerica Corp., 4 Columbus Ave., San Francisco, Calif.	do	NYC 86816	5	$2.00	do	do	F-28-22917-D-12.

[F. R. Doc. 50-10065; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15384]

MRS. GEORGETTE KURTH

In re: Rights of Mrs. Georgette Kurth, nee Betrand, under insurance contract. File No. F-28-8930-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Mrs. Georgette Kurth, nee Bertrand, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That one-half of the net proceeds due or to become due under a contract of insurance evidenced by policy No. 1,679,467, issued by The Travelers Insurance Company, Hartford, Connecticut, to Kathie Ziegler and Hermann Ziegler, together with the right to demand, receive and collect one-half of said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 27, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10066; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15417]

CLARA J. AUCHTER

In re: Rights of Clara J. Auchter under insurance contract. File No. F-28-30176 H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Clara J. Auchter, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 76777058 issued by the Metropolitan Life Insurance Company, New York, New York to Clara J. Auchter, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10067; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15418]

OLGA ARNS

In re: Rights of Olga Arns under insurance contract. File No. F-28-26721-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Olga Arns, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 208427, issued by the West Coast Life Insurance Company, San Francisco, California, to Ernst Heinrich Arns, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10068; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15419]

DOROTHY LUDEMANN BEITZ

In re: Rights of Dorothy Ludemann Beitz under insurance contracts. Files: F-28-30624 H-1 and H-2.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Dorothy Ludemann Beitz, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under contracts of insurance evidenced by policies No. 818,861,M and 101,398,695, issued by the Metropolitan Life Insurance Company, New York, New York, to Dorothy Ludemann Beitz, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10069; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15420]

ELSIE ZENGEL BENDIXEN

In re: Rights of Elsie Zengel Bendixen, also known as Elisabeth Bendixen, nee Zengel, under insurance contract. File No. F-28-91-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Elsie Zengel Bendixen, also known as Elisabeth Bendixen, nee Zengel, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 11 306 168, issued by the New York Life Insurance Company, New York, New York, to Mathias Bendixen, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10070; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15421]

MARGARET L. DRENOWATZ

In re: Rights of Margaret L. Drenowatz under insurance contract. File No. F-28-24744-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Margaret L. Drenowatz, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by Policy No. 4 022 645 A issued by the Metropolitan Life Insurance Company, New York, New York, to Gustav A. Drenowatz, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as

a national of a designated enemy country (Germany),

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10071; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15422]

GERTRUDE ECKHARDT

In re: Rights of Gertrude Eckhardt, now Loth, under insurance contract. File No. F-28-30175 H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Gertrude Eckhardt, now Loth, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 1 430 491 M issued by the Metropolitan Life Insurance Company, New York, New York, to Gertrude Eckhardt, now Loth, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10072; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15423]

PAUL EUCHNER

In re: Rights of Paul Euchner under insurance contract. File No. F-28-24735-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Paul Euchner, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 92239245, issued by the Metropolitan Life Insurance Company, New York, New York, to Paul Euchner, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10073; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15424]

ALFRED FALKENBERG

In re: Rights of Alfred Falkenberg under insurance contracts. File Nos. F-28-24729-H-1, H-2.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Alfred Falkenberg, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under contracts of insurance evidenced by policies No. 104 402 635 and 106 187 867, issued by the Metropolitan Life Insurance Company, New York, New York, to Alfred Falkenberg, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10074; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15425]

RUDOLPH FESSEL

In re: Rights of Rudolph Fessel under insurance contract. File No. F-28-24726-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Rudolph Fessel, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 93 817 737, issued by the Metropolitan Life Insurance Company, New York, New York, to Rudolph Fessel, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10075; Filed, Nov. 9, 1950; 8:51 a. m.]

[Vesting Order 15426]

ELSIE FISCHER

In re: Rights of Elsie Fischer under insurance contract. File No. F-28-24723-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Elsie Fischer, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 4 818 051 C, issued by the Metropolitan Life Insurance Company, New York, New York, to Elsie Fischer, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10076; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15427]

MARGARET FUETTEREK

In re: Rights of Margaret Fuetterek under insurance contract. File No. F-28-24756-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Margaret Fuetterek, whose last known address is Germany, is a resident of Germany and a national of a designated country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 72 764 359, issued by the Metropolitan Life Insurance Company, New York, New York, to Margaret Fuetterek, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10077; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15428]

MARGA FRANZ

In re: Rights of Marga Franz under insurance contract. File No. D-28-10936-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Marga Franz, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. certificate 75—group policy 612-G, issued by the John Hancock Mutual Life Insurance Company, Boston, Massachusetts, to Paul Franz, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10078; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15438]

GUSTAV MAUS AND UNION TRUST CO. OF PITTSBURGH

In re: Trust agreement dated June 27, 1925, between Gustav Maus, grantor, and The Union Trust Company of Pittsburgh, trustee, and amendments thereto. File No. D-28-2469 G-2.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Gustav Maus, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the property described as follows:

a. All right, title, interest and claim of any kind or character whatsoever of the person named in subparagraph 1 hereof in and to and arising out of or under that certain trust agreement dated June 27, 1925, by and between Gustav Maus, grantor, and The Union Trust Company of Pittsburgh, trustee, and amendments thereto, dated July 19, 1934, and June 15, 1936, including particularly but not limited to the right of Gustav Maus to revoke the trust agreement, presently being administered by Mellon National Bank and Trust Company, as trustee, Post Office Box 926, Pittsburgh 30, Pennsylvania, and

b. All property in the possession, custody or control of the Mellon National Bank and Trust Company, Post Office Box 926, Pittsburgh 30, Pennsylvania, as trustee of the trust created pursuant to that certain trust agreement dated June 27, 1925, by and between Gustav Maus, grantor and The Union Trust Company of Pittsburgh, trustee, and amendments thereto, dated July 19, 1934, and June 15, 1936,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above subject to all lawful fees and disbursements of the Mellon National Bank and Trust Company, Post Office Box 926, Pittsburgh 30, Pennsylvania, as trustee under that certain trust agreement dated June 27, 1925, by and between Gustav Maus, grantor, and The Union Trust Company of Pittsburgh, trustee, and amendments thereto dated July 19, 1934, and June 15, 1936.

All such property so vested shall be held, used, administered, liquidated, sold, or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10081; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15429]

HILDA FRUBWALD

In re: Rights of Hilda Frubwald under insurance contract. File No. F-28-30529 H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Hilda Frubwald, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the net proceeds due or to become due under contract of insurance evidenced by policy No. 78619440, issued by The Prudential Insurance Company of America, Newark, New Jersey, to Hilda Frubwald, together with the right to demand, receive and collect said net proceeds,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10079; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15430]

HERMANN W. GEHRLING ET AL.

In re: Rights of Hermann W. Gehrling et al. under insurance contract. File No. F-28-24769-H-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Hermann W. Gehrling, whose last known address is Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the domiciliary personal representatives, heirs, next of kin, legatees and distributees, names unknown, of Hermann W. Gehrling, who there is reasonable cause to believe are residents of Germany, are nationals of a designated enemy country (Germany);

3. That the net proceeds due or to become due under a contract of insurance evidenced by policy No. 114132, issued by The Prudential Insurance Company of America, Newark, New Jersey, to Hermann W. Gehrling, together with the right to demand, receive and collect said net proceeds, is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, Hermann W. Gehrling or the domiciliary personal representatives, heirs, next of kin, legatees and distributees, names unknown, of Hermann W. Gehrling, the aforesaid nationals of a designated enemy country (Germany);

and it is hereby determined:

4. That to the extent that the person named in subparagraph 1 hereof and the domiciliary personal representatives, heirs, next of kin, legatees and distributees, names unknown, of Hermann W. Gehrling, are not within a designated enemy country, the national interest of the United States requires that such persons be treated as nationals of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10080; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15450]

MARGARET YURT

In re: Estate of Margaret Yurt, deceased. File D 28-7755 E. T. sec. 8354.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Josef Vasold, George Vasold, Maria Wagner, nee Vasold, Johann George Vasold, Hans Harrer and Anna Harrer, whose last known address is Germany, are residents of Germany and nationals of a designated enemy country (Germany);

2. That the domiciliary personal representatives, heirs, next of kin, legatees and distributees, names unknown, of Jacob Vasold, also known as J. Reisold, deceased, and of Margi Reisold, nee Hanle, deceased, who there is reasonable cause to believe are residents of Germany, are nationals of a designated enemy country (Germany);

3. That all right, title, interest and claim of any kind or character whatsoever of the persons identified in subparagraphs 1 and 2 hereof, and each of them, in and to the estate of Margaret Yurt, deceased, is property payable or deliverable to, or claimed by the aforesaid nationals of a designated enemy country (Germany);

4. That such property is in the process of administration by Lawrence J. Mackey, Administrator with the will annexed, acting under the judicial supervision of the County Court of Jefferson County, State of Kentucky;

and it is hereby determined:

5. That to the extent that the persons named in subparagraph 1 hereof and the domiciliary personal representatives, heirs, next of kin, legatees and distributees, names unknown, of Jacob Vasold, also known as J. Reisold, deceased, and of Margi Reisold, nee Hanle, deceased, are not within a designated enemy country, the national interest of the United States requires that such persons be treated as nationals of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10082; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15451]

HEINRICH ALBERT

In re: Debt owing to Heinrich Albert, also known as Heinrich F. Albert. F-28-911.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Heinrich Albert, also known as Heinrich F. Albert, whose last known address is 18 Flensburger Strasse, Berlin-Zehlendorf-West, Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the property described as follows: That certain debt or other obligation owing to Heinrich Albert, also known as Heinrich F. Albert, by Cravath, Swaine & Moore, 15 Broad Street, New York 5, New York, in the amount of $205.75, as of November 9, 1949, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10083; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15452]

F. B. BROAD AND SON

In re: Debt owing to F. B. Broad & Son. F-39-267-C-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That F. B. Broad & Son, the last known address of which is No. 1-2 Chome Makicho, Kyobashi-ku, Tokyo, Japan, is a corporation, partnership, association or other business organization, organized under the laws of Japan, and which has or, since the effective date of Executive Order 8389, as amended, has had its principal place of business in Tokyo, Japan, and is a national of a designated enemy country (Japan);

2. That the property described as follows: That certain debt or other obligation owing to F. B. Broad & Son, by Iron & Ore Corporation of America, 122 East 42d Street, New York, New York, representing sales commissions, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Japan);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Japan).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10084; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15454]

HAMBURGER ALLGEMEINE VERSICH AKTIENGESELLSCHAFT

In re: Debt owing to Hamburger Allgemeine Versich Aktiengesellschaft. F-28-8190-C-2.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Hamburger Allgemeine Versich Aktiengesellschaft, the last known address of which is Adolphsplatz 8, Hamburg 11, Germany, is a corporation, organized under the laws of Germany, and which has or, since the effective date of Executive Order 8389, as amended, has had its principal place of business in Germany and is a national of a designated enemy country (Germany);

2. That the property described as follows: That certain debt or other obligation of Johnson & Higgins, 63 Wall Street, New York 5, New York, in the amount of $110.94, as of December 31, 1945, representing a balance from General Average deposits made by various consignees of cargo as security for payment of General Average Charges on shipments on Str. Santa Rita accident (February 1938) and balance from allowances to cargo under Interest Nos. 644, 644a, 644b, and 644c, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, Hamburger Allgemeine Versich Aktiengesellschaft, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10085; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15461]

FELIX ROEFFS

In re: Bank account owned by Felix Roeffs. F-28-2464; E-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Felix Roeffs, who there is reasonable cause to believe is a resident of Germany, is a national of a designated enemy country (Germany);

2. That the property described as follows: That certain debt or other obligation owing to Felix Roeffs by The Chase National Bank of the City of New York, 20 Pine Street, New York, N. Y., arising out of a checking account, entitled Felix Roeffs, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10087; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15460]

REWALD AND WEISSBERG

In re: Debt owing to Rewald & Weissberg. F-28-8819-C-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Rewald & Weissberg, the last known address of which is Johannisstrasse 18/19, Berlin N. 4, Germany, is a corporation, partnership, association or other business organization, organized under the laws of Germany, and which has or, since the effective date of Executive Order 8389, as amended, has had its principal place of business in Germany, and is a national of a designated enemy country (Germany);

2. That the property described as follows: That certain debt or other obligation owing to Rewald & Weissberg, by The Spiegel Brothers Corporation, 806 Broadway, New York, New York, representing amounts due for merchandise purchased together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States, owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10086; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15463]

SANSHO CO.

In re: Debt owing to Sansho Company, also known as Sansho Goshi Kaisha. F-39-826 C-1.

Under the authority of the Trading With the Enemy Act, as amended, Ex-

ecutive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Sansho Company, also known as Sansho Goshi Kaisha, the last known address of which is Kitahama 2 Chome, Higashiki, Osaka, Japan, is a partnership, organized under the laws of Japan, and which has or, since the effective date of Executive Order 8389, as amended, has had its principal place of business in Osaka, Japan and is a national of a designated enemy country (Japan);

2. That the property described as follows: That certain debt or other obligation owing to Sansho Company, also known as Sanisho Goshi Kaisha, by Harriss and Vose, 60 Beaver Street, New York 4, New York, representing a balance from transactions in cotton futures, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Japan);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Japan).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10068; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15467]

DR. MARTA UNGER

In re: Debt owing to Dr. Marta Unger. F-28-12459-A-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Dr. Marta Unger, whose last known address is Schloss-Wolfsbrunnenweg 56, Heidelberg - Schlierbach, Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the property described as follows: That certain debt or other obligation of The Cleveland Trust Company, Cleveland, Ohio, in the amount of $564.08 representing a portion of funds on deposit in a savings account, Account No. 146208, entitled Mary E. Unger, Clarence P. Bill Attorney, maintained at the branch office of the aforesaid bank located at Euclid Avenue and East 105th Street, Cleveland 6, Ohio, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10090; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15465]

SIEMENS AND HALSKE AKTIENGESELLSCHAFT

In re: Debt owing to and bank account owned by Siemens & Halske Aktiengesellschaft, also known as Siemens & Halske, A. G. F-28-1851; C-1; E-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Siemens & Halske Aktiengesellschaft, also known as Siemens & Halske, A. G., the last known address of which is Berlin-Siemensstadt, Germany, is a corporation organized under the laws of Germany, and which has or, since the effective date of Executive Order 8389, as amended, has had its principal place of business in Berlin-Siemensstadt, Germany, and is a national of a designated enemy country (Germany);

2. That the property described as follows:

a. That certain debt or other obligation owing to Siemens & Halske Aktiengesellschaft, also known as Siemens & Halske, A. G., by International Standard Electric Corporation, 67 Broad Street, New York, New York, in the sum of $34,664.97, as of September 20, 1950, together with all rights to demand, enforce and collect the aforesaid debt or other obligation, and

b. That certain debt or other obligation of the Chase National Bank of the City of New York, 20 Pine Street, New York, New York, arising out of an Old Checks Outstanding Account, entitled "Siemens & Halske, A. G.", maintained with the aforesaid bank, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of or on account of, or owing to, or which is evidence of ownership or control by, Siemens & Halske Aktiengesellschaft, also known as Siemens & Halske, A. G., the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

EXHIBIT A—AGREEMENTS EXECUTED BY STANDARD TELEPHONES & CABLES, LIMITED, FOR AND ON BEHALF OF INTERNATIONAL STANDARD ELECTRIC CORPORATION

Parties to Agreement and Nature of Agreement

Western Electric Co., Limited (now called Standard Telephones & Cables, Limited), and Siemens & Halske, Aktiengesellschaft: Telod Agreement of July 25, 1924, and amendments.

Standard Telephones & Cables, Limited, and Siemens & Halske, Aktiengesellschaft: Carrier Current Agreement of December 1, 1932.

Standard Telephones & Cables, Limited, and Siemens & Halske, Aktiengesellschaft: Printing Telegraph Business Agreement of December 2, 1935, and amendments.

Standard Telephones & Cables, Ltd., and Siemens & Halske, Aktiengesellschaft, and Telefonaktiebolaget L. M. Ericsson: Open Wire Carrier Agreement of December 20, 1937.

Standard Telephones & Cables, Ltd., and Siemens & Halske, Aktiengesellschaft: Wire Transmission Agreement of April 4, 1938, and amendments.

[F. R. Doc. 50-10089; Filed, Nov. 9, 1950; 8:52 a. m.]

[Vesting Order 15468]

GUSTAV ZIEGLER

In re: Debt owing to Gustav Ziegler. F-28-25878-C-1.

Under the authority of the Trading With the Enemy Act, as amended, Executive Order 9193, as amended, and Executive Order 9788, and pursuant to law, after investigation, it is hereby found:

1. That Gustav Ziegler, whose last known address is Trostbrucke 1, Hamburg 11, Germany, is a resident of Germany and a national of a designated enemy country (Germany);

2. That the property described as follows:

a. That certain debt or other obligation of Johnson & Higins, 63 Wall Street, New York 5, New York, in the amount of $27.96, as of April 25, 1946, arising out of a refund of a General Average Deposit, paid as security for payment of charges due on a shipment, said shipment designated as Interest No. 642, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same, and

b. That certain debt or other obligation of Johnson & Higgins, 63 Wall Street, New York 5, New York, in the amount of $280.43, as of April 25, 1946, representing a balance from allowance to cargo for loss on a shipment, said shipment designated as Interest No. 722, together with any and all accruals thereto, and any and all rights to demand, enforce and collect the same,

is property within the United States owned or controlled by, payable or deliverable to, held on behalf of, or on account of, or owing to, or which is evidence of ownership or control by, the aforesaid national of a designated enemy country (Germany);

and it is hereby determined:

3. That to the extent that the person named in subparagraph 1 hereof is not within a designated enemy country, the national interest of the United States requires that such person be treated as a national of a designated enemy country (Germany).

All determinations and all action required by law, including appropriate consultation and certification, having been made and taken, and, it being deemed necessary in the national interest,

There is hereby vested in the Attorney General of the United States the property described above, to be held, used, administered, liquidated, sold or otherwise dealt with in the interest of and for the benefit of the United States.

The terms "national" and "designated enemy country" as used herein shall have the meanings prescribed in section 10 of Executive Order 9193, as amended.

Executed at Washington, D. C., on October 30, 1950.

For the Attorney General.

[SEAL] HAROLD I. BAYNTON,
Assistant Attorney General,
Director, Office of Alien Property.

[F. R. Doc. 50-10091; Filed, Nov. 9, 1950; 8:52 a. m.]

[Return Order 787]

FRANCESCO DE BENEDETTI

Having considered the claim set forth below and having issued a determination allowing the claim, which is incorporated by reference herein and filed herewith,

It is ordered, That the claimed property, described below and in the determination, including all royalties accrued thereunder and all damages and profits recoverable for past infringement thereof, be returned after adequate provision for taxes and conservatory expenses:

Claimant, Claim Number, Notice of Intention To Return Published, and Property

Francesco De Benedetti, Milan, Italy; Claim No. 41350; September 30, 1950 (15 F. R. 6823); property described in Vesting Order No. 2246 (8 F. R. 14020, October 14, 1943), relating to United States Letters Patent No. 2,156,584.

Appropriate documents and papers effectuating this order will issue.

Executed at Washington, D. C., on November 6, 1950.

For the Attorney General.

[SEAL] PAUL V. MYRON,
Deputy Director,
Office of Alien Property.

[F. R. Doc. 50-10092; Filed, Nov. 9, 1950; 8:52 a. m.]

[Return Order 790]

MARIE JOSEPHINE SPIESSER ET AL.

Having considered the claim set forth below and having issued a determination allowing the claim, which is incorporated by reference herein and filed herewith,

It is ordered, That the claimed property, described below and in the determination, be returned, subject to any increase or decrease resulting from the administration thereof prior to return, and after adequate provision for taxes and conservatory expenses:

Claimant, Claim No., Notice of Intention To Return Published, and Property

Marie Josephine Spiesser, Strasbourg, France, and August Hoerdt, Chicago, Ill.; the latter acting for himself and as conservator of the estate of Fred Hoerdt, an incompetent; Claim No. 42482; September 27, 1950 (15 F. R. 6528); $1,200 in the Treasury of the United States and all interest of Sophie Brejcha in the trust estate created under the will of Leonhard Hoerdt, deceased. The trust is under the judicial supervision of the Probate Court of Cook County, Illinois. Of the property 40% is returnable to Marie Josephine Spiesser, 30% to August Hoerdt, individually, and 30% to him as Conservator of the Estate of Fred Hoerdt, an incompetent.

Appropriate documents and papers effectuating this order will issue.

Executed at Washington, D. C., on November 6, 1950.

For the Attorney General.

[SEAL] PAUL V. MYRON,
Deputy Director,
Office of Alien Property.

[F. R. Doc. 50-10093; Filed, Nov. 9, 1950; 8:52 a. m.]